Experimentos WDB

Preprocesamiento

In [1]:
```
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from pathlib import Path

from tensorflow.keras.models import load_model, Model
from tensorflow.keras import layers
from matplotlib.lines import Line2D
from sklearn.decomposition import PCA, KernelPCA
from sklearn.manifold import TSNE
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, DBSCAN, SpectralClustering
from sklearn.metrics import silhouette_score

from ipywidgets import interact, widgets

import seaborn as sns
from sklearn.metrics.cluster import contingency_matrix
```

In [2]:
```
project_dir = Path().resolve().parent

model_path = os.path.join(project_dir, r"models\model_4C_3FC.h5")
data_path = os.path.join(project_dir, r"data\raw\sdss_dat_files")
labels_path = os.path.join(project_dir, r"data\raw\label_data")

print(project_dir)
print(model_path)
print(data_path)
print(labels_path)
```

```
C:\Users\javip\OneDrive - Universidad Técnica Federico Santa María\6to año\elo308\ML
_NPF
C:\Users\javip\OneDrive - Universidad Técnica Federico Santa María\6to año\elo308\ML
_NPF\models\model_4C_3FC.h5
C:\Users\javip\OneDrive - Universidad Técnica Federico Santa María\6to año\elo308\ML
_NPF\data\raw\sdss_dat_files
C:\Users\javip\OneDrive - Universidad Técnica Federico Santa María\6to año\elo308\ML
_NPF\data\raw\label_data
```

Función process_spectrum

In [3]:
```
def process_spectrum(file_path):
    try:
```

```
        sed = np.loadtxt(file_path, unpack=True)
        flux = sed[1, 48:-400]   # 4200 puntos

        # Ajuste a 3750 puntos, como espera la red
        if len(flux) > 3750:
            diff = len(flux) - 3750
            start = diff // 2
            flux = flux[start:start + 3750]
        elif len(flux) < 3750:
            flux = np.pad(flux, (0, 3750 - len(flux)), mode='constant')

        norm_val = flux[2100]
        if norm_val == 0 or np.isnan(norm_val):
            norm_val = np.mean(flux)

        if norm_val == 0 or np.isnan(norm_val):
            return None

        flux = flux / norm_val

        if np.isnan(flux).any():
            return None

        return flux.astype(np.float32)

    except Exception:
        return None
```

In [4]:
```
fields = ['File Name', 'MJD', 'Target ID', 'DB ID', 'Classification', 'Data Quality
raw_labels_df = pd.DataFrame()

for filename in os.listdir(labels_path):
    if filename.endswith('.csv'):
        file_path = os.path.join(labels_path, filename)
        df = pd.read_csv(file_path, usecols=fields)
        raw_labels_df = pd.concat([raw_labels_df, df], ignore_index=True)

print(raw_labels_df.shape)
print(raw_labels_df['Classification'].value_counts().head(20))
```

```
(35253, 6)
Classification
WDA        13748
STAR        8749
WD+MS       2555
DUNNO       2086
UNCLASS     1441
WDC         1353
WDB         1186
WDELM       1069
sdX          744
WD           743
WDZ          383
CV           356
EXGAL        301
WDH          272
WDO          184
WDQ           81
WELM           1
DA             1
Name: count, dtype: int64
```

In [5]:
```
raw_labels_df
```

Out[5]:

	Classification	DB ID	Data Quality	File Name	MJD	Target ID
0	WDA	21899	NaN	spec-15249-59265-04601916346-21899.png	59265	4601916346
1	sdX	23360	BLEND	spec-15301-59338-04602344336-23360.png	59338	4602344336
2	WDA	18900	SNR	spec-15113-59217-04538755416-18900.png	59217	4538755416
3	WDA	18575	NaN	spec-15086-59267-04474143209-18575.png	59267	4474143209
4	WD	20528	SNR	spec-15198-59269-04351438385-20528.png	59269	4351438385
...	...	...	...	...	...	...
35248	WDA	15895	SNR	spec-15011-59222-04399876473-15895.png	59222	4399876473
35249	WDA	20605	NaN	spec-15200-59324-04589320996-20605.png	59324	4589320996
35250	WDA	19388	NaN	spec-15193-59243-04594923329-19388.png	59243	4594923329
35251	DUNNO	19161	SNR	spec-15121-59212-04342114705-19161.png	59212	4342114705
35252	WDB	24506	FC	spec-15336-59294-04524328661-24506.png	59294	4524328661

35253 rows × 6 columns

In [6]:
```
file_dict = {}

for folder in os.listdir(data_path):
    folder_full_path = os.path.join(data_path, folder)
    if os.path.isdir(folder_full_path):
        for filename in os.listdir(folder_full_path):
            if filename.endswith('.dat'):
                key = filename.split('-')[-2] + '_' + filename.split('_')[-1]
                file_dict[key] = os.path.join(folder, filename)

print(f"Archivos .dat indexados: {len(file_dict)}")
```

```
Archivos .dat indexados: 33698
```

In [7]:
```
full_model = load_model(model_path)

dummy_input = np.zeros((1, 3750, 1), dtype=np.float32)
_ = full_model.predict(dummy_input, verbose=0)

full_model.summary()
```

```
c:\Users\javip\OneDrive - Universidad Técnica Federico Santa María\6to año\elo308\ML_NPF\.venv\lib\site-packages\keras\src\layers\convolutional\base_conv.py:113: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
c:\Users\javip\OneDrive - Universidad Técnica Federico Santa María\6to año\elo308\ML_NPF\.venv\lib\site-packages\keras\src\optimizers\base_optimizer.py:86: UserWarning: Argument `decay` is no longer supported and will be ignored.
  warnings.warn(
WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile_metrics` will be empty until you train or evaluate the model.
WARNING:absl:Error in loading the saved optimizer state. As a result, your model is starting with a freshly initialized optimizer.
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv1d (Conv1D)	(None, 1874, 128)	640
max_pooling1d (MaxPooling1D)	(None, 937, 128)	0
conv1d_1 (Conv1D)	(None, 467, 64)	32,832
max_pooling1d_1 (MaxPooling1D)	(None, 233, 64)	0
conv1d_2 (Conv1D)	(None, 115, 32)	8,224
max_pooling1d_2 (MaxPooling1D)	(None, 57, 32)	0
conv1d_3 (Conv1D)	(None, 27, 16)	2,064
max_pooling1d_3 (MaxPooling1D)	(None, 13, 16)	0
flatten (Flatten)	(None, 208)	0
dense (Dense)	(None, 128)	26,752
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 64)	8,256
dropout_1 (Dropout)	(None, 64)	0
dense_2 (Dense)	(None, 32)	2,080
dense_3 (Dense)	(None, 12)	396

Total params: 81,246 (317.37 KB)

Trainable params: 81,244 (317.36 KB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 2 (12.00 B)

Construir extractores de capas

In [8]:
```
def build_extractor_until(layer_name, full_model):
    input_layer = layers.Input(shape=(3750, 1))
    x = input_layer

    for layer in full_model.layers:
        x = layer(x)
        if layer.name == layer_name:
            break

    extractor = Model(inputs=input_layer, outputs=x)
    return extractor

flatten_extractor = build_extractor_until("flatten", full_model)
dense_extractor   = build_extractor_until("dense", full_model)
dense1_extractor  = build_extractor_until("dense_1", full_model)
dense2_extractor  = build_extractor_until("dense_2", full_model)

print("extrat listos")
```

```
extrat listos
```

In [9]:
```
def build_class_dataframe(class_name):
    star_df = raw_labels_df[raw_labels_df['Classification'] == class_name].copy()

    spectrum_list = []
    for _, row in star_df.iterrows():
        filekey = str(row['MJD']) + '_' + str(row['Target ID']) + '.dat'
        if filekey in file_dict:
            spectrum_list.append({
                'filename': file_dict[filekey],
                'classID': row['Classification']
            })

    spectrum_df = pd.DataFrame(spectrum_list)
    return spectrum_df
```

In [10]:
```
# fc que carga espectros de una clase, los preprocesa y los deja listo para entrar
def load_class_spectra(class_name, n_samples=1100, random_state=42):
    spectrum_df = build_class_dataframe(class_name)
    print(f"{class_name}: disponibles con match a .dat = {len(spectrum_df)}")

    if len(spectrum_df) < n_samples:
        raise ValueError(f"No hay suficientes espectros para {class_name}. Disponib

    selected_df = spectrum_df.sample(n=n_samples, random_state=random_state).reset_

    X_list = []
    valid_rows = []

    for _, row in selected_df.iterrows():
        f_path = os.path.join(data_path, row['filename'])
        spec = process_spectrum(f_path)
```

```
        if spec is not None:
            X_list.append(spec)
            valid_rows.append(row)

    X_final = np.array(X_list, dtype=np.float32)
    X_input = X_final.reshape((X_final.shape[0], X_final.shape[1], 1))
    valid_df = pd.DataFrame(valid_rows).reset_index(drop=True)

    print(f"{class_name}: espectros válidos finales = {X_input.shape[0]}")
    return X_input, valid_df
```

In [11]:
```
# fc que carga los espectros y los conserva en formado 2D para poder graficarlos de
# retorna X_final, que son los espectros originales preprocesados, listos para graf
def load_class_spectra_DRACULA(class_name, n_samples=1100, random_state=42):
    spectrum_df = build_class_dataframe(class_name)
    print(f"{class_name}: disponibles con match a .dat = {len(spectrum_df)}")

    if len(spectrum_df) < n_samples:
        raise ValueError(f"No hay suficientes espectros para {class_name}. Disponib

    selected_df = spectrum_df.sample(n=n_samples, random_state=random_state).reset_

    X_list = []
    valid_rows = []

    for _, row in selected_df.iterrows():
        f_path = os.path.join(data_path, row['filename'])
        spec = process_spectrum(f_path)
        if spec is not None:
            X_list.append(spec)
            valid_rows.append(row)

    X_final = np.array(X_list, dtype=np.float32)
    X_input = X_final.reshape((X_final.shape[0], X_final.shape[1], 1))
    valid_df = pd.DataFrame(valid_rows).reset_index(drop=True)

    print(f"{class_name}: espectros válidos finales = {X_input.shape[0]}")
    return X_final, X_input, valid_df
```

In [12]:
```
# fc que carga espectros, conserva el flujo (eje y) normalizado y conserva el eje r
def load_class_spectra_DRACULA_with_wavelength(class_name, n_samples=1100, random_s
    spectrum_df = build_class_dataframe(class_name)
    print(f"{class_name}: disponibles con match a .dat = {len(spectrum_df)}")

    if len(spectrum_df) < n_samples:
        raise ValueError(f"No hay suficientes espectros para {class_name}. Disponib

    selected_df = spectrum_df.sample(n=n_samples, random_state=random_state).reset_

    wave_list = []
    flux_list = []
    valid_rows = []

    for _, row in selected_df.iterrows():
        f_path = os.path.join(data_path, row['filename'])
```

```
        wave, flux = process_spectrum_with_wavelength(f_path)

        if wave is not None and flux is not None:
            wave_list.append(wave)
            flux_list.append(flux)
            valid_rows.append(row)

    W_final = np.array(wave_list, dtype=np.float32)
    X_final = np.array(flux_list, dtype=np.float32)

    X_input = X_final.reshape((X_final.shape[0], X_final.shape[1], 1))
    valid_df = pd.DataFrame(valid_rows).reset_index(drop=True)

    print(f"{class_name}: espectros válidos finales = {X_input.shape[0]}")
    print(f"{class_name}: shape wavelength = {W_final.shape}")
    print(f"{class_name}: shape flux = {X_final.shape}")

    return W_final, X_final, X_input, valid_df

'''
Uso de esta fc -> mostrar un espectro / muestra
W_wda, X_wda_spec, X_wda_input, df_wda_DRAC = load_class_spectra_DRACULA_with_wavel
    "WDA",
    n_samples=1100,
    random_state=42
)

sample_idx = 0

plt.figure(figsize=(10, 5))
plt.plot(W_wda[sample_idx], X_wda_spec[sample_idx])
plt.xlabel("Wavelength [Å]")
plt.ylabel("Normalized flux")
plt.title(f"WDA - sample {sample_idx}")
plt.tight_layout()
plt.show()
'''

'''
Uso de esta fc -> mostrar el promedio + lo anterior
mean_wave = np.nanmean(W_wda, axis=0)
mean_flux = np.nanmean(X_wda_spec, axis=0)

plt.figure(figsize=(10, 5))
plt.plot(mean_wave, mean_flux, label="Total mean spectrum")
plt.xlabel("Wavelength [Å]")
plt.ylabel("Normalized flux")
plt.title("WDA - Total mean spectrum")
plt.legend()
plt.tight_layout()
plt.show()
'''
```

Out[12]:
```
'\nUso de esta fc -> mostrar el promedio + lo anterior\nmean_wave = np.nanmean(W_w
da, axis=0)\nmean_flux = np.nanmean(X_wda_spec, axis=0)\n\nplt.figure(figsize=(10,
5))\nplt.plot(mean_wave, mean_flux, label="Total mean spectrum")\nplt.xlabel("Wave
length [Å]")\nplt.ylabel("Normalized flux")\nplt.title("WDA - Total mean spectru
m")\nplt.legend()\nplt.tight_layout()\nplt.show()\n'
```

In [13]:
```
def extract_layer_features(X_input): # fc que extrae representaciones internas de l
    flatten_features = flatten_extractor.predict(X_input, batch_size=64, verbose=1)
    dense_features   = dense_extractor.predict(X_input, batch_size=64, verbose=1)
    dense1_features  = dense1_extractor.predict(X_input, batch_size=64, verbose=1)
    dense2_features  = dense2_extractor.predict(X_input, batch_size=64, verbose=1)

    print("flatten:", flatten_features.shape)
    print("dense:", dense_features.shape)
    print("dense_1:", dense1_features.shape)
    print("dense_2:", dense2_features.shape)

    return flatten_features, dense_features, dense1_features, dense2_features
```

In [14]:
```
def reduce_all_to_32(flatten_features, dense_features, dense1_features, dense2_feat
    flatten_32 = PCA(n_components=32, random_state=42).fit_transform(flatten_featur
    dense_32   = PCA(n_components=32, random_state=42).fit_transform(dense_features
    dense1_32  = PCA(n_components=32, random_state=42).fit_transform(dense1_feature
    dense2_32  = dense2_features.copy()  # ya está en 32

    print("flatten_32:", flatten_32.shape)
    print("dense_32:", dense_32.shape)
    print("dense1_32:", dense1_32.shape)
    print("dense2_32:", dense2_32.shape)

    return {
        "flatten": flatten_32,
        "dense": dense_32,
        "dense_1": dense1_32,
        "dense_2": dense2_32
    }
```

Funciones para graficar

In [15]:
```
# fc que grafica los features_nd en 2D
def plot_internal_structure_2d(
    features_nd,
    class_name,
    layer_name,
    method="pca",
    xlim=None,
    ylim=None,
    kpca_kernel="rbf",
    kpca_gamma=None,
    kpca_degree=3,
    kpca_coef0=1,
    scale_before_pca=False,
    scale_before_tsne=False,
```

```
    scale_before_kpca=True
):
    method = method.lower()

    if method == "pca":
        X_plotPCA = features_nd.copy()

        if scale_before_pca:
            X_plotPCA = StandardScaler().fit_transform(X_plotPCA)

        emb_2d = PCA(n_components=2, random_state=42).fit_transform(X_plotPCA)

    elif method == "tsne":
        X_plotTSNE = features_nd.copy()

        if scale_before_tsne:
            X_plotTSNE = StandardScaler().fit_transform(X_plotTSNE)

        emb_2d = TSNE(
            n_components=2,
            random_state=42,
            perplexity=30,
            init='pca'
        ).fit_transform(X_plotTSNE)

    elif method == "kpca":
        X_kpca = features_nd.copy()

        if scale_before_kpca:
            X_kpca = StandardScaler().fit_transform(X_kpca)

        emb_2d = KernelPCA(
            n_components=2,
            kernel=kpca_kernel,
            gamma=kpca_gamma,
            degree=kpca_degree,
            coef0=kpca_coef0
        ).fit_transform(X_kpca)

    else:
        raise ValueError("method debe ser 'pca', 'tsne' o 'kpca'")

    plt.figure(figsize=(6, 5))
    plt.scatter(emb_2d[:, 0], emb_2d[:, 1], s=10, alpha=0.65)

    if method == "kpca":
        extra = " + scaled" if scale_before_kpca else ""
        n_samples = features_nd.shape[0]
        plt.title(f"{class_name} - {layer_name} - estructura interna (KPCA-{kpca_ke

    elif method == "pca":
        extra = " + scaled" if scale_before_pca else ""
        n_samples = features_nd.shape[0]
        plt.title(f"{class_name} - {layer_name} - estructura interna (PCA{extra}, n

    elif method == "tsne":
```

```
        extra = " + scaled" if scale_before_tsne else ""
        n_samples = features_nd.shape[0]
        plt.title(f"{class_name} - {layer_name} - estructura interna (TSNE{extra},

    plt.xlabel("Comp. 1")
    plt.ylabel("Comp. 2")

    if xlim is not None:
        plt.xlim(xlim)
    if ylim is not None:
        plt.ylim(ylim)

    plt.tight_layout()
    plt.show()
```

In [16]:

```
# funcion para graficar el espectro promedio total
def plot_total_mean_spectrum(
    spectra_2d,
    title="Total mean spectrum",
    wavelength=None,
    figsize=(10, 5),
    save_path=None
):
    """
    Grafica el espectro promedio total de todas las muestras.

    Parameters
    ----------
    spectra_2d : np.ndarray
        Matriz shape (n_samples, n_wavelengths)
    title : str
        Título del gráfico
    wavelength : np.ndarray or None
        Eje x. Si None, usa índice espectral
    figsize : tuple
    save_path : str or None
        Si se entrega, guarda la figura
    """

    spectra_2d = np.asarray(spectra_2d)

    if spectra_2d.ndim != 2:
        raise ValueError(f"spectra_2d debe ser 2D. Shape actual: {spectra_2d.shape}

    mean_spec = spectra_2d.mean(axis=0)

    if wavelength is None:
        wavelength = np.arange(spectra_2d.shape[1])

    plt.figure(figsize=figsize)
    plt.plot(wavelength, mean_spec, label="Total mean spectrum")
    plt.xlabel("Spectral index")
    plt.ylabel("Normalized flux")
    plt.title(title)
    plt.legend()
    plt.tight_layout()
```

```
    if save_path is not None:
        plt.savefig(save_path, dpi=300, bbox_inches="tight")

    plt.show()
```

In [17]:

```
# función para graficar una muestra individual
def plot_single_spectrum(
    spectra_2d,
    sample_idx,
    valid_df=None,
    title=None,
    wavelength=None,
    figsize=(10, 5),
    save_path=None
):
    """
    Grafica un espectro individual a partir de su índice.

    Parameters
    ----------
    spectra_2d : np.ndarray
        Matriz shape (n_samples, n_wavelengths)
    sample_idx : int
        Índice de la muestra a graficar
    valid_df : pd.DataFrame or None
        Metadata opcional para mostrar info adicional en el título
    title : str or None
        Si None, construye uno automático
    wavelength : np.ndarray or None
        Eje x. Si None, usa índice espectral
    figsize : tuple
    save_path : str or None
        Si se entrega, guarda la figura
    """

    spectra_2d = np.asarray(spectra_2d)

    if spectra_2d.ndim != 2:
        raise ValueError(f"spectra_2d debe ser 2D. Shape actual: {spectra_2d.shape}

    if sample_idx < 0 or sample_idx >= spectra_2d.shape[0]:
        raise IndexError(f"sample_idx fuera de rango. Debe estar entre 0 y {spectra

    spectrum = spectra_2d[sample_idx]

    if wavelength is None:
        wavelength = np.arange(spectra_2d.shape[1])

    if title is None:
        title = f"Single spectrum - sample {sample_idx}"

        if valid_df is not None and sample_idx < len(valid_df):
            row = valid_df.iloc[sample_idx]

            extra_parts = []
```

```
            if "filename" in row.index:
                extra_parts.append(f"file: {row['filename']}")
            if "classID" in row.index:
                extra_parts.append(f"class: {row['classID']}")

            if len(extra_parts) > 0:
                title += "\n" + " | ".join(extra_parts)

    plt.figure(figsize=figsize)
    plt.plot(wavelength, spectrum, label=f"Sample {sample_idx}")
    plt.xlabel("Spectral index")
    plt.ylabel("Normalized flux")
    plt.title(title)
    plt.legend()
    plt.tight_layout()

    if save_path is not None:
        plt.savefig(save_path, dpi=300, bbox_inches="tight")

    plt.show()
```

In [18]:

```
def plot_multiple_spectra(
    spectra_2d,
    sample_indices,
    valid_df=None,
    title="Selected spectra",
    wavelength=None,
    figsize=(10, 5),
    alpha=0.9,
    save_path=None
):
    """
    Grafica varias muestras individuales en la misma figura.

    Parameters
    ----------
    spectra_2d : np.ndarray
        Matriz shape (n_samples, n_wavelengths)
    sample_indices : list or array-like
        Índices de las muestras a graficar
    valid_df : pd.DataFrame or None
    title : str
    wavelength : np.ndarray or None
    figsize : tuple
    alpha : float
    save_path : str or None
    """

    spectra_2d = np.asarray(spectra_2d)

    if spectra_2d.ndim != 2:
        raise ValueError(f"spectra_2d debe ser 2D. Shape actual: {spectra_2d.shape}")

    if wavelength is None:
        wavelength = np.arange(spectra_2d.shape[1])
```

```
    plt.figure(figsize=figsize)

    for idx in sample_indices:
        if idx < 0 or idx >= spectra_2d.shape[0]:
            raise IndexError(f"sample_idx {idx} fuera de rango.")

        label = f"sample {idx}"

        if valid_df is not None and idx < len(valid_df):
            row = valid_df.iloc[idx]
            if "filename" in row.index:
                label += f" | {row['filename']}"

        plt.plot(wavelength, spectra_2d[idx], alpha=alpha, label=label)

    plt.xlabel("Spectral index")
    plt.ylabel("Normalized flux")
    plt.title(title)
    plt.legend(fontsize=8)
    plt.tight_layout()

    if save_path is not None:
        plt.savefig(save_path, dpi=300, bbox_inches="tight")

    plt.show()
```

Espectro representativo de clase — diagnóstico de normalización

In [19]:

```
def plot_class_mean_spectrum(
        spectra_2d, wavelength, class_name,
        use_median=True,
        show_individuals=False,
        n_individuals=10,
        percentile_band=(10, 90),
        flux_clip=(-0.5, 5.0),
        figsize=(13, 5)
):
    """
    Muestra el espectro promedio (o mediana) de una clase completa.

    Parametros
    ----------
    spectra_2d       : array (n x 3750) con los espectros normalizados
    wavelength       : array de longitudes de onda
    class_name       : str, solo para el titulo
    use_median       : True -> usa mediana (mas robusta a outliers)
                       False -> usa media
    show_individuals : si True, grafica n_individuals espectros individuales
    n_individuals    : cuantos espectros individuales mostrar
    percentile_band  : tupla (p_low, p_high) para la banda de dispersion
    flux_clip        : (min, max) para detectar espectros anomalos.
                       None = no filtrar
```

```
    """
    spectra_2d = np.asarray(spectra_2d, dtype=np.float32)
    wavelength = np.asarray(wavelength)
    n_total    = spectra_2d.shape[0]

    # ── Diagnostico de espectros anomalos ──────────────────────────
    flux_min = spectra_2d.min(axis=1)   # minimo de cada espectro
    flux_max = spectra_2d.max(axis=1)   # maximo de cada espectro

    print(f'\n=== Diagnostico {class_name} (n={n_total}) ===')
    print(f'  Flux minimo por espectro:  media={flux_min.mean():.3f}  '
          f'p5={np.percentile(flux_min,5):.3f}  '
          f'p95={np.percentile(flux_min,95):.3f}')
    print(f'  Flux maximo por espectro:  media={flux_max.mean():.3f}  '
          f'p5={np.percentile(flux_max,5):.3f}  '
          f'p95={np.percentile(flux_max,95):.3f}')

    if flux_clip is not None:
        clip_lo, clip_hi = flux_clip
        bad_mask = (flux_min < clip_lo) | (flux_max > clip_hi)
        n_bad    = bad_mask.sum()
        pct_bad  = 100 * n_bad / n_total
        print(f'  Espectros fuera de [{clip_lo}, {clip_hi}]: '
              f'{n_bad} ({pct_bad:.1f}%)')
        spectra_clean = spectra_2d[~bad_mask]
        label_clean   = f'n={spectra_clean.shape[0]} (sin anomalos)'
    else:
        spectra_clean = spectra_2d
        label_clean   = f'n={n_total}'

    # ── Calcular curvas representativas ────────────────────────────
    if use_median:
        central = np.median(spectra_clean, axis=0)
        central_label = f'Mediana {class_name} ({label_clean})'
    else:
        central = np.mean(spectra_clean, axis=0)
        central_label = f'Media {class_name} ({label_clean})'

    p_lo = np.percentile(spectra_clean, percentile_band[0], axis=0)
    p_hi = np.percentile(spectra_clean, percentile_band[1], axis=0)

    # tambien con todos (sin filtrar) para comparar
    if flux_clip is not None:
        central_all = (np.median(spectra_2d, axis=0) if use_median
                       else np.mean(spectra_2d, axis=0))
    else:
        central_all = None

    # ── Grafico ────────────────────────────────────────────────────
    fig, ax = plt.subplots(figsize=figsize)

    # Espectros individuales (opcional)
    if show_individuals:
        idx_sample = np.random.choice(spectra_clean.shape[0],
                                      size=min(n_individuals, spectra_clean.shape[0
                                      replace=False)
```

```
        for i in idx_sample:
            ax.plot(wavelength, spectra_clean[i], color='steelblue',
                    linewidth=0.4, alpha=0.3)

    # Banda de percentiles
    ax.fill_between(wavelength, p_lo, p_hi,
                    alpha=0.15, color='steelblue',
                    label=f'P{percentile_band[0]}-P{percentile_band[1]}')

    # Curva representativa SIN filtrar (si se filtro)
    if central_all is not None:
        ax.plot(wavelength, central_all, linewidth=1.2, color='gray',
                linestyle='--', alpha=0.6, label='Todos (con anomalos)')

    # Curva representativa limpia
    ax.plot(wavelength, central, linewidth=2.2, color='steelblue',
            label=central_label)

    # Linea en flux=1 (punto de normalizacion)
    ax.axhline(1.0, linestyle=':', linewidth=1.0, color='black',
               alpha=0.5, label='flux = 1 (punto normaliz. ~6158 A)')

    add_balmer_lines(ax)

    ax.set_xlabel('Wavelength [A]')
    ax.set_ylabel('Normalized flux  (norm. en ~6158 A)')
    ax.set_title(f'Espectro representativo de clase {class_name}')
    ax.legend(fontsize=9)
    plt.tight_layout()
    plt.show()

    return {
        'central'       : central,
        'p_lo'          : p_lo,
        'p_hi'          : p_hi,
        'spectra_clean' : spectra_clean,
        'n_bad'         : int(bad_mask.sum()) if flux_clip is not None else 0
    }
def plot_class_mean_spectrum(
        spectra_2d, wavelength, class_name,
        use_median=True,
        show_individuals=False,
        n_individuals=10,
        percentile_band=(10, 90),
        flux_clip=(-0.5, 5.0),
        figsize=(13, 5)
):
    """
    Muestra el espectro promedio (o mediana) de una clase completa.

    Parametros
    ----------
    spectra_2d       : array (n x 3750) con los espectros normalizados
    wavelength       : array de longitudes de onda
    class_name       : str, solo para el titulo
    use_median       : True -> usa mediana (mas robusta a outliers)
```

```
                    False -> usa media
show_individuals : si True, grafica n_individuals espectros individuales
n_individuals    : cuantos espectros individuales mostrar
percentile_band  : tupla (p_low, p_high) para la banda de dispersion
flux_clip        : (min, max) para detectar espectros anomalos.
                   None = no filtrar
"""
spectra_2d = np.asarray(spectra_2d, dtype=np.float32)
wavelength = np.asarray(wavelength)
n_total    = spectra_2d.shape[0]

# ── Diagnostico de espectros anomalos ──────────────────────────
flux_min = spectra_2d.min(axis=1)   # minimo de cada espectro
flux_max = spectra_2d.max(axis=1)   # maximo de cada espectro

print(f'\n=== Diagnostico {class_name} (n={n_total}) ===')
print(f'  Flux minimo por espectro:  media={flux_min.mean():.3f}  '
      f'p5={np.percentile(flux_min,5):.3f}  '
      f'p95={np.percentile(flux_min,95):.3f}')
print(f'  Flux maximo por espectro:  media={flux_max.mean():.3f}  '
      f'p5={np.percentile(flux_max,5):.3f}  '
      f'p95={np.percentile(flux_max,95):.3f}')

if flux_clip is not None:
    clip_lo, clip_hi = flux_clip
    bad_mask = (flux_min < clip_lo) | (flux_max > clip_hi)
    n_bad    = bad_mask.sum()
    pct_bad  = 100 * n_bad / n_total
    print(f'  Espectros fuera de [{clip_lo}, {clip_hi}]: '
          f'{n_bad} ({pct_bad:.1f}%)')
    spectra_clean = spectra_2d[~bad_mask]
    label_clean   = f'n={spectra_clean.shape[0]} (sin anomalos)'
else:
    spectra_clean = spectra_2d
    label_clean   = f'n={n_total}'

# ── Calcular curvas representativas ────────────────────────────
if use_median:
    central = np.median(spectra_clean, axis=0)
    central_label = f'Mediana {class_name} ({label_clean})'
else:
    central = np.mean(spectra_clean, axis=0)
    central_label = f'Media {class_name} ({label_clean})'

p_lo = np.percentile(spectra_clean, percentile_band[0], axis=0)
p_hi = np.percentile(spectra_clean, percentile_band[1], axis=0)

# tambien con todos (sin filtrar) para comparar
if flux_clip is not None:
    central_all = (np.median(spectra_2d, axis=0) if use_median
                   else np.mean(spectra_2d, axis=0))
else:
    central_all = None

# ── Grafico ────────────────────────────────────────────────────
fig, ax = plt.subplots(figsize=figsize)
```

```
    # Espectros individuales (opcional)
    if show_individuals:
        idx_sample = np.random.choice(spectra_clean.shape[0],
                                      size=min(n_individuals, spectra_clean.shape[0
                                      replace=False)
        for i in idx_sample:
            ax.plot(wavelength, spectra_clean[i], color='steelblue',
                    linewidth=0.4, alpha=0.3)

    # Banda de percentiles
    ax.fill_between(wavelength, p_lo, p_hi,
                    alpha=0.15, color='steelblue',
                    label=f'P{percentile_band[0]}-P{percentile_band[1]}')

    # Curva representativa SIN filtrar (si se filtro)
    if central_all is not None:
        ax.plot(wavelength, central_all, linewidth=1.2, color='gray',
                linestyle='--', alpha=0.6, label='Todos (con anomalos)')

    # Curva representativa limpia
    ax.plot(wavelength, central, linewidth=2.2, color='steelblue',
            label=central_label)

    # Linea en flux=1 (punto de normalizacion)
    ax.axhline(1.0, linestyle=':', linewidth=1.0, color='black',
               alpha=0.5, label='flux = 1 (punto normaliz. ~6158 A)')

    add_balmer_lines(ax)

    ax.set_xlabel('Wavelength [A]')
    ax.set_ylabel('Normalized flux  (norm. en ~6158 A)')
    ax.set_title(f'Espectro representativo de clase {class_name}')
    ax.legend(fontsize=9)
    plt.tight_layout()
    plt.show()

    return {
        'central'       : central,
        'p_lo'          : p_lo,
        'p_hi'          : p_hi,
        'spectra_clean' : spectra_clean,
        'n_bad'         : int(bad_mask.sum()) if flux_clip is not None else 0
    }
```

Función que agrega líneas de Balmer (mientras no tengo el paquete)

In [20]:
```
def add_balmer_lines(ax, y_text=None, alpha=0.6, fontsize=9):
    """
    Agrega líneas verticales de Balmer al eje.
    """
    balmer_lines = {
```

```
        "Hε": 3970.1,
        "Hδ": 4101.7,
        "Hγ": 4340.5,
        "Hβ": 4861.3,
        "Hα": 6562.8
    }

    ymin, ymax = ax.get_ylim()

    if y_text is None:
        y_text = ymax - 0.03 * (ymax - ymin)

    for name, lam in balmer_lines.items():
        ax.axvline(lam, linestyle="--", alpha=alpha, linewidth=1.0)
        ax.text(
            lam,
            y_text,
            name,
            rotation=90,
            va="top",
            ha="right",
            fontsize=fontsize
        )
```

Función que genera archivos de salida

In [21]:

```
# ============================================================
# FUNCIÓN PRINCIPAL PARA GENERAR OUTPUTS ESPECTRALES -> ESPECTROS POR CLUSTER
# ============================================================
def generate_cluster_spectra_outputs(
    spectra_2d,              # shape (n_samples, n_wavelengths)
    cluster_labels,          # shape (n_samples,)
    valid_df,                # metadata alineada con spectra_2d
    class_name,
    layer_name,
    method_name,
    output_dir="dracula_outputs",
    ignore_noise=False,
    wavelength=None
):
    """
    Genera salidas para análisis espectral por cluster.

    Guarda:
    - sample_cluster_map.csv
    - cluster_summary.csv
    - spectra.dat general
    - clusters.dat general
    - wavelength.dat
    - spectra individuales por cluster
    - mean por cluster
    - std por cluster

    Parameters
```

```
    ----------
    spectra_2d : np.ndarray
        Matriz de espectros originales, shape (n_samples, n_wavelengths)
    cluster_labels : np.ndarray
        Etiquetas de clustering, shape (n_samples,)
    valid_df : pd.DataFrame
        Metadata alineada con spectra_2d
    class_name : str
    layer_name : str
    method_name : str
    output_dir : str
    ignore_noise : bool
        Si True, elimina cluster -1 (útil para DBSCAN)
    wavelength : np.ndarray or None
        Vector de longitudes de onda. Si None, usa 0..n_wavelengths-1

    Returns
    -------
    dict con rutas y dataframes útiles
    """

    spectra_2d = np.asarray(spectra_2d)
    cluster_labels = np.asarray(cluster_labels)

    # ------------------------------
    # Validaciones
    # ------------------------------
    if spectra_2d.ndim != 2:
        raise ValueError(f"spectra_2d debe ser 2D. Shape actual: {spectra_2d.shape}

    if len(cluster_labels) != spectra_2d.shape[0]:
        raise ValueError(
            f"cluster_labels y spectra_2d no coinciden: "
            f"{len(cluster_labels)} vs {spectra_2d.shape[0]}"
        )

    if len(valid_df) != spectra_2d.shape[0]:
        raise ValueError(
            f"valid_df y spectra_2d no coinciden: "
            f"{len(valid_df)} vs {spectra_2d.shape[0]}"
        )

    if wavelength is None:
        wavelength = np.arange(spectra_2d.shape[1], dtype=np.float32)

    # ------------------------------
    # Crear carpeta base
    # ------------------------------
    base_name = f"{class_name}_{layer_name}_{method_name}"
    base_dir = os.path.join(output_dir, class_name, layer_name, method_name)
    os.makedirs(base_dir, exist_ok=True)

    # ------------------------------
    # Crear dataframe muestra -> cluster
    # ------------------------------
    cluster_df = valid_df.copy().reset_index(drop=True)
```

```
    cluster_df["sample_idx"] = np.arange(len(cluster_df))
    cluster_df["cluster"] = cluster_labels

    # ------------------------------
    # Manejo de ruido DBSCAN (-1)
    # ------------------------------
    if ignore_noise:
        mask = cluster_df["cluster"] != -1
        cluster_df = cluster_df.loc[mask].reset_index(drop=True)
        spectra_used = spectra_2d[mask.to_numpy()]
        labels_used = cluster_df["cluster"].to_numpy()
    else:
        spectra_used = spectra_2d
        labels_used = cluster_labels

    unique_clusters = np.unique(labels_used)

    # ------------------------------
    # Guardar mapa muestra -> cluster
    # ------------------------------
    cluster_map_csv = os.path.join(base_dir, f"{base_name}_sample_cluster_map.csv")
    cluster_df.to_csv(cluster_map_csv, index=False)

    # ------------------------------
    # Guardar archivos generales
    # ------------------------------
    spectra_dat = os.path.join(base_dir, f"{base_name}_spectra.dat")
    labels_dat = os.path.join(base_dir, f"{base_name}_clusters.dat")
    wavelength_dat = os.path.join(base_dir, f"{base_name}_wavelength.dat")

    np.savetxt(spectra_dat, spectra_used, fmt="%.8e")
    np.savetxt(labels_dat, labels_used.astype(int), fmt="%d")
    np.savetxt(wavelength_dat, wavelength, fmt="%.8e")

    # ------------------------------
    # Guardar resumen y archivos por cluster
    # ------------------------------
    summary_rows = []

    for c in unique_clusters:
        mask_c = labels_used == c
        cluster_spectra = spectra_used[mask_c]

        mean_spec = cluster_spectra.mean(axis=0)
        std_spec = cluster_spectra.std(axis=0)

        n_cluster = int(mask_c.sum())

        summary_rows.append({
            "cluster": int(c),
            "n_samples": n_cluster
        })

        # metadata del cluster
        cluster_meta = cluster_df.loc[cluster_df["cluster"] == c].copy()
        cluster_meta_csv = os.path.join(base_dir, f"{base_name}_cluster_{c}_metadat
```

```
        cluster_meta.to_csv(cluster_meta_csv, index=False)

        # todos los espectros del cluster
        cluster_specs_path = os.path.join(base_dir, f"{base_name}_cluster_{c}_spect
        np.savetxt(cluster_specs_path, cluster_spectra, fmt="%.8e")

        # media del cluster
        mean_path = os.path.join(base_dir, f"{base_name}_cluster_{c}_mean.dat")
        np.savetxt(mean_path, mean_spec, fmt="%.8e")

        # std del cluster
        std_path = os.path.join(base_dir, f"{base_name}_cluster_{c}_std.dat")
        np.savetxt(std_path, std_spec, fmt="%.8e")

    summary_df = pd.DataFrame(summary_rows).sort_values("cluster").reset_index(drop
    summary_csv = os.path.join(base_dir, f"{base_name}_cluster_summary.csv")
    summary_df.to_csv(summary_csv, index=False)

    return {
        "base_dir": base_dir,
        "cluster_map_csv": cluster_map_csv,
        "spectra_dat": spectra_dat,
        "labels_dat": labels_dat,
        "wavelength_dat": wavelength_dat,
        "summary_csv": summary_csv,
        "summary_df": summary_df,
        "cluster_df": cluster_df
    }
```

In [22]:

```
# fc que automatiza la generación de archivos de salida para todos los experimentos
def generate_all_outputs_for_class(
    spectra_2d,
    valid_df,
    class_name,
    labels_dict_for_class,
    output_dir="dracula_outputs",
    ignore_noise_for_dbscan=False,
    wavelength=None
):
    """
    Recorre todas las capas y métodos de una clase y genera outputs espectrales.

    Parameters
    ----------
    spectra_2d : np.ndarray
        Espectros originales shape (n_samples, n_wavelengths)
    valid_df : pd.DataFrame
    class_name : str
    labels_dict_for_class : dict
        Ejemplo:
        cluster_labels_all["WDA"]
    output_dir : str
    ignore_noise_for_dbscan : bool
    wavelength : np.ndarray or None

    Returns
```

```
    -------
    dict
    """

    all_outputs = {}

    for layer_name, methods_dict in labels_dict_for_class.items():
        all_outputs[layer_name] = {}

        for method_name, labels in methods_dict.items():
            ignore_noise = (method_name == "dbscan" and ignore_noise_for_dbscan)

            result = generate_cluster_spectra_outputs(
                spectra_2d=spectra_2d,
                cluster_labels=labels,
                valid_df=valid_df,
                class_name=class_name,
                layer_name=layer_name,
                method_name=method_name,
                output_dir=output_dir,
                ignore_noise=ignore_noise,
                wavelength=wavelength
            )

            all_outputs[layer_name][method_name] = result

    return all_outputs
```

Contenedores -> diccionarios anidados que contienen array de enteros donde cada posición es una muestra y el valor es el cluster al que fue asignado.

Tiene más o menos esta forma: cluster_labels_all -> "WDA" -> "flatten" -> "spectral" -> np.array([0,1,2,0,2,1,...]) shape (n_muestras,)

In [23]:
```
cluster_labels_all = {
    "WDA": {},
    "WDB": {}
}

cluster_labels_all_n2 = {
    "WDA": {},
    "WDB": {}
}
```

In [24]:
```
#guarda labels dentro del diccionario general cluster_labels_all.
def save_cluster_labels(container, class_name, layer_name, method_name, labels):
    """
    estructura final:
    cluster_labels_all["WDA"]["dense_1"]["spectral"] = labels
```

```
    """

    if class_name not in container:
        container[class_name] = {}

    if layer_name not in container[class_name]:
        container[class_name][layer_name] = {}

    container[class_name][layer_name][method_name] = np.asarray(labels).copy()
```

Funciones que consideran las longitudes de onda

In [25]:
```
def process_spectrum_with_wavelength(file_path):
    try:
        sed = np.loadtxt(file_path, unpack=True)

        wave = sed[0, 48:-400]
        flux = sed[1, 48:-400]

        # Ajuste a 3750 puntos, como espera la red
        if len(flux) > 3750:
            diff = len(flux) - 3750
            start = diff // 2

            flux = flux[start:start + 3750]
            wave = wave[start:start + 3750]

        elif len(flux) < 3750:
            pad_len = 3750 - len(flux)

            flux = np.pad(flux, (0, pad_len), mode='constant')

            # Para wavelength, mejor evitar inventar demasiado.
            # Extendemos linealmente usando el paso medio.
            if len(wave) > 1:
                dw = np.nanmedian(np.diff(wave))
                extra_wave = wave[-1] + dw * np.arange(1, pad_len + 1)
                wave = np.concatenate([wave, extra_wave])
            else:
                wave = np.pad(wave, (0, pad_len), mode='constant')

        norm_val = flux[2100]

        if norm_val == 0 or np.isnan(norm_val):
            norm_val = np.nanmean(flux)

        if norm_val == 0 or np.isnan(norm_val):
            return None, None

        flux = flux / norm_val

        if np.isnan(flux).any() or np.isnan(wave).any():
```

```
            return None, None

        return wave.astype(np.float32), flux.astype(np.float32)

    except Exception:
        return None, None
```

In [26]:
```
def load_class_spectra_DRACULA_with_wavelength(class_name, n_samples=1100, random_s
    spectrum_df = build_class_dataframe(class_name)
    print(f"{class_name}: disponibles con match a .dat = {len(spectrum_df)}")

    if len(spectrum_df) < n_samples:
        raise ValueError(f"No hay suficientes espectros para {class_name}. Disponib

    selected_df = spectrum_df.sample(n=n_samples, random_state=random_state).reset_

    wave_list = []
    flux_list = []
    valid_rows = []

    for _, row in selected_df.iterrows():
        f_path = os.path.join(data_path, row['filename'])

        wave, flux = process_spectrum_with_wavelength(f_path)

        if wave is not None and flux is not None:
            wave_list.append(wave)
            flux_list.append(flux)
            valid_rows.append(row)

    W_final = np.array(wave_list, dtype=np.float32)
    X_final = np.array(flux_list, dtype=np.float32)

    X_input = X_final.reshape((X_final.shape[0], X_final.shape[1], 1))
    valid_df = pd.DataFrame(valid_rows).reset_index(drop=True)

    print(f"{class_name}: espectros válidos finales = {X_input.shape[0]}")
    print(f"{class_name}: shape wavelength = {W_final.shape}")
    print(f"{class_name}: shape flux = {X_final.shape}")

    return W_final, X_final, X_input, valid_df
```

Función para evaluación de K con Silhouette Score

In [27]:
```
def evaluate_kmeans_k(features_32, class_name, layer_name, k_values=range(2, 9)):
    Xs = StandardScaler().fit_transform(features_32)

    #inertias = []
    silhouettes = []

    for k in k_values:
        km = KMeans(n_clusters=k, random_state=42, n_init=10)
```

```
        labels = km.fit_predict(Xs)

        #inertias.append(km.inertia_)
        silhouettes.append(silhouette_score(Xs, labels))

    plt.figure(figsize=(12,4))

    '''
    plt.subplot(1,2,1)
    plt.plot(list(k_values), inertias, marker='o')
    plt.title(f"{class_name} - {layer_name} - Elbow")
    plt.xlabel("k")
    plt.ylabel("Inercia")
    '''
    #plt.subplot(1,2,2)
    n_samples = features_32.shape[0]
    plt.plot(list(k_values), silhouettes, marker='o')
    plt.title(f"{class_name} - {layer_name} - Silhouette(n={n_samples})")
    #plt.legend([f"n={n_samples}"])
    plt.xlabel("k")
    plt.ylabel("Silhouette")

    plt.tight_layout()
    plt.show()
```

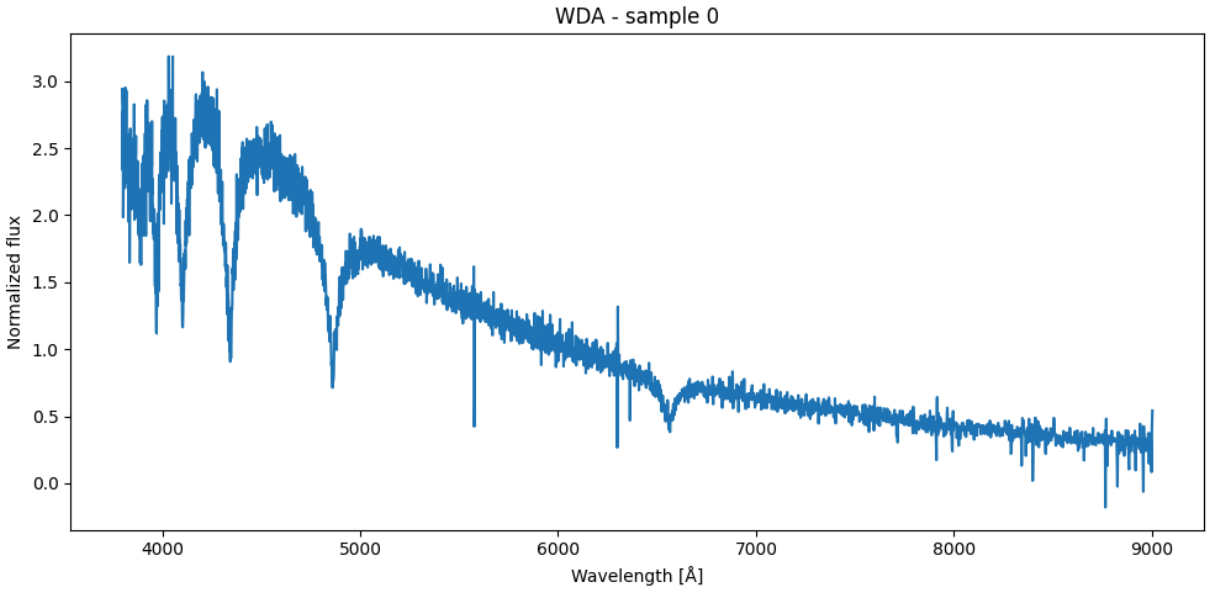
Función para hacer matrices (de contingencia) y analizar cambios de k para Spectral Clustering

In [28]:
```
def plot_migration_matrix(labels_a, labels_b, ka, kb, title=None, figsize=(6, 5)):
    cm = contingency_matrix(labels_a, labels_b)

    rows_labels = [f'C{i}' for i in range(cm.shape[0])]
    cols_labels = [f'C{j}' for j in range(cm.shape[1])]

    fig, ax = plt.subplots(figsize=figsize)
    sns.heatmap(
        cm,
        annot=True,
        fmt='d',
        cmap='Blues',
        xticklabels=cols_labels,
        yticklabels=rows_labels,
        ax=ax
    )
    ax.set_xlabel(f'Clústeres k={kb}', fontsize=11)
    ax.set_ylabel(f'Clústeres k={ka}', fontsize=11)
    _title = title if title else f'Migración k={ka} → k={kb}  (Flatten, WDA)'
    ax.set_title(_title, fontsize=12, fontweight='bold')
    plt.tight_layout()
    plt.show()
```

Visualización de un espectro con la longitud de onda correcta

In [29]:
```
W_wda, X_wda_spec, X_wda_input, df_wda_DRAC = load_class_spectra_DRACULA_with_wavel
    "WDA",
    n_samples=1100,
    random_state=42
)

sample_idx = 0

plt.figure(figsize=(10, 5))
plt.plot(W_wda[sample_idx], X_wda_spec[sample_idx])
plt.xlabel("Wavelength [Å]")
plt.ylabel("Normalized flux")
plt.title(f"WDA - sample {sample_idx}")
plt.tight_layout()
plt.show()
```

```
WDA: disponibles con match a .dat = 13723
WDA: espectros válidos finales = 1100
WDA: shape wavelength = (1100, 3750)
WDA: shape flux = (1100, 3750)
```

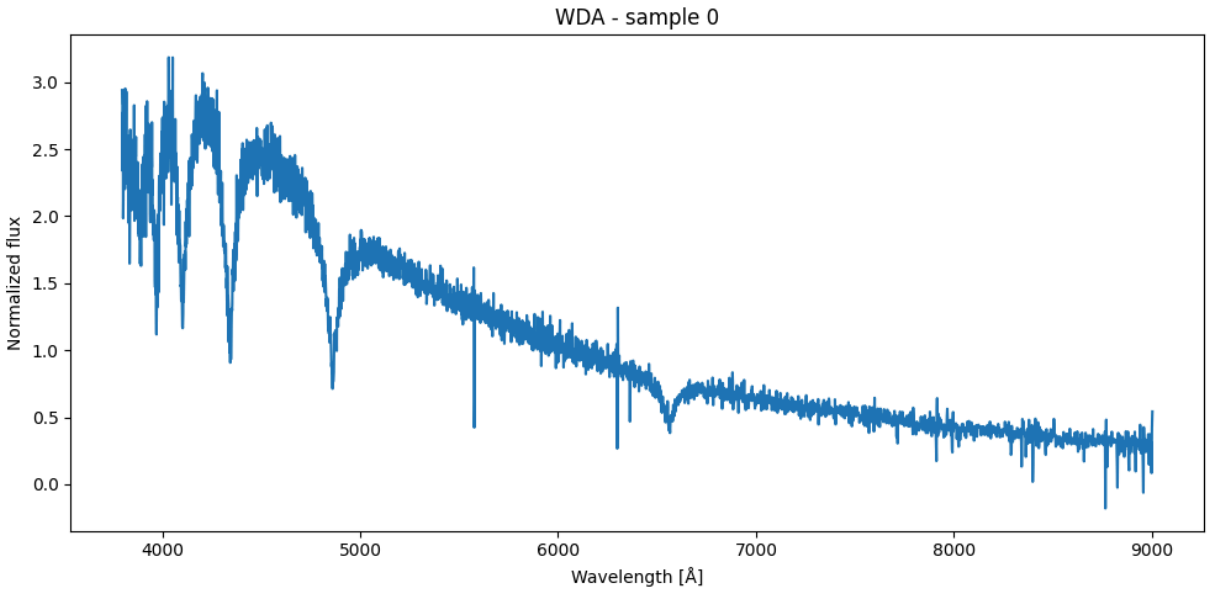


Anteriormente se consideran n=1100 muestras, ahora usaré n=13000 muestras (casi el total de muestras de WDA, que son 13723) -> los espectros válidos finales pueden ser menos que el n seleccionado, debido a que si en la función $process\_spectrum$ un espectro falla al cargarse, contiene valores nulos o tiene un valor de normalización 0, esta función retorna $None$ y el espectro es descartado (no se agrega al conjunto final X_list).

In [30]:
```
W_wda_n2, X_wda_spec_n2, X_wda_input_n2, df_wda_DRAC_n2 = load_class_spectra_DRACUL
    "WDA",
    n_samples=13723,
```

```
    random_state=42
)

sample_idx = 0

plt.figure(figsize=(10, 5))
plt.plot(W_wda_n2[sample_idx], X_wda_spec_n2[sample_idx])
plt.xlabel("Wavelength [Å]")
plt.ylabel("Normalized flux")
plt.title(f"WDA - sample {sample_idx}")
plt.tight_layout()
plt.show()
```

```
WDA: disponibles con match a .dat = 13723
WDA: espectros válidos finales = 13711
WDA: shape wavelength = (13711, 3750)
WDA: shape flux = (13711, 3750)
```



Funciones que hacen cluster y plot según método, y generan espectros a partir de los clusters encontrados en cada método. Se implementa interact

In [31]:
```
def _add_cluster_legend(ax, labels, cmap=plt.cm.viridis):
    unique_labels = np.unique(labels)
    label_min, label_max = unique_labels.min(), unique_labels.max()
    handles = []
    for lab in unique_labels:
        count = int(np.sum(labels == lab))
        if lab == -1:
            color, text = "black", f"ruido (-1): n={count}"
        else:
            norm = 0.5 if label_max == label_min else (lab - label_min) / (label_ma
            color, text = cmap(norm), f"cluster {lab}: n={count}"
        handles.append(Line2D([0], [0], marker='o', color='w',
```

```
                                markerfacecolor=color, markersize=8, label=text))
    ax.legend(handles=handles, title="Clusters", loc="best", fontsize=9)
```

In [32]:

```
def _print_metrics_box(class_name, layer_name, method, param_str, metrics, dbcv=Non
    k     = metrics['n_clusters']
    noise = metrics['n_noise']
    sil   = f"{metrics['silhouette']:.4f}"        if metrics['silhouette']        i
    db    = f"{metrics['davies_bouldin']:.4f}"    if metrics['davies_bouldin']    i
    ch    = f"{metrics['calinski_harabasz']:.2f}" if metrics['calinski_harabasz'] i
    sep   = chr(9472) * 64
    print(f'\n  \u250c{sep}\u2510')
    print(f'  \u2502  {class_name} | {layer_name} | {method} | {param_str}')
    print(f'  \u2502  clusters={k}  noise={noise}  sil={sil}  DB={db}  CH={ch}')
    if dbcv is not None:
        dbcv_str = f'{dbcv:.4f}' if dbcv == dbcv else 'N/A'
        print(f'  \u2502  DBCV = {dbcv_str}  (rango [-1,1], especifico DBSCAN)')
    print(f'  \u2514{sep}\u2518\n')


def cluster_plot_KMeans(features_32, class_name, layer_name,
                        kmeans_k=3, random_state=42,
                        vis_method="pca", xlim=None, ylim=None,
                        container=None):

    Xs = StandardScaler().fit_transform(features_32)

    if vis_method.lower() == "pca":
        emb_2d = PCA(n_components=2, random_state=42).fit_transform(Xs)
    elif vis_method.lower() == "tsne":
        emb_2d = TSNE(n_components=2, random_state=42,
                      perplexity=30, init='pca').fit_transform(Xs)
    else:
        raise ValueError("vis_method debe ser 'pca' o 'tsne'")

    km_labels = KMeans(n_clusters=kmeans_k, random_state=random_state,
                       n_init=10).fit_predict(Xs)

    if container is not None:
        save_cluster_labels(container, class_name, layer_name, "kmeans", km_labels)

    fig, ax = plt.subplots(figsize=(9, 6))
    ax.scatter(emb_2d[:, 0], emb_2d[:, 1], c=km_labels, s=30, alpha=0.6, cmap='viri
    ax.set_title(f"{class_name} \u2014 {layer_name} \u2014 KMeans "
                 f"(k={kmeans_k}, n={features_32.shape[0]})", fontsize=15)
    _add_cluster_legend(ax, km_labels)
    ax.set_xlabel("Comp. 1"); ax.set_ylabel("Comp. 2")
    if xlim: ax.set_xlim(xlim)
    if ylim: ax.set_ylim(ylim)
    plt.tight_layout(); plt.show()

    metrics = compute_cluster_metrics(Xs, km_labels)
    _print_metrics_box(class_name, layer_name, "kmeans", f"k={kmeans_k}", metrics)
    return {"kmeans": km_labels, "metrics": metrics}


def cluster_plot_Spect(features_32, class_name, layer_name,
```

```
                      spectral_k=3, random_state=42,
                      vis_method="pca", xlim=None, ylim=None,
                      container=None):

    Xs = StandardScaler().fit_transform(features_32)

    if vis_method.lower() == "pca":
        emb_2d = PCA(n_components=2, random_state=42).fit_transform(Xs)
    elif vis_method.lower() == "tsne":
        emb_2d = TSNE(n_components=2, random_state=42,
                      perplexity=30, init='pca').fit_transform(Xs)
    else:
        raise ValueError("vis_method debe ser 'pca' o 'tsne'")

    sc_labels = SpectralClustering(n_clusters=spectral_k, random_state=random_state
                                   affinity='nearest_neighbors').fit_predict(Xs)

    if container is not None:
        save_cluster_labels(container, class_name, layer_name, "spectral", sc_label

    fig, ax = plt.subplots(figsize=(9, 6))
    ax.scatter(emb_2d[:, 0], emb_2d[:, 1], c=sc_labels, s=30, alpha=0.6, cmap='viri
    ax.set_title(f"{class_name} \u2014 {layer_name} \u2014 Spectral "
                 f"(k={spectral_k}, n={features_32.shape[0]})", fontsize=15)
    _add_cluster_legend(ax, sc_labels)
    ax.set_xlabel("Comp. 1"); ax.set_ylabel("Comp. 2")
    if xlim: ax.set_xlim(xlim)
    if ylim: ax.set_ylim(ylim)
    plt.tight_layout(); plt.show()

    metrics = compute_cluster_metrics(Xs, sc_labels)
    _print_metrics_box(class_name, layer_name, "spectral", f"k={spectral_k}", metri
    return {"spectral": sc_labels, "metrics": metrics}


def cluster_plot_DBSCAN(features_32, class_name, layer_name,
                        dbscan_eps=1.5, dbscan_min_samples=10,
                        vis_method="pca", xlim=None, ylim=None,
                        container=None):

    Xs = StandardScaler().fit_transform(features_32)

    if vis_method.lower() == "pca":
        emb_2d = PCA(n_components=2, random_state=42).fit_transform(Xs)
    elif vis_method.lower() == "tsne":
        emb_2d = TSNE(n_components=2, random_state=42,
                      perplexity=30, init='pca').fit_transform(Xs)
    else:
        raise ValueError("vis_method debe ser 'pca' o 'tsne'")

    db_labels = DBSCAN(eps=dbscan_eps,
                       min_samples=dbscan_min_samples).fit_predict(Xs)

    if container is not None:
        save_cluster_labels(container, class_name, layer_name, "dbscan", db_labels)
```

```
    fig, ax = plt.subplots(figsize=(9, 6))
    ax.scatter(emb_2d[:, 0], emb_2d[:, 1], c=db_labels, s=30, alpha=0.3, cmap='viri
    ax.set_title(f"{class_name} \u2014 {layer_name} \u2014 DBSCAN "
                 f"(eps={dbscan_eps}, n={features_32.shape[0]})", fontsize=15)
    _add_cluster_legend(ax, db_labels)
    ax.set_xlabel("Comp. 1"); ax.set_ylabel("Comp. 2")
    if xlim: ax.set_xlim(xlim)
    if ylim: ax.set_ylim(ylim)
    plt.tight_layout(); plt.show()

    metrics = compute_cluster_metrics(Xs, db_labels)
    n_pts   = int(np.sum(db_labels != -1))
    dbcv    = dbcv_score(Xs, db_labels) if n_pts <= 5000 else None
    if n_pts > 5000:
        print(f'  [DBCV omitido: {n_pts} puntos no-ruido > 5000]')
    _print_metrics_box(class_name, layer_name, "dbscan", f"eps={dbscan_eps}", metri
    return {"dbscan": db_labels, "metrics": metrics, "dbcv": dbcv}
```

In [33]:

```
def _plot_cluster_spectra_base(
    spectra_2d, labels, wavelength,
    class_name, layer_name, method_name,
    linewidth_mean=2.4,
    figsize=(11, 5),
    mark_balmer=True,
    robust_ylim=True,
    ignore_noise=False,
    diff_percentiles=(1, 99),
    show_std=False,
    smooth_sigma=3             # suavizado Gaussian para el plot de diferencia (pix
):
    spectra_2d = np.asarray(spectra_2d)
    labels     = np.asarray(labels)
    wavelength = np.asarray(wavelength)

    if ignore_noise:
        mask       = labels != -1
        spectra_2d = spectra_2d[mask]
        labels     = labels[mask]

    unique_clusters = np.sort(np.unique(labels))
    total_mean      = np.mean(spectra_2d, axis=0)
    group_means     = {int(c): np.mean(spectra_2d[labels == c], axis=0)
                       for c in unique_clusters}

    n_clusters = len(unique_clusters)
    cmap       = plt.cm.get_cmap("tab10", n_clusters)

    # promedio de cada cluster por separado
    for i, c in enumerate(unique_clusters):
        group_spectra = spectra_2d[labels == c]
        group_mean    = group_means[int(c)]
        n             = int(group_spectra.shape[0])
        #color         = cmap(i)

        if c == -1:
            color = "gray"
```

```
            label_mean = f"Ruido (-1) mean (n={n})"
            title_tag = f"Ruido (-1) (n={n})"
        else:
            color = cmap(i)
            label_mean = f"Cluster {c} mean (n={n})"
            title_tag = f"Cluster {c} (n={n})"

        fig, ax = plt.subplots(figsize=figsize)

        if show_std:
            std = np.std(group_spectra, axis=0)
            ax.fill_between(wavelength,
                            group_mean - std,
                            group_mean + std,
                            alpha=0.18, color=color, label="±1σ")

        ax.plot(wavelength, group_mean, linewidth=linewidth_mean,
                color=color, label=f"Cluster {c} mean (n={n})")

        ax.set_xlabel("Wavelength [Å]")
        ax.set_ylabel("Normalized flux")
        ax.set_title(f"{class_name} — {layer_name} — {method_name}\n{title_tag}")

        if robust_ylim:
            p_low, p_high = np.percentile(group_mean, [1, 99])
            margin = 0.10 * (p_high - p_low)
            ax.set_ylim(p_low - margin, p_high + margin)

        if mark_balmer:
            add_balmer_lines(ax)

        ax.legend()
        plt.tight_layout()
        plt.show()

    # promedio total ——
    fig, ax = plt.subplots(figsize=figsize)
    ax.plot(wavelength, total_mean, linewidth=linewidth_mean,
            color="black", label=f"Total mean (n={spectra_2d.shape[0]})")

    ax.set_xlabel("Wavelength [Å]")
    ax.set_ylabel("Normalized flux")
    ax.set_title(f"{class_name} — {layer_name} — {method_name}\n"
                 f"Total mean spectrum")

    if robust_ylim:
        p_low, p_high = np.percentile(total_mean, [1, 99])
        margin = 0.10 * (p_high - p_low)
        ax.set_ylim(p_low - margin, p_high + margin)

    if mark_balmer:
        add_balmer_lines(ax)

    ax.legend()
    plt.tight_layout()
    plt.show()
```

```
    # diferencia: promedio cluster - promedio total
    from scipy.ndimage import gaussian_filter1d
    for i, c in enumerate(unique_clusters):
        diff        = group_means[int(c)] - total_mean
        diff_smooth = gaussian_filter1d(diff, sigma=smooth_sigma)
    # ruido dbscan
        if c == -1:
            color     = "gray"
            diff_label = "Ruido (-1) - total mean"
            diff_title = "Diff: Ruido (-1) mean - total mean"
        else:
            color      = cmap(i)
            diff_label = f"Cluster {c} - total mean"
            diff_title = f"Diff: Cluster {c} mean - total mean"

        fig, ax = plt.subplots(figsize=figsize)
        ax.plot(wavelength, diff,        linewidth=0.8, color=color,
                alpha=0.25, label="raw")
        ax.plot(wavelength, diff_smooth, linewidth=linewidth_mean, color=color,
                alpha=1.0,  label=f"{diff_label} (σ={smooth_sigma}px)")
        ax.axhline(0.0, linestyle="--", linewidth=1.2, color="gray", alpha=0.7)

        ax.set_xlabel("Wavelength [Å]")
        ax.set_ylabel("Δ Normalized flux")
        ax.set_title(f"{class_name} — {layer_name} — {method_name}\n{diff_title}")

        if robust_ylim:
            p_low, p_high = np.percentile(diff_smooth, diff_percentiles)
            margin = 0.15 * (p_high - p_low + 1e-8)
            ax.set_ylim(p_low - margin, p_high + margin)

        if mark_balmer:
            add_balmer_lines(ax)

        ax.legend()
        plt.tight_layout()
        plt.show()

    # todos los promedios juntos
    fig, ax = plt.subplots(figsize=figsize)
    all_curves = []

    for i, c in enumerate(unique_clusters):
        group_spectra = spectra_2d[labels == c]
        mean_spec     = group_means[int(c)]
        n_in_cluster  = int(np.sum(labels == c))
        #color         = cmap(i)
        if c == -1:
            color = "gray"
            leg   = f"Ruido (-1) (n={n_in_cluster})"
        else:
            color = cmap(i)
            leg   = f"Cluster {c} mean (n={n_in_cluster})"

        if show_std:
```

```
            std = np.std(spectra_2d[labels==c], axis=0)
            ax.fill_between(wavelength,
                            mean_spec - std,
                            mean_spec + std,
                            alpha=0.18, color=color)

        ax.plot(wavelength, mean_spec, linewidth=2.2,
                color=color, label=leg)
        all_curves.append(mean_spec)

    ax.plot(wavelength, total_mean, linewidth=2.4, linestyle="--",
            color="black", label=f"Total mean (n={spectra_2d.shape[0]})")

    ax.set_xlabel("Wavelength [Å]")
    ax.set_ylabel("Normalized flux")
    ax.set_title(f"{class_name} — {layer_name} — {method_name}\n"
                 f"All cluster means")

    if robust_ylim:
        stacked = np.vstack([*all_curves, total_mean])
        p_low, p_high = np.percentile(stacked, [1, 99])
        margin = 0.10 * (p_high - p_low + 1e-8)
        ax.set_ylim(p_low - margin, p_high + margin)

    if mark_balmer:
        add_balmer_lines(ax)

    ax.legend()
    plt.tight_layout()
    plt.show()

    return {
        "total_mean"     : total_mean,
        "group_means"    : group_means,
        "unique_clusters": unique_clusters
    }
```

In [34]:

```
def plot_spectra_kmeans(spectra_2d, labels, wavelength, class_name, layer_name,
                        linewidth_mean=2.4, figsize=(11,5),
                        mark_balmer=True, robust_ylim=True, show_std=False,
                        smooth_sigma=3):
    return _plot_cluster_spectra_base(
        spectra_2d, labels, wavelength,
        class_name, layer_name, method_name="kmeans",
        linewidth_mean=linewidth_mean, figsize=figsize,
        mark_balmer=mark_balmer, robust_ylim=robust_ylim,
        ignore_noise=False, show_std=show_std, smooth_sigma=smooth_sigma
    )

def plot_spectra_spectral(spectra_2d, labels, wavelength, class_name, layer_name,
                          linewidth_mean=2.4, figsize=(11,5),
                          mark_balmer=True, robust_ylim=True, show_std=False,
                          smooth_sigma=3):
    return _plot_cluster_spectra_base(
        spectra_2d, labels, wavelength,
        class_name, layer_name, method_name="spectral",
```

```
        linewidth_mean=linewidth_mean, figsize=figsize,
        mark_balmer=mark_balmer, robust_ylim=robust_ylim,
        ignore_noise=False, show_std=show_std, smooth_sigma=smooth_sigma
    )

def plot_spectra_dbscan(spectra_2d, labels, wavelength, class_name, layer_name,
                        ignore_noise=False, linewidth_mean=2.4, figsize=(11,5),
                        mark_balmer=True, robust_ylim=True, show_std=False,
                        smooth_sigma=3):
    return _plot_cluster_spectra_base(
        spectra_2d, labels, wavelength,
        class_name, layer_name, method_name="dbscan",
        linewidth_mean=linewidth_mean, figsize=figsize,
        mark_balmer=mark_balmer, robust_ylim=robust_ylim,
        ignore_noise=ignore_noise, show_std=show_std, smooth_sigma=smooth_sigma
    )
```

In [35]:

```
# Widget interactivo para explorar los espectros promedio por cluster
def interactive_plot_cluster_spectra(
    spectra_2d,
    wavelength,
    class_name,
    container,
    linewidth_mean=2.4,
    figsize=(13, 5),
    mark_balmer=True
):
    """
    Controles:
    - Capa       : flatten / dense / dense_1 / dense_2
    - Método     : kmeans / spectral / dbscan
    - Mostrar ±1σ: activa/desactiva la banda de desviación estándar
    """
    spectra_2d = np.asarray(spectra_2d)
    wavelength = np.asarray(wavelength)

    # leer capas y métodos disponibles desde el container
    available_layers  = list(container[class_name].keys())
    available_methods = list(next(iter(container[class_name].values())).keys())

    layer_dropdown = widgets.Dropdown(
        options=available_layers,
        value=available_layers[0],
        description="Capa:"
    )
    method_dropdown = widgets.Dropdown(
        options=available_methods,
        value=available_methods[0],
        description="Método:"
    )
    std_toggle = widgets.ToggleButtons(
        options=["No", "Sí"],
        value="No",
        description="Mostrar ±1σ:",
        button_style=""
    )
```

```
    def _plot(layer_name, method_name, mostrar_std):
        if method_name not in container[class_name].get(layer_name, {}):
            print(f"Sin labels para {class_name} — {layer_name} — {method_name}")
            return

        labels          = np.asarray(container[class_name][layer_name][method_name]
        unique_clusters = np.sort(np.unique(labels))
        total_mean      = np.mean(spectra_2d, axis=0)
        n_clusters      = len(unique_clusters)
        cmap            = plt.cm.get_cmap("tab10", n_clusters)
        show_std        = mostrar_std == "Sí"

        fig, ax = plt.subplots(figsize=figsize)

        for i, c in enumerate(unique_clusters):
            mask          = labels == c
            group_spectra = spectra_2d[mask]
            group_mean    = np.mean(group_spectra, axis=0)
            n_in_cluster  = int(mask.sum())
            color         = cmap(i)

            if show_std:
                std = np.std(group_spectra, axis=0)
                ax.fill_between(wavelength,
                                group_mean - std,
                                group_mean + std,
                                alpha=0.18, color=color)

            ax.plot(wavelength, group_mean, linewidth=linewidth_mean,
                    color=color, label=f"Cluster {c} (n={n_in_cluster})")

        ax.plot(wavelength, total_mean, linewidth=2.0, linestyle="--",
                color="black", label=f"Total mean (n={spectra_2d.shape[0]})")

        ax.set_xlabel("Wavelength [Å]")
        ax.set_ylabel("Normalized flux")
        ax.set_title(f"{class_name} — {layer_name} — {method_name}"
                     + ("   [±1σ]" if show_std else ""))

        if mark_balmer:
            add_balmer_lines(ax)

        ax.legend()
        plt.tight_layout()
        plt.show()

    interact(
        _plot,
        layer_name=layer_dropdown,
        method_name=method_dropdown,
        mostrar_std=std_toggle
    )
```

Función para el cálculo de métricas Davies Bouldin Score, Calinsi-Harabasz Score y Silhouette Score para un clustering dado

In [36]:
```
from sklearn.metrics import davies_bouldin_score, calinski_harabasz_score, silhouet


def compute_cluster_metrics(Xs, labels):
    """
    Calcula DB, CH y Silhouette sobre el espacio escalado.
    Excluye ruido (-1) automaticamente.
    Retorna dict con metricas, o None en cada metrica si hay menos de 2 clusters.
    """
    mask         = labels != -1
    Xs_clean     = Xs[mask]
    labels_clean = labels[mask]

    n_clusters = len(np.unique(labels_clean))
    n_noise    = int(np.sum(~mask))

    if n_clusters < 2:
        return {
            "n_clusters"        : n_clusters,
            "n_noise"           : n_noise,
            "silhouette"        : None,
            "davies_bouldin"    : None,
            "calinski_harabasz" : None
        }

    return {
        "n_clusters"        : n_clusters,
        "n_noise"           : n_noise,
        "silhouette"        : round(silhouette_score(Xs_clean, labels_clean), 6),
        "davies_bouldin"    : round(davies_bouldin_score(Xs_clean, labels_clean), 6
        "calinski_harabasz" : round(calinski_harabasz_score(Xs_clean, labels_clean)
    }
```

In [37]:
```
# censity based clustering validation (DBCV)
from scipy.spatial.distance import cdist
from scipy.sparse.csgraph import minimum_spanning_tree
from scipy.sparse import csr_matrix


def dbcv_score(X, labels, noise_id=-1):
    """
    rango [-1, 1]. Mas alto = mejor.
    los puntos de ruido (noise_id=-1) se excluyen automaticamente.
    complejidad O(n^2). Recomendado para n <= 5000.
    """
    X      = np.asarray(X, dtype=np.float64)
    labels = np.asarray(labels)
```

```
    mask   = labels != noise_id
    X      = X[mask]
    labels = labels[mask]

    unique = np.unique(labels)
    if len(unique) < 2:
        return np.nan

    n = X.shape[0]

    # matriz de distancias completa
    D = cdist(X, X)
    np.fill_diagonal(D, np.inf)

    # distancia core: media armonica de distancias dentro del cluster
    # d_core(x_i) = (|C|-1) / sum_{j in C, j!=i} (1/d(x_i,x_j))
    core_dists = np.full(n, np.inf)
    for label in unique:
        idx = np.where(labels == label)[0]
        if len(idx) <= 1:
            continue
        D_sub = D[np.ix_(idx, idx)]
        with np.errstate(divide='ignore'):
            inv_d = np.where(np.isinf(D_sub), 0.0, 1.0 / D_sub)
        harmonic_core = (len(idx) - 1) / inv_d.sum(axis=1)
        for k, i in enumerate(idx):
            core_dists[i] = harmonic_core[k]

    # Mutual reachability distance: mrd(i,j) = max(core_i, core_j, d(i,j))
    D_fin = np.where(np.isinf(D), 0.0, D)
    mrd   = np.maximum(
        np.maximum(core_dists[:, None], core_dists[None, :]),
        D_fin
    )

    # Arbol de expansion minima sobre mrd
    mst     = minimum_spanning_tree(csr_matrix(mrd)).toarray()
    mst_sym = np.maximum(mst, mst.T)

    # DSC: max arista interna del MST por cluster (dispersion interna)
    dsc = {}
    for label in unique:
        idx      = np.where(labels == label)[0]
        internal = mst_sym[np.ix_(idx, idx)]
        dsc[label] = float(internal.max()) if internal.size > 0 else 0.0

    # DSPC: min MRD entre clusters distintos (separacion entre clusters)
    dspc = {}
    for i, li in enumerate(unique):
        for j, lj in enumerate(unique):
            if i >= j:
                continue
            ix_i = np.where(labels == li)[0]
            ix_j = np.where(labels == lj)[0]
            val  = float(mrd[np.ix_(ix_i, ix_j)].min())
            dspc[(li, lj)] = dspc[(lj, li)] = val
```

```
    # Validez por cluster: (separacion - dispersion) / max(separacion, dispersion)
    validity = {}
    for label in unique:
        min_sep = min(dspc[(label, o)] for o in unique if o != label)
        denom   = max(min_sep, dsc[label])
        validity[label] = 0.0 if denom == 0 else (min_sep - dsc[label]) / denom

    # Score global: promedio ponderado por tamano de cluster
    score = sum(
        (np.sum(labels == label) / n) * validity[label]
        for label in unique
    )
    return round(float(score), 6)
```

In [38]:
```
# tabla de métricas
def build_metrics_table(
    representations_32,
    class_name,
    spectral_k_values    = [2, 3, 4, 5],
    dbscan_eps_values    = [0.5, 1.0, 1.5, 2.0, 2.5, 3.0],
    dbscan_min_samples   = 10,
    random_state         = 42
):
    """
    Itera sobre todas las combinaciones capa × método × parámetro
    y devuelve una tabla con las métricas de clustering.

    Solo considera Spectral Clustering y DBSCAN.

    Parameters
    ----------
    representations_32 : dict {layer_name: np.ndarray shape (n, 32)}
    class_name         : str
    spectral_k_values  : list de k a probar para Spectral
    dbscan_eps_values  : list de eps a probar para DBSCAN
    dbscan_min_samples : int

    Returns
    -------
    pd.DataFrame con columnas:
        class | layer | method | param | param_value |
        n_clusters | n_noise | silhouette | davies_bouldin | calinski_harabasz
    """
    rows = []

    for layer_name, feats in representations_32.items():
        Xs = StandardScaler().fit_transform(feats)

        # ── Spectral Clustering: barre sobre k ──
        for k in spectral_k_values:
            labels = SpectralClustering(
                n_clusters    = k,
                affinity      = "nearest_neighbors",
                assign_labels = "kmeans",
                random_state  = random_state
```

```
            ).fit_predict(Xs)

            metrics = compute_cluster_metrics(Xs, labels)

            rows.append({
                "class"             : class_name,
                "layer"             : layer_name,
                "method"            : "spectral",
                "param"             : "k",
                "param_value"       : k,
                **metrics
            })

        # ── DBSCAN: barre sobre eps ──
        for eps in dbscan_eps_values:
            labels = DBSCAN(
                eps         = eps,
                min_samples = dbscan_min_samples
            ).fit_predict(Xs)

            metrics = compute_cluster_metrics(Xs, labels)

            rows.append({
                "class"             : class_name,
                "layer"             : layer_name,
                "method"            : "dbscan",
                "param"             : "eps",
                "param_value"       : eps,
                **metrics
            })

    df = pd.DataFrame(rows, columns=[
        "class", "layer", "method", "param", "param_value",
        "n_clusters", "n_noise",
        "silhouette", "davies_bouldin", "calinski_harabasz"
    ])

    return df
```

In [39]:

```
# identificar las mejores configuraciones
def get_best_configs(metrics_df, top_n=5, min_clusters=2):
    """
    Rankea las configuraciones usando las tres métricas combinadas.

    Criterios:
        silhouette        → mayor es mejor
        davies_bouldin    → menor es mejor
        calinski_harabasz → mayor es mejor

    Se filtran filas con métricas None o n_clusters < min_clusters.

    Parameters
    ----------
    metrics_df  : output de build_metrics_table
    top_n       : cuántas configuraciones mostrar
    min_clusters: mínimo de clusters para considerar válida la fila
```

```
    Returns
    -------
    pd.DataFrame con las top_n configuraciones y su score_compuesto
    """
    df = metrics_df.dropna(subset=["silhouette", "davies_bouldin",
                                   "calinski_harabasz"]).copy()
    df = df[df["n_clusters"] >= min_clusters].reset_index(drop=True)

    # Ranking por cada métrica (1 = mejor)
    df["rank_sil"] = df["silhouette"].rank(ascending=False)
    df["rank_db"]  = df["davies_bouldin"].rank(ascending=True)   # menor es mejor
    df["rank_ch"]  = df["calinski_harabasz"].rank(ascending=False)

    # Score compuesto: promedio de los tres rankings (menor = mejor)
    df["score_compuesto"] = (df["rank_sil"] + df["rank_db"] + df["rank_ch"]) / 3

    top = (df.sort_values("score_compuesto")
             .head(top_n)
             .reset_index(drop=True))

    cols_show = [
        "class", "layer", "method", "param_value", "n_clusters", "n_noise",
        "silhouette", "davies_bouldin", "calinski_harabasz", "score_compuesto"
    ]

    return top[cols_show]
```

In [40]:

```
# visualización tabla
def plot_metrics_heatmap(metrics_df, metric="silhouette", class_name=None):
    """
    Heatmap de una métrica para todas las combinaciones capa × configuración.

    Parameters
    ----------
    metric : "silhouette" | "davies_bouldin" | "calinski_harabasz"
    """
    if class_name:
        df = metrics_df[metrics_df["class"] == class_name].copy()
    else:
        df = metrics_df.copy()

    df["config"] = df["method"] + " " + df["param"] + "=" + df["param_value"].astyp

    pivot = df.pivot_table(index="layer", columns="config",
                           values=metric, aggfunc="first")

    # Para DB, invertir la escala visualmente (menor = mejor = más oscuro)
    ascending = metric != "davies_bouldin"

    fig, ax = plt.subplots(figsize=(max(10, len(pivot.columns)), 4))
    sns.heatmap(
        pivot,
        annot=True, fmt=".4f",
        cmap="RdYlGn" if ascending else "RdYlGn_r",
        ax=ax,
```

```
        linewidths=0.5,
        cbar_kws={"label": metric}
    )
    title_note = "(↑ mejor)" if ascending else "(↓ mejor)"
    ax.set_title(f"{class_name or ''} — {metric} {title_note}", fontsize=13)
    ax.set_xlabel("")
    ax.set_ylabel("Capa")
    plt.xticks(rotation=30, ha="right")
    plt.tight_layout()
    plt.show()
```

Caracterización espectral — profundidad de líneas Balmer por cluster

In [41]:
```
BALMER_LINES = {
    'Hε': 3970,
    'Hδ': 4101,
    'Hγ': 4340,
    'Hβ': 4861,
    'Hα': 6563,
}


def _measure_line(flux, wave, center,
                  line_hw=20, cont_offset=50, cont_hw=25):
    wave = np.asarray(wave)
    flux = np.asarray(flux)

    line_mask = np.abs(wave - center) <= line_hw
    cont_mask = (
        ((wave >= center - cont_offset - cont_hw) & (wave <= center - cont_offset))
        ((wave >= center + cont_offset)           & (wave <= center + cont_offset +
    )

    if line_mask.sum() == 0 or cont_mask.sum() == 0:
        return np.nan, np.nan

    F_cont = np.mean(flux[cont_mask])
    if F_cont == 0:
        return np.nan, np.nan

    F_line = flux[line_mask]
    W_line = wave[line_mask]
    depth  = F_cont - np.min(F_line)
    d_lam  = np.abs(np.gradient(W_line)).mean()
    ew     = np.sum(np.clip(1.0 - F_line / F_cont, 0, None)) * d_lam
    return round(float(depth), 6), round(float(ew), 4)


def compute_balmer_depths(group_means, wavelength,
                          class_name, layer_name, method_name,
                          label_counts=None):
    """
```

```
    Calcula profundidad y pseudo-EW de lineas Balmer para cada cluster.

    group_means  : dict {cluster_id: array espectro medio}  <- viene de _plot_clust
    label_counts : dict {cluster_id: n_muestras}  (opcional)

    Retorna pd.DataFrame  filas=clusters, columnas=profundidad/EW por linea
    """
    wavelength = np.asarray(wavelength)
    rows = []

    for cid in sorted(group_means.keys()):
        mean_spec = np.asarray(group_means[cid])
        row = {
            'class'  : class_name,
            'layer'  : layer_name,
            'method' : method_name,
            'cluster': cid,
        }
        if label_counts is not None:
            row['n'] = label_counts.get(cid, None)

        for line_name, center in BALMER_LINES.items():
            if center < wavelength.min() or center > wavelength.max():
                row[f'{line_name}_depth'] = np.nan
                row[f'{line_name}_ew']    = np.nan
                continue
            depth, ew = _measure_line(mean_spec, wavelength, center)
            row[f'{line_name}_depth'] = depth
            row[f'{line_name}_ew']    = ew

        rows.append(row)

    return pd.DataFrame(rows)


def plot_balmer_comparison(balmer_df, metric='depth', figsize=(10, 4)):
    """
    Barras agrupadas: profundidad (o EW) de cada linea Balmer por cluster.
    metric : 'depth' o 'ew'
    """
    cols = [f'{ln}_{metric}' for ln in BALMER_LINES if f'{ln}_{metric}' in balmer_d
    n_clusters = len(balmer_df)
    x          = np.arange(len(cols))
    width      = 0.8 / n_clusters
    cmap       = plt.cm.get_cmap('tab10', n_clusters)

    fig, ax = plt.subplots(figsize=figsize)
    for i, (_, row) in enumerate(balmer_df.iterrows()):
        vals  = [row[c] for c in cols]
        label = f"Cluster {int(row['cluster'])}"
        if 'n' in row and row['n'] is not None:
            label += f" (n={int(row['n'])})"
        ax.bar(x + i * width, vals, width=width, color=cmap(i), alpha=0.85, label=l

    ax.set_xticks(x + width * (n_clusters - 1) / 2)
    ax.set_xticklabels([c.replace(f'_{metric}', '') for c in cols], fontsize=11)
```

```
    ax.set_ylabel('Profundidad (flux units)' if metric == 'depth' else 'Pseudo-EW [
    row0 = balmer_df.iloc[0]
    ax.set_title(f"Balmer {metric} por cluster — {row0['class']} | {row0['layer']}
    ax.legend()
    plt.tight_layout(); plt.show()
```

Estabilidad de clusters — ARI entre múltiples corridas

In [42]:
```
from sklearn.metrics import adjusted_rand_score
from itertools import combinations as _combinations


def _pairwise_ari(all_labels):
    pairs = list(_combinations(range(len(all_labels)), 2))
    return [adjusted_rand_score(all_labels[i], all_labels[j]) for i, j in pairs]


def test_kmeans_stability(features_32, k, class_name, layer_name,
                          n_runs=30, random_state_base=0):
    """
    Corre KMeans n_runs veces con semillas distintas.
    ARI ~ 1.0 => muy estable.  ARI < 0.6 => inestable.
    """
    Xs = StandardScaler().fit_transform(features_32)
    all_labels = [
        KMeans(n_clusters=k, n_init=10, random_state=seed).fit_predict(Xs)
        for seed in range(random_state_base, random_state_base + n_runs)
    ]
    aris = _pairwise_ari(all_labels)
    mu, sd = np.mean(aris), np.std(aris)
    print(f"[{class_name} | {layer_name} | KMeans k={k}]  "
          f"ARI = {mu:.3f} +/- {sd:.3f}  "
          f"(min={min(aris):.3f}, max={max(aris):.3f})  [{n_runs} corridas]")
    return {'mean_ari': mu, 'std_ari': sd, 'all_aris': aris}


def test_spectral_stability(features_32, k, class_name, layer_name,
                            n_runs=20, random_state_base=0):
    """
    Corre SpectralClustering n_runs veces con semillas distintas.
    """
    Xs = StandardScaler().fit_transform(features_32)
    all_labels = [
        SpectralClustering(
            n_clusters=k, affinity='nearest_neighbors',
            assign_labels='kmeans', random_state=seed
        ).fit_predict(Xs)
        for seed in range(random_state_base, random_state_base + n_runs)
    ]
    aris = _pairwise_ari(all_labels)
    mu, sd = np.mean(aris), np.std(aris)
    print(f"[{class_name} | {layer_name} | Spectral k={k}]  "
```

```
          f"ARI = {mu:.3f} +/- {sd:.3f}  "
          f"(min={min(aris):.3f}, max={max(aris):.3f})  [{n_runs} corridas]")
    return {'mean_ari': mu, 'std_ari': sd, 'all_aris': aris}


def test_subsample_stability(features_32, cluster_func, class_name, layer_name,
                             n_runs=30, subsample_frac=0.80, random_state=42):
    """
    Estabilidad por submuestreo sin reposicion (recomendado para DBSCAN).
    cluster_func(Xs) -> labels  (Xs ya escalado).
    """
    rng   = np.random.default_rng(random_state)
    Xs    = StandardScaler().fit_transform(features_32)
    n     = Xs.shape[0]
    n_sub = int(n * subsample_frac)

    runs = []
    for _ in range(n_runs):
        idx    = np.sort(rng.choice(n, size=n_sub, replace=False))
        labels = cluster_func(Xs[idx])
        runs.append((idx, labels))

    aris = []
    for (idx_i, lbl_i), (idx_j, lbl_j) in _combinations(runs[:20], 2):
        common = np.intersect1d(idx_i, idx_j)
        if len(common) < 20:
            continue
        pos_i = np.searchsorted(idx_i, common)
        pos_j = np.searchsorted(idx_j, common)
        aris.append(adjusted_rand_score(lbl_i[pos_i], lbl_j[pos_j]))

    if not aris:
        print(f"[{class_name} | {layer_name}] Sin pares comparables.")
        return {}

    mu, sd = np.mean(aris), np.std(aris)
    print(f"[{class_name} | {layer_name} | subsample {subsample_frac*100:.0f}%]  "
          f"ARI = {mu:.3f} +/- {sd:.3f}  "
          f"(min={min(aris):.3f}, max={max(aris):.3f})  [{n_runs} corridas]")
    return {'mean_ari': mu, 'std_ari': sd, 'all_aris': aris}


def plot_ari_distribution(stability_result, title='Distribucion ARI', figsize=(7, 3
    aris = stability_result.get('all_aris', [])
    if not aris:
        return
    fig, ax = plt.subplots(figsize=figsize)
    ax.hist(aris, bins=20, color='steelblue', edgecolor='white', alpha=0.85)
    ax.axvline(np.mean(aris), color='red', linestyle='--',
               label=f'Media = {np.mean(aris):.3f}')
    ax.set_xlabel('ARI'); ax.set_ylabel('Frecuencia')
    ax.set_title(title); ax.legend()
    plt.tight_layout(); plt.show()
```

Funciones Interactive

In [43]:
```
# interactive de Spectral Clustering (tSNE + PCA)

from IPython.display import display

# Almacen de experimentos hechos via el widget:
#   clave = (layer_name, k)  ->  {labels, emb_tsne, emb_pca, metrics}
spectral_store = {}


def _run_spectral_cached(features_32, k, random_state=42):
    """Corre SpectralClustering + embeddings tSNE y PCA una sola vez y los cachea."
    Xs       = StandardScaler().fit_transform(features_32)
    emb_tsne = TSNE(n_components=2, random_state=42,
                    perplexity=30, init='pca').fit_transform(Xs)
    emb_pca  = PCA(n_components=2, random_state=42).fit_transform(Xs)
    labels   = SpectralClustering(n_clusters=k, random_state=random_state,
                                  affinity='nearest_neighbors').fit_predict(Xs)
    metrics  = compute_cluster_metrics(Xs, labels)
    return {"labels": labels, "emb_tsne": emb_tsne,
            "emb_pca": emb_pca, "metrics": metrics}


def _plot_spectral_views(data, class_name, layer_name, k):
    """Dibuja los scatter tSNE y PCA a partir de datos cacheados."""
    labels = data["labels"]
    views = [("tsne", data["emb_tsne"], None,      None),
             ("pca",  data["emb_pca"],  (-10, 10), (-10, 10))]
    for vis_name, emb, xlim, ylim in views:
        fig, ax = plt.subplots(figsize=(9, 6))
        ax.scatter(emb[:, 0], emb[:, 1], c=labels, s=30, alpha=0.6, cmap='viridis')
        ax.set_title(f"{class_name} — {layer_name} — Spectral "
                     f"(k={k}, vis={vis_name}, n={len(labels)})", fontsize=14)
        _add_cluster_legend(ax, labels)
        ax.set_xlabel("Comp. 1"); ax.set_ylabel("Comp. 2")
        if xlim: ax.set_xlim(xlim)
        if ylim: ax.set_ylim(ylim)
        plt.tight_layout(); plt.show()
    _print_metrics_box(class_name, layer_name, "spectral", f"k={k}", data["metrics"


def spectral_explorer(features_dict, class_name, suffix,
                      store=None,
                      layers_avail=("flatten", "dense", "dense_1", "dense_2"),
                      k_values=range(2, 13)):

    if store is None:
        store = spectral_store

    def _show(capa, k, recalcular):
        k   = int(k)
        key = (capa, k)
```

```
        hechos = sorted(f"{l}/k={kk}" for (l, kk) in store)
        print("Experimentos guardados:",
              ", ".join(hechos) if hechos else "ninguno")
        print("-" * 60)

        if key not in store or recalcular:
            print(f"Calculando {class_name} | {capa} | k={k} ...")
            store[key] = _run_spectral_cached(features_dict[capa], k)
            save_cluster_labels(cluster_labels_all, class_name,
                                capa, "spectral", store[key]["labels"])
            varname = f"labels_{capa}_k{k}_{suffix}"
            globals()[varname] = store[key]["labels"]
            print(f"Guardado en store[('{capa}', {k})], en "
                  f"cluster_labels_all['{class_name}']['{capa}']['spectral'] "
                  f"y en la variable global `{varname}`")
        else:
            print(f"Mostrando experimento cacheado: {class_name} | {capa} | k={k}")
        print("-" * 60)

        _plot_spectral_views(store[key], class_name, capa, k)

    interact(
        _show,
        capa=widgets.Dropdown(options=list(layers_avail), description="Capa:"),
        k=widgets.Dropdown(options=list(k_values), value=3, description="k:"),
        recalcular=widgets.Checkbox(value=False, description="Forzar recalculo"),
    )


def get_spectral_labels(layer, k, store=None):
    """Devuelve las etiquetas de un experimento guardado por el explorador."""
    if store is None:
        store = spectral_store
    key = (layer, int(k))
    if key not in store:
        raise KeyError(f"No existe el experimento ({layer}, k={k}). "
                       f"Correlo primero en el explorador.")
    return store[key]["labels"]
```

In [44]:
```
# explorador interactivo de DBSCAN (tSNE + PCA)
dbscan_store = {}   # cache: (layer, eps, min_samples) -> {labels, emb_tsne, emb_pc

def _run_dbscan_cached(features_32, eps, min_samples):
    """Corre DBSCAN + embeddings tSNE y PCA y los cachea."""
    Xs       = StandardScaler().fit_transform(features_32)
    emb_tsne = TSNE(n_components=2, random_state=42,
                    perplexity=30, init='pca').fit_transform(Xs)
    emb_pca  = PCA(n_components=2, random_state=42).fit_transform(Xs)
    labels   = DBSCAN(eps=eps, min_samples=min_samples).fit_predict(Xs)
    metrics  = compute_cluster_metrics(Xs, labels)
    n_pts    = int(np.sum(labels != -1))
    dbcv     = dbcv_score(Xs, labels) if 0 < n_pts <= 5000 else None
    if n_pts > 5000:
        print(f'  [DBCV omitido: {n_pts} puntos no-ruido > 5000]')
    return {"labels": labels, "emb_tsne": emb_tsne,
```

```
            "emb_pca": emb_pca, "metrics": metrics, "dbcv": dbcv}


def _plot_dbscan_views(data, class_name, layer_name, eps, min_samples):
    """Dibuja scatter tSNE y PCA a partir de datos cacheados."""
    labels = data["labels"]
    views  = [("tsne", data["emb_tsne"]), ("pca", data["emb_pca"])]
    for vis_name, emb in views:
        fig, ax = plt.subplots(figsize=(9, 6))
        ax.scatter(emb[:, 0], emb[:, 1], c=labels,
                   s=30, alpha=0.4, cmap='viridis')
        ax.set_title(
            f"{class_name} \u2014 {layer_name} \u2014 DBSCAN "
            f"(eps={eps}, ms={min_samples}, vis={vis_name}, n={len(labels)})",
            fontsize=14)
        _add_cluster_legend(ax, labels)
        ax.set_xlabel("Comp. 1"); ax.set_ylabel("Comp. 2")
        plt.tight_layout(); plt.show()
    _print_metrics_box(class_name, layer_name, "dbscan",
                       f"eps={eps} ms={min_samples}",
                       data["metrics"], dbcv=data["dbcv"])


def dbscan_explorer(features_dict, class_name, suffix,
                    store=None,
                    layers_avail=("flatten", "dense", "dense_1", "dense_2"),
                    eps_values=None,
                    ms_values=None):

    if store is None:
        store = dbscan_store
    if eps_values is None:
        eps_values = [0.3, 0.5, 0.7, 1.0, 1.2, 1.5, 2.0, 2.5, 3.0, 4.0]
    if ms_values is None:
        ms_values = [3, 5, 8, 10, 12, 15, 20, 25, 30]

    def _show(capa, eps, min_samples, recalcular):
        key    = (capa, eps, min_samples)
        hechos = sorted(f"{l}/eps={e}/ms={m}" for (l, e, m) in store)
        print("Experimentos guardados:",
              ", ".join(hechos) if hechos else "ninguno")
        print("-" * 60)

        if key not in store or recalcular:
            print(f"Calculando {class_name} | {capa} | eps={eps} ms={min_samples} .
            store[key] = _run_dbscan_cached(features_dict[capa], eps, min_samples)
            save_cluster_labels(cluster_labels_all, class_name,
                                capa, "dbscan", store[key]["labels"])
            eps_str = str(eps).replace(".", "p")
            varname = f"labels_dbs_{capa}_e{eps_str}_ms{min_samples}_{suffix}"
            globals()[varname] = store[key]["labels"]
            print(f"Guardado en store[{key}], en "
                  f"cluster_labels_all['{class_name}']['{capa}']['dbscan'] "
                  f"y en la variable global `{varname}`")
        else:
            print(f"Mostrando cacheado: {class_name} | {capa} | eps={eps} ms={min_s
```

```
        print("-" * 60)
        _plot_dbscan_views(store[key], class_name, capa, eps, min_samples)

    interact(
        _show,
        capa=widgets.Dropdown(options=list(layers_avail), description="Capa:"),
        eps=widgets.Dropdown(options=eps_values, value=1.5, description="eps:"),
        min_samples=widgets.Dropdown(options=ms_values, value=10, description="min_
        recalcular=widgets.Checkbox(value=False, description="Forzar recalculo"),
    )


def get_dbscan_labels(layer, eps, min_samples, store=None):
    """Devuelve etiquetas de un experimento guardado por el explorador DBSCAN."""
    if store is None:
        store = dbscan_store
    key = (layer, float(eps), int(min_samples))
    if key not in store:
        raise KeyError(
            f"No existe el experimento ({layer}, eps={eps}, ms={min_samples}). "
            f"Correlo primero en el explorador DBSCAN.")
    return store[key]["labels"]
```

In [45]:

```
# explorador de espectros por cluster (Spectral + DBSCAN)
def _plot_cluster_spectra_interactive(
    spectra_2d, labels, wavelength, total_mean,
    class_name, layer_name, method_name, param_str,
    show_media=True, show_balmer=False, show_std=False,
    figsize=(11, 5), linewidth_mean=2.4
):
    """Dibuja graficos de espectro por cluster + grafico combinado."""
    labels     = np.asarray(labels)
    unique_cls = np.sort(np.unique(labels))
    n_clusters = len(unique_cls)
    cmap       = plt.cm.get_cmap("tab10", n_clusters)

    # --- graficos individuales por cluster ---
    for i, c in enumerate(unique_cls):
        mask   = labels == c
        g_spec = spectra_2d[mask]
        g_mean = np.mean(g_spec, axis=0)
        n      = int(mask.sum())
        color  = "gray" if c == -1 else cmap(i)
        lbl    = "Ruido (-1)" if c == -1 else f"Cluster {c}"

        fig, ax = plt.subplots(figsize=figsize)
        if show_std:
            std = np.std(g_spec, axis=0)
            ax.fill_between(wavelength, g_mean - std, g_mean + std,
                            alpha=0.18, color=color, label="\u00b11\u03c3")
        ax.plot(wavelength, g_mean, linewidth=linewidth_mean,
                color=color, label=f"{lbl} mean (n={n})")
        ax.set_xlabel("Wavelength [\u00c5]")
        ax.set_ylabel("Normalized flux")
        ax.set_title(
            f"{class_name} \u2014 {layer_name} \u2014 "
```

```
            f"{method_name} ({param_str})\n{lbl} (n={n})")
        if show_balmer:
            add_balmer_lines(ax)
        ax.legend()
        plt.tight_layout()
        plt.show()

    # --- grafico combinado ---
    fig, ax    = plt.subplots(figsize=figsize)
    all_curves = []
    for i, c in enumerate(unique_cls):
        mask   = labels == c
        g_spec = spectra_2d[mask]
        g_mean = np.mean(g_spec, axis=0)
        n      = int(mask.sum())
        color  = "gray" if c == -1 else cmap(i)
        leg    = "Ruido (-1)" if c == -1 else f"Cluster {c} (n={n})"
        if show_std:
            std = np.std(g_spec, axis=0)
            ax.fill_between(wavelength, g_mean - std, g_mean + std,
                            alpha=0.12, color=color)
        ax.plot(wavelength, g_mean, linewidth=2.2, color=color, label=leg)
        all_curves.append(g_mean)

    if show_media and len(all_curves) > 0:
        ax.plot(wavelength, total_mean, linewidth=2.4, linestyle="--",
                color="black", label=f"Media clase (n={len(labels)})")
        all_curves.append(total_mean)

    ax.set_xlabel("Wavelength [\u00c5]")
    ax.set_ylabel("Normalized flux")
    ax.set_title(
        f"{class_name} \u2014 {layer_name} \u2014 "
        f"{method_name} ({param_str})\nAll cluster means")

    if all_curves:
        stacked    = np.vstack(all_curves)
        p_lo, p_hi = np.percentile(stacked, [1, 99])
        margin     = 0.10 * (p_hi - p_lo + 1e-8)
        ax.set_ylim(p_lo - margin, p_hi + margin)

    if show_balmer:
        add_balmer_lines(ax)
    ax.legend()
    plt.tight_layout()
    plt.show()


def spectra_viewer(spectra_2d, wavelength, class_name, suffix,
                   spectral_store_ref=None, dbscan_store_ref=None):

    if spectral_store_ref is None:
        spectral_store_ref = spectral_store
    if dbscan_store_ref is None:
        dbscan_store_ref = dbscan_store
```

```
spectra_2d = np.asarray(spectra_2d)
wavelength = np.asarray(wavelength)
total_mean = np.mean(spectra_2d, axis=0)

W = widgets.Layout

metodo_dd = widgets.Dropdown(
    options=["spectral", "dbscan"], description="Metodo:",
    layout=W(width="190px"))
capa_dd = widgets.Dropdown(
    options=["flatten", "dense", "dense_1", "dense_2"],
    description="Capa:", layout=W(width="200px"))
k_dd = widgets.Dropdown(
    options=list(range(2, 13)), value=3,
    description="k:", layout=W(width="140px"))
eps_dd = widgets.Dropdown(
    options=[0.3, 0.5, 0.7, 1.0, 1.2, 1.5, 2.0, 2.5, 3.0, 4.0],
    value=1.5, description="eps:", layout=W(width="150px"))
ms_dd = widgets.Dropdown(
    options=[3, 5, 8, 10, 12, 15, 20, 25, 30],
    value=10, description="min_s:", layout=W(width="160px"))

media_btn  = widgets.ToggleButton(
    value=True,  description="Media clase",
    button_style="info", layout=W(width="150px"))
balmer_btn = widgets.ToggleButton(
    value=False, description="Balmer",
    button_style="", layout=W(width="110px"))
std_btn    = widgets.ToggleButton(
    value=False, description="\u00b11\u03c3 individual",
    button_style="", layout=W(width="150px"))

out = widgets.Output()

def _update_visibility(change=None):
    is_sp = metodo_dd.value == "spectral"
    k_dd.layout.display   = ""     if is_sp else "none"
    eps_dd.layout.display = "none" if is_sp else ""
    ms_dd.layout.display  = "none" if is_sp else ""

metodo_dd.observe(_update_visibility, names="value")
_update_visibility()

def _render(change=None):
    with out:
        out.clear_output(wait=True)
        metodo = metodo_dd.value
        capa   = capa_dd.value

        if metodo == "spectral":
            k   = int(k_dd.value)
            key = (capa, k)
            if key not in spectral_store_ref:
                print(f"No hay experimento spectral ({capa}, k={k}).\n"
                      "Correlo primero en el explorador Spectral Clustering.")
                return
```

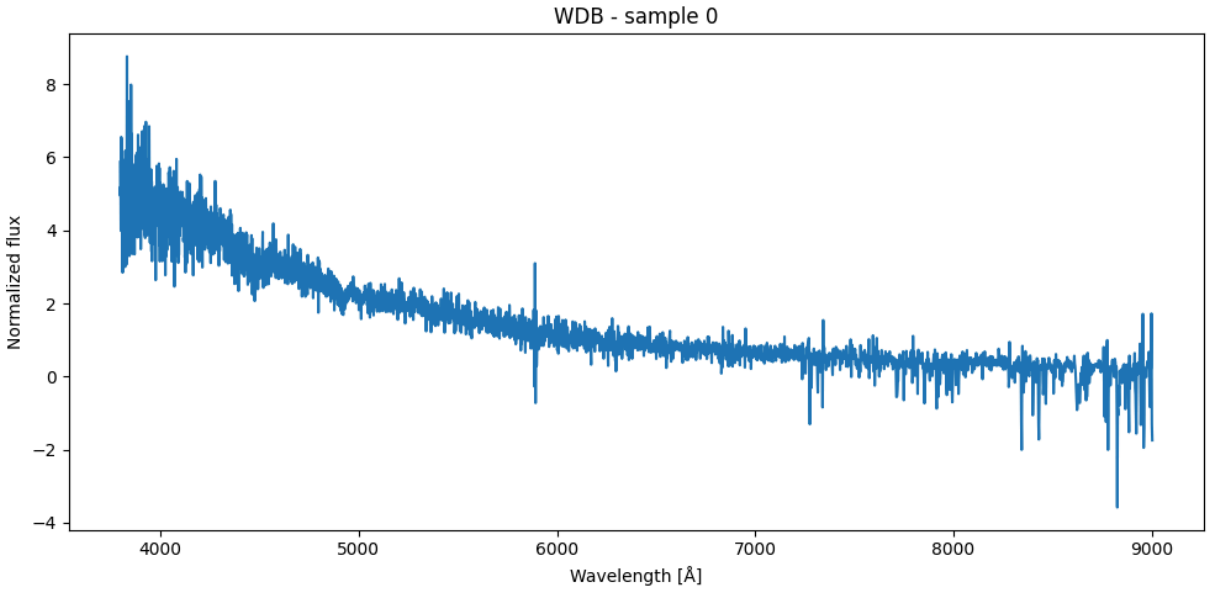
```
                labels    = spectral_store_ref[key]["labels"]
                param_str = f"k={k}"
            else:
                eps = float(eps_dd.value)
                ms  = int(ms_dd.value)
                key = (capa, eps, ms)
                if key not in dbscan_store_ref:
                    print(f"No hay experimento DBSCAN ({capa}, eps={eps}, ms={ms}).
                          "Correlo primero en el explorador DBSCAN.")
                    return
                labels    = dbscan_store_ref[key]["labels"]
                param_str = f"eps={eps} ms={ms}"

            _plot_cluster_spectra_interactive(
                spectra_2d, labels, wavelength, total_mean,
                class_name, capa, metodo, param_str,
                show_media  = media_btn.value,
                show_balmer = balmer_btn.value,
                show_std    = std_btn.value,
            )

    for w in [metodo_dd, capa_dd, k_dd, eps_dd, ms_dd,
              media_btn, balmer_btn, std_btn]:
        w.observe(_render, names="value")

    row1 = widgets.HBox([metodo_dd, capa_dd, k_dd, eps_dd, ms_dd])
    row2 = widgets.HBox([media_btn, balmer_btn, std_btn])
    display(widgets.VBox([row1, row2, out]))
    _render()
```

Experimentos WDB

Visualización de un espectro con longitud de onda

In [46]:
```
W_wdb, X_wdb_spec, X_wdb_input, df_wdb_DRAC = load_class_spectra_DRACULA_with_wavel
    "WDB",
    n_samples=1100,
    random_state=42
)

sample_idx = 0

plt.figure(figsize=(10, 5))
plt.plot(W_wdb[sample_idx], X_wdb_spec[sample_idx])
plt.xlabel("Wavelength [Å]")
plt.ylabel("Normalized flux")
plt.title(f"WDB - sample {sample_idx}")
plt.tight_layout()
plt.show()
```

```
WDB: disponibles con match a .dat = 1184
WDB: espectros válidos finales = 1099
WDB: shape wavelength = (1099, 3750)
WDB: shape flux = (1099, 3750)
```



In [47]:
```
# Experimento WDB para n=1100
X_wdb_spec, X_wdb_input, df_wdb_DRAC = load_class_spectra_DRACULA("WDB", n_samples=

flatten_wdb_DRAC, dense_wdb_DRAC, dense1_wdb_DRAC, dense2_wdb_DRAC = extract_layer_

repr_wdb_32 = reduce_all_to_32(flatten_wdb_DRAC, dense_wdb_DRAC, dense1_wdb_DRAC, d
```

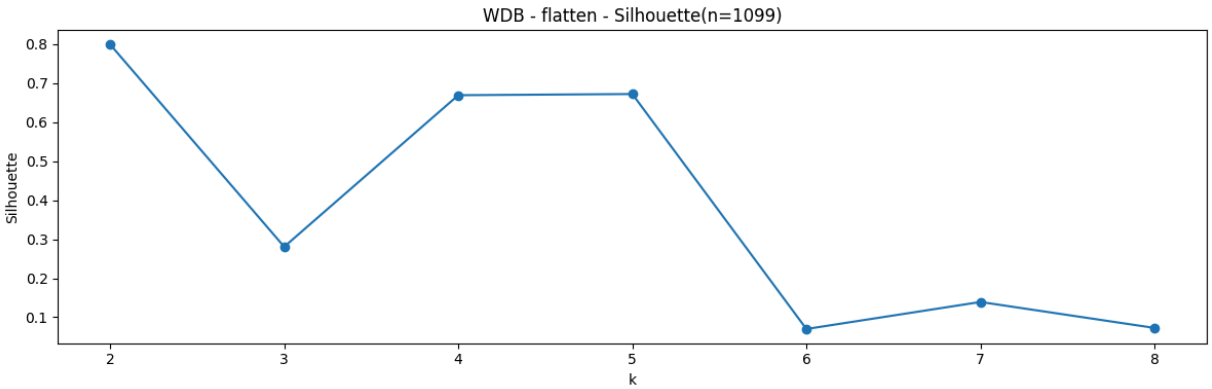
```
WDB: disponibles con match a .dat = 1184
WDB: espectros válidos finales = 1099
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 25ms/step
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 25ms/step
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 25ms/step
18/18 ━━━━━━━━━━━━━━━━━━━━ 1s 26ms/step
flatten: (1099, 208)
dense: (1099, 128)
dense_1: (1099, 64)
dense_2: (1099, 32)
flatten_32: (1099, 32)
dense_32: (1099, 32)
dense1_32: (1099, 32)
dense2_32: (1099, 32)
```

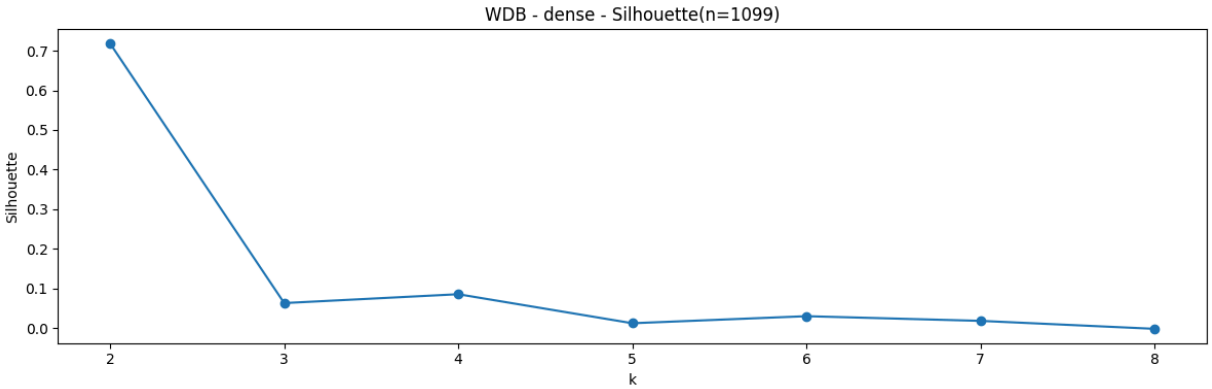
In [48]:
```
# Experimento WDB para n=1184
X_wdb_spec_n2, X_wdb_input_n2, df_wdb_n2 = load_class_spectra_DRACULA("WDB", n_samp

flatten_wdb_n2, dense_wdb_n2, dense1_wdb_n2, dense2_wdb_n2 = extract_layer_features

repr_wdb_32_n2 = reduce_all_to_32(flatten_wdb_n2, dense_wdb_n2, dense1_wdb_n2, dens
```

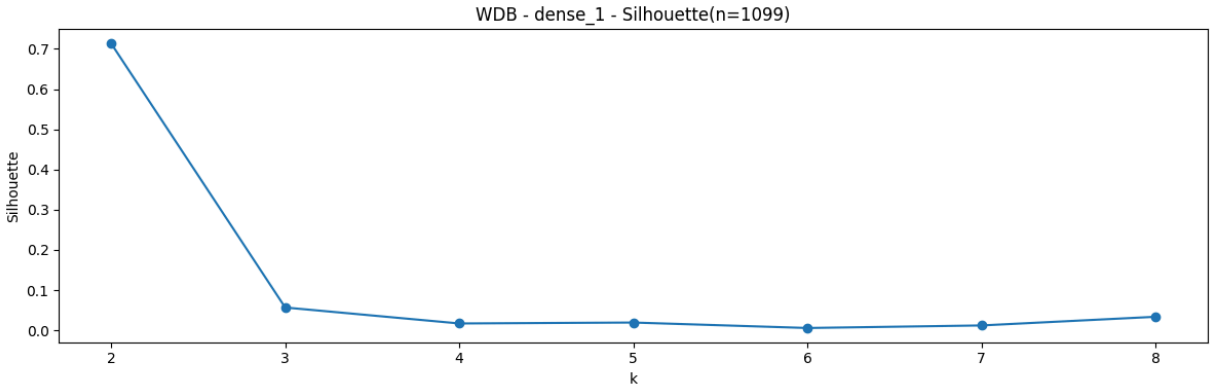
```
WDB: disponibles con match a .dat = 1184
WDB: espectros válidos finales = 1183
19/19 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step
19/19 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step
19/19 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step
19/19 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step
flatten: (1183, 208)
dense: (1183, 128)
dense_1: (1183, 64)
dense_2: (1183, 32)
flatten_32: (1183, 32)
dense_32: (1183, 32)
dense1_32: (1183, 32)
dense2_32: (1183, 32)
```

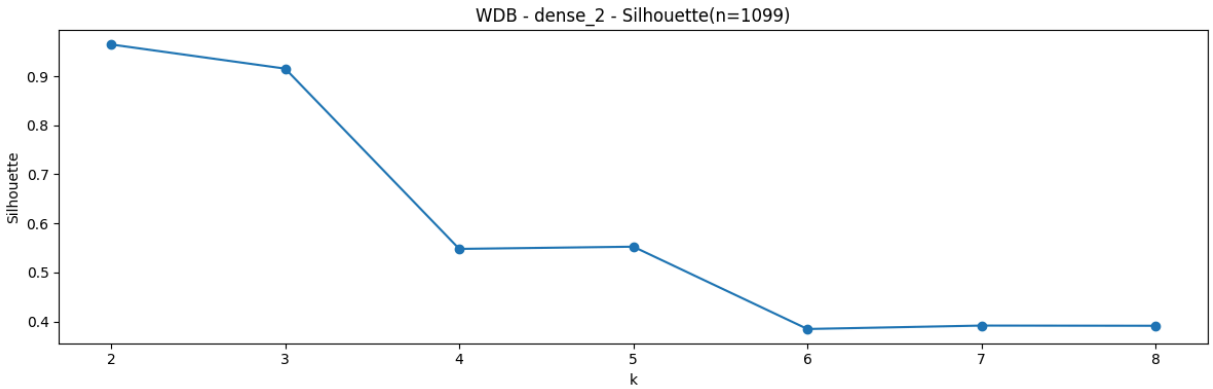
Evaluación K para n = 1100

In [49]:
```
evaluate_kmeans_k(repr_wdb_32["flatten"], "WDB", "flatten")
evaluate_kmeans_k(repr_wdb_32["dense"], "WDB", "dense")
evaluate_kmeans_k(repr_wdb_32["dense_1"], "WDB", "dense_1")
evaluate_kmeans_k(repr_wdb_32["dense_2"], "WDB", "dense_2")
```



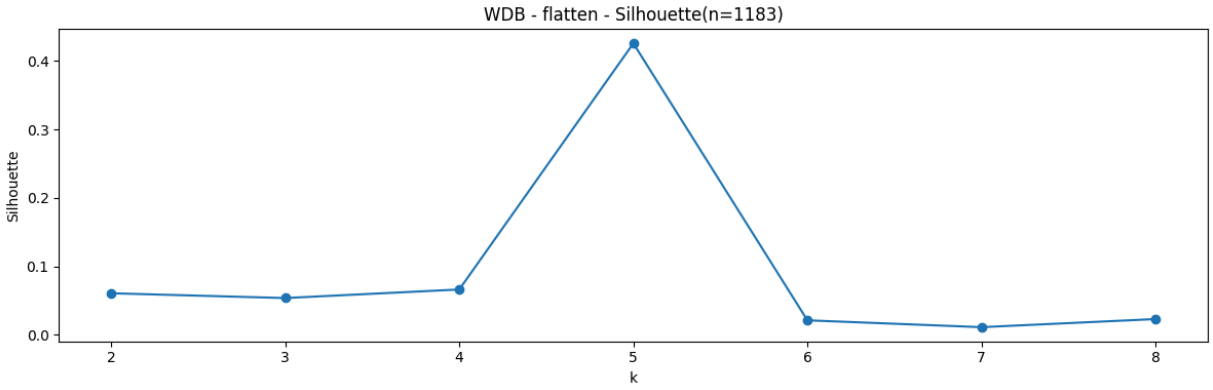


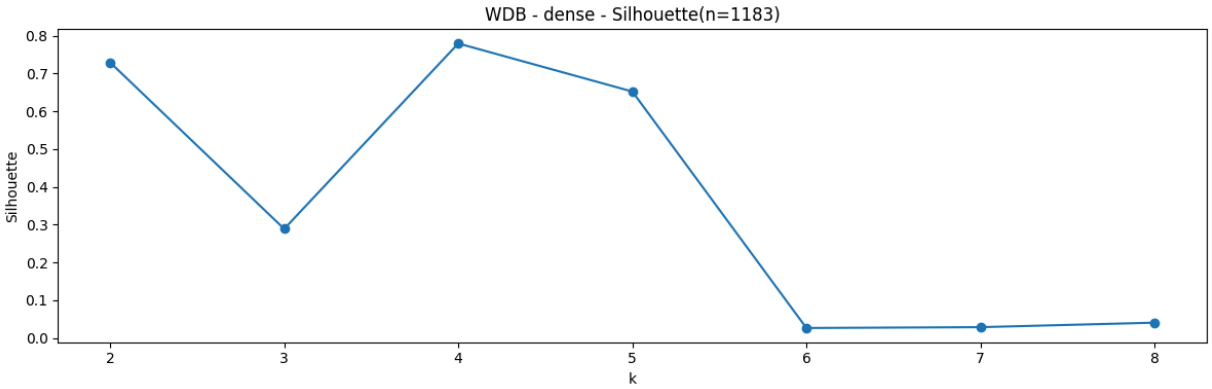


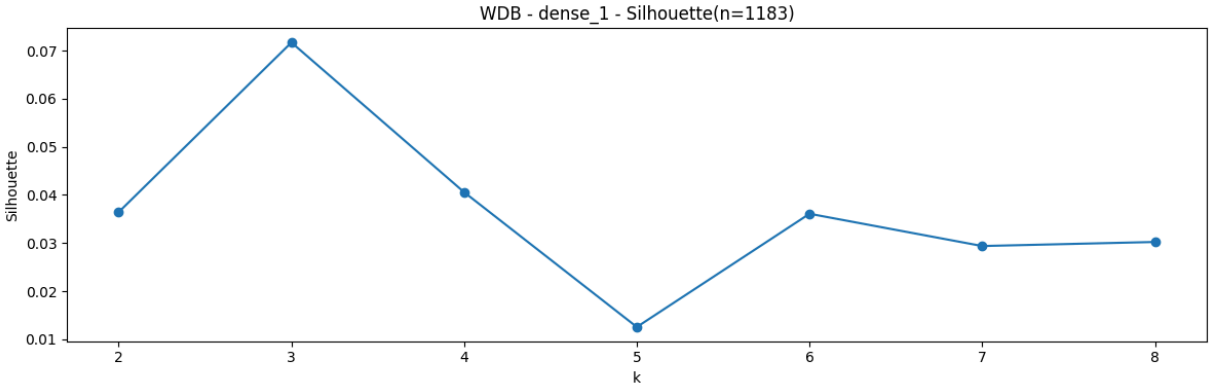


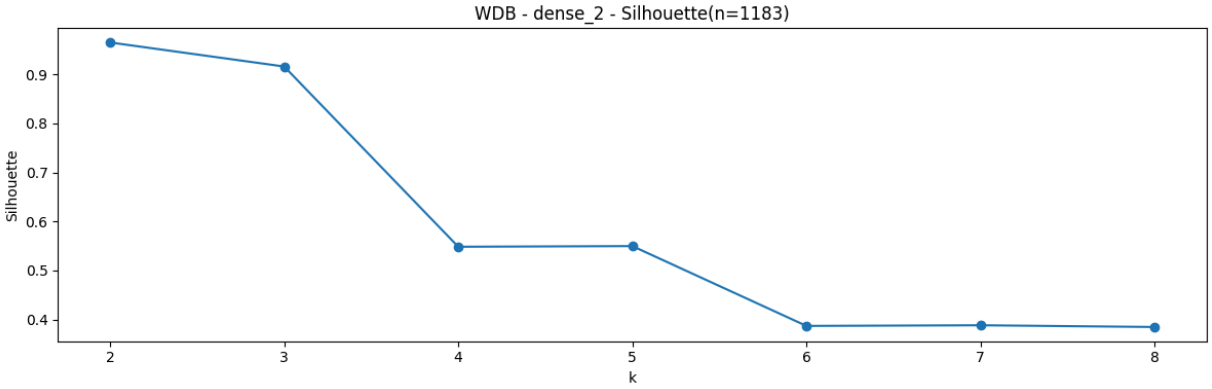
Evaluación para n = 1184

In [50]:
```
evaluate_kmeans_k(repr_wdb_32_n2["flatten"], "WDB", "flatten")
evaluate_kmeans_k(repr_wdb_32_n2["dense"], "WDB", "dense")
evaluate_kmeans_k(repr_wdb_32_n2["dense_1"], "WDB", "dense_1")
evaluate_kmeans_k(repr_wdb_32_n2["dense_2"], "WDB", "dense_2")
```









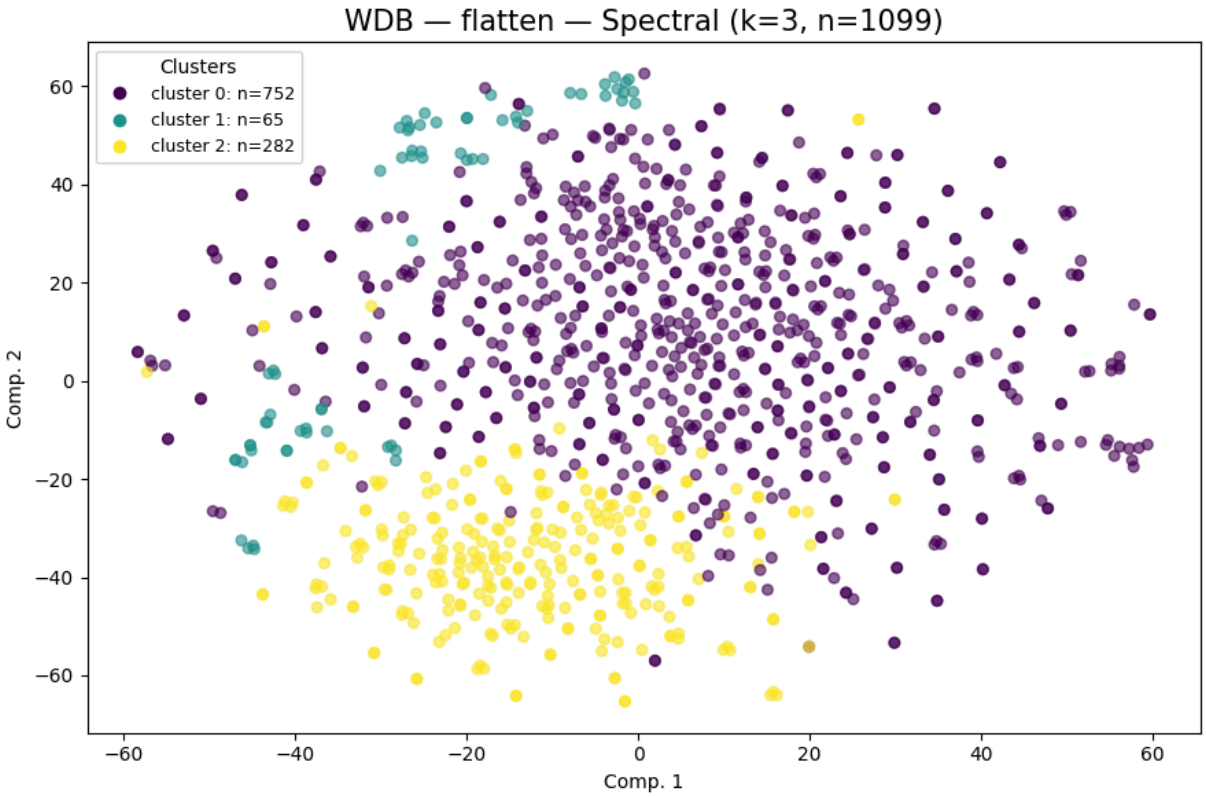
Gráficos t-SNE y PCA + Clustering

Resultados `capa flatten`, con Spectral Clustering + tSNE y PCA para n=1100

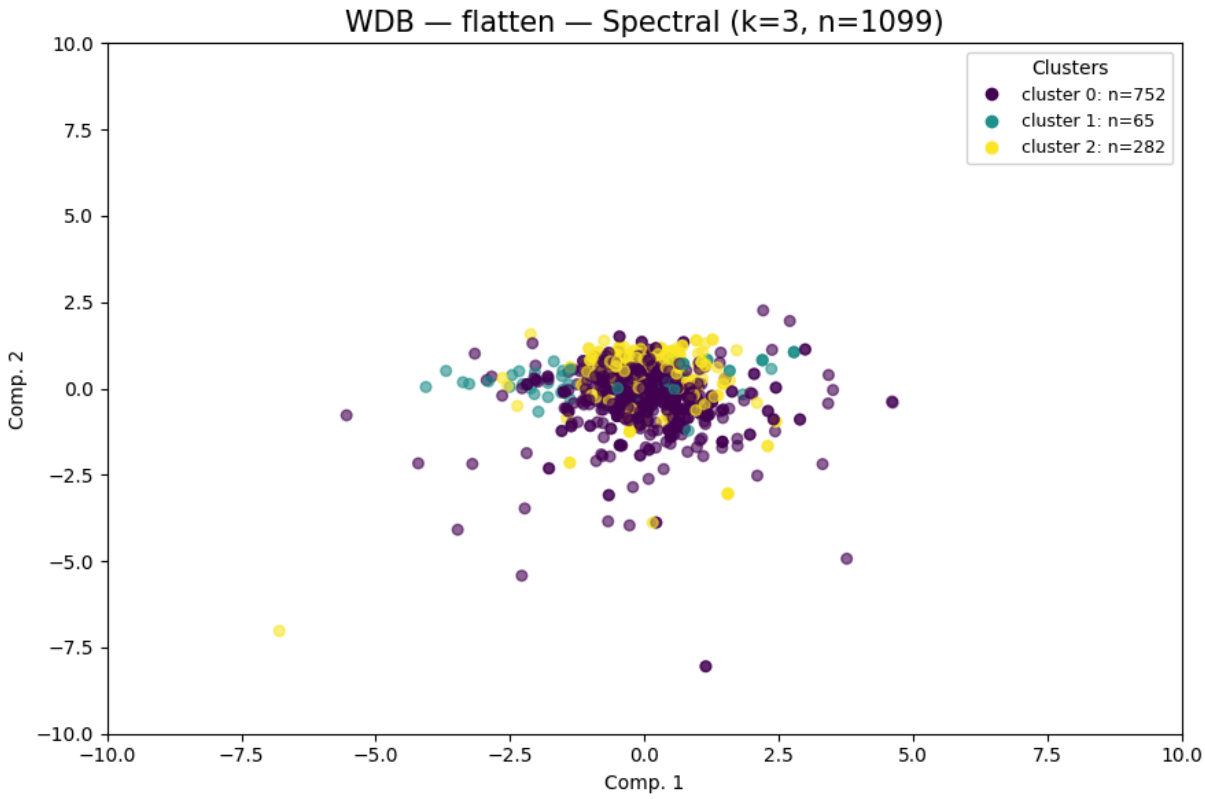
Spectral Clustering k = 3

In [51]:
```
flatten_wdb_k3=cluster_plot_Spect(
            repr_wdb_32["flatten"],
            "WDB",
            "flatten",
            spectral_k=3,
            vis_method="tsne",
            container=cluster_labels_all
        )
```

```
labels_flatten_k3_wdb = flatten_wdb_k3["spectral"]
cluster_plot_Spect(
            repr_wdb_32["flatten"],
            "WDB",
            "flatten",
            spectral_k=3,
            vis_method="pca",
            xlim=(-10,10), ylim=(-10,10)
        )
```




```
┌──────────────────────────────────────────────────────────────────────┐
│  WDB | flatten | spectral | k=3                                      │
│  clusters=3  noise=0  sil=0.0081  DB=4.0160  CH=20.00                │
└──────────────────────────────────────────────────────────────────────┘
```



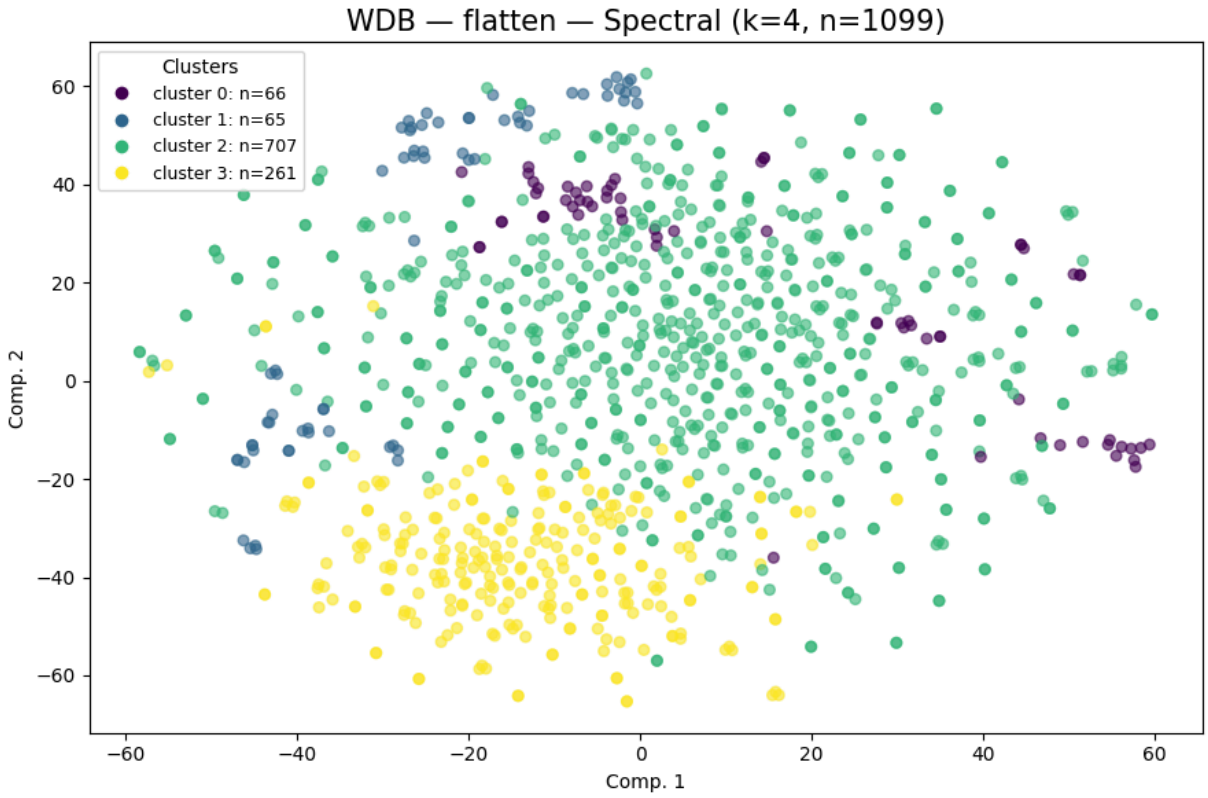

```
┌──────────────────────────────────────────────────────────────────────┐
│  WDB | flatten | spectral | k=3                                      │
│  clusters=3  noise=0  sil=0.0081  DB=4.0160  CH=20.00                │
└──────────────────────────────────────────────────────────────────────┘
```

Out[51]:
```
{'spectral': array([0, 2, 0, ..., 0, 1, 0]),
 'metrics': {'n_clusters': 3,
  'n_noise': 0,
  'silhouette': 0.008147,
  'davies_bouldin': 4.015992,
  'calinski_harabasz': 20.005}}
```

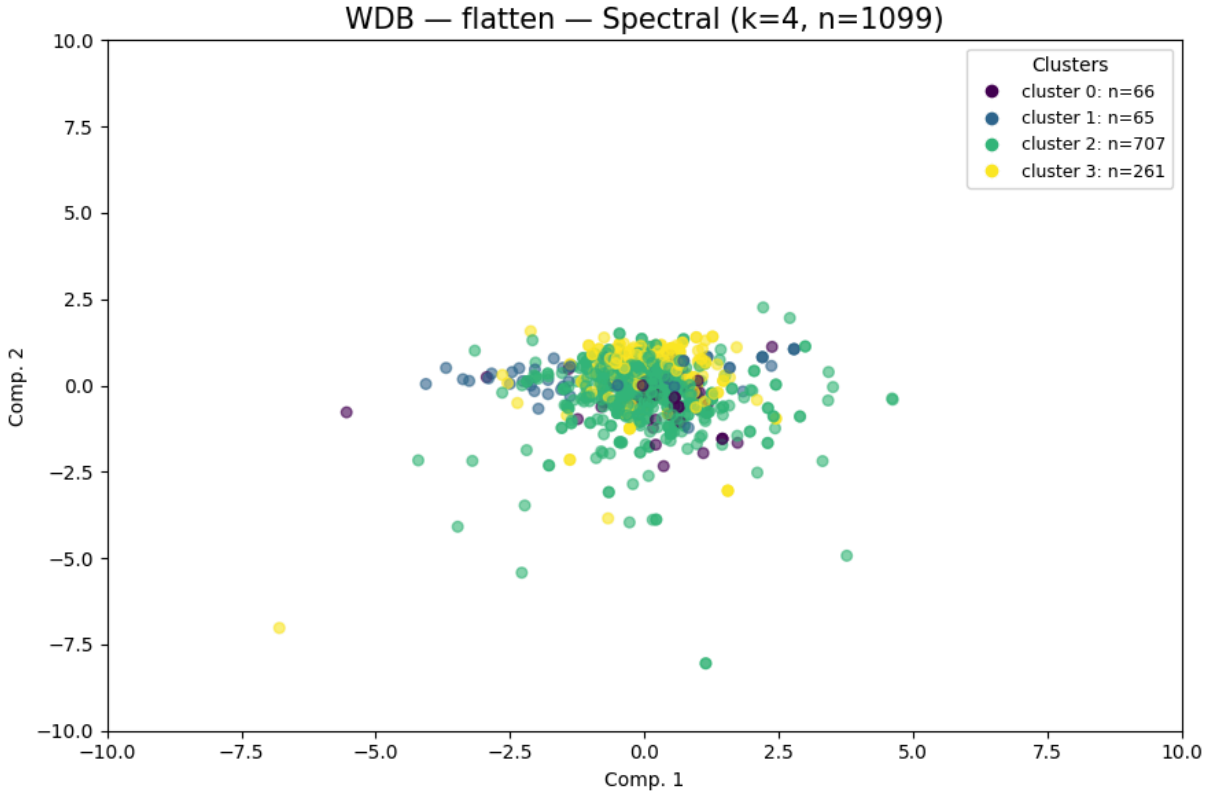
Spectral Clustering k = 4

In [52]:
```
flatten_wdb_k4=cluster_plot_Spect(
            repr_wdb_32["flatten"],
            "WDB",
            "flatten",
            spectral_k=4,
            vis_method="tsne",
            container=cluster_labels_all
        )

labels_flatten_k4_wdb = flatten_wdb_k4["spectral"]
cluster_plot_Spect(
            repr_wdb_32["flatten"],
            "WDB",
            "flatten",
            spectral_k=4,
            vis_method="pca",
```

```
            xlim=(-10,10), ylim=(-10,10)
        )
```




```
┌──────────────────────────────────────────────────────────────────────┐
│  WDB | flatten | spectral | k=4                                      │
│  clusters=4  noise=0  sil=0.0134  DB=3.7985  CH=18.00                │
└──────────────────────────────────────────────────────────────────────┘
```

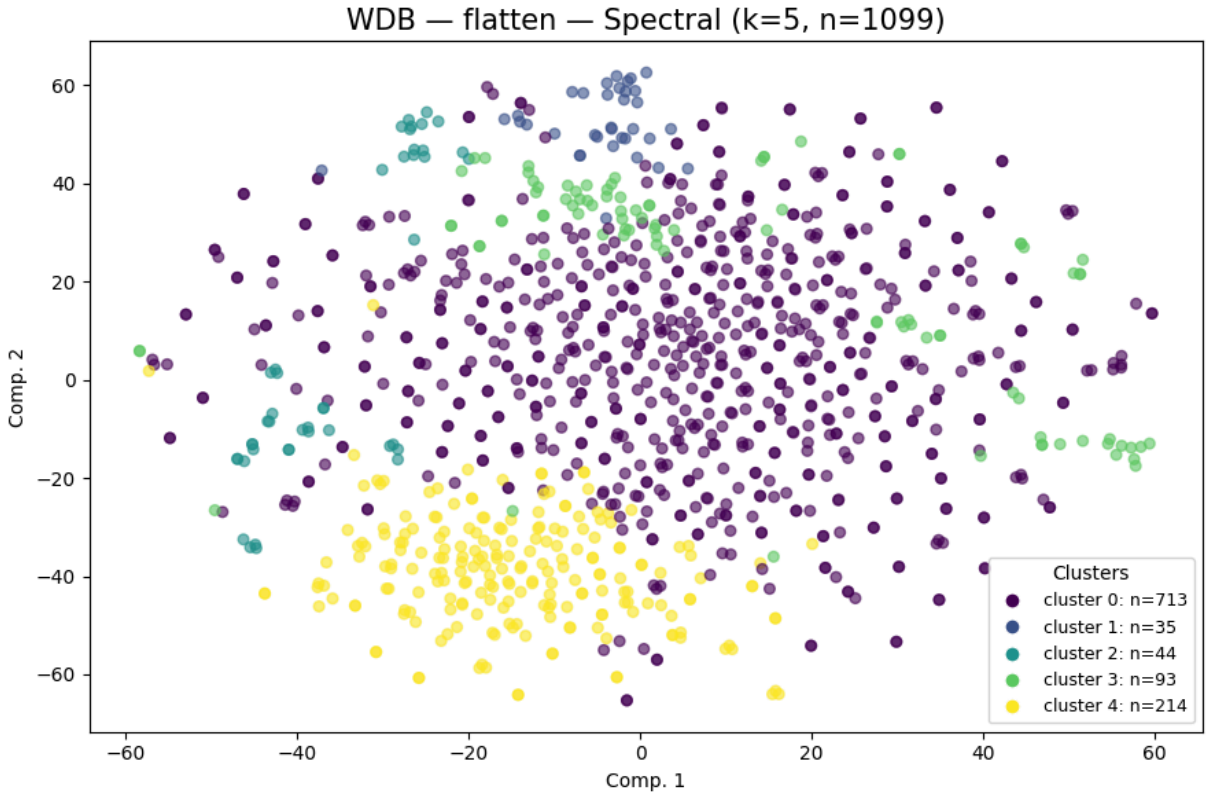


```
┌──────────────────────────────────────────────────────────────────────────┐
│  WDB | flatten | spectral | k=4
│  clusters=4  noise=0  sil=0.0134  DB=3.7985  CH=18.00
└──────────────────────────────────────────────────────────────────────────┘
```

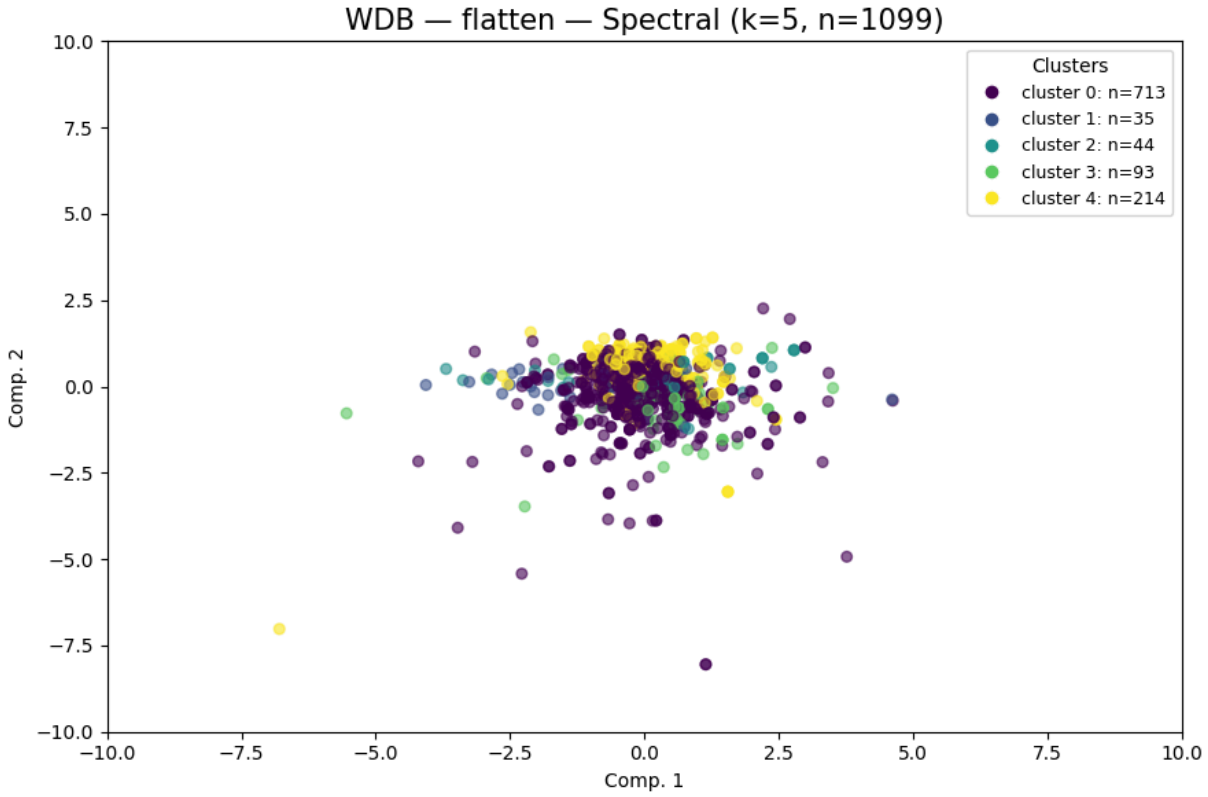
Out[52]:
```
{'spectral': array([2, 3, 2, ..., 2, 1, 2]),
 'metrics': {'n_clusters': 4,
  'n_noise': 0,
  'silhouette': 0.013448,
  'davies_bouldin': 3.798504,
  'calinski_harabasz': 18.0022}}
```

Spectral Clustering k = 5

In [53]:
```
flatten_wdb_k5=cluster_plot_Spect(
            repr_wdb_32["flatten"],
            "WDB",
            "flatten",
            spectral_k=5,
            vis_method="tsne",
            container=cluster_labels_all
        )

labels_flatten_k5_wdb = flatten_wdb_k5["spectral"]
cluster_plot_Spect(
            repr_wdb_32["flatten"],
            "WDB",
            "flatten",
            spectral_k=5,
            vis_method="pca",
            xlim=(-10,10), ylim=(-10,10)
        )
```




```
┌──────────────────────────────────────────────────────────────┐
│  WDB | flatten | spectral | k=5
│  clusters=5  noise=0  sil=0.0045  DB=3.4836  CH=16.80
└──────────────────────────────────────────────────────────────┘
```

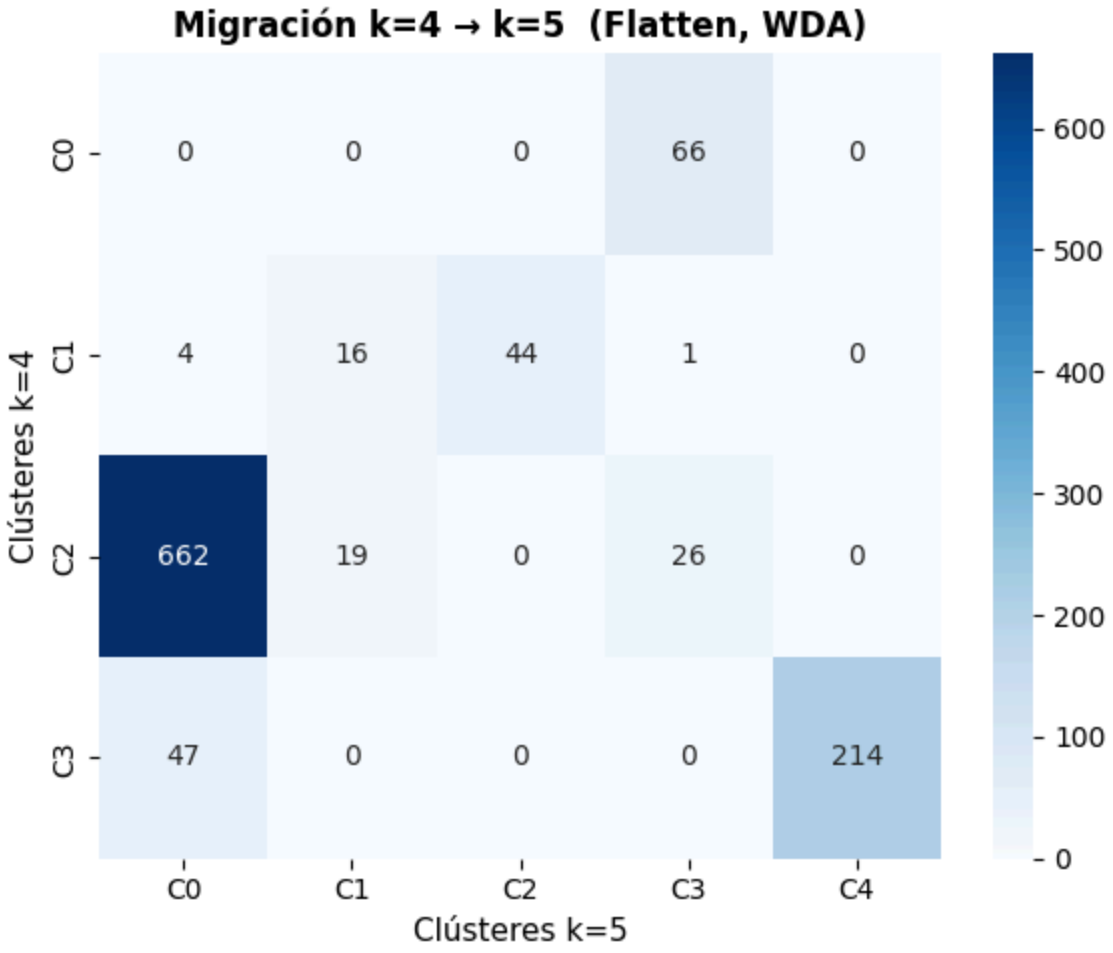


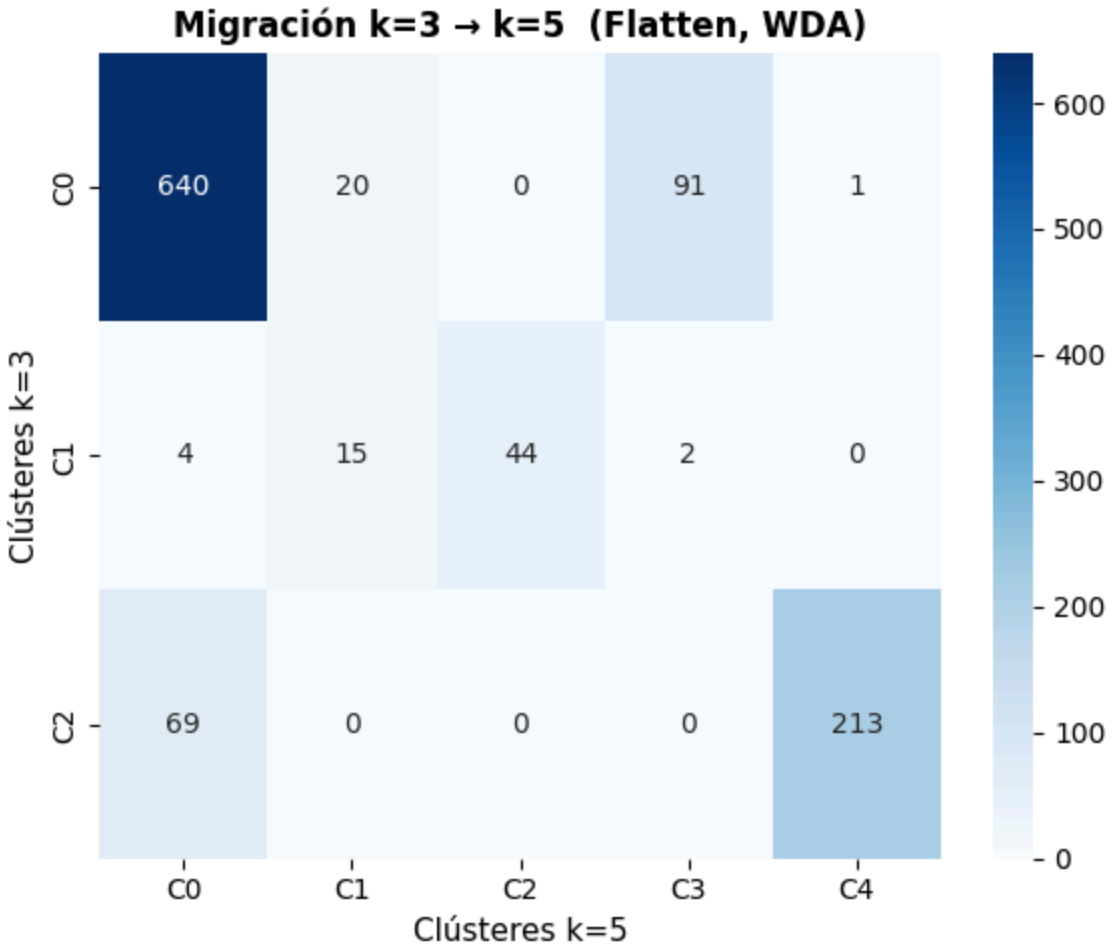
```
┌──────────────────────────────────────────────────────────────────────────┐
│  WDB | flatten | spectral | k=5                                          │
│  clusters=5  noise=0  sil=0.0045  DB=3.4836  CH=16.80                    │
└──────────────────────────────────────────────────────────────────────────┘
```

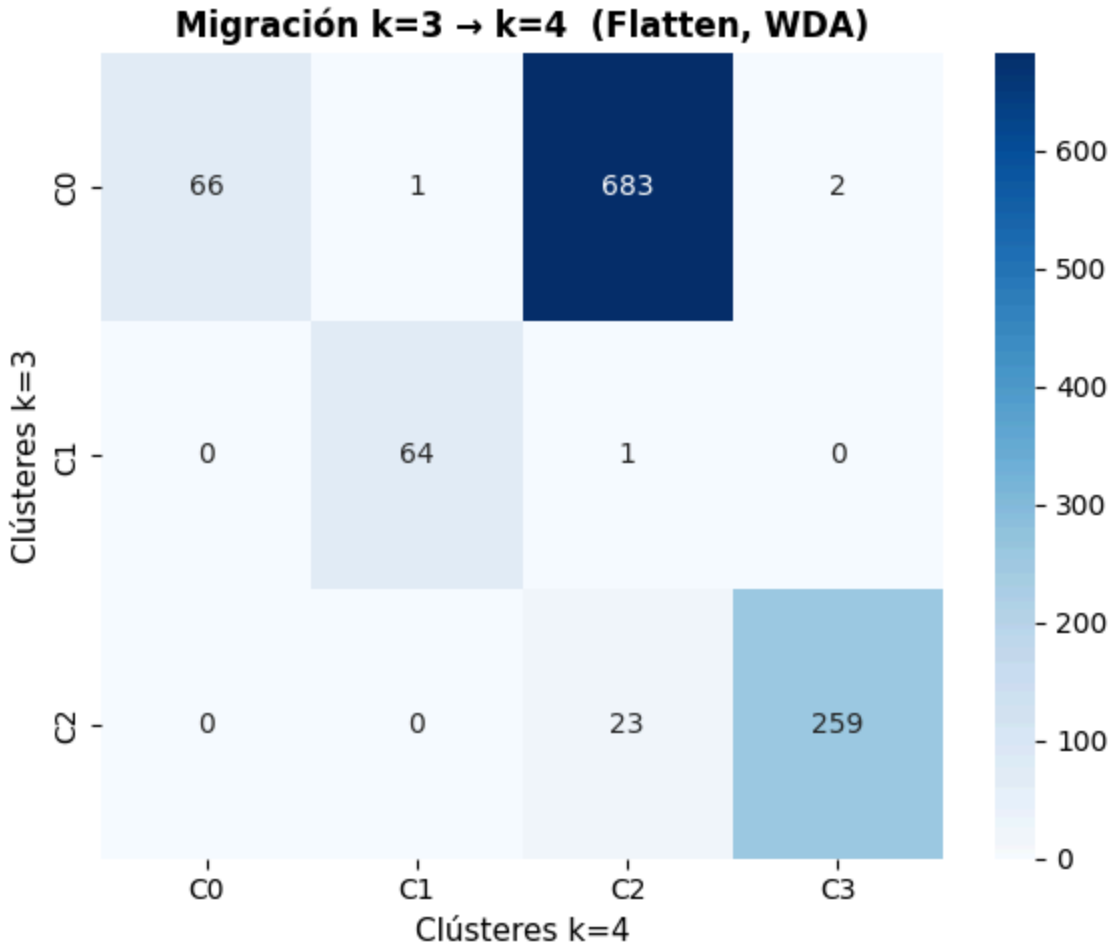
Out[53]:
```
{'spectral': array([1, 4, 0, ..., 0, 2, 0]),
 'metrics': {'n_clusters': 5,
  'n_noise': 0,
  'silhouette': 0.004542,
  'davies_bouldin': 3.483588,
  'calinski_harabasz': 16.7954}}
```

Análisis de migración de clústeres

In [54]:
```
plot_migration_matrix(labels_flatten_k4_wdb, labels_flatten_k5_wdb, ka=4, kb=5)
plot_migration_matrix(labels_flatten_k3_wdb, labels_flatten_k5_wdb, ka=3, kb=5)
plot_migration_matrix(labels_flatten_k3_wdb, labels_flatten_k4_wdb, ka=3, kb=4)
```



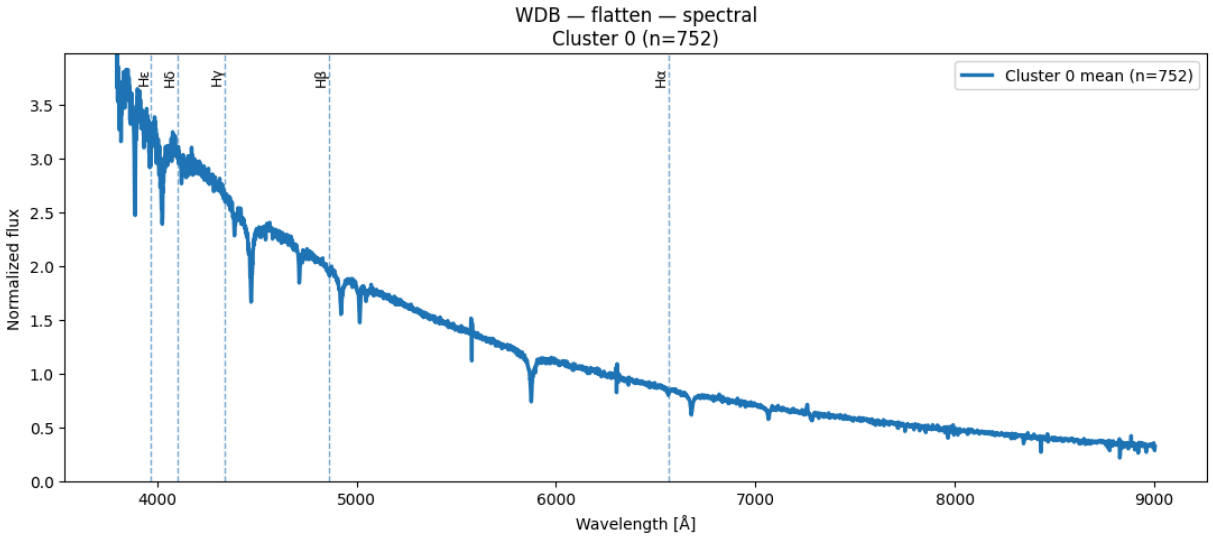

Migración k=3 → k=5 (Flatten, WDA)
Clústeres k=3
C0
C1
C2
640
20
0
91
1
4
15
44
2
0
69
0
0
0
213
C0
C1
C2
C3
C4
Clústeres k=5
600
500
400
300
200
100
0

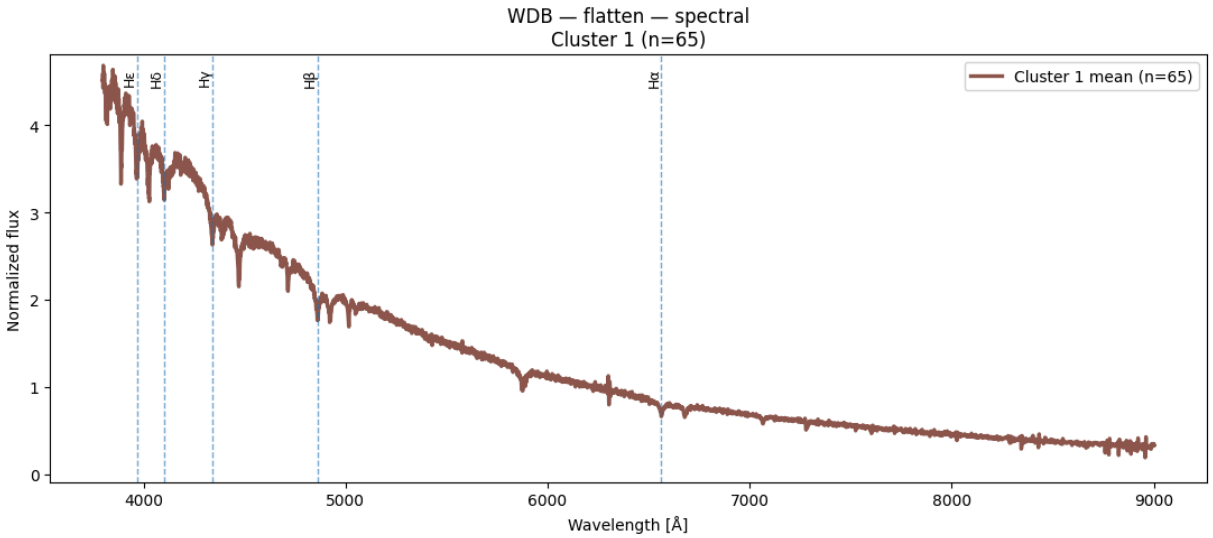


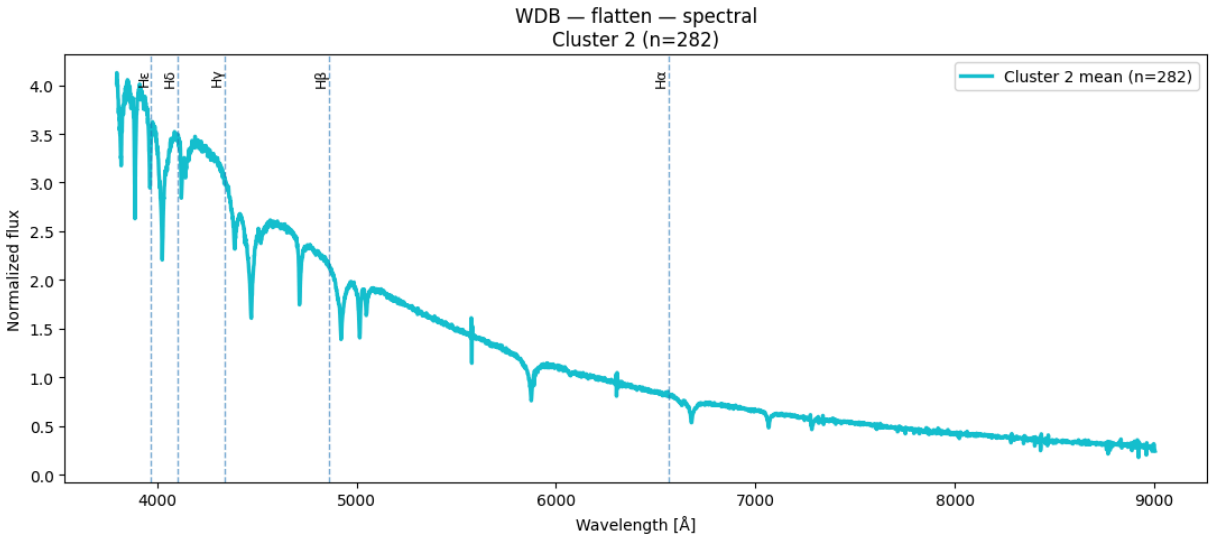
Espectros obtenidos de Spectral Clustering para k=3 y n=1100

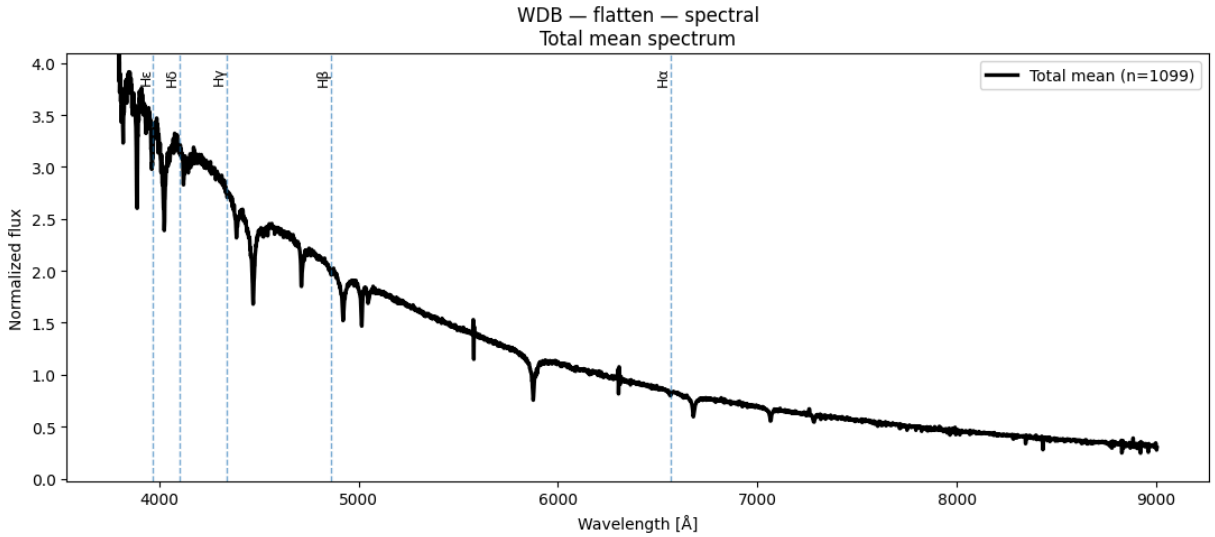
In [55]:
```
result = plot_spectra_spectral(X_wdb_spec, labels_flatten_k3_wdb, W_wdb[0],
                               class_name="WDB", layer_name="flatten")
```

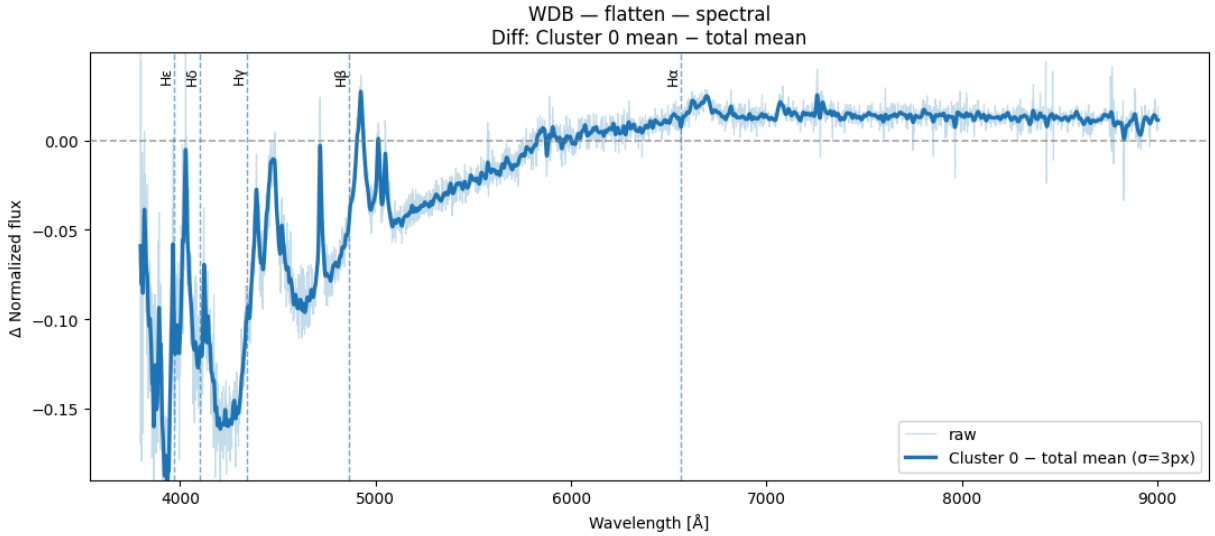
```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```

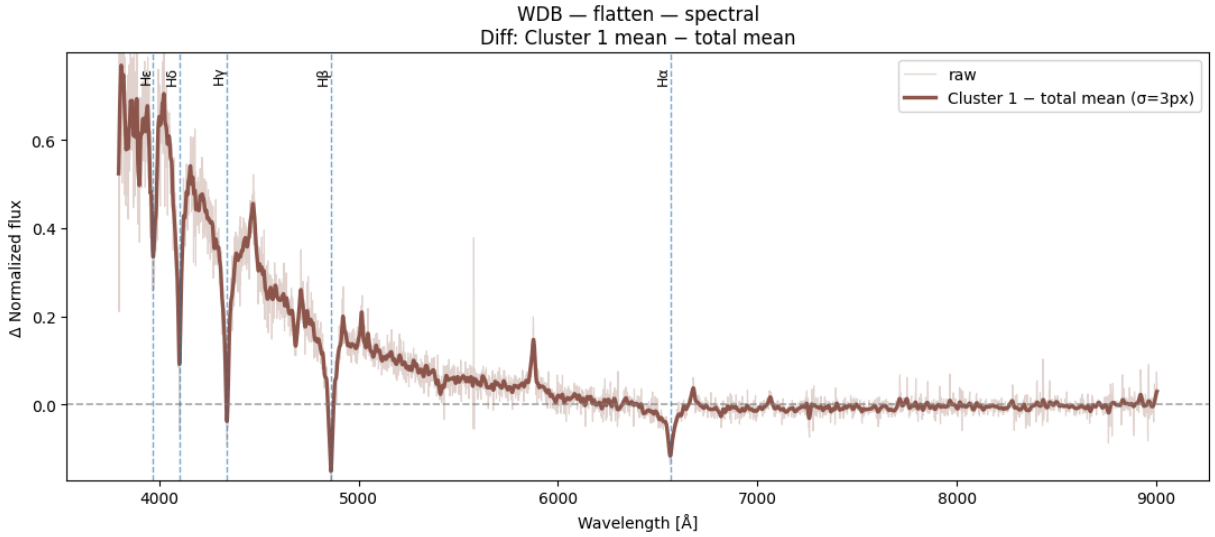

WDB — flatten — spectral
Cluster 0 (n=752)
Cluster 0 mean (n=752)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

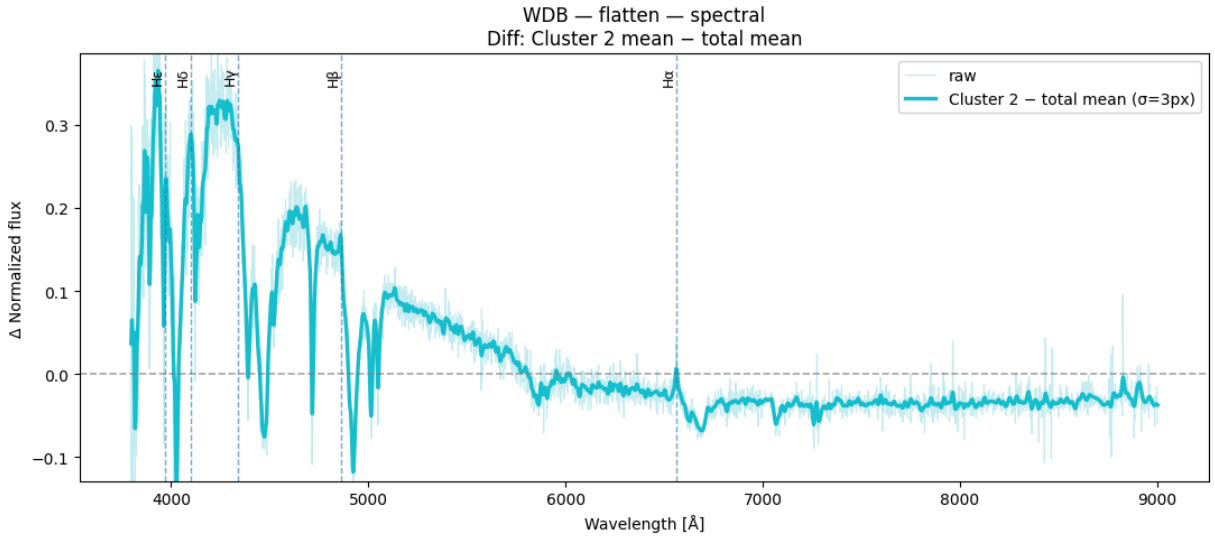

WDB — flatten — spectral
Cluster 1 (n=65)
Cluster 1 mean (n=65)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

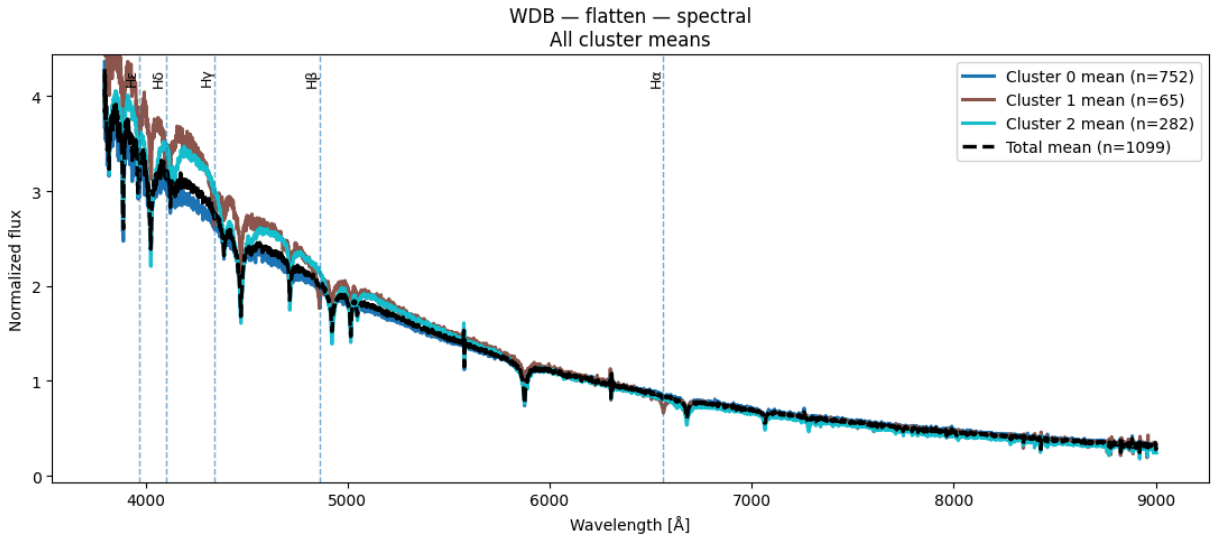

WDB — flatten — spectral
Cluster 2 (n=282)
Cluster 2 mean (n=282)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — flatten — spectral
Total mean spectrum
Total mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — flatten — spectral
Diff: Cluster 0 mean − total mean
raw
Cluster 0 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]


WDB — flatten — spectral
Diff: Cluster 1 mean − total mean
raw
Cluster 1 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]

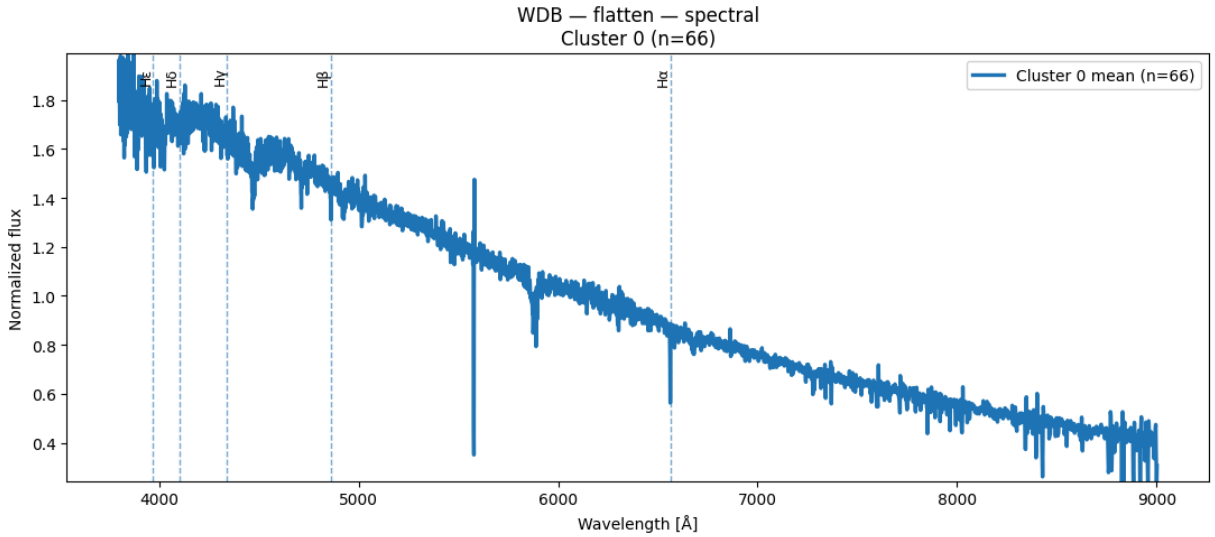


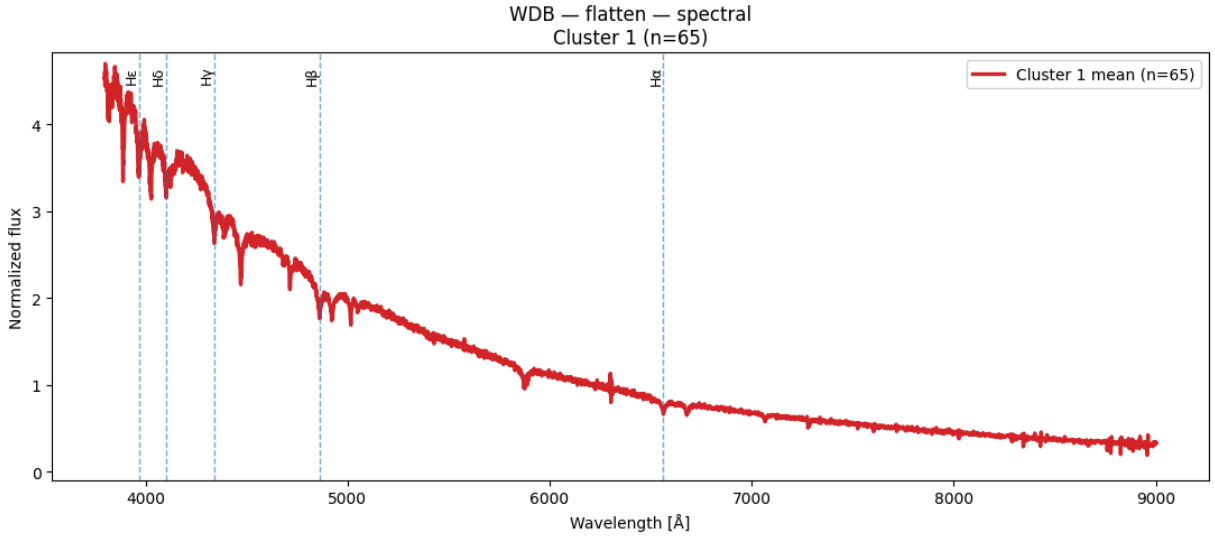


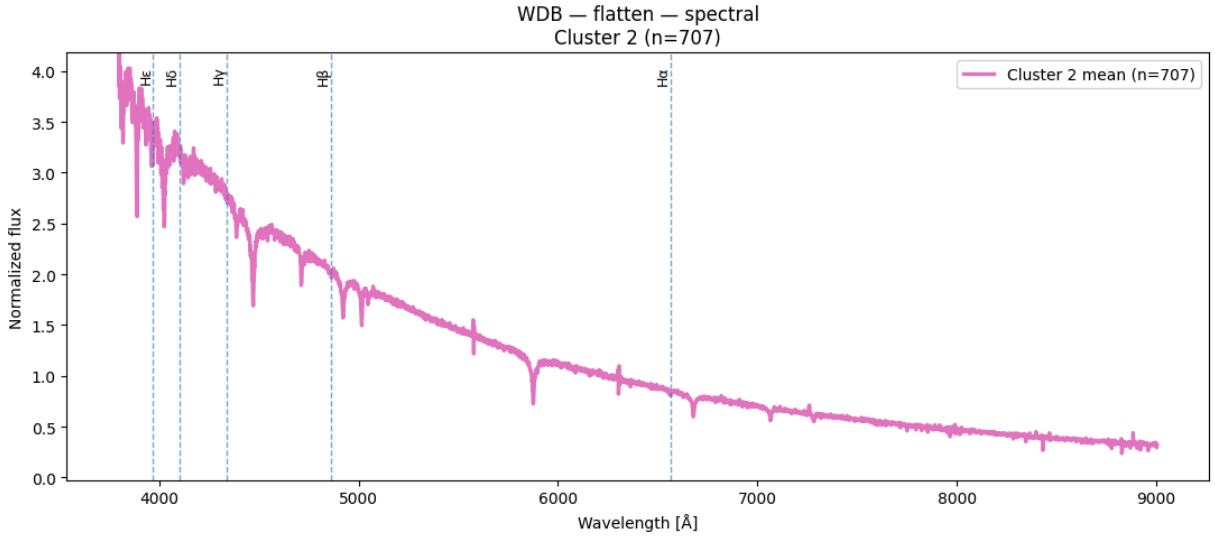
Espectros obtenidos de Spectral Clustering para k = 4

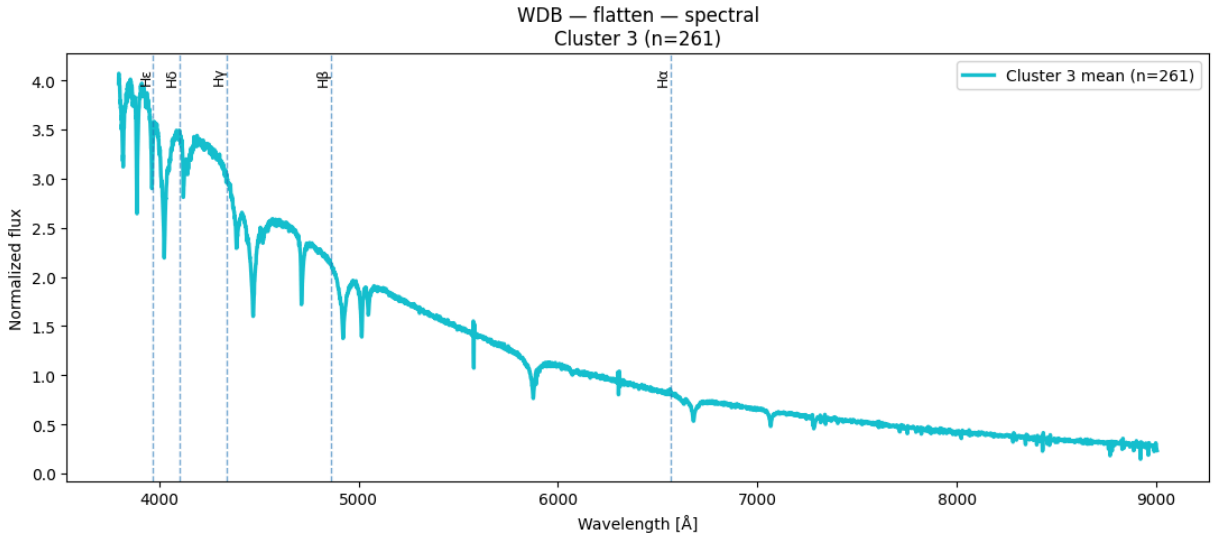
In [56]:
```
result = plot_spectra_spectral(X_wdb_spec, labels_flatten_k4_wdb, W_wdb[0],
                               class_name="WDB", layer_name="flatten")
```

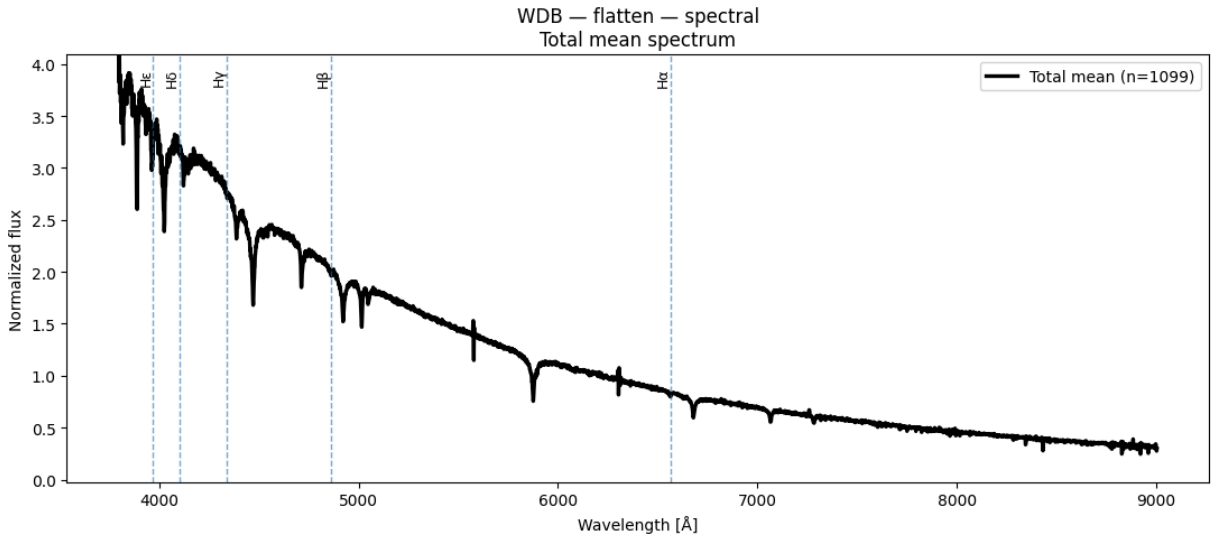
```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```

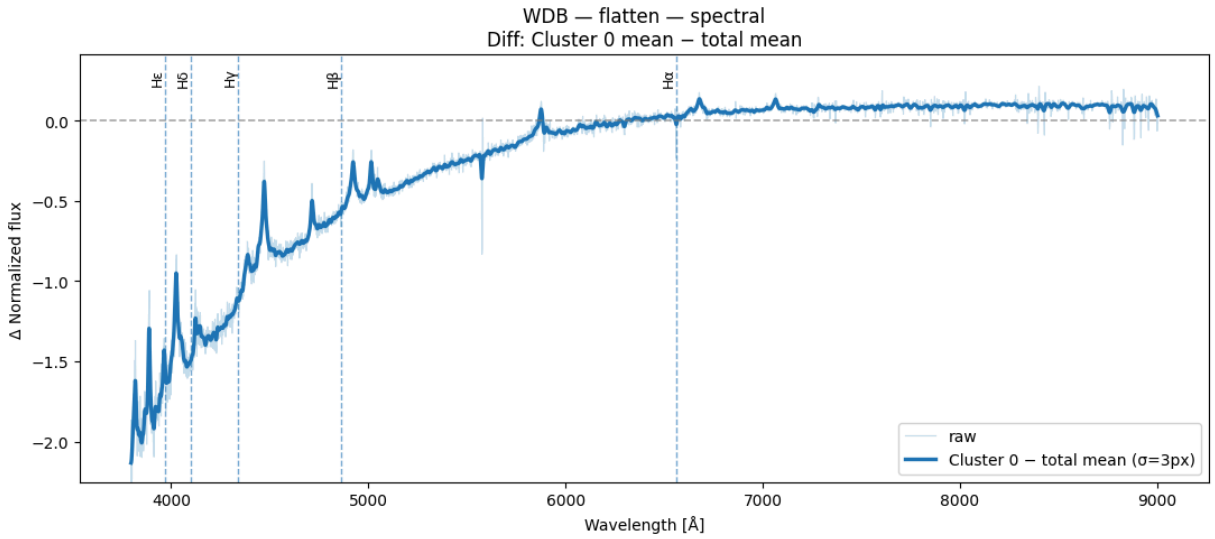

WDB — flatten — spectral
Cluster 0 (n=66)
Cluster 0 mean (n=66)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

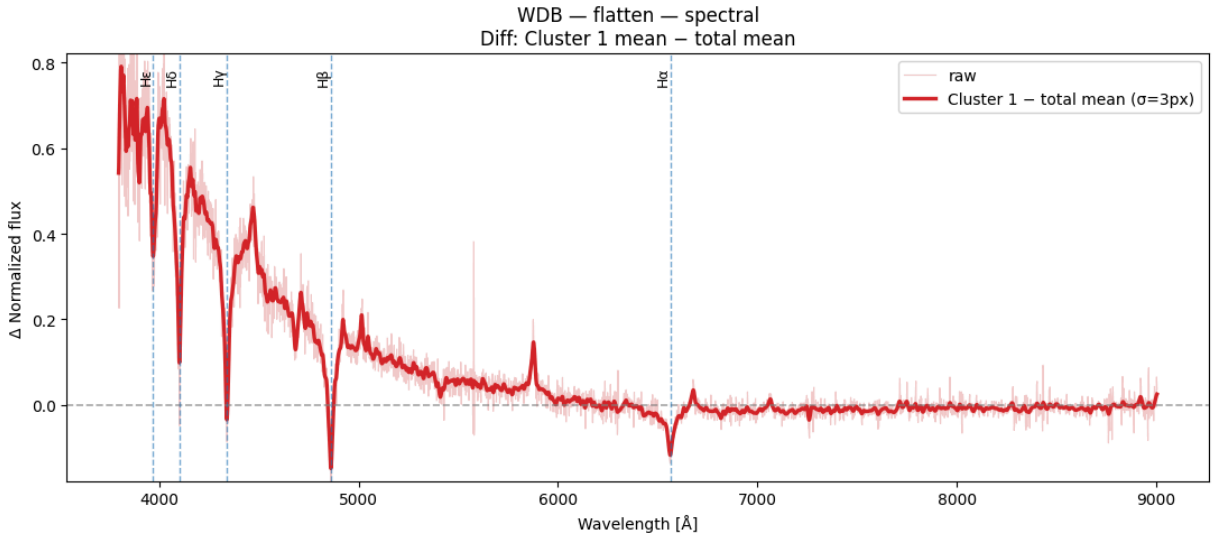

WDB — flatten — spectral
Cluster 1 (n=65)
Cluster 1 mean (n=65)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

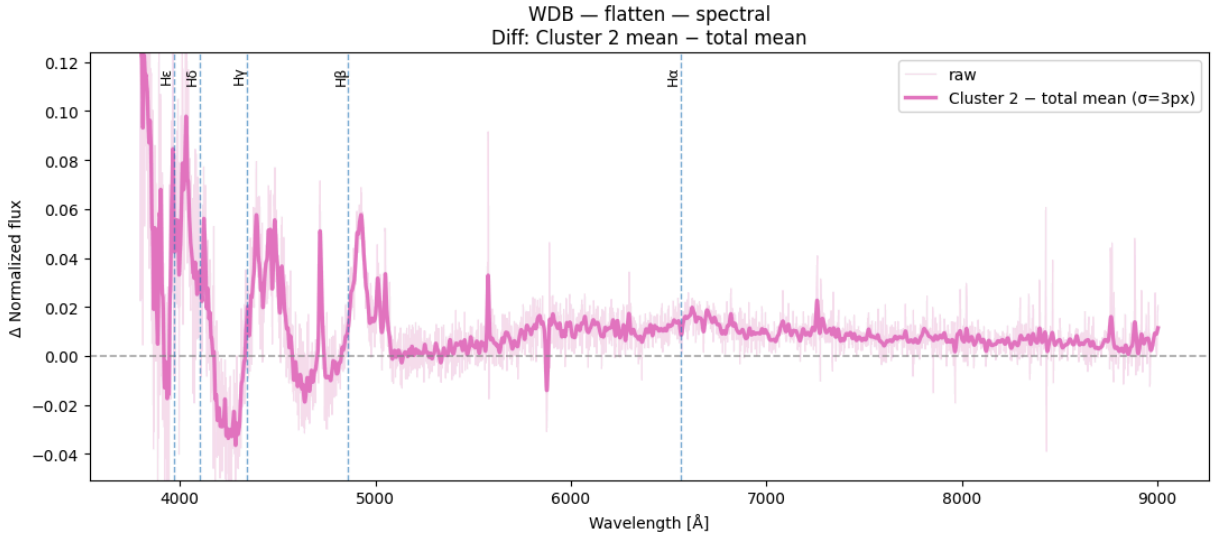

WDB — flatten — spectral
Cluster 2 (n=707)
Cluster 2 mean (n=707)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

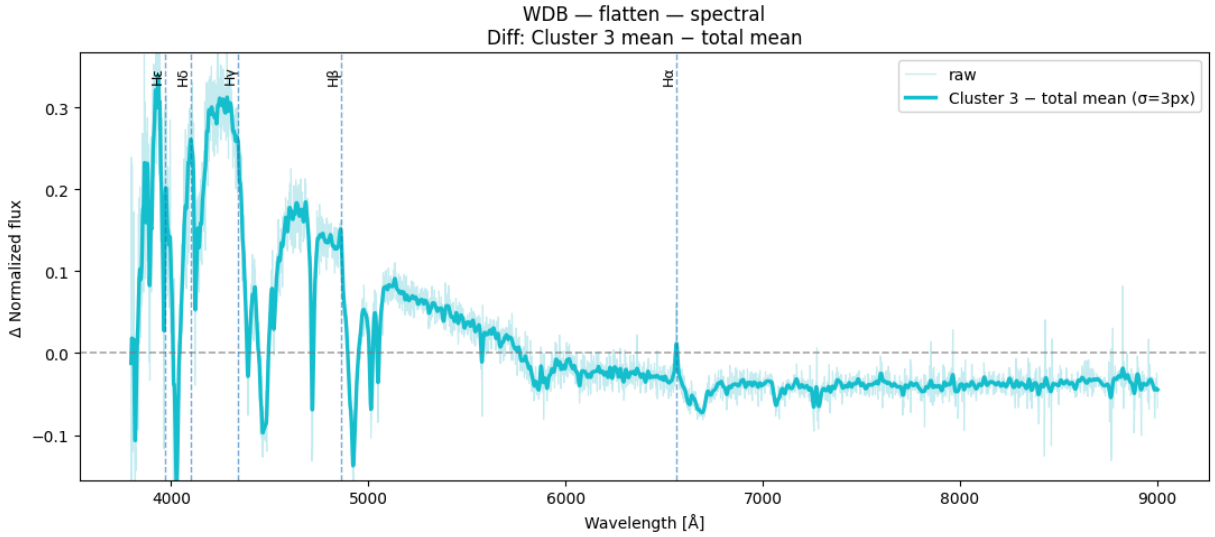

WDB — flatten — spectral
Cluster 3 (n=261)
Cluster 3 mean (n=261)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

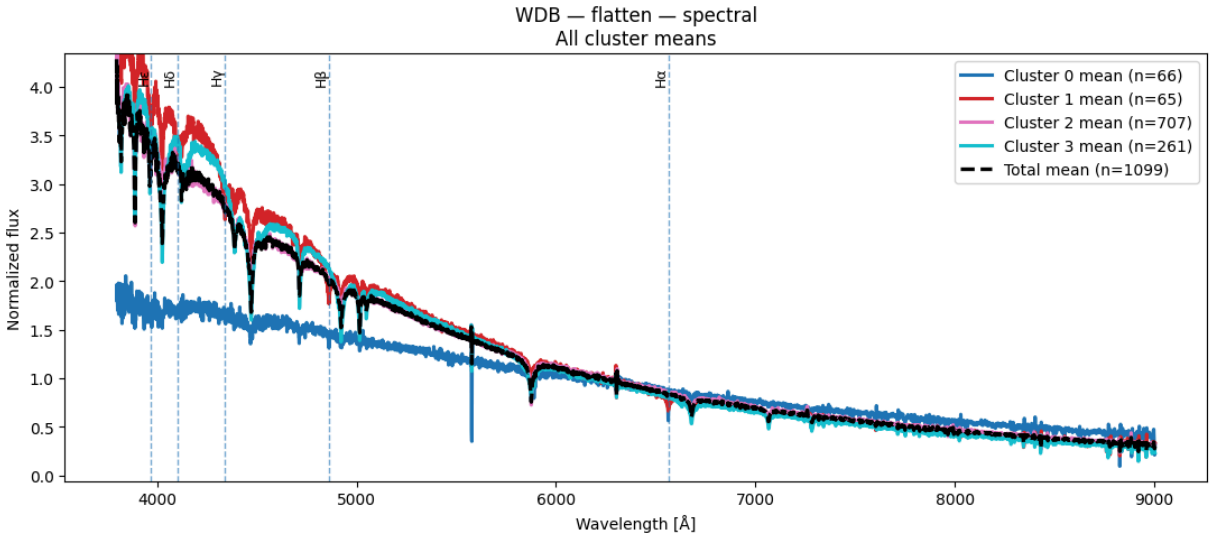

WDB — flatten — spectral
Total mean spectrum
Total mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — flatten — spectral
Diff: Cluster 0 mean − total mean
raw
Cluster 0 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]


WDB — flatten — spectral
Diff: Cluster 1 mean − total mean
raw
Cluster 1 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]


WDB — flatten — spectral
Diff: Cluster 2 mean − total mean
raw
Cluster 2 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]

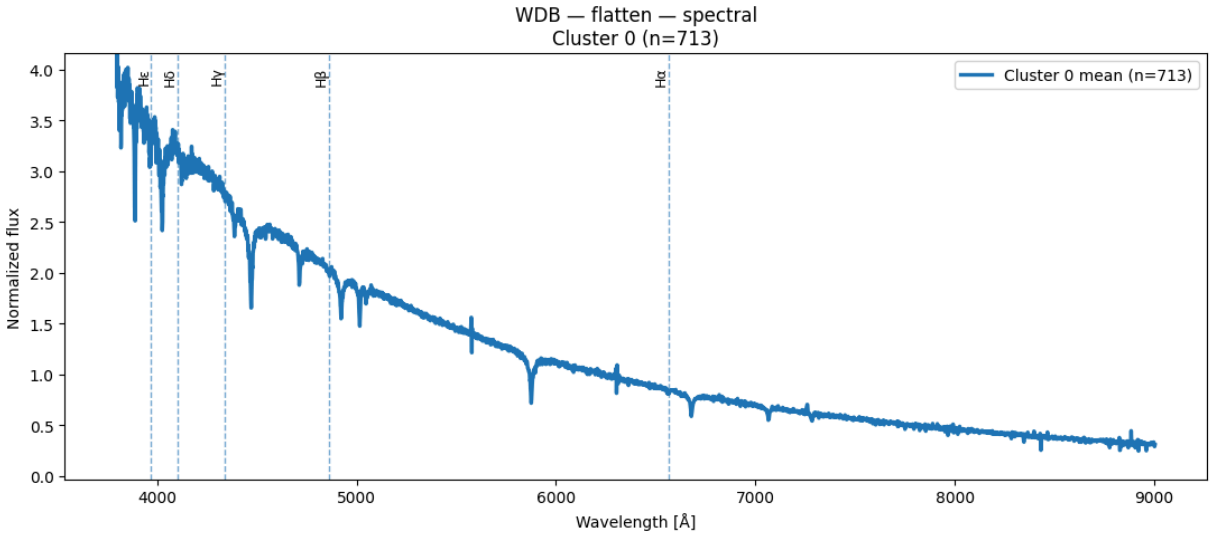

WDB — flatten — spectral
Diff: Cluster 3 mean − total mean
raw
Cluster 3 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]

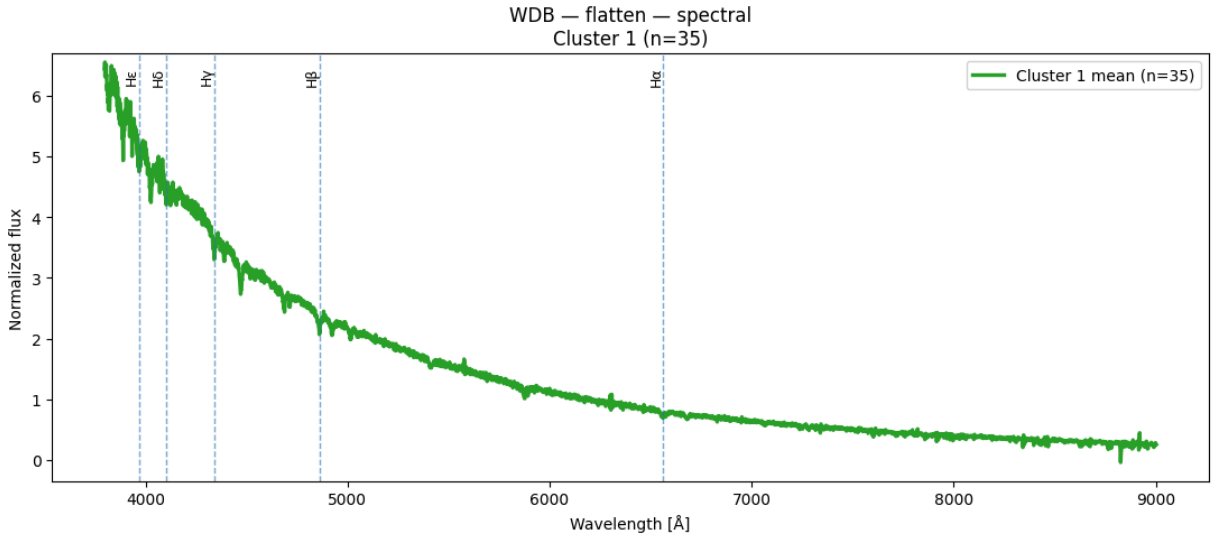


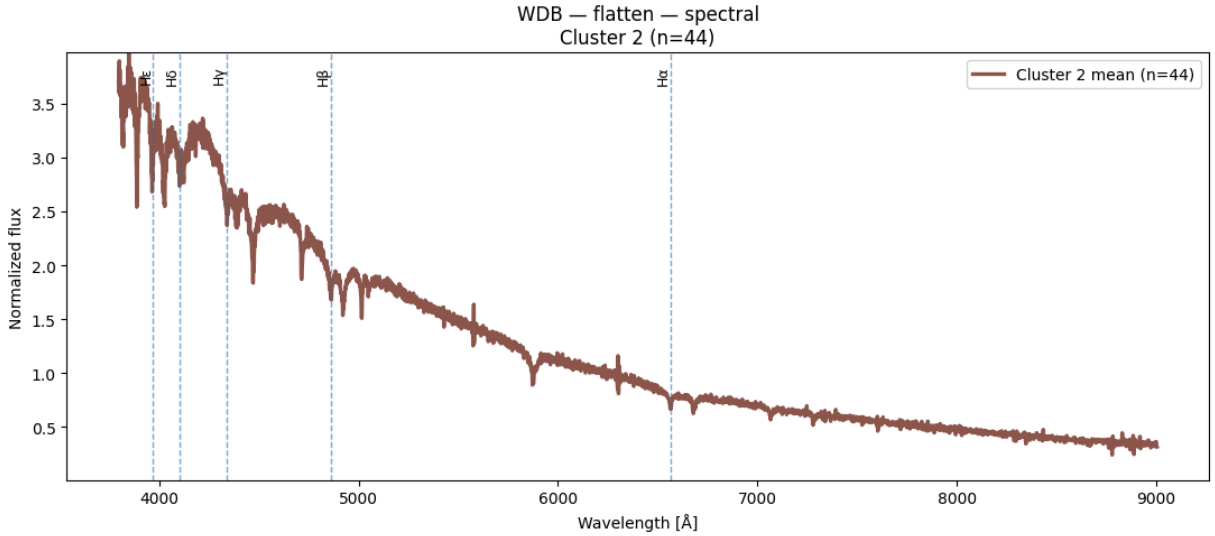
Espectros obtenidos de Spectral Clustering para k = 5 y n = 1100

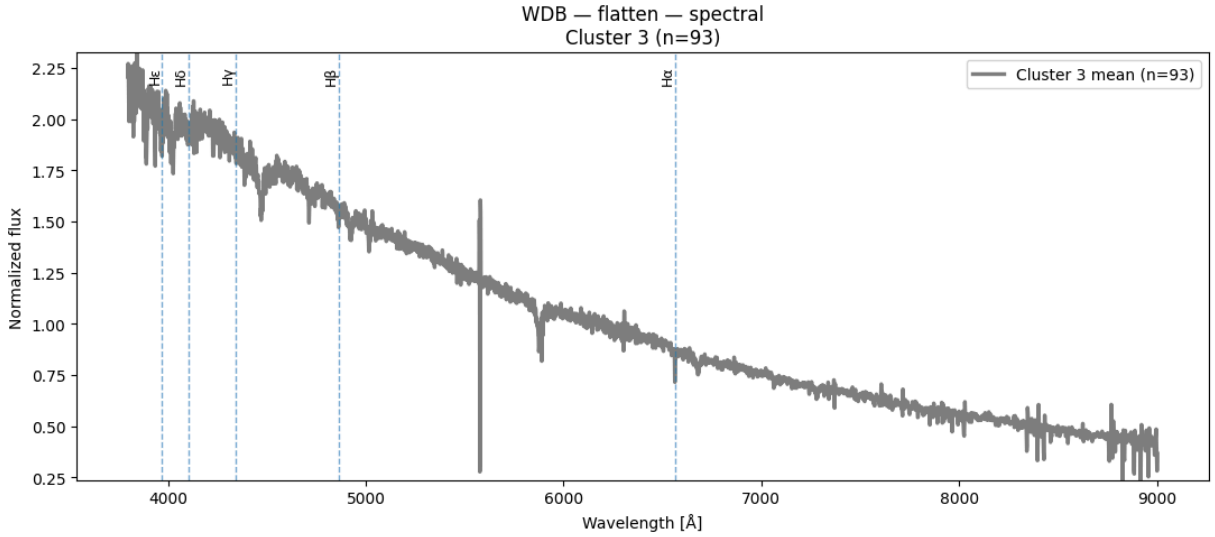
In [57]:
```
result = plot_spectra_spectral(X_wdb_spec, labels_flatten_k5_wdb, W_wdb[0],
                               class_name="WDB", layer_name="flatten")
```

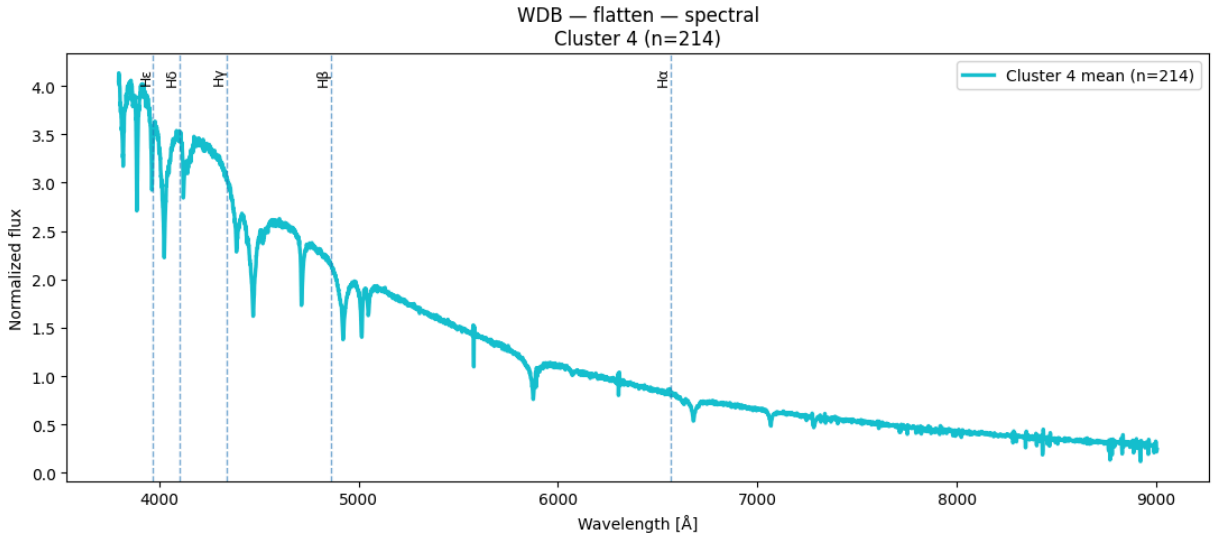
```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```

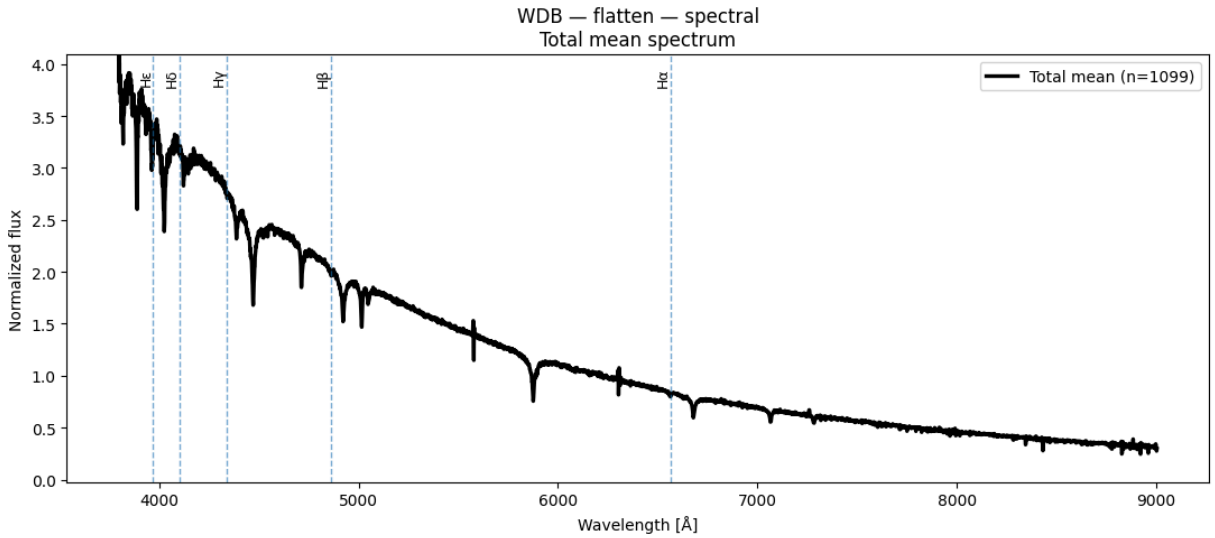


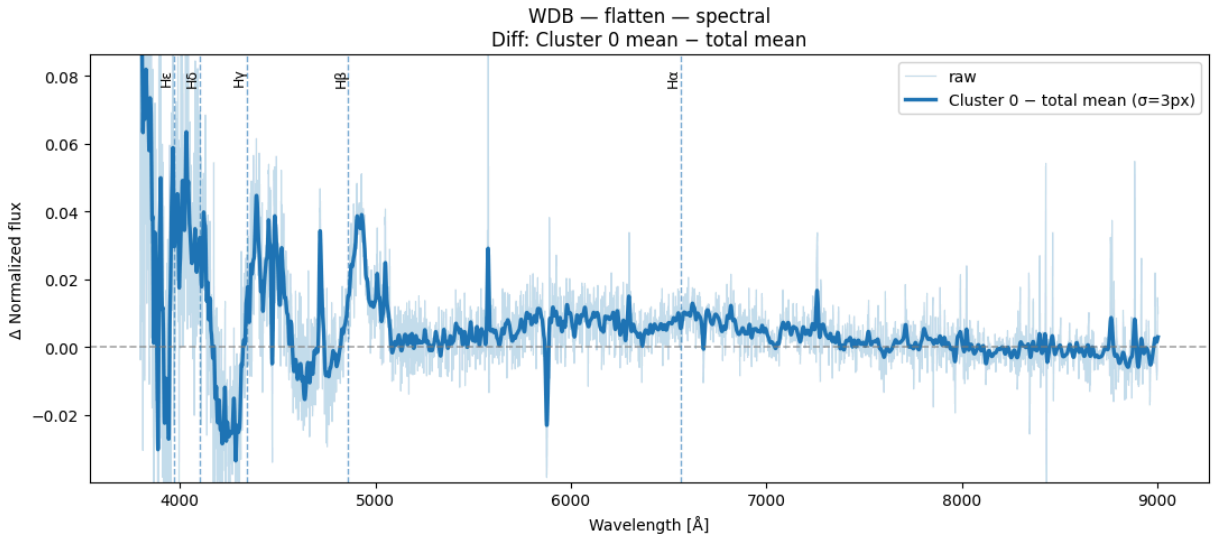

WDB — flatten — spectral
Cluster 1 (n=35)
Cluster 1 mean (n=35)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

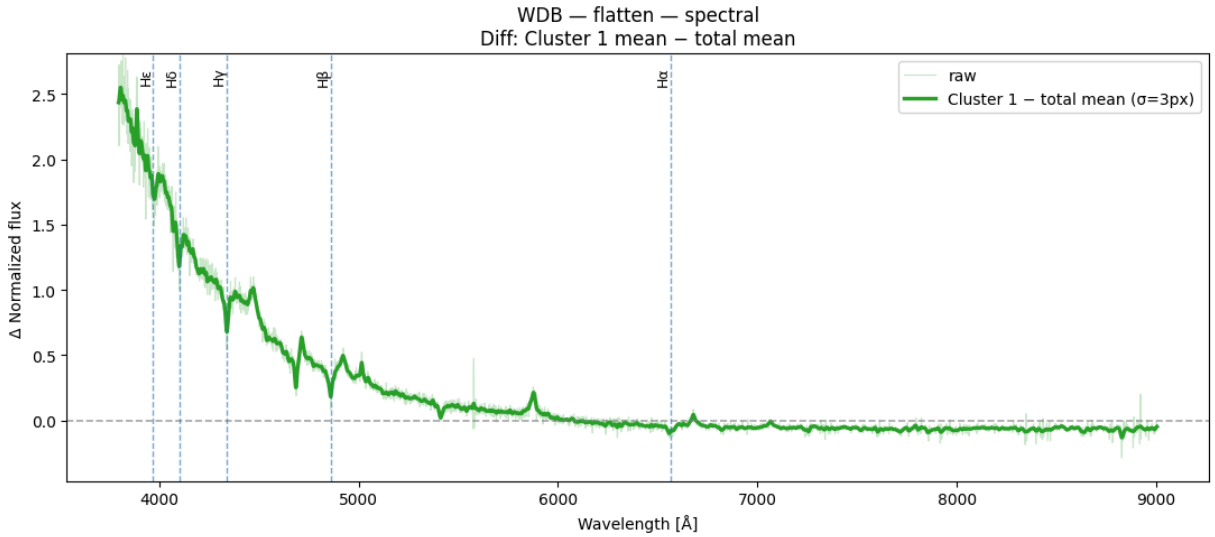

WDB — flatten — spectral
Cluster 2 (n=44)
Cluster 2 mean (n=44)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

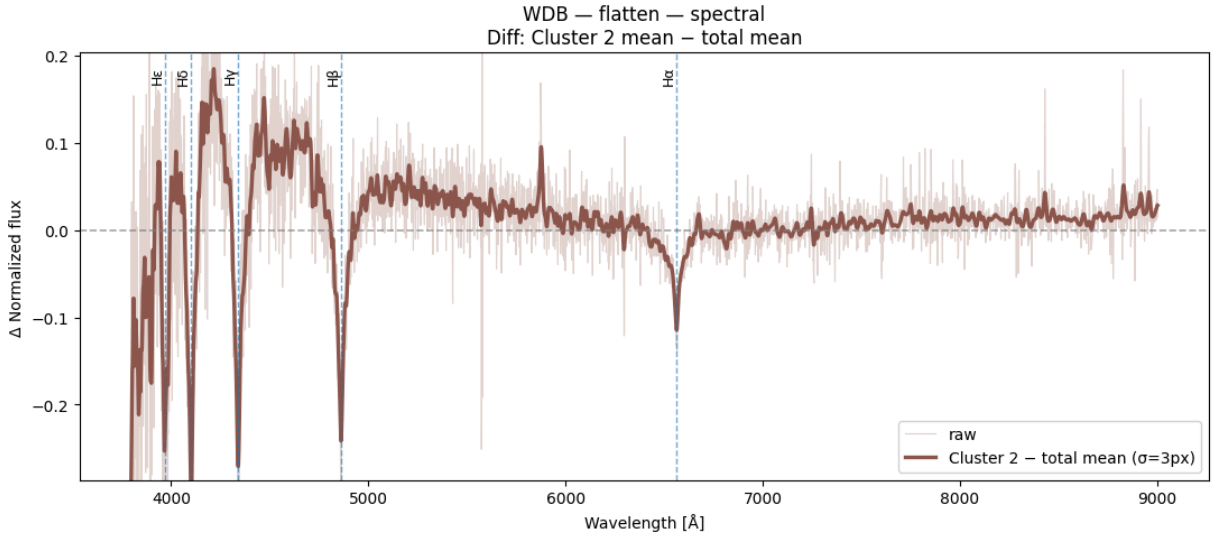

WDB — flatten — spectral
Cluster 3 (n=93)
Cluster 3 mean (n=93)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

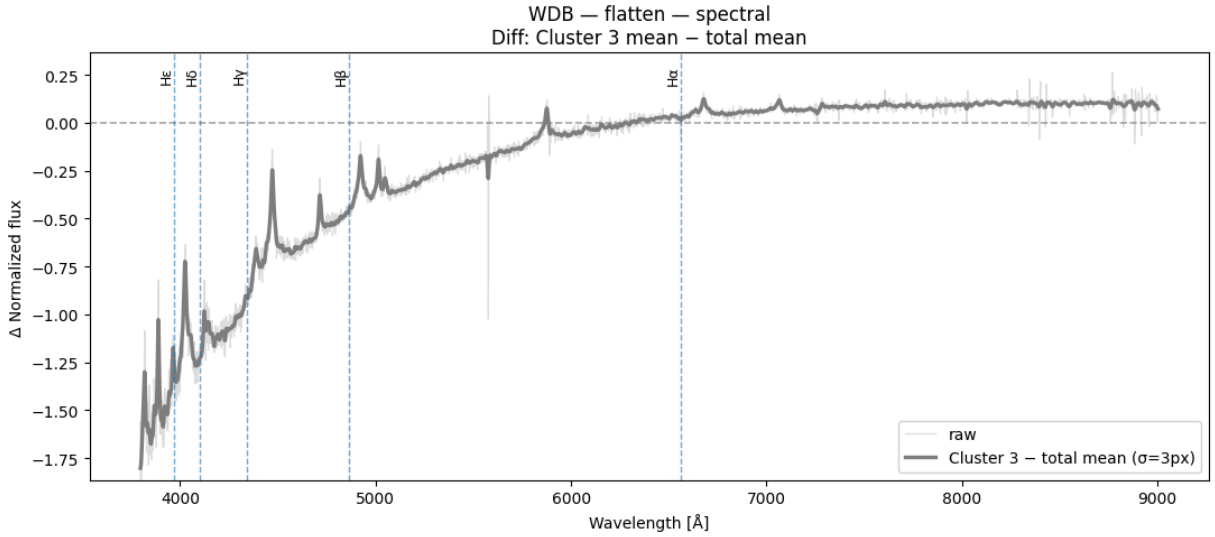

WDB — flatten — spectral
Cluster 4 (n=214)
Cluster 4 mean (n=214)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

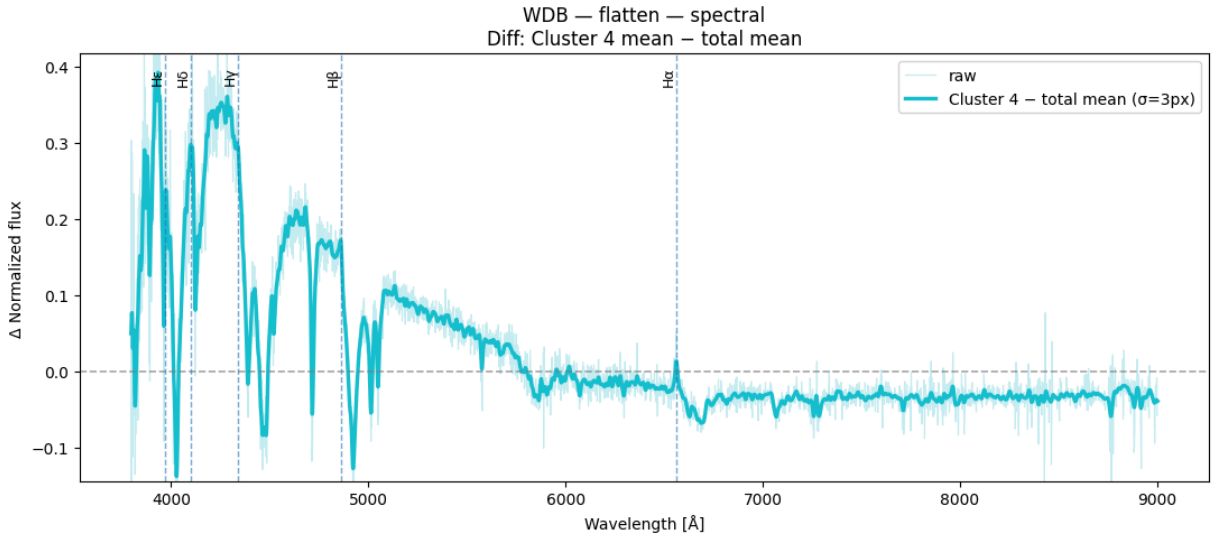

WDB — flatten — spectral
Total mean spectrum
Total mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

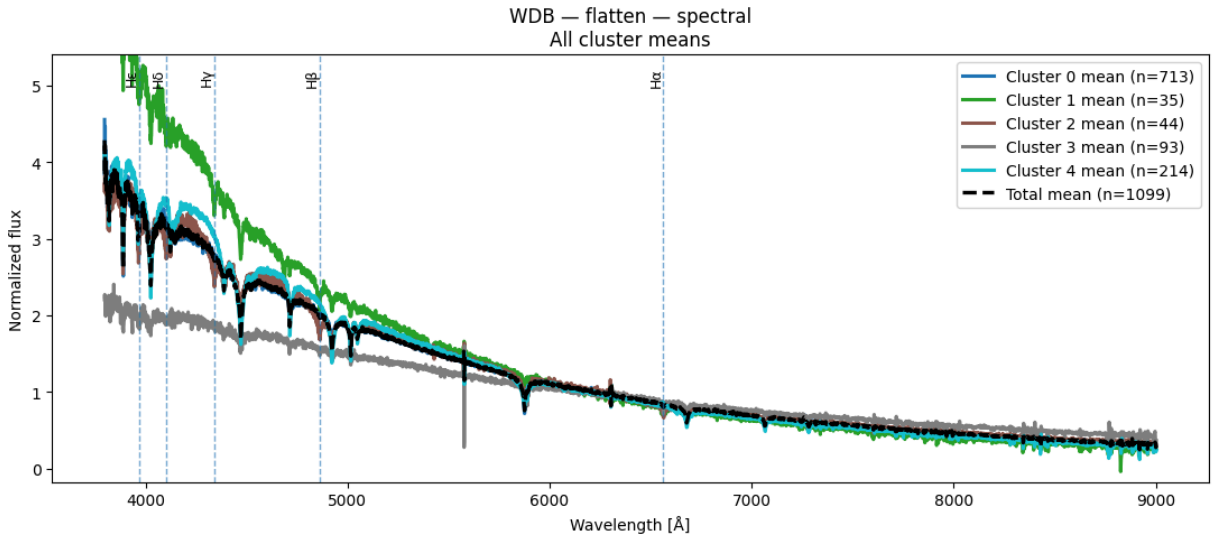

WDB — flatten — spectral
Diff: Cluster 0 mean − total mean
raw
Cluster 0 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]


WDB — flatten — spectral
Diff: Cluster 1 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
raw
Cluster 1 − total mean (σ=3px)
Δ Normalized flux
Wavelength [Å]


WDB — flatten — spectral
Diff: Cluster 2 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
raw
Cluster 2 − total mean (σ=3px)
Δ Normalized flux
Wavelength [Å]


WDB — flatten — spectral
Diff: Cluster 3 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
raw
Cluster 3 − total mean (σ=3px)
Δ Normalized flux
Wavelength [Å]

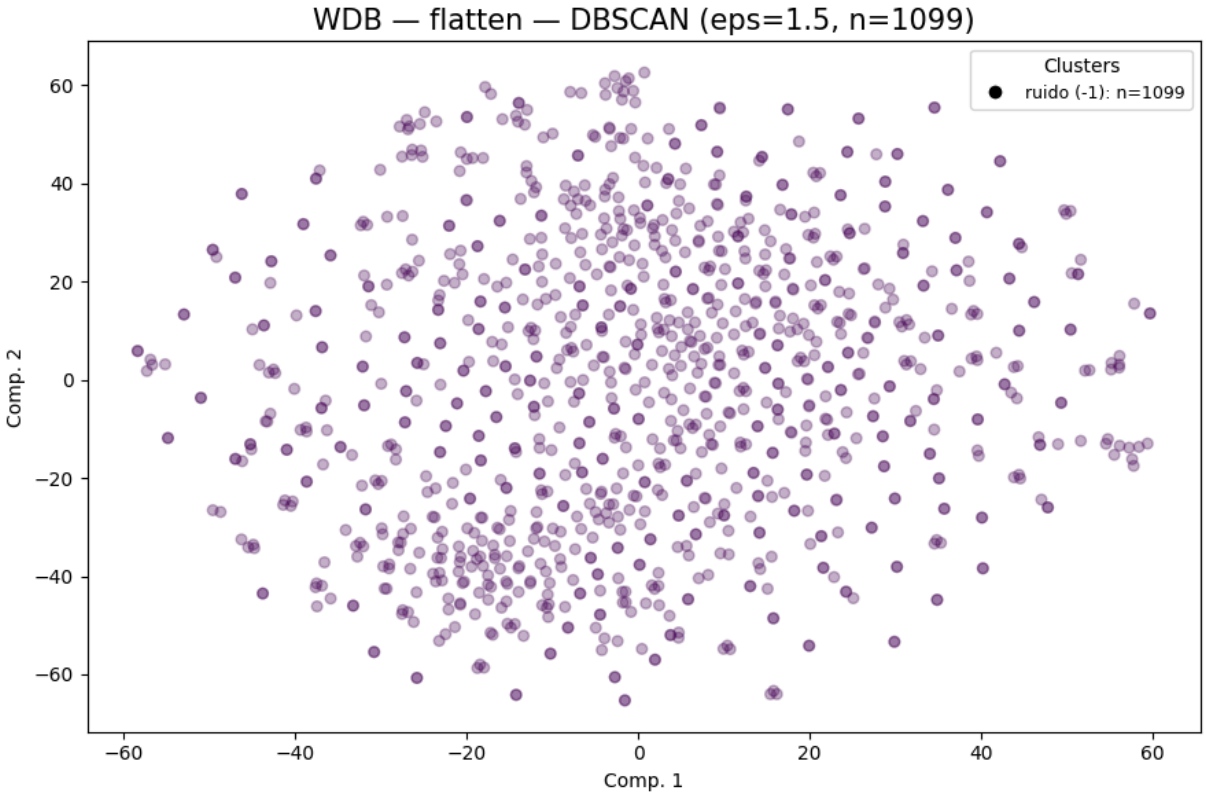




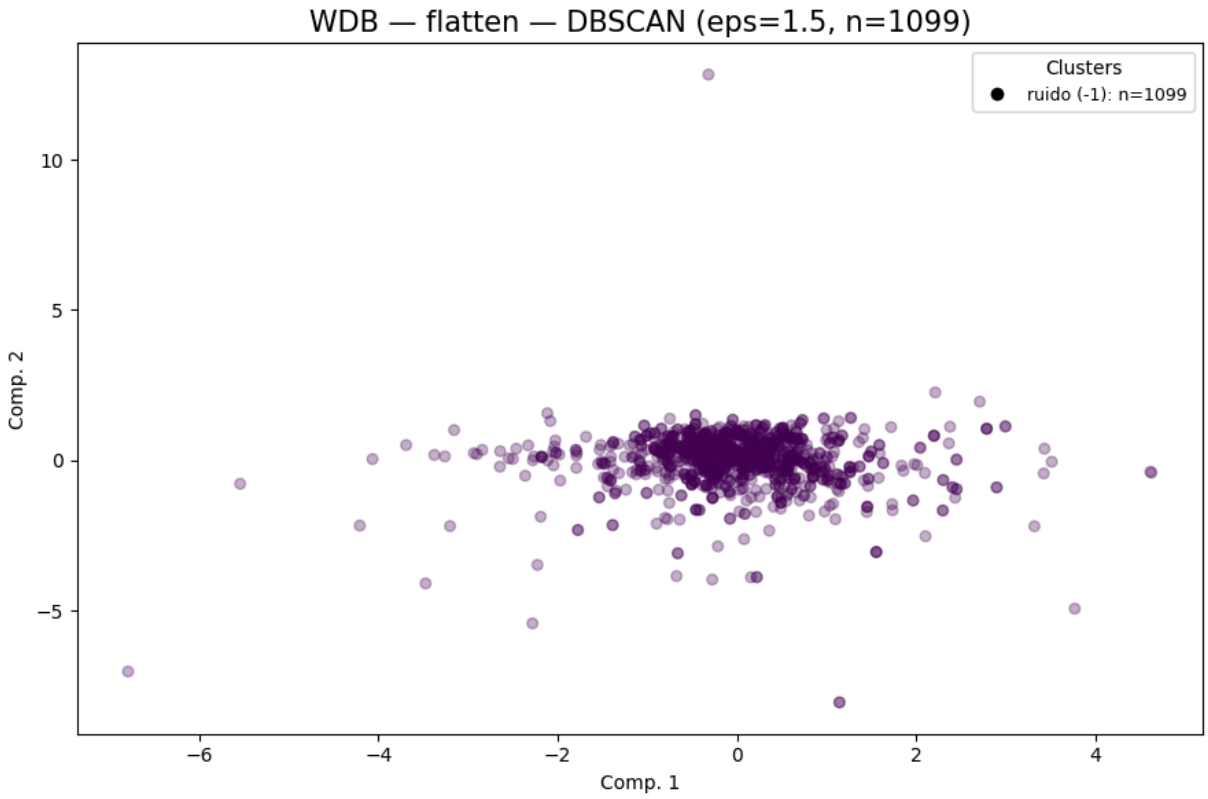
DBSCAN

In [58]:
```
dbs_flatten_wdb = cluster_plot_DBSCAN(
            repr_wdb_32["flatten"],
            "WDB",
            "flatten",
            dbscan_eps=1.5,
            dbscan_min_samples=10,
            vis_method="tsne",
            xlim=None,
            ylim=None,
            container=cluster_labels_all
        )
labels_dbs_flatten_wdb = dbs_flatten_wdb["dbscan"]
cluster_plot_DBSCAN(
            repr_wdb_32["flatten"],
            "WDB",
            "flatten",
            dbscan_eps=1.5,
            dbscan_min_samples=10,
            vis_method="pca",
            xlim=None,
```

```
            ylim=None,
            container=cluster_labels_all
        )
```




```
┌──────────────────────────────────────────────────────────────────────────────┐
│  WDB | flatten | dbscan | eps=1.5
│  clusters=0  noise=1099  sil=N/A  DB=N/A  CH=N/A
│  DBCV = N/A  (rango [-1,1], especifico DBSCAN)
└──────────────────────────────────────────────────────────────────────────────┘
```



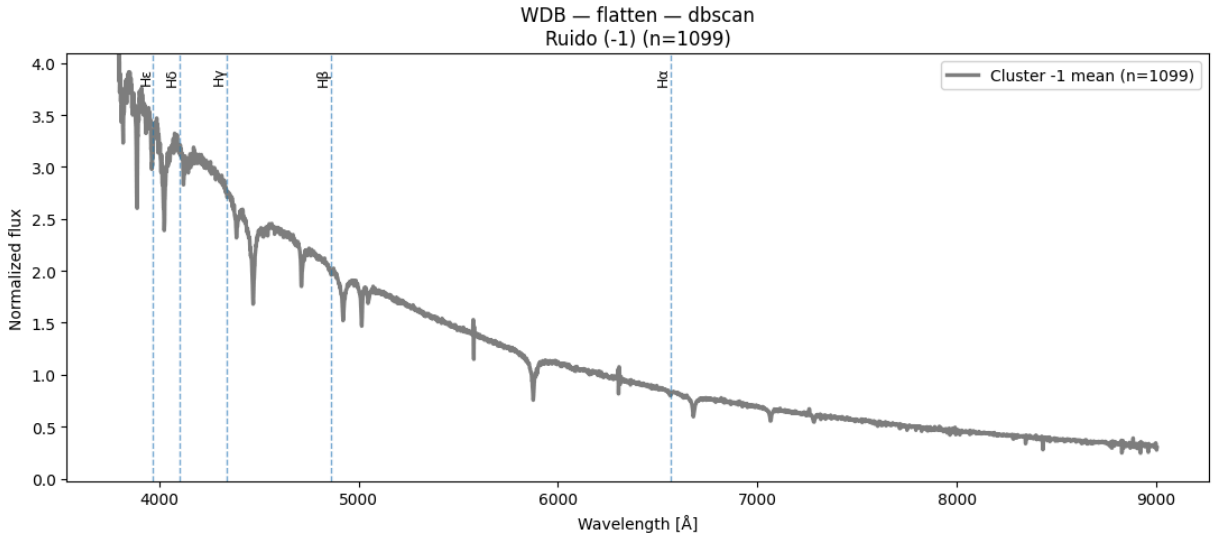

```
┌──────────────────────────────────────────────────────────────┐
│  WDB | flatten | dbscan | eps=1.5                            │
│  clusters=0  noise=1099  sil=N/A  DB=N/A  CH=N/A             │
│  DBCV = N/A  (rango [-1,1], especifico DBSCAN)               │
└──────────────────────────────────────────────────────────────┘
```

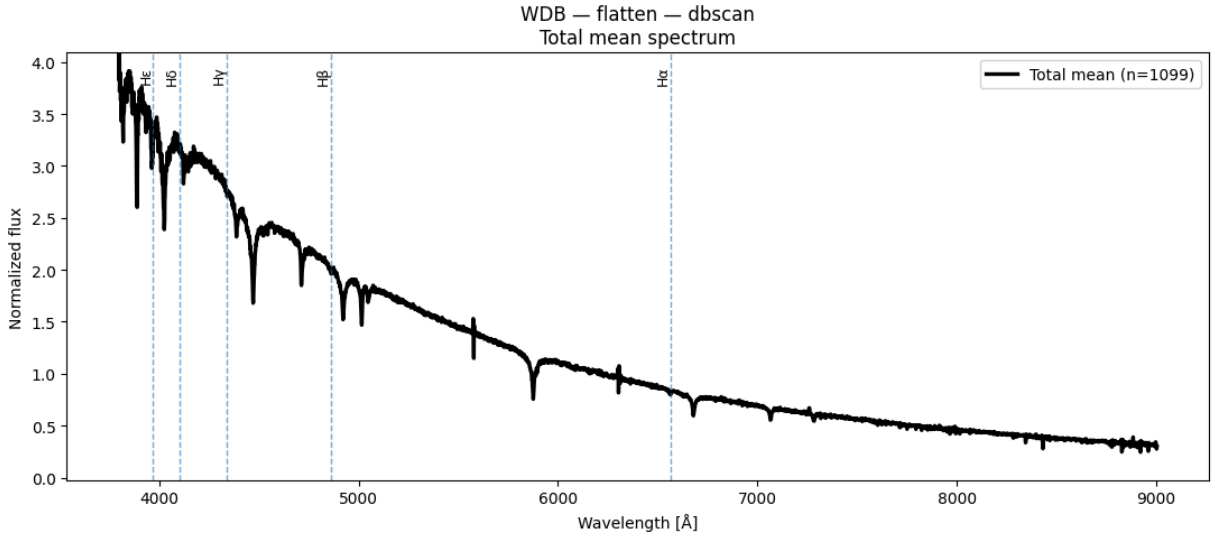
Out[58]:
```
{'dbscan': array([-1, -1, -1, ..., -1, -1, -1], dtype=int64),
 'metrics': {'n_clusters': 0,
  'n_noise': 1099,
  'silhouette': None,
  'davies_bouldin': None,
  'calinski_harabasz': None},
 'dbcv': nan}
```

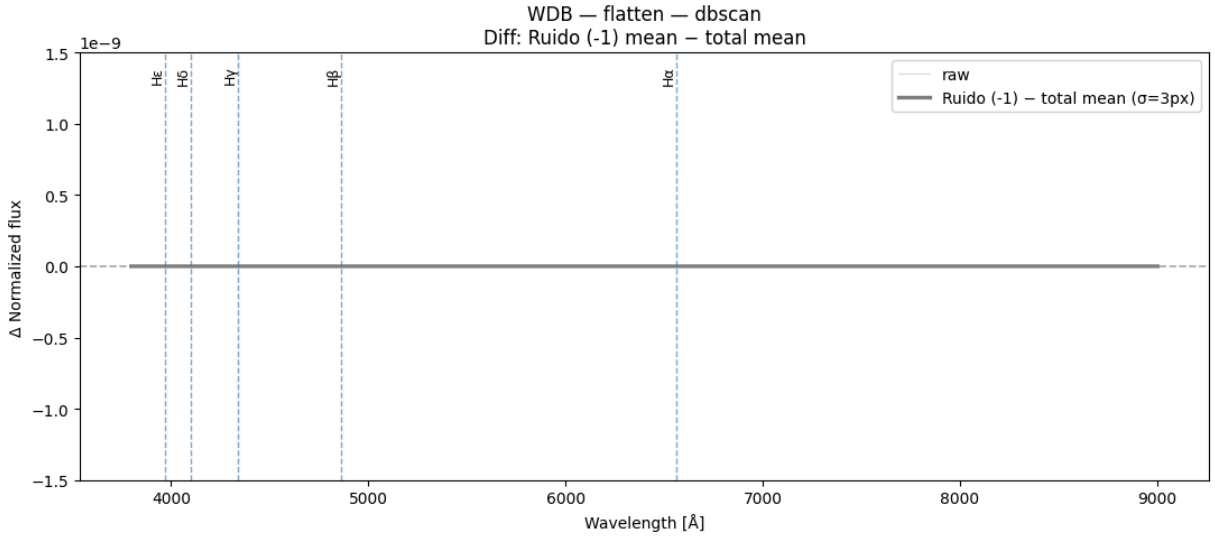
Espectros obtenidos de DBSCAN n = 1100

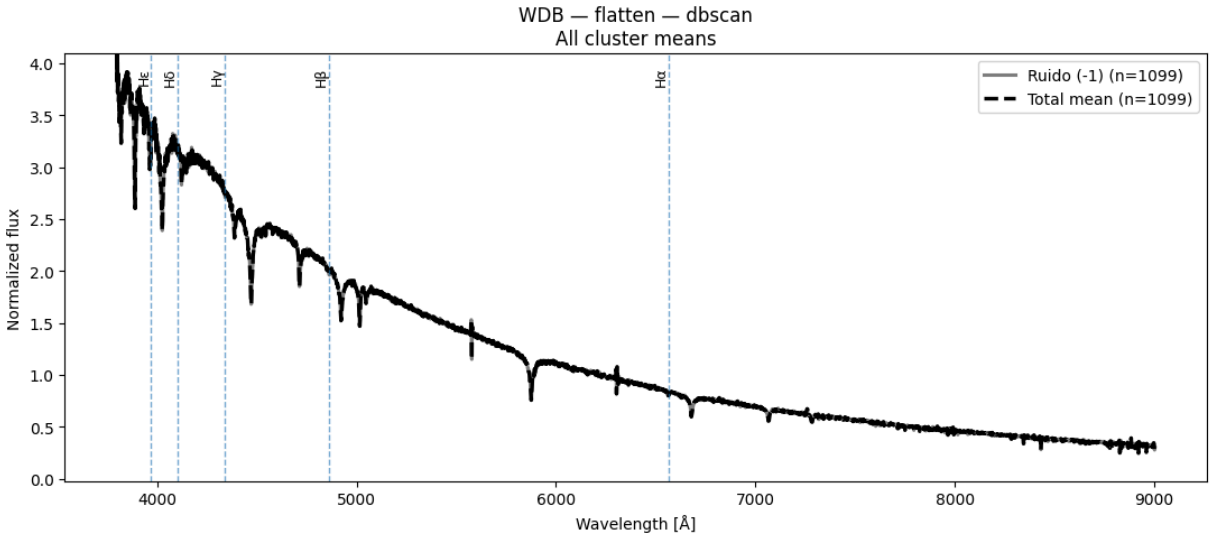
In [59]:
```
plot_spectra_dbscan(
    X_wdb_spec,
    labels_dbs_flatten_wdb,
    W_wdb[0],
    class_name="WDB",
    layer_name="flatten"
)
```

```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```


WDB — flatten — dbscan
Ruido (-1) (n=1099)
Cluster -1 mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — flatten — dbscan
Total mean spectrum
Total mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — flatten — dbscan
Diff: Ruido (-1) mean − total mean
1e−9
raw
Ruido (-1) − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]

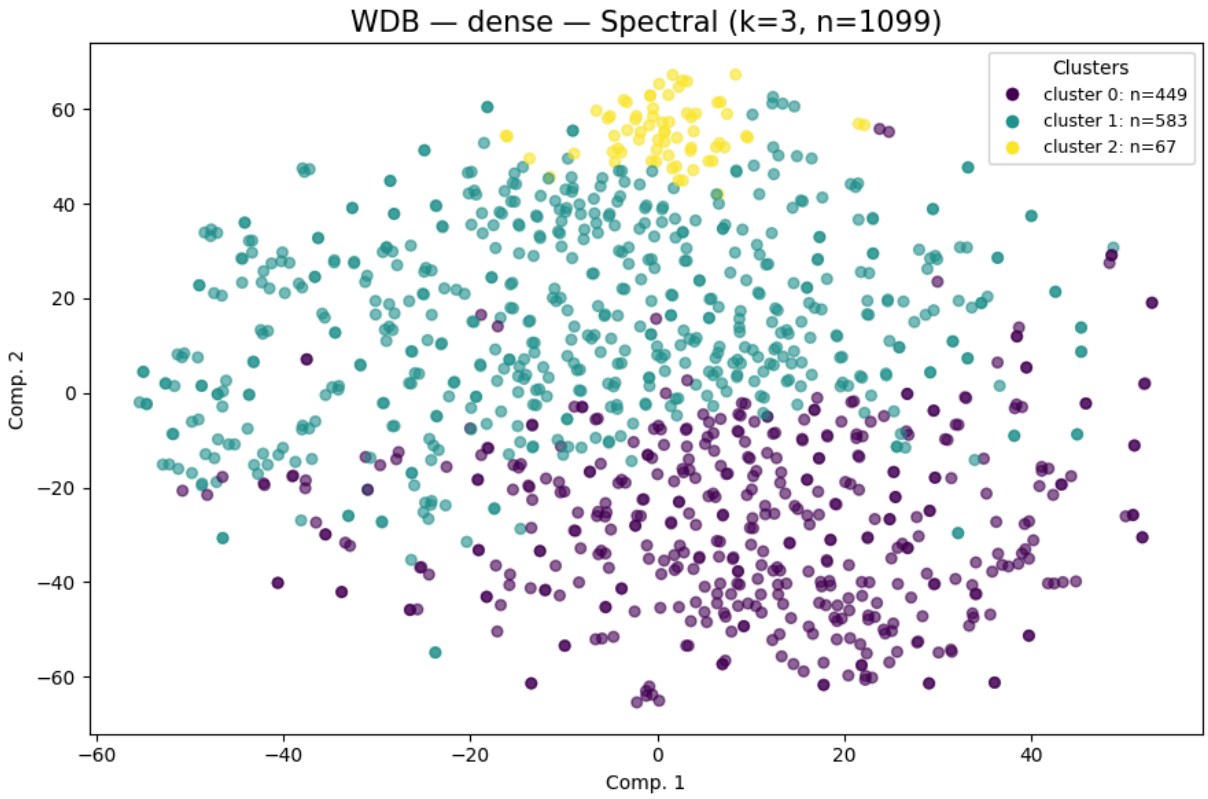


Out[59]:
{'total_mean': array([4.2840786 , 3.8304825 , 3.8275013 , ..., 0.3190189 , 0.27551222,
0.2985266], dtype=float32),
'group_means': {-1: array([4.2840786 , 3.8304825 , 3.8275013 , ..., 0.3190189 , 0.27551222,
0.2985266], dtype=float32)},
'unique_clusters': array([-1], dtype=int64)}

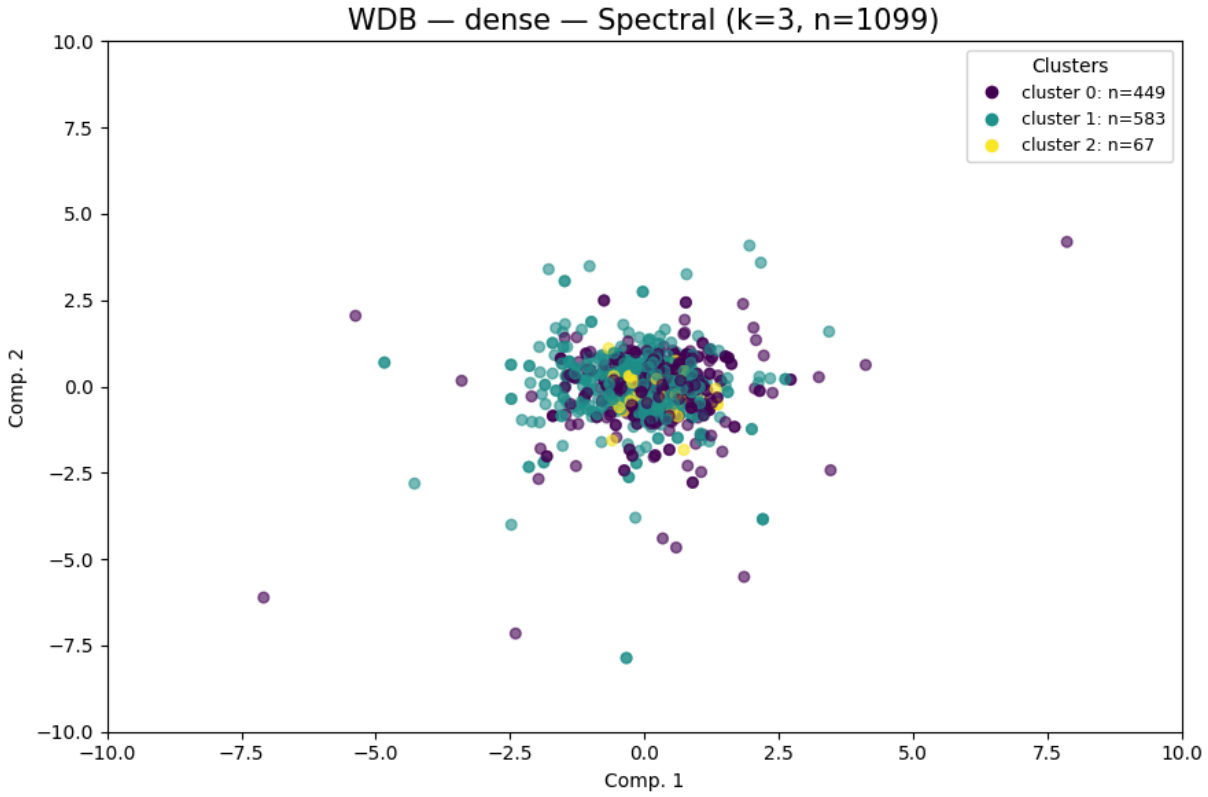
Resultados capa dense, con Spectral Clustering + tSNE y PCA para n=1100

Spectral Clustering k = 3

In [60]:
```
dense_wdb_k3=cluster_plot_Spect(
            repr_wdb_32["dense"],
            "WDB",
            "dense",
            spectral_k=3,
            vis_method="tsne",
            container=cluster_labels_all
        )

labels_dense_k3_wdb = dense_wdb_k3["spectral"]
cluster_plot_Spect(
            repr_wdb_32["dense"],
            "WDB",
            "dense",
            spectral_k=3,
            vis_method="pca",
            xlim=(-10,10), ylim=(-10,10)
        )
```




```
┌──────────────────────────────────────────────────────────────────────┐
│  WDB | dense | spectral | k=3
│  clusters=3  noise=0  sil=-0.0482  DB=4.7460  CH=19.04
└──────────────────────────────────────────────────────────────────────┘
```

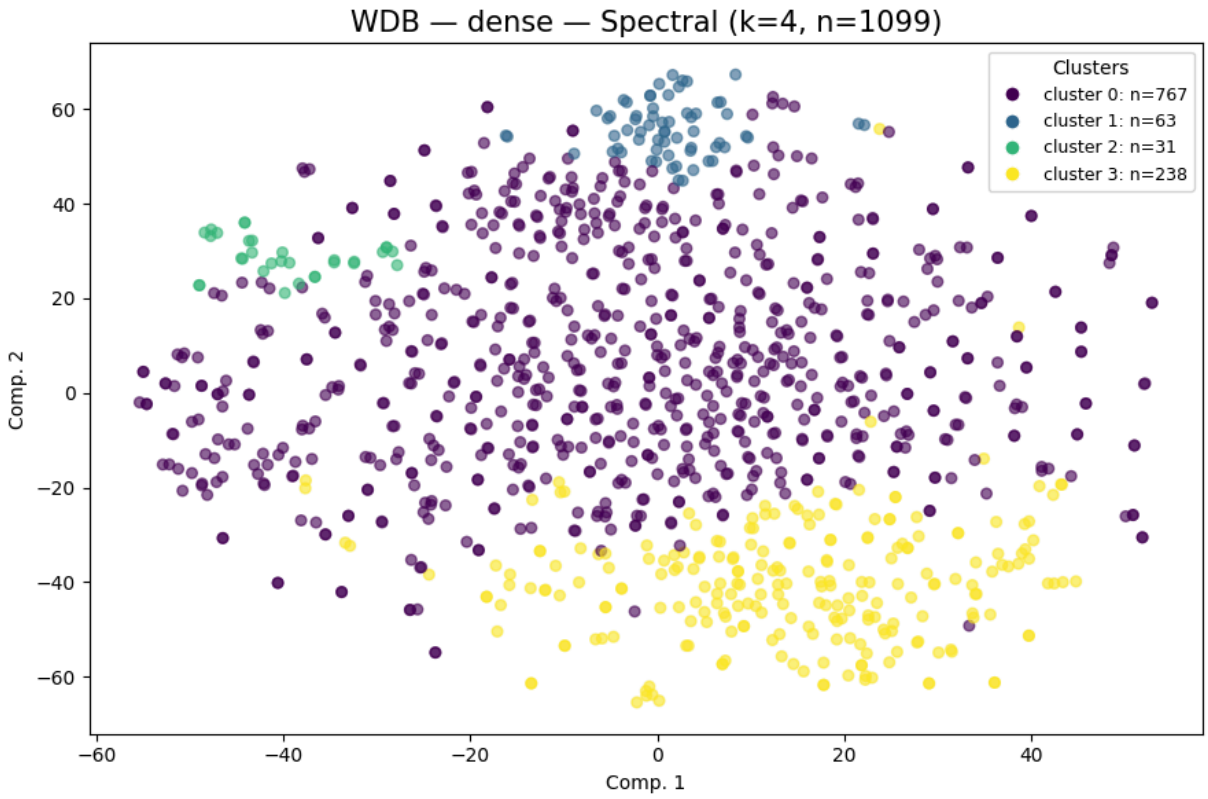


```
┌──────────────────────────────────────────────────────────────────────────────┐
│  WDB | dense | spectral | k=3
│  clusters=3  noise=0  sil=-0.0482  DB=4.7460  CH=19.04
└──────────────────────────────────────────────────────────────────────────────┘
```

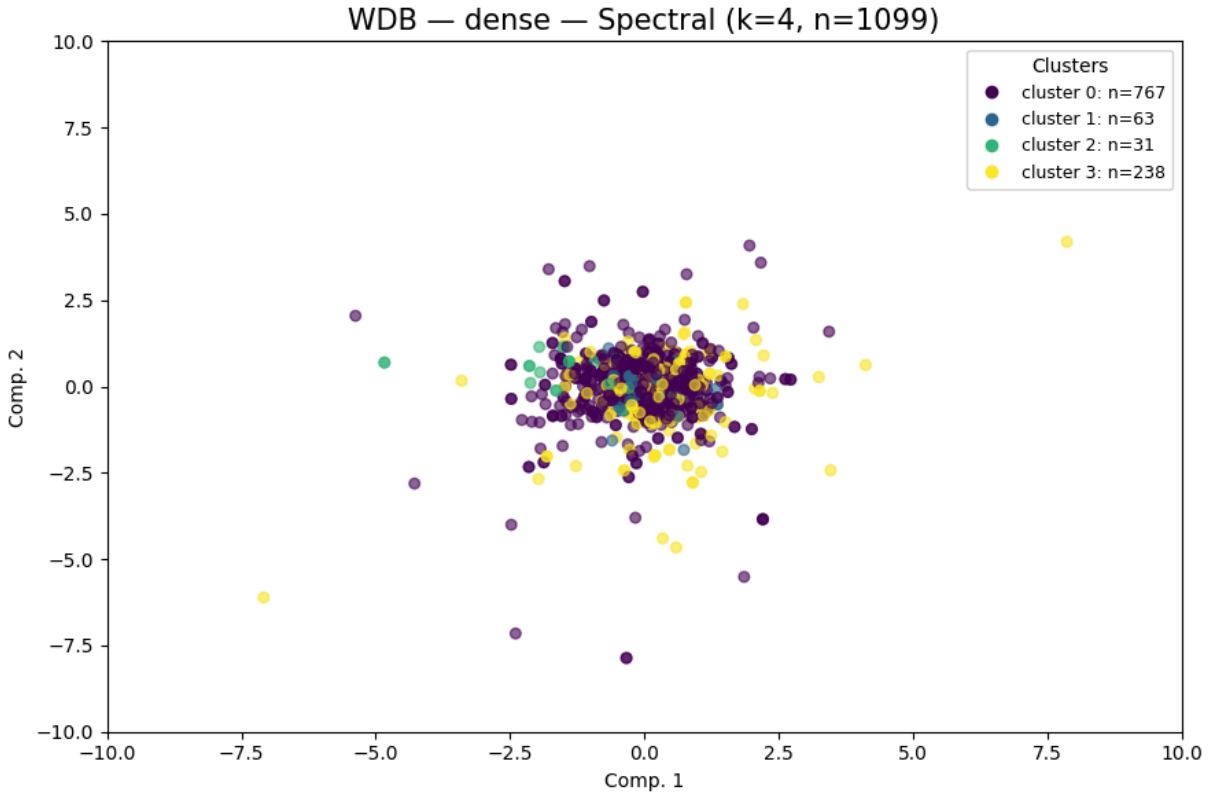
Out[60]:
```
{'spectral': array([0, 1, 1, ..., 1, 1, 1]),
 'metrics': {'n_clusters': 3,
  'n_noise': 0,
  'silhouette': -0.048228,
  'davies_bouldin': 4.746015,
  'calinski_harabasz': 19.0417}}
```

Spectral Clustering k = 4

In [61]:
```
dense_wdb_k4=cluster_plot_Spect(
            repr_wdb_32["dense"],
            "WDB",
            "dense",
            spectral_k=4,
            vis_method="tsne",
            container=cluster_labels_all
        )

labels_dense_k4_wdb = dense_wdb_k4["spectral"]
cluster_plot_Spect(
            repr_wdb_32["dense"],
            "WDB",
            "dense",
            spectral_k=4,
            vis_method="pca",
            xlim=(-10,10), ylim=(-10,10)
        )
```




```
┌──────────────────────────────────────────────────────────────────────────┐
│  WDB | dense | spectral | k=4
│  clusters=4  noise=0  sil=-0.0339  DB=3.5764  CH=18.81
└──────────────────────────────────────────────────────────────────────────┘
```

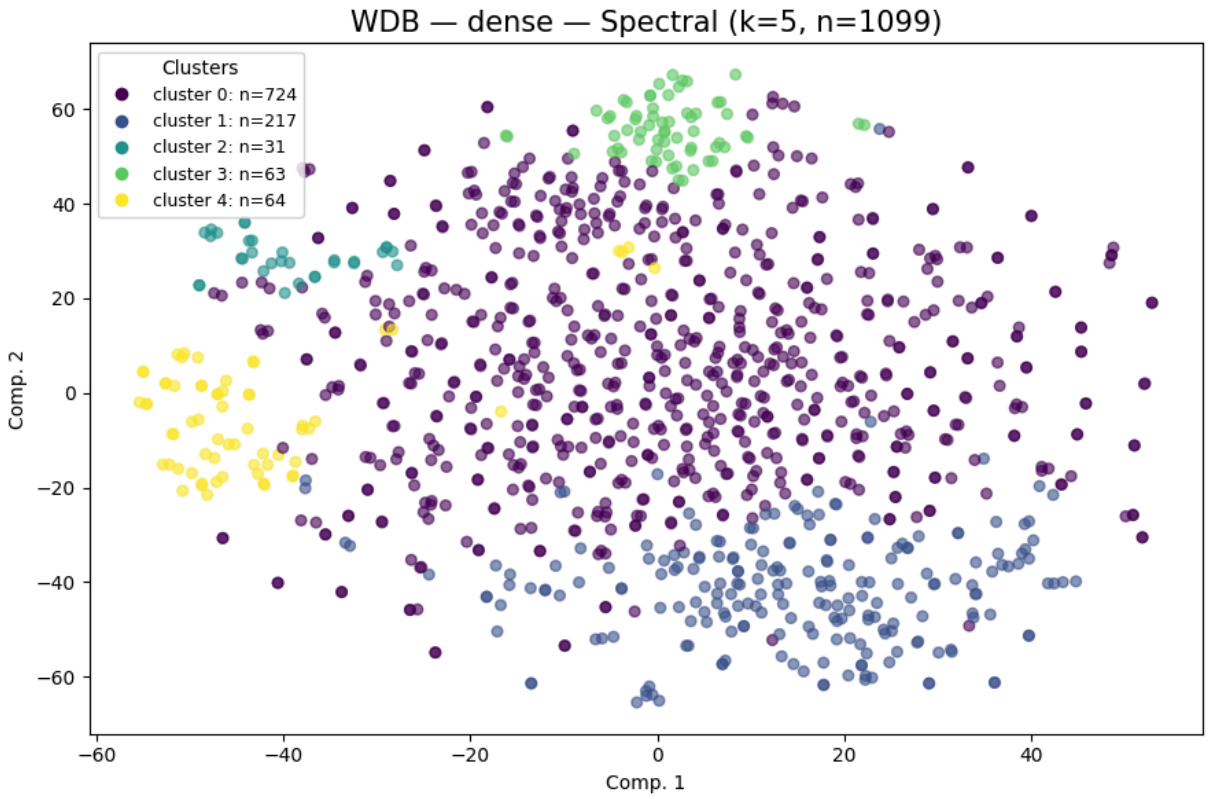


```
┌──────────────────────────────────────────────────────────────────────────┐
│  WDB | dense | spectral | k=4                                            │
│  clusters=4  noise=0  sil=-0.0339  DB=3.5764  CH=18.81                   │
└──────────────────────────────────────────────────────────────────────────┘
```

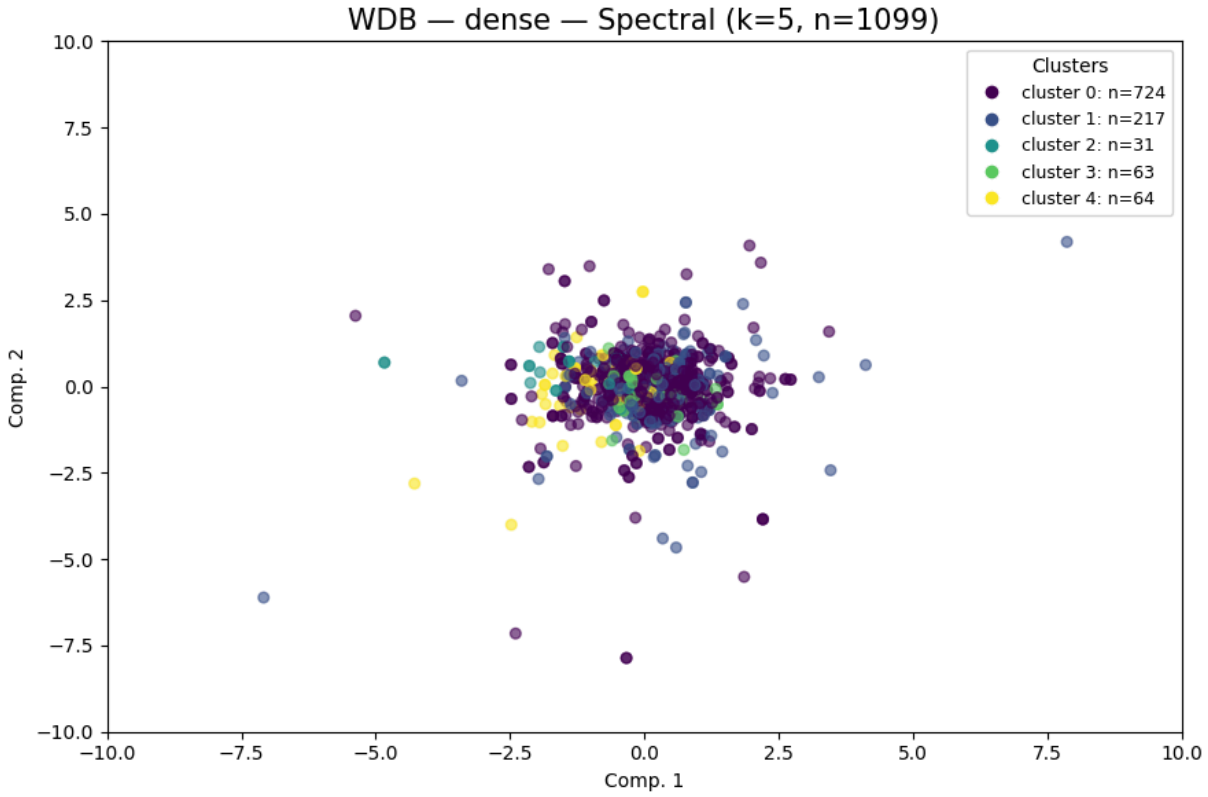
Out[61]:
```
{'spectral': array([0, 0, 0, ..., 0, 0, 0]),
 'metrics': {'n_clusters': 4,
  'n_noise': 0,
  'silhouette': -0.033931,
  'davies_bouldin': 3.57644,
  'calinski_harabasz': 18.8101}}
```

Spectral Clustering k = 5

In [62]:
```
dense_wdb_k5=cluster_plot_Spect(
            repr_wdb_32["dense"],
            "WDB",
            "dense",
            spectral_k=5,
            vis_method="tsne",
            container=cluster_labels_all
        )

labels_dense_k5_wdb = dense_wdb_k5["spectral"]
cluster_plot_Spect(
            repr_wdb_32["dense"],
            "WDB",
            "dense",
            spectral_k=5,
            vis_method="pca",
            xlim=(-10,10), ylim=(-10,10)
        )
```




```
┌──────────────────────────────────────────────────────────────────────────┐
│  WDB | dense | spectral | k=5
│  clusters=5  noise=0  sil=-0.0242  DB=3.3592  CH=18.79
└──────────────────────────────────────────────────────────────────────────┘
```

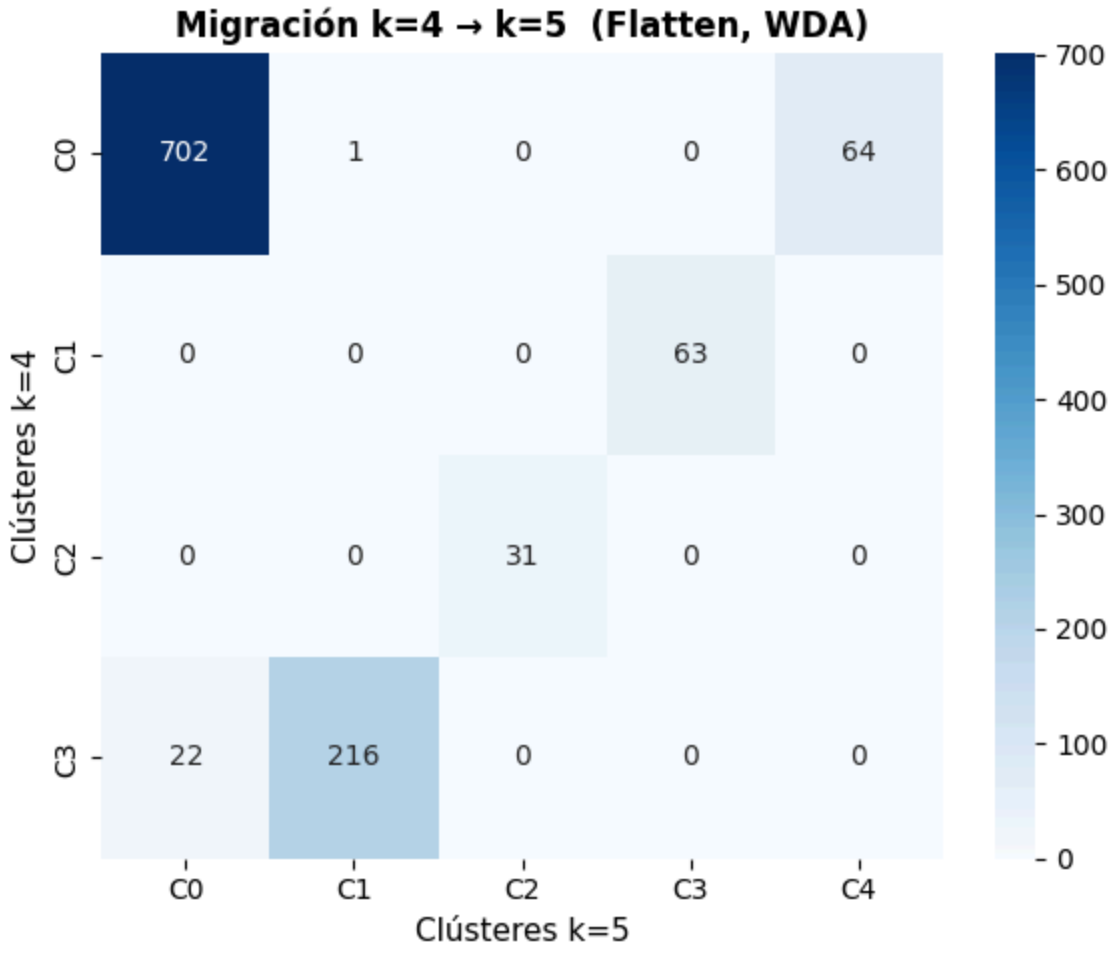


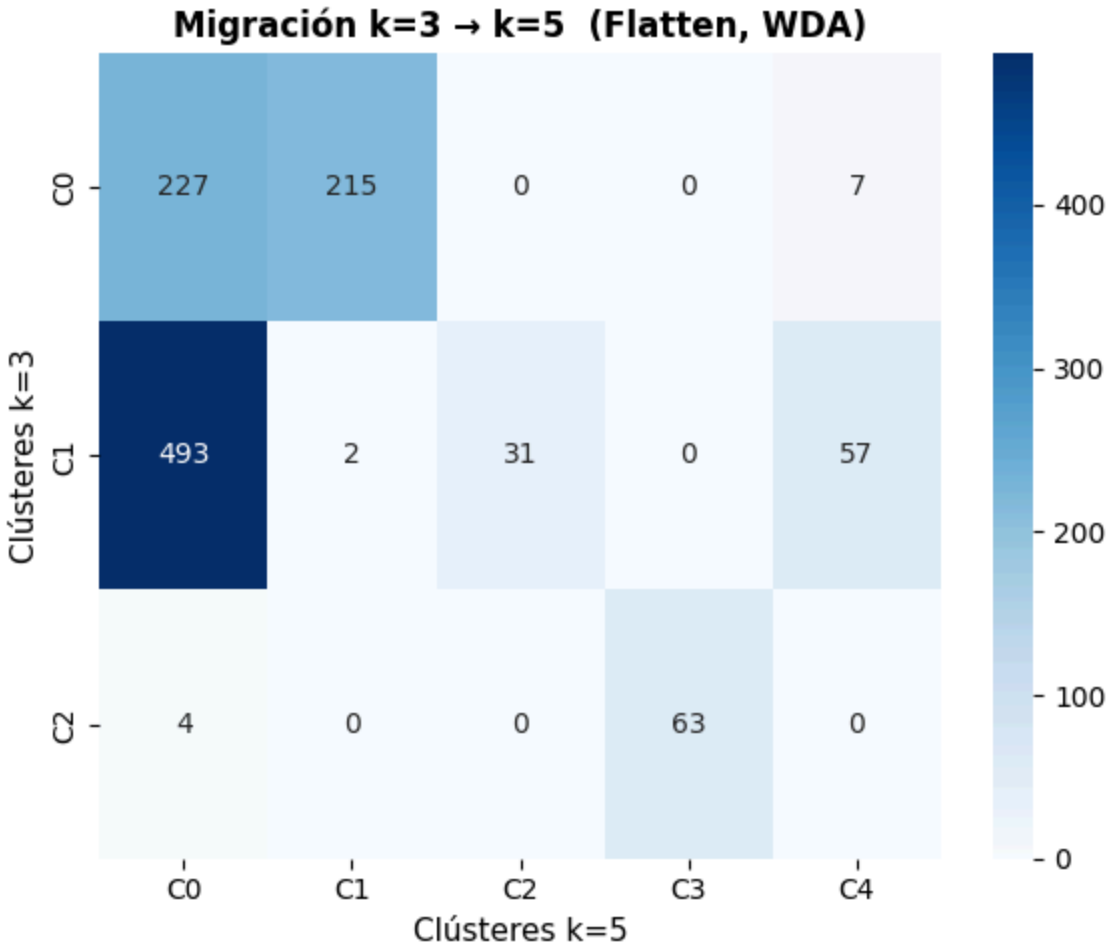
```
┌──────────────────────────────────────────────────────────────────────┐
│  WDB | dense | spectral | k=5                                        │
│  clusters=5  noise=0  sil=-0.0242  DB=3.3592  CH=18.79               │
└──────────────────────────────────────────────────────────────────────┘
```

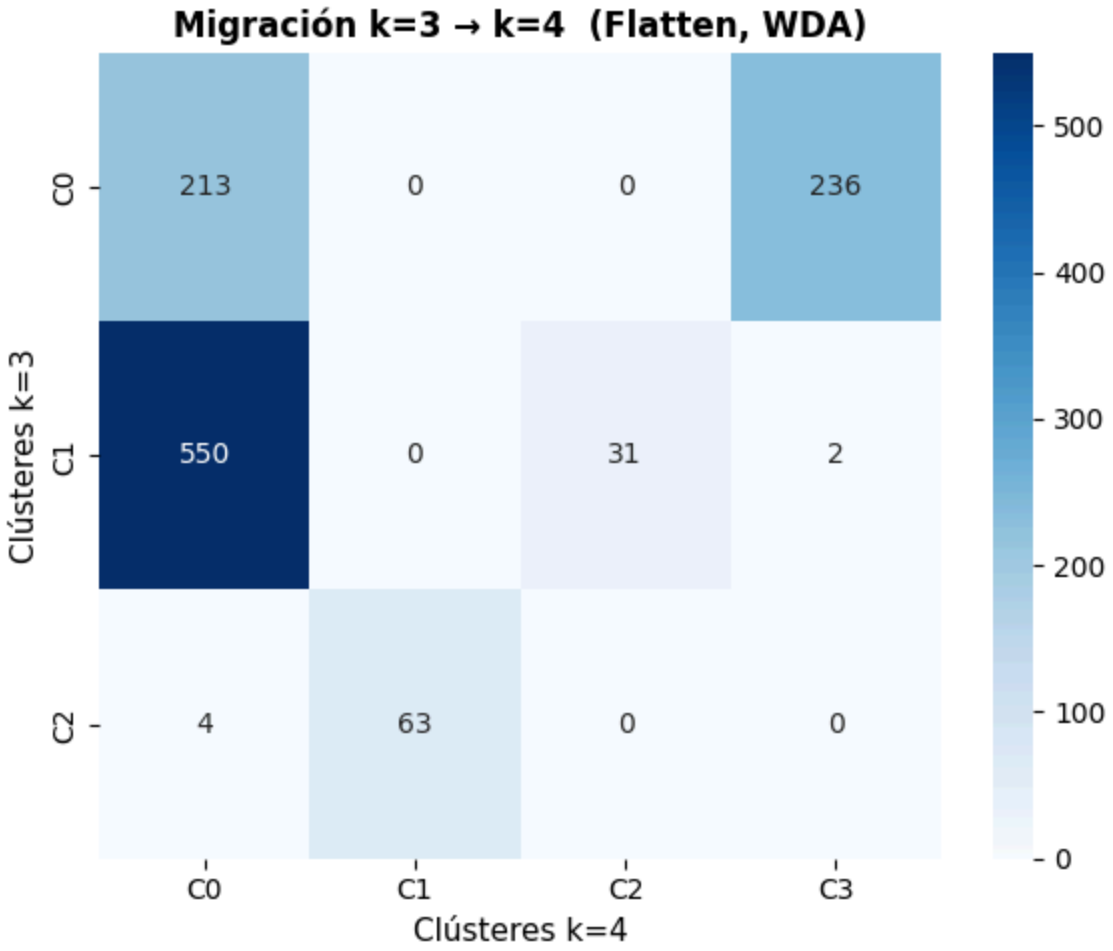
Out[62]:
```
{'spectral': array([0, 0, 0, ..., 0, 4, 0]),
 'metrics': {'n_clusters': 5,
  'n_noise': 0,
  'silhouette': -0.024244,
  'davies_bouldin': 3.359229,
  'calinski_harabasz': 18.7864}}
```

Análisis de migración de clústeres

In [63]:
```
plot_migration_matrix(labels_dense_k4_wdb, labels_dense_k5_wdb, ka=4, kb=5)
plot_migration_matrix(labels_dense_k3_wdb, labels_dense_k5_wdb, ka=3, kb=5)
plot_migration_matrix(labels_dense_k3_wdb, labels_dense_k4_wdb, ka=3, kb=4)
```



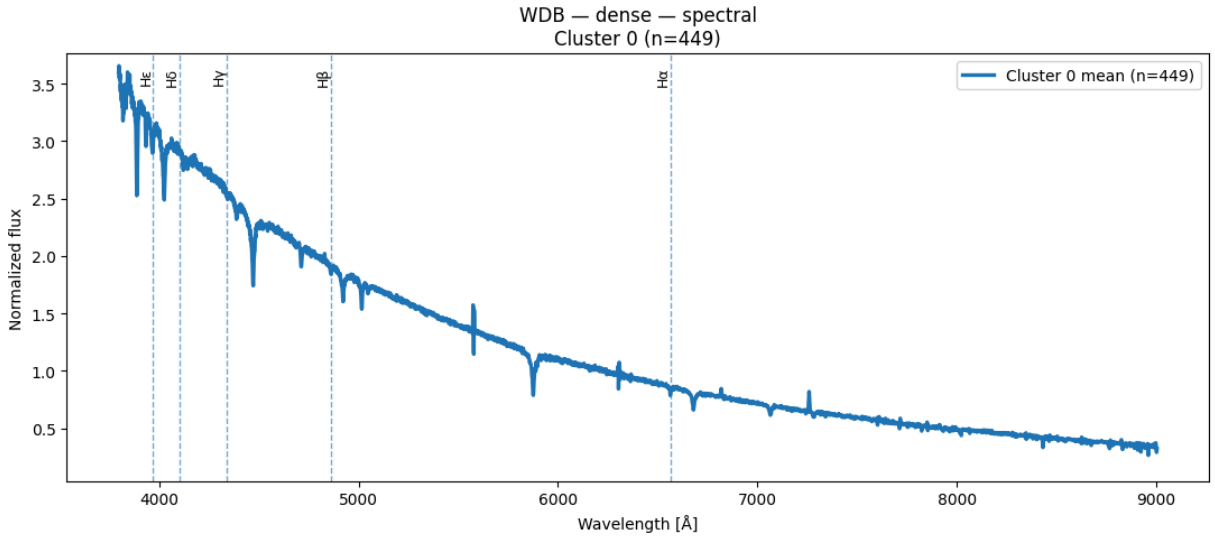

Migración k=3 → k=5 (Flatten, WDA)
Clústeres k=3
C0
C1
C2
227
215
0
0
7
493
2
31
0
57
4
0
0
63
0
C0
C1
C2
C3
C4
Clústeres k=5
400
300
200
100
0

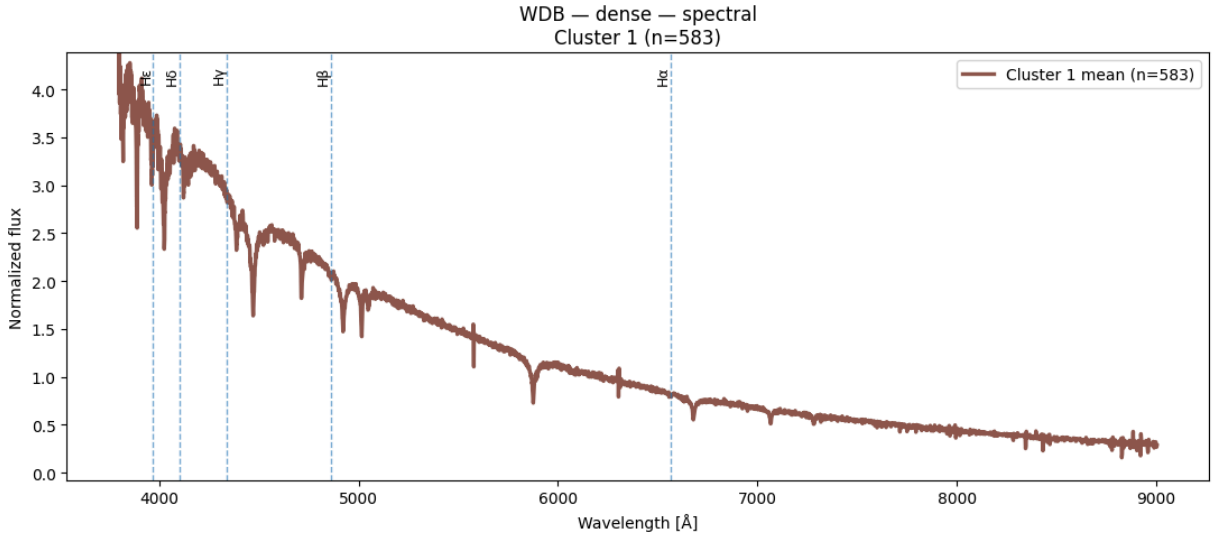


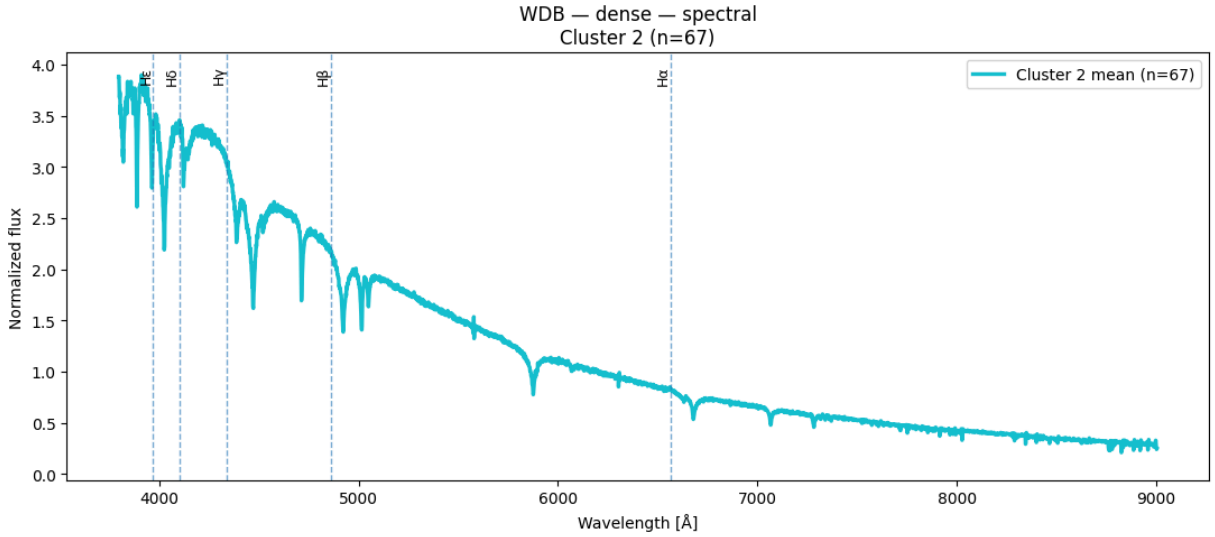
Espectros obtenidos de Spectral Clustering k = 3 y n = 1100

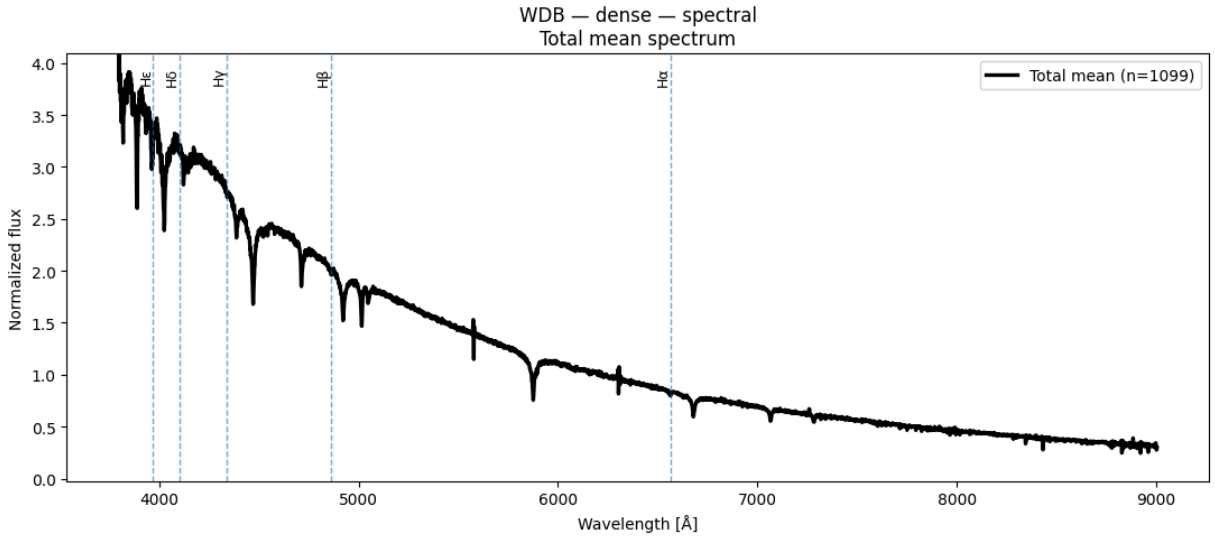
In [64]:
```
result = plot_spectra_spectral(X_wdb_spec, labels_dense_k3_wdb, W_wdb[0],
                               class_name="WDB", layer_name="dense")
```

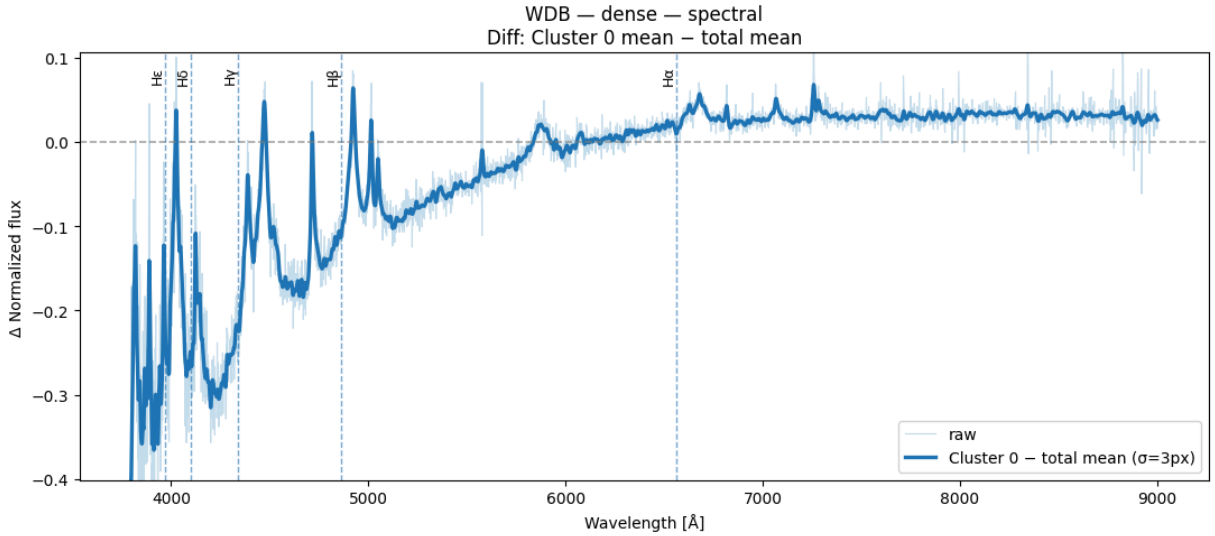
```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```

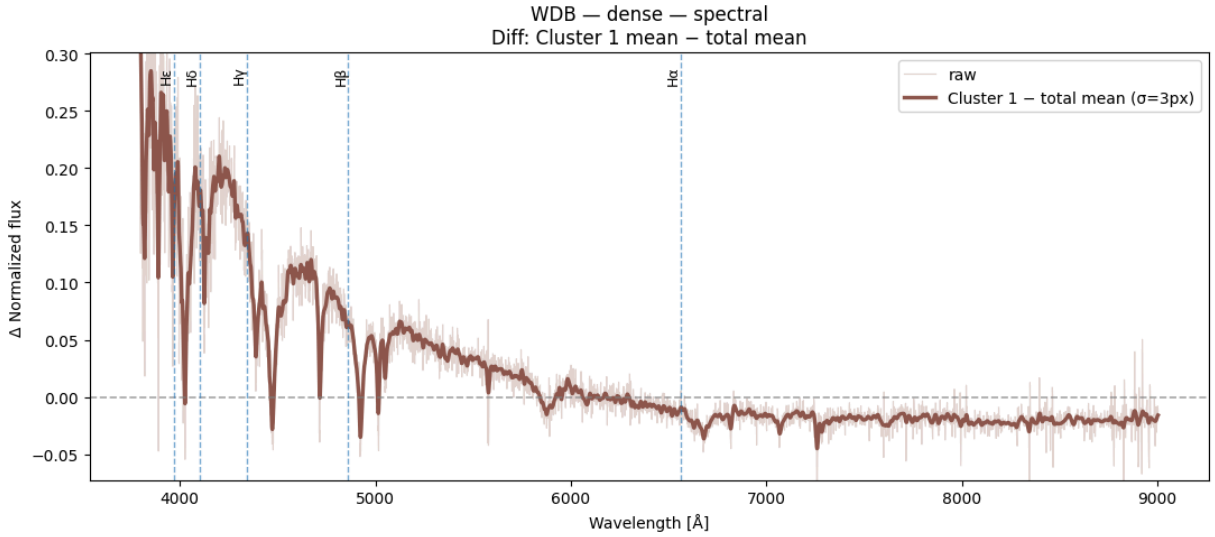

WDB — dense — spectral
Cluster 0 (n=449)
Cluster 0 mean (n=449)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

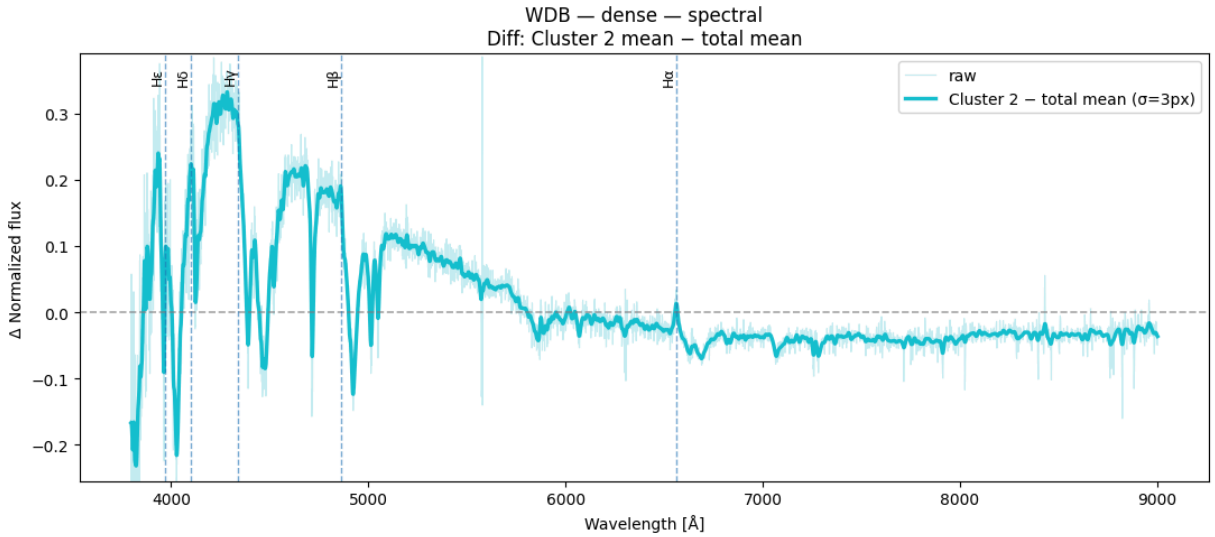

WDB — dense — spectral
Cluster 1 (n=583)
Cluster 1 mean (n=583)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

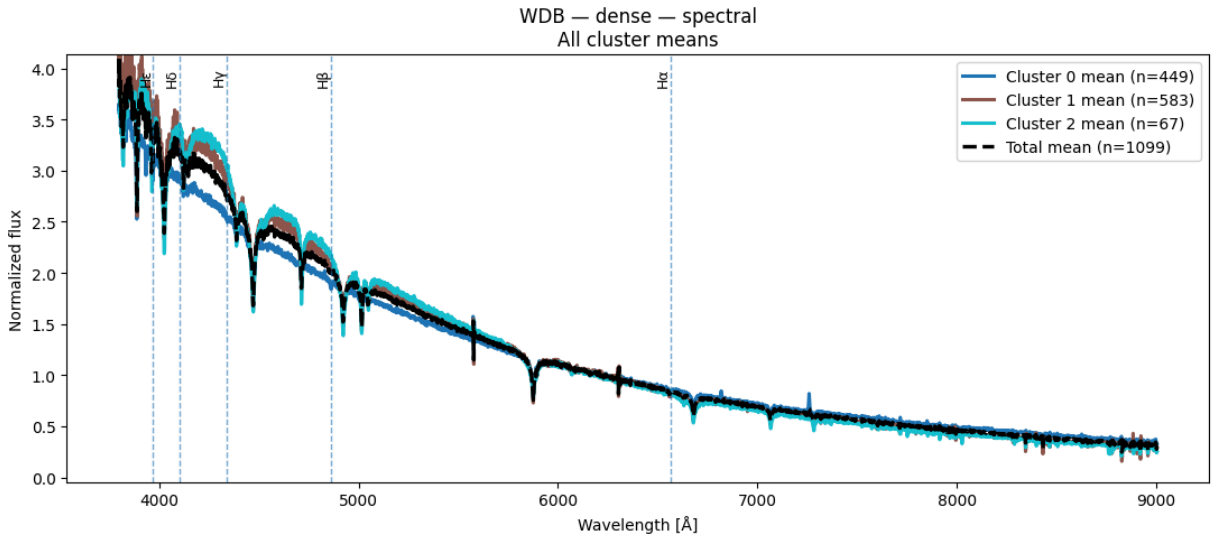

WDB — dense — spectral
Cluster 2 (n=67)
Cluster 2 mean (n=67)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — dense — spectral
Total mean spectrum
Total mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — dense — spectral
Diff: Cluster 0 mean − total mean
raw
Cluster 0 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]


WDB — dense — spectral
Diff: Cluster 1 mean − total mean
raw
Cluster 1 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]

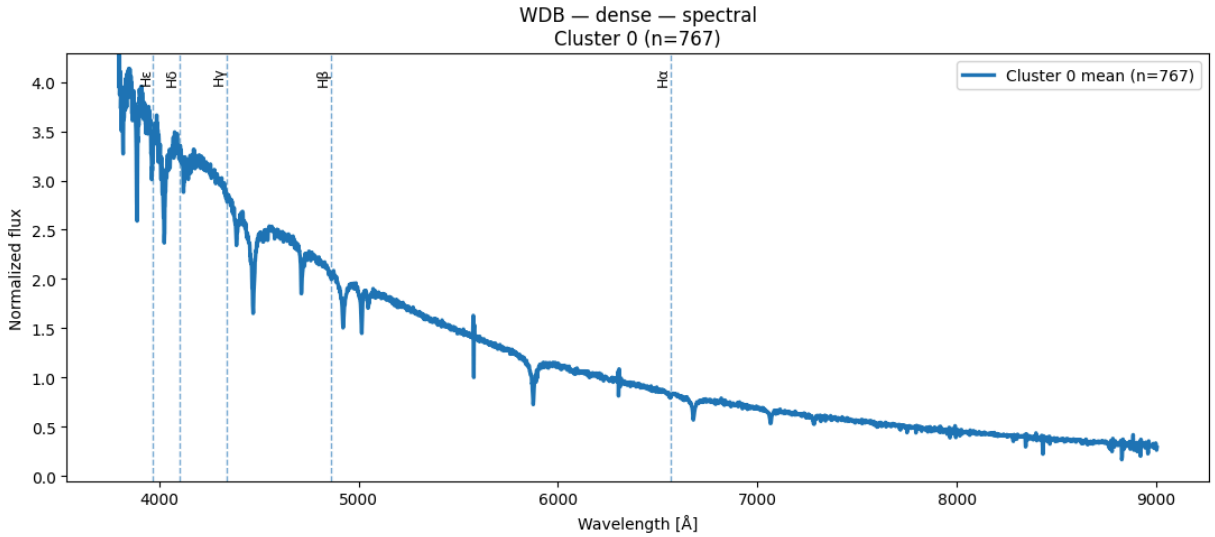


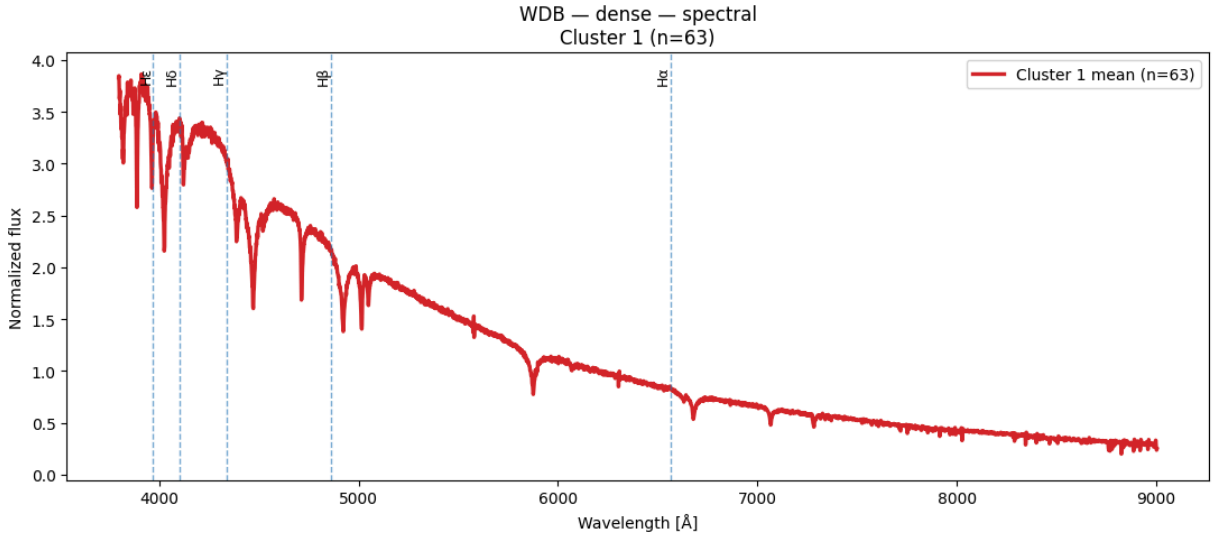


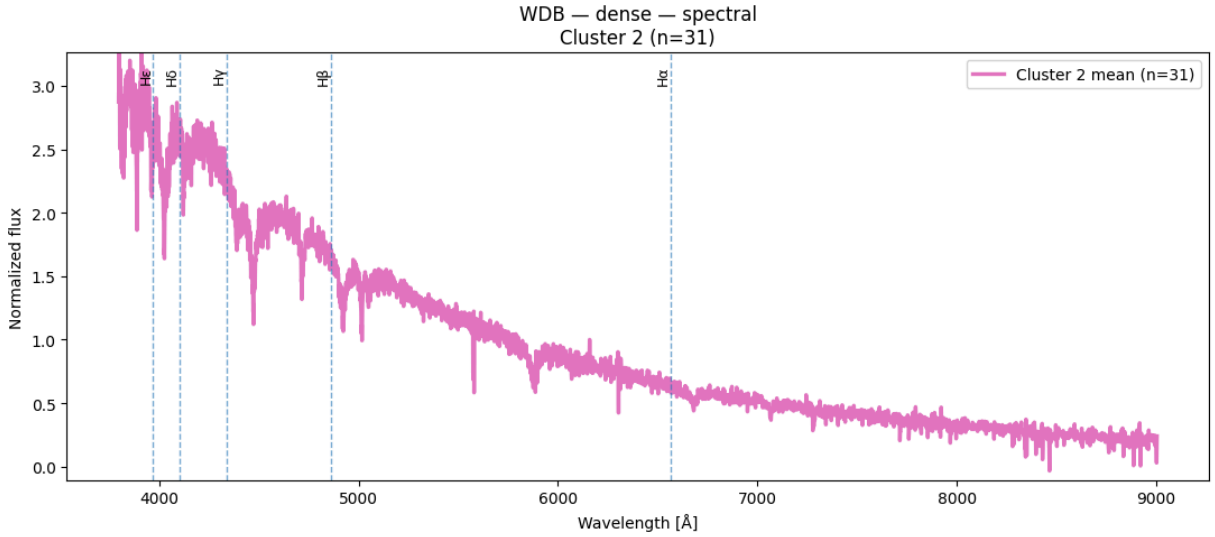
Espectros obtenidos de Spectral Clustering para k = 4 y n = 1100

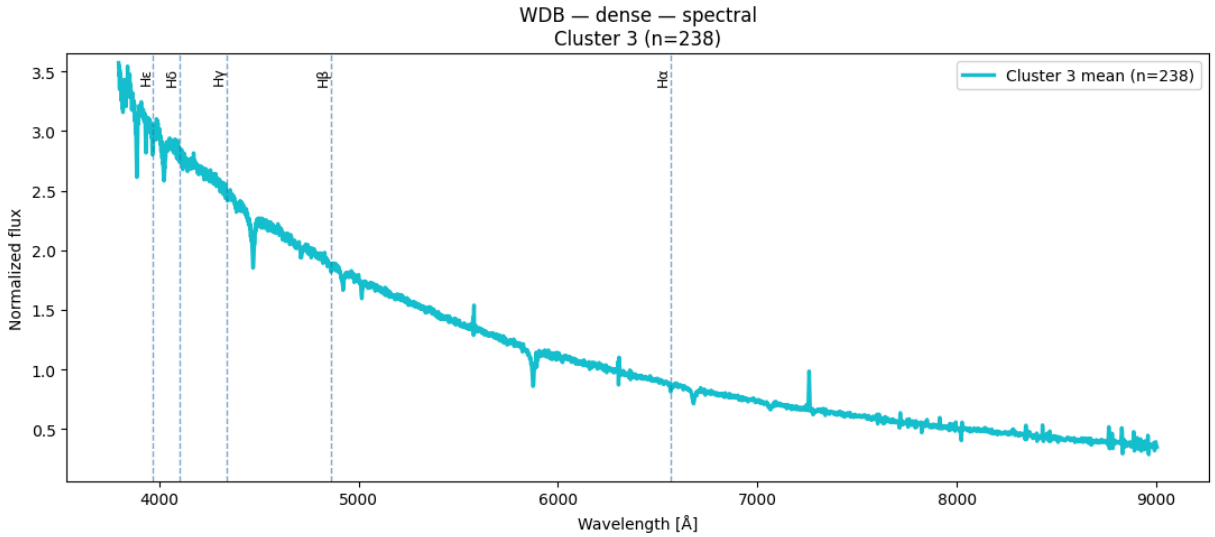
In [65]:
```
result = plot_spectra_spectral(X_wdb_spec, labels_dense_k4_wdb, W_wdb[0],
                               class_name="WDB", layer_name="dense")
```

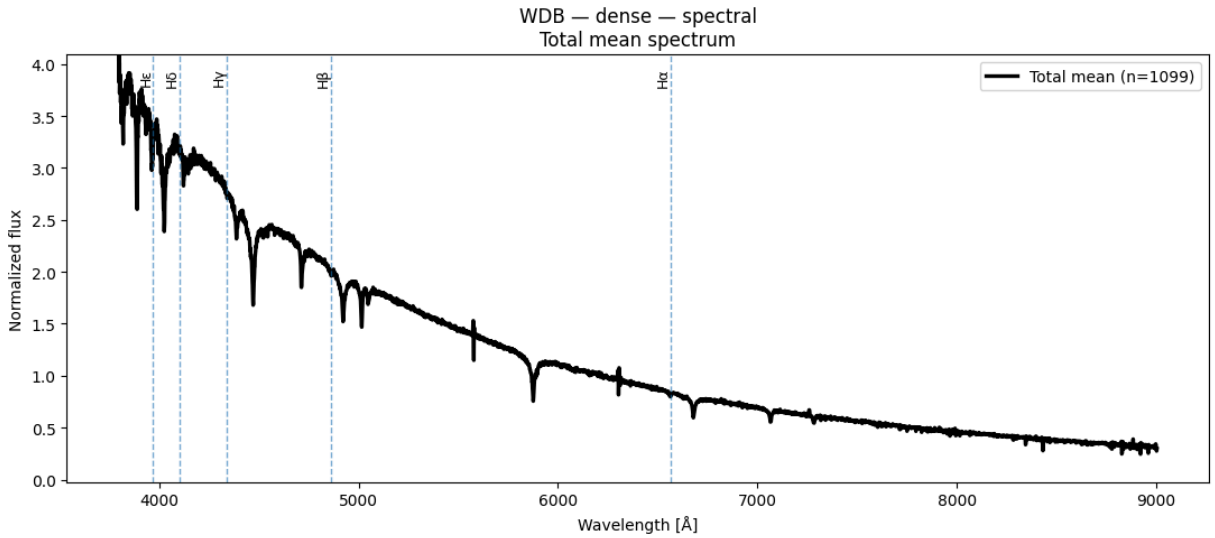
```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```

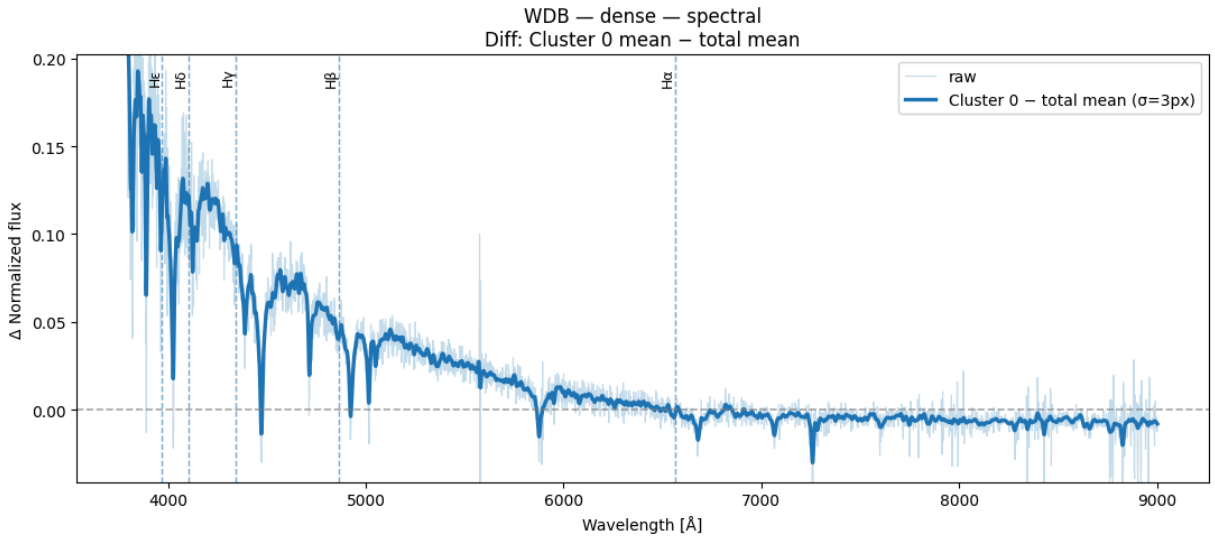

WDB — dense — spectral
Cluster 0 (n=767)
Cluster 0 mean (n=767)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

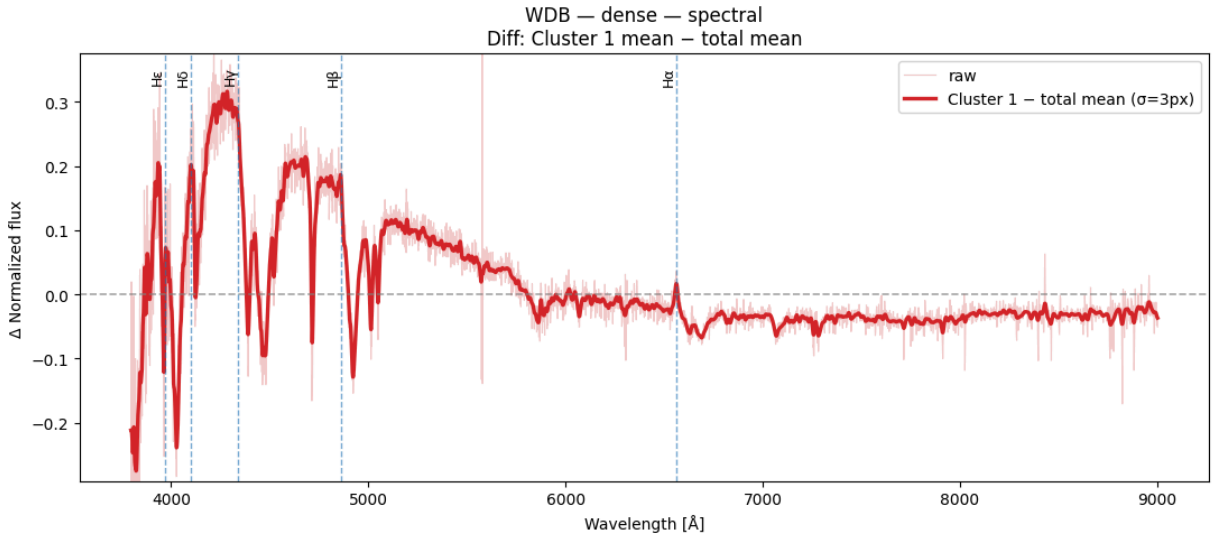

WDB — dense — spectral
Cluster 1 (n=63)
Cluster 1 mean (n=63)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

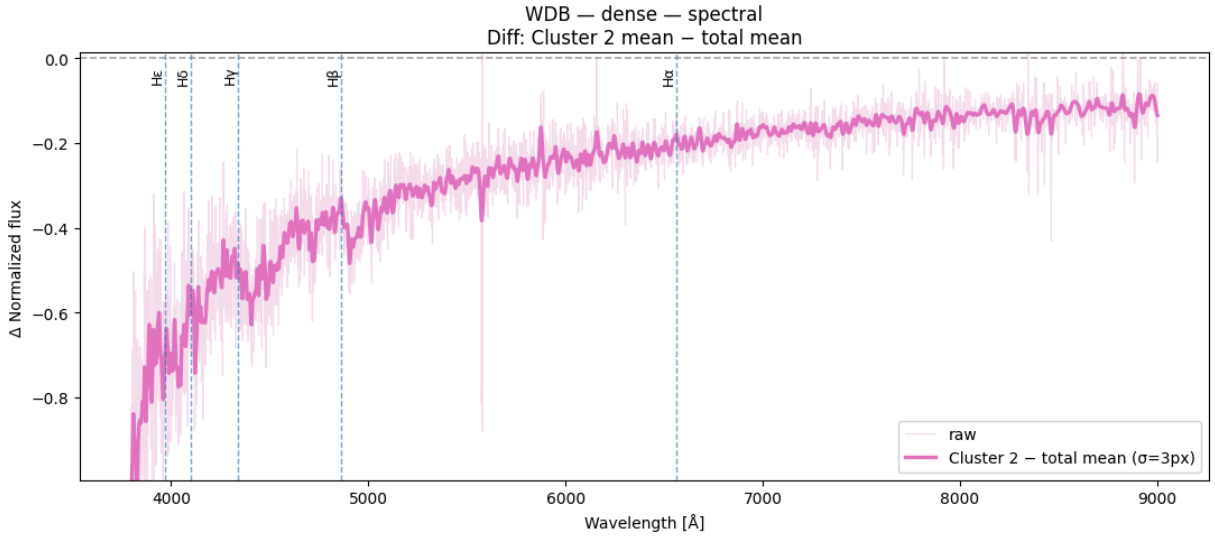

WDB — dense — spectral
Cluster 2 (n=31)
Cluster 2 mean (n=31)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

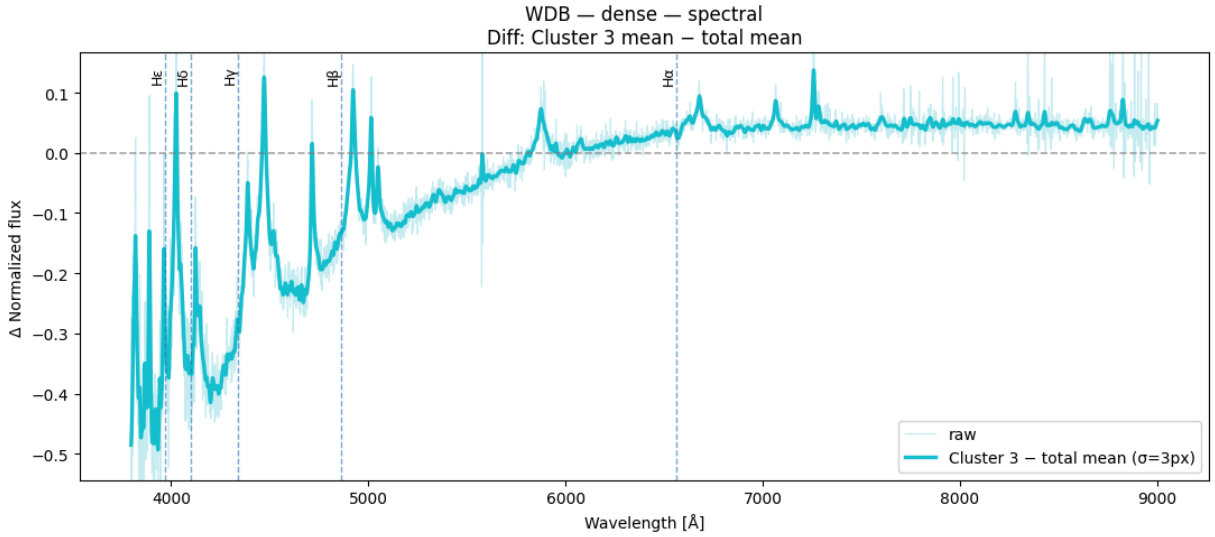

WDB — dense — spectral
Cluster 3 (n=238)
Cluster 3 mean (n=238)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

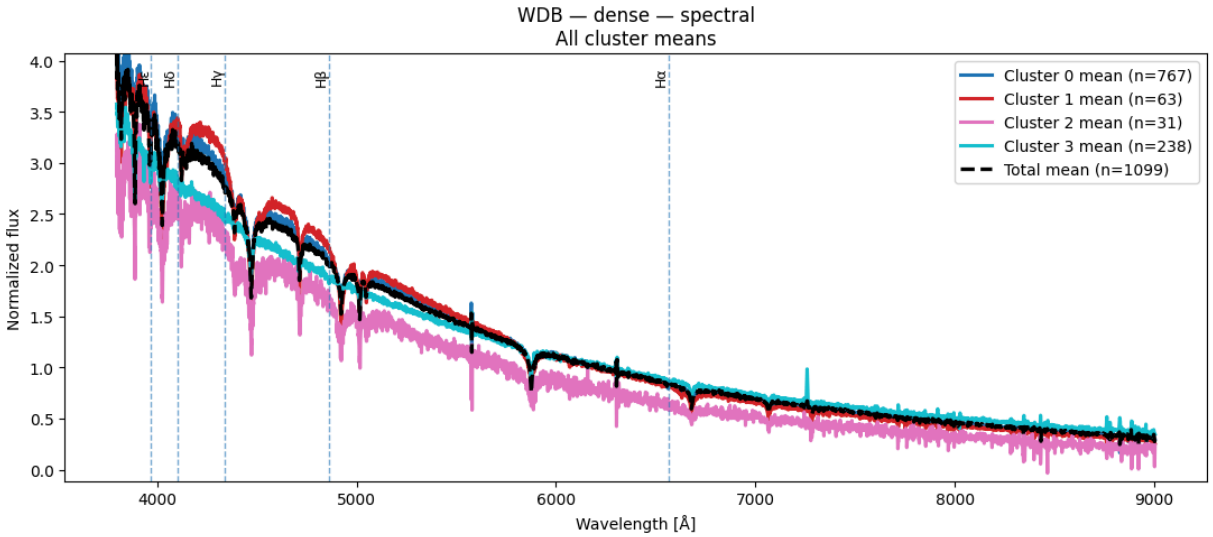

WDB — dense — spectral
Total mean spectrum
Total mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — dense — spectral
Diff: Cluster 0 mean − total mean
raw
Cluster 0 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]


WDB — dense — spectral
Diff: Cluster 1 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
raw
Cluster 1 − total mean (σ=3px)
Δ Normalized flux
Wavelength [Å]


WDB — dense — spectral
Diff: Cluster 2 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
raw
Cluster 2 − total mean (σ=3px)
Δ Normalized flux
Wavelength [Å]

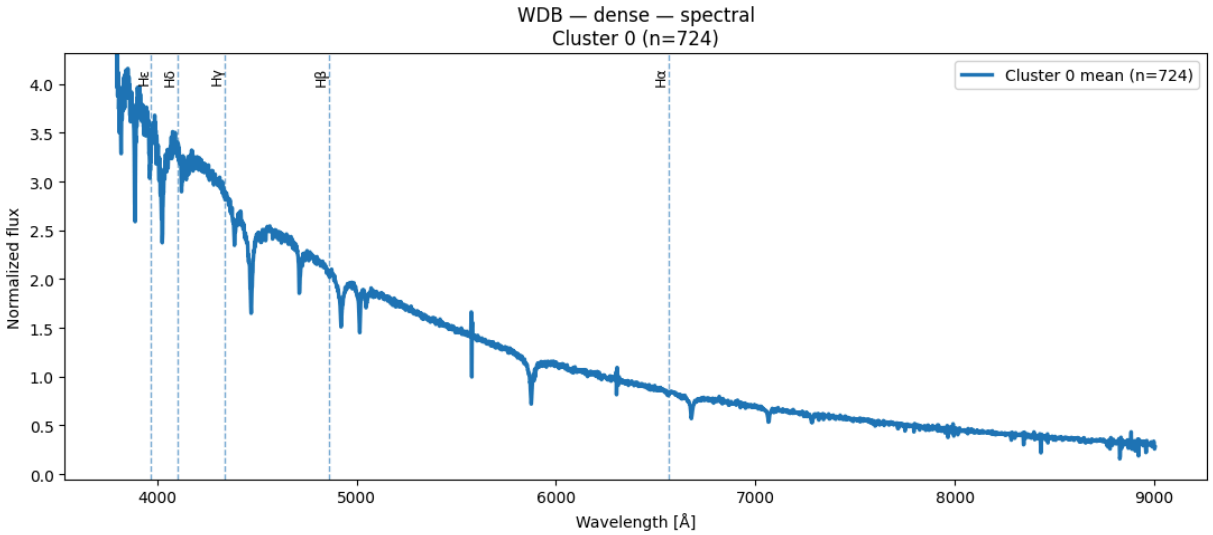

WDB — dense — spectral
Diff: Cluster 3 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
raw
Cluster 3 − total mean (σ=3px)
Δ Normalized flux
Wavelength [Å]

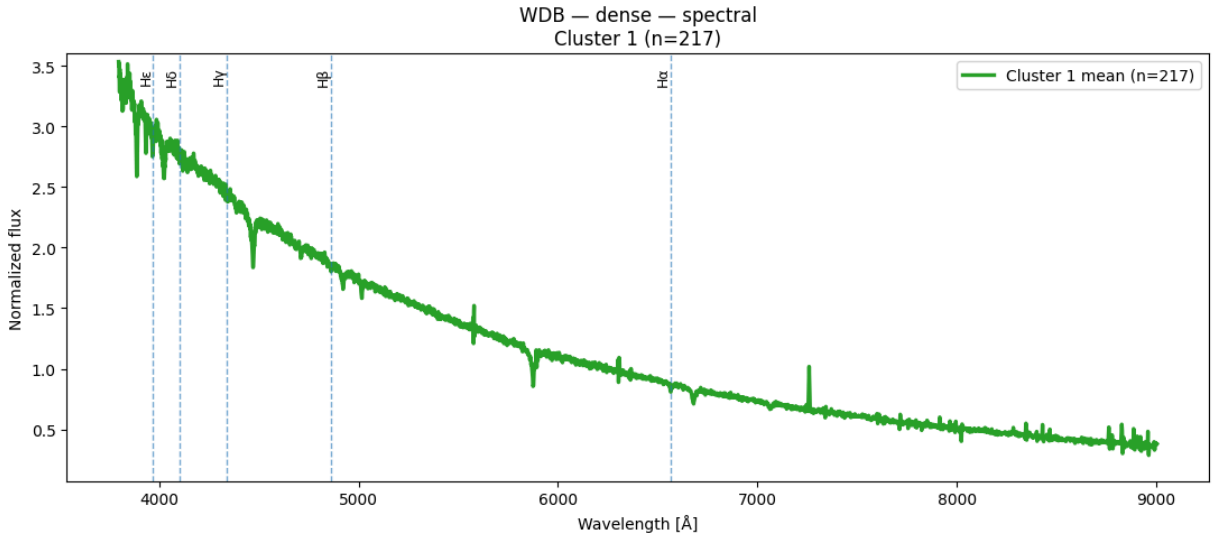


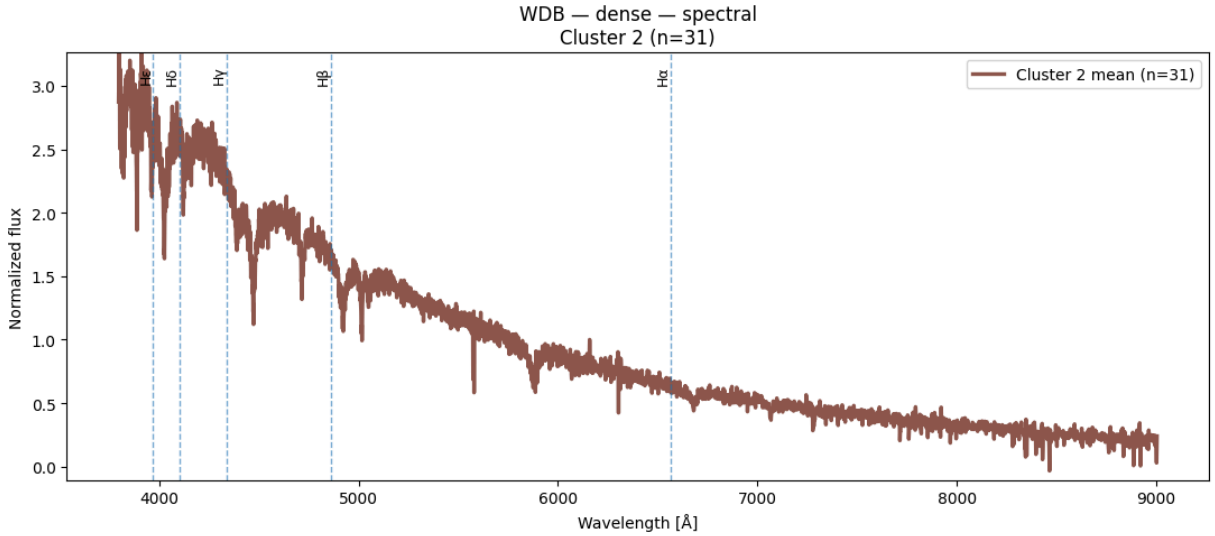
Espectros obtenidos de Spectral Clustering k = 5 y n = 1100

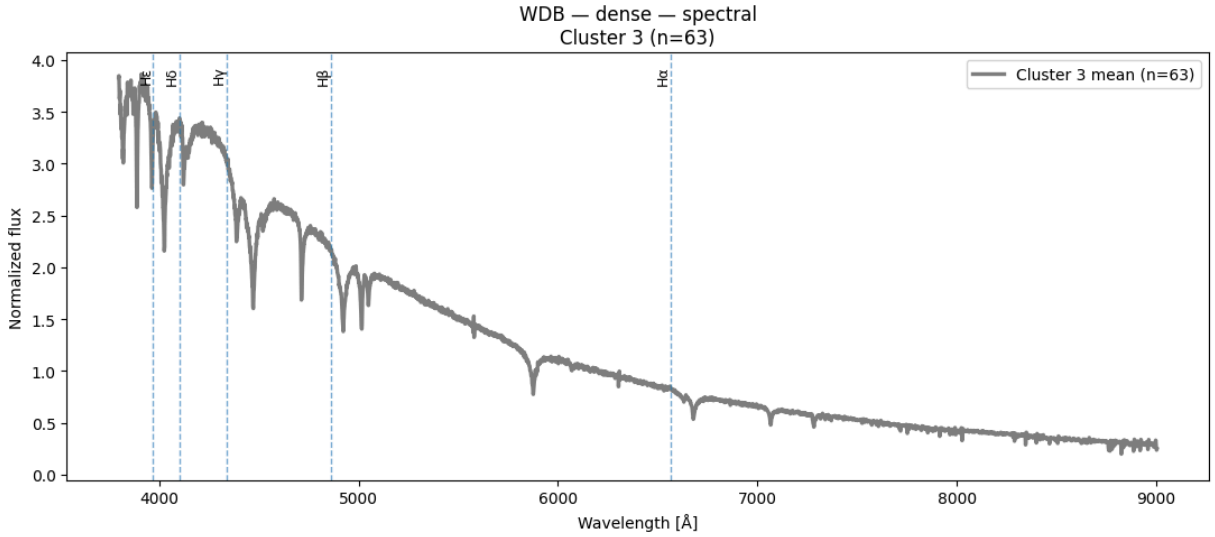
In [66]:
```
result = plot_spectra_spectral(X_wdb_spec, labels_dense_k5_wdb, W_wdb[0],
                               class_name="WDB", layer_name="dense")
```

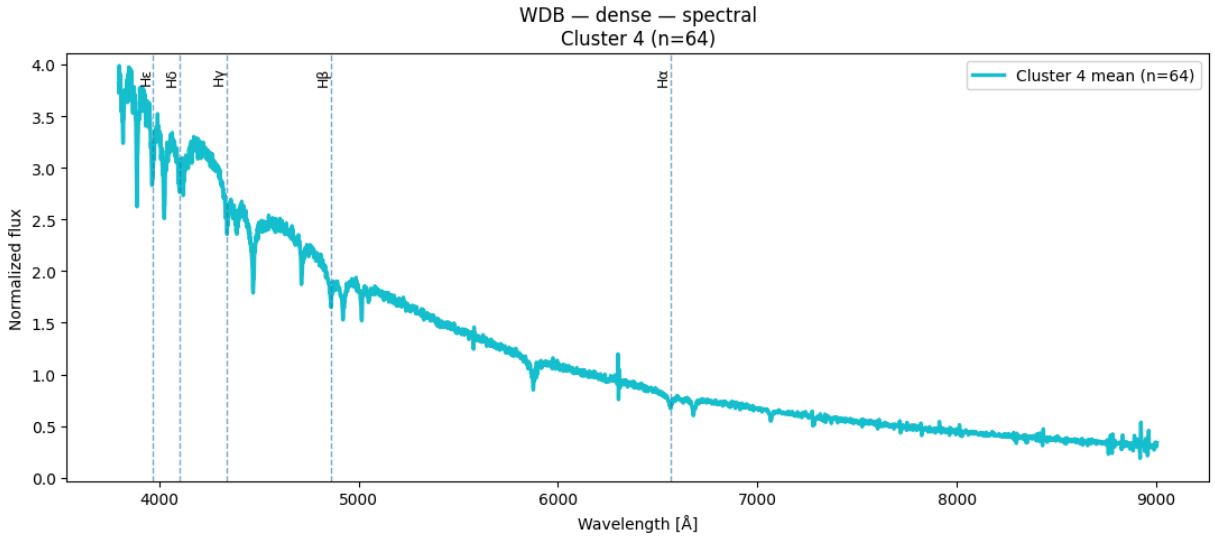
```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```

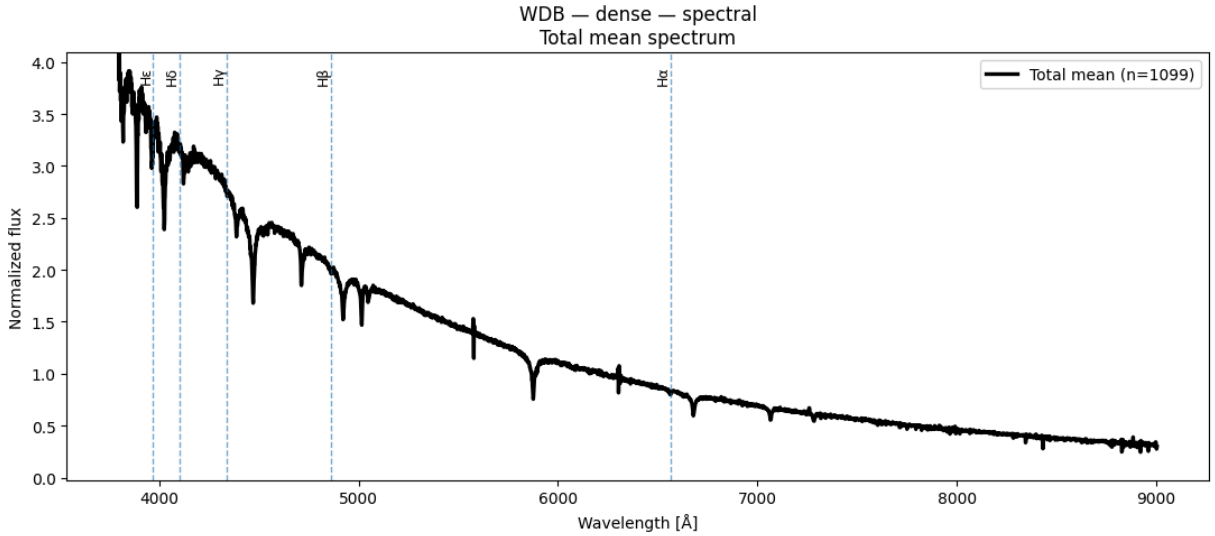


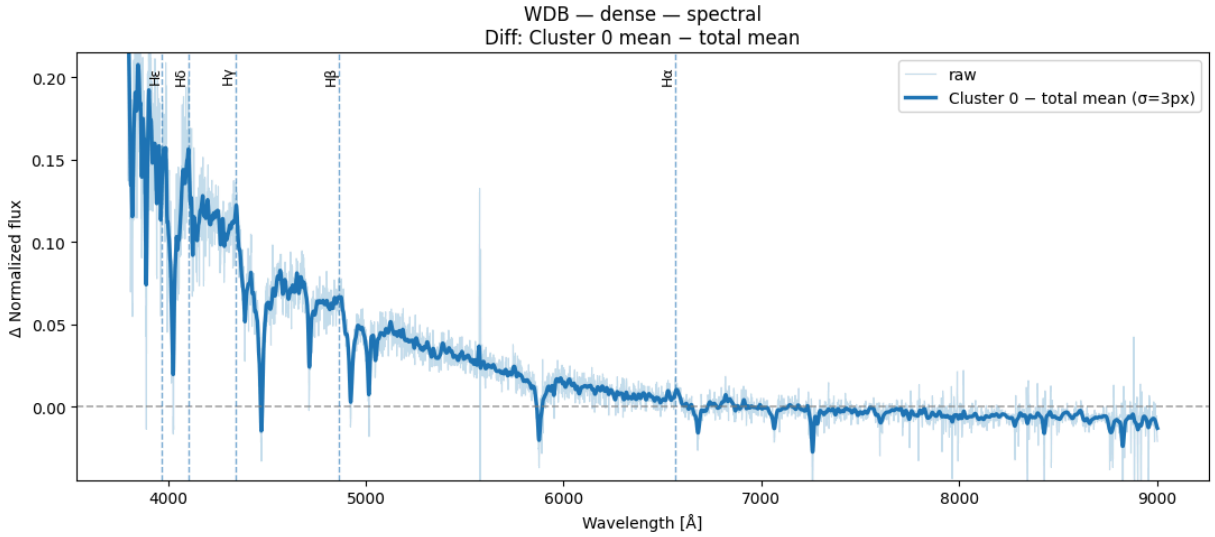

WDB — dense — spectral
Cluster 1 (n=217)
Cluster 1 mean (n=217)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

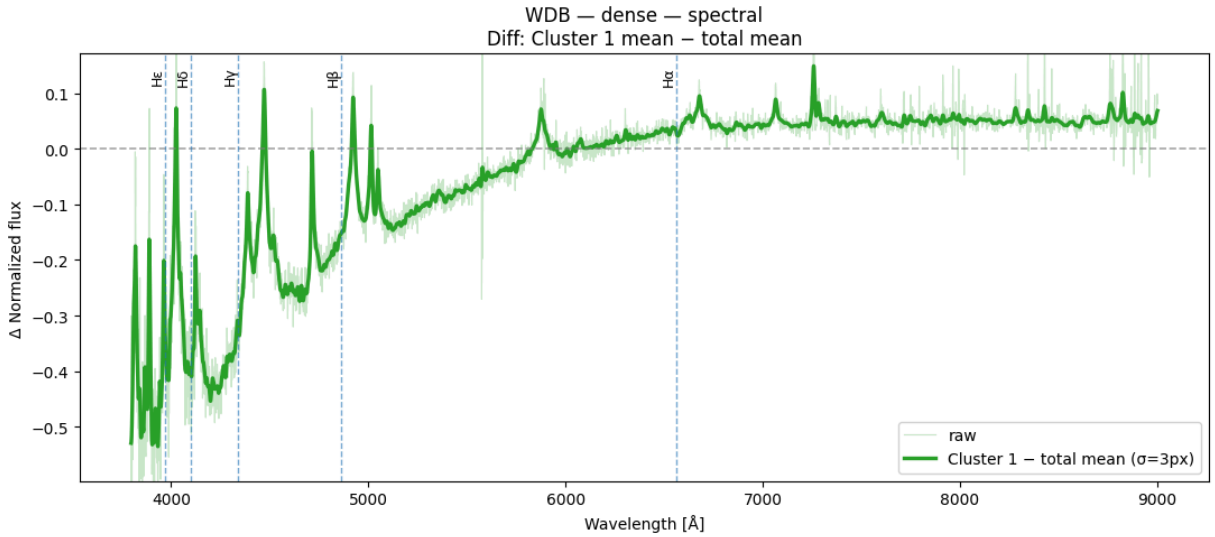

WDB — dense — spectral
Cluster 2 (n=31)
Cluster 2 mean (n=31)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

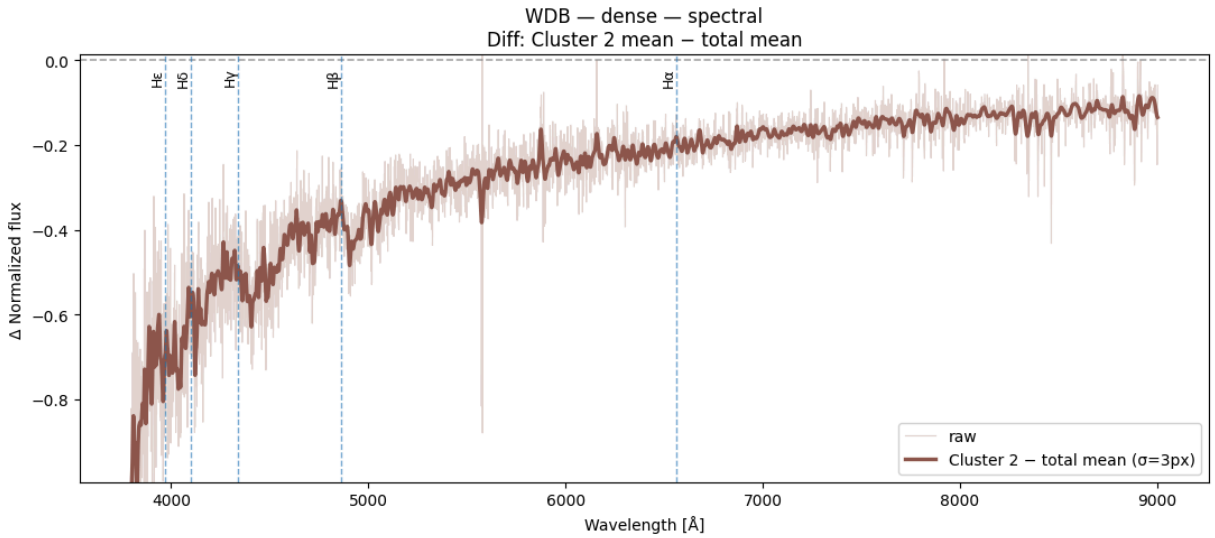

WDB — dense — spectral
Cluster 3 (n=63)
Cluster 3 mean (n=63)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

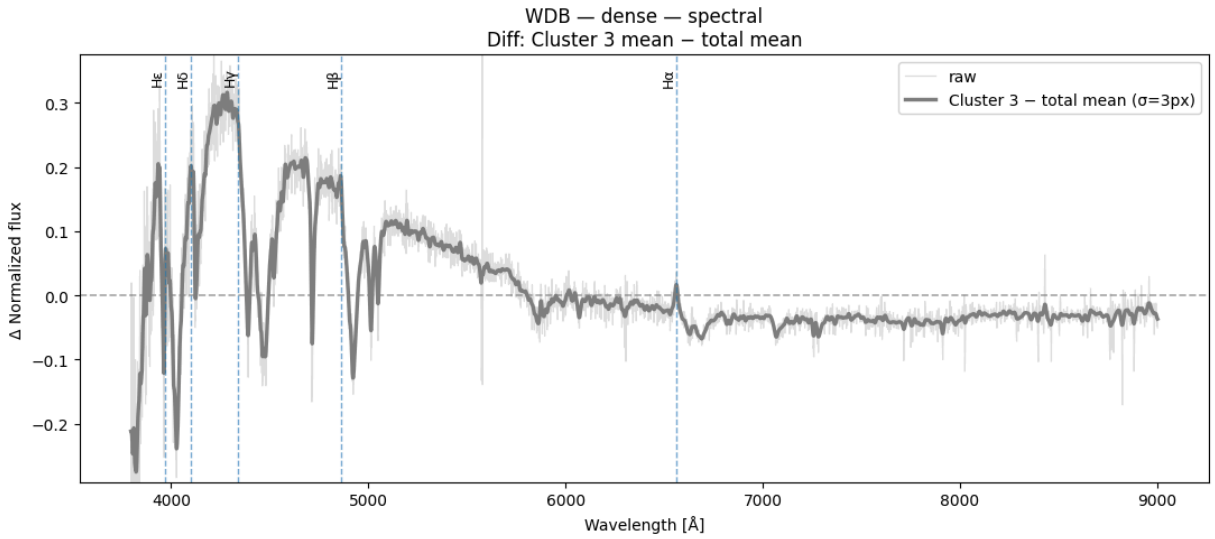

WDB — dense — spectral
Cluster 4 (n=64)
Cluster 4 mean (n=64)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

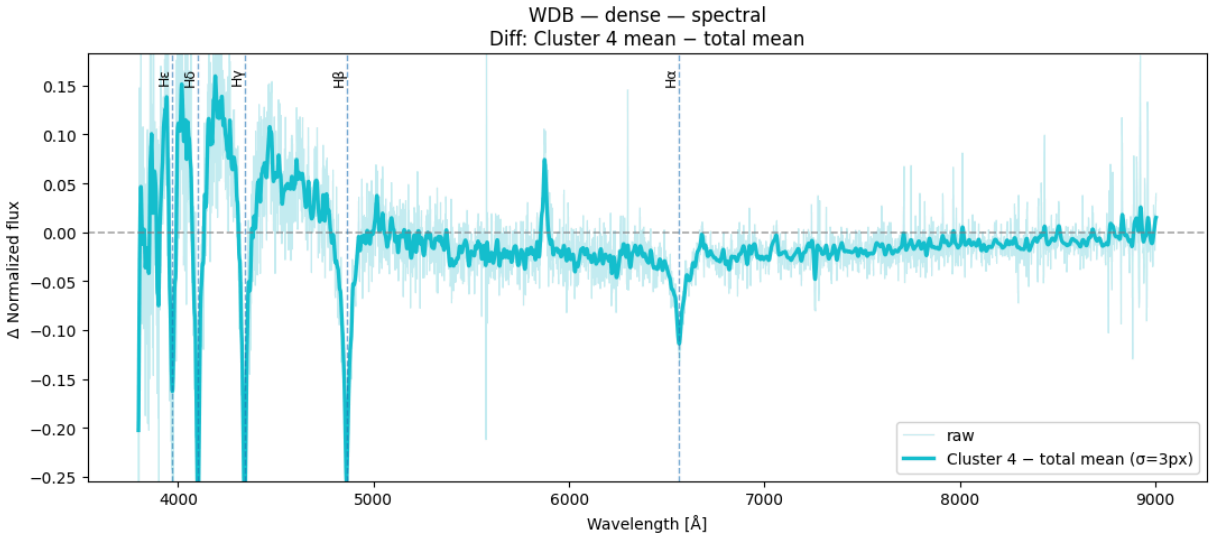

WDB — dense — spectral
Total mean spectrum
Total mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

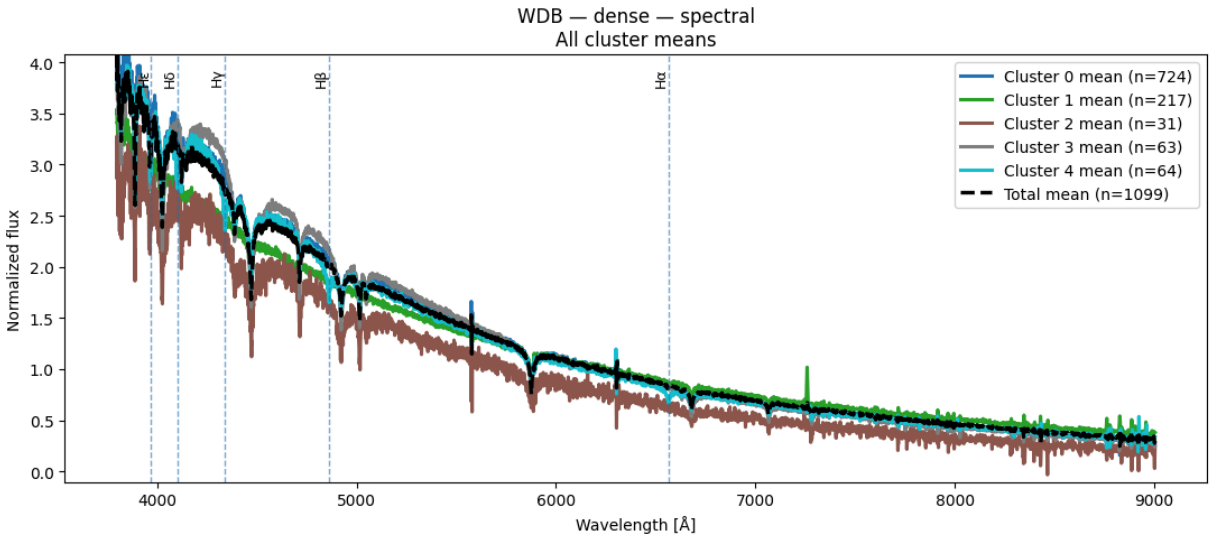

WDB — dense — spectral
Diff: Cluster 0 mean − total mean
raw
Cluster 0 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]


WDB — dense — spectral
Diff: Cluster 1 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]
raw
Cluster 1 − total mean (σ=3px)


WDB — dense — spectral
Diff: Cluster 2 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]
raw
Cluster 2 − total mean (σ=3px)


WDB — dense — spectral
Diff: Cluster 3 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]
raw
Cluster 3 − total mean (σ=3px)

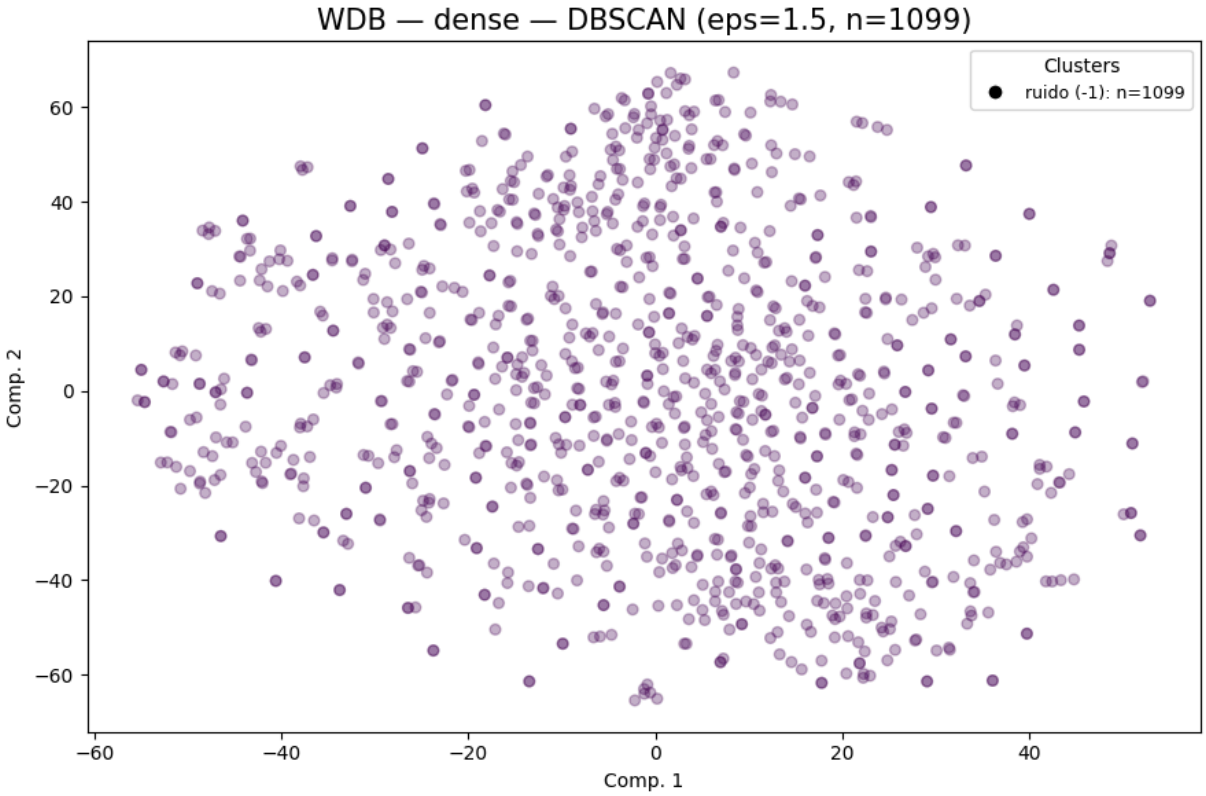




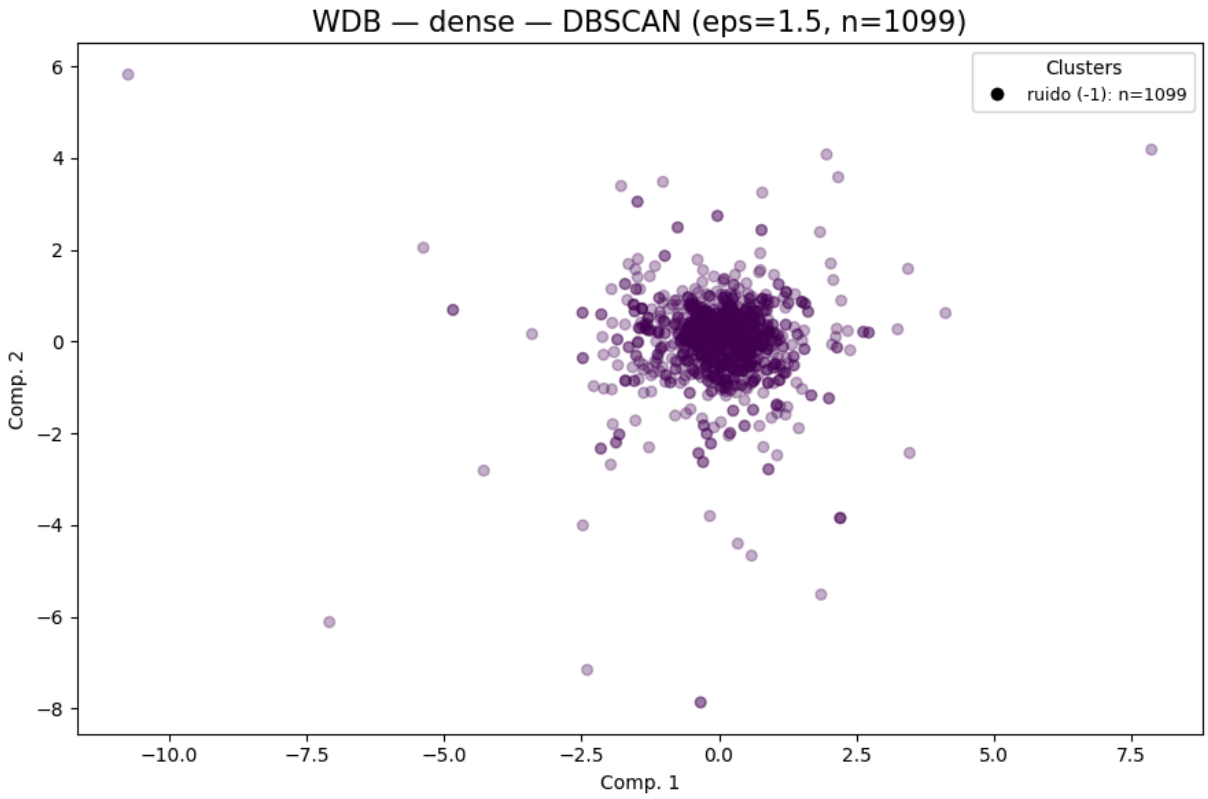
DBSCAN

In [67]:
```
dbs_dense_wdb = cluster_plot_DBSCAN(
            repr_wdb_32["dense"],
            "WDB",
            "dense",
            dbscan_eps=1.5,
            dbscan_min_samples=10,
            vis_method="tsne",
            xlim=None,
            ylim=None,
            container=cluster_labels_all
        )
labels_dbs_dense_wdb = dbs_dense_wdb["dbscan"]
cluster_plot_DBSCAN(
            repr_wdb_32["dense"],
            "WDB",
            "dense",
            dbscan_eps=1.5,
            dbscan_min_samples=10,
            vis_method="pca",
            xlim=None,
```

```
            ylim=None,
            container=cluster_labels_all
        )
```




```
┌──────────────────────────────────────────────────────────────────────────────┐
│  WDB | dense | dbscan | eps=1.5
│  clusters=0  noise=1099  sil=N/A  DB=N/A  CH=N/A
│  DBCV = N/A  (rango [-1,1], especifico DBSCAN)
└──────────────────────────────────────────────────────────────────────────────┘
```



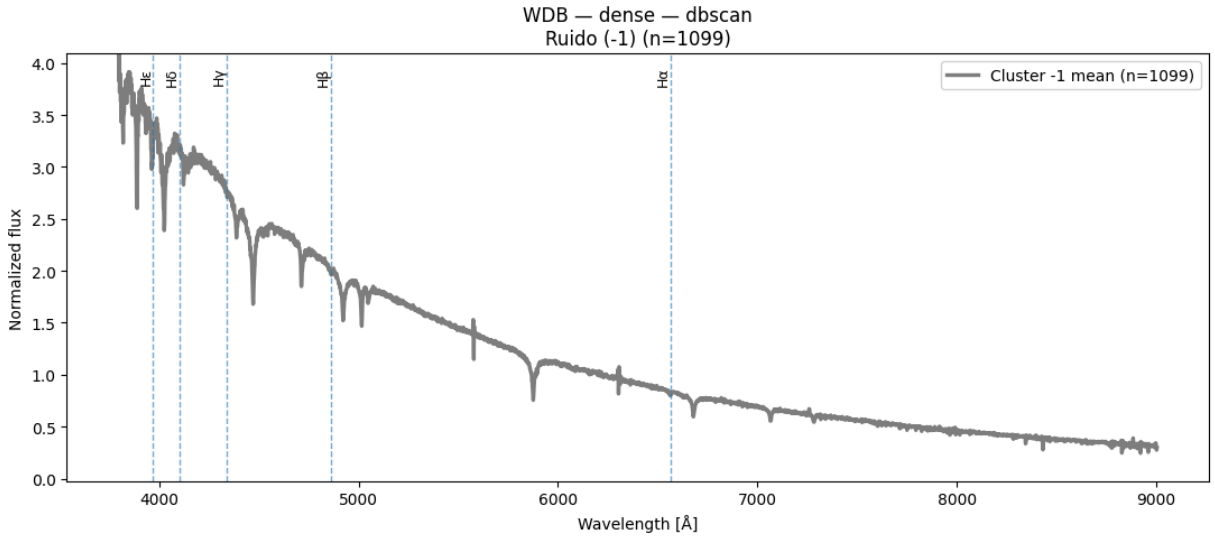

```
┌──────────────────────────────────────────────────────────────┐
│  WDB | dense | dbscan | eps=1.5                              │
│  clusters=0  noise=1099  sil=N/A  DB=N/A  CH=N/A             │
│  DBCV = N/A  (rango [-1,1], especifico DBSCAN)               │
└──────────────────────────────────────────────────────────────┘
```

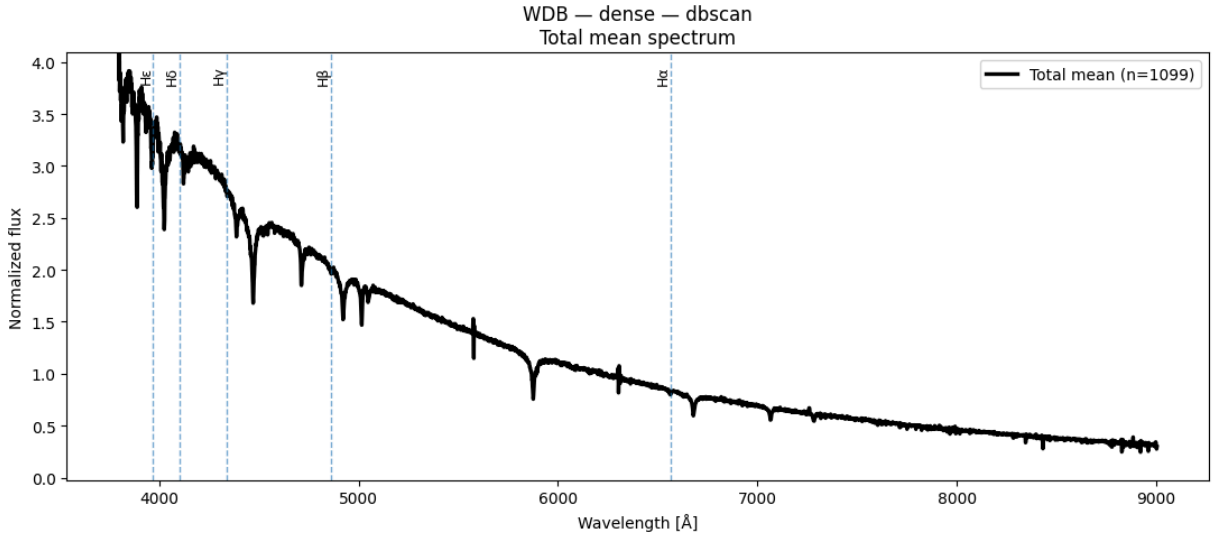
Out[67]:
```
{'dbscan': array([-1, -1, -1, ..., -1, -1, -1], dtype=int64),
 'metrics': {'n_clusters': 0,
  'n_noise': 1099,
  'silhouette': None,
  'davies_bouldin': None,
  'calinski_harabasz': None},
 'dbcv': nan}
```

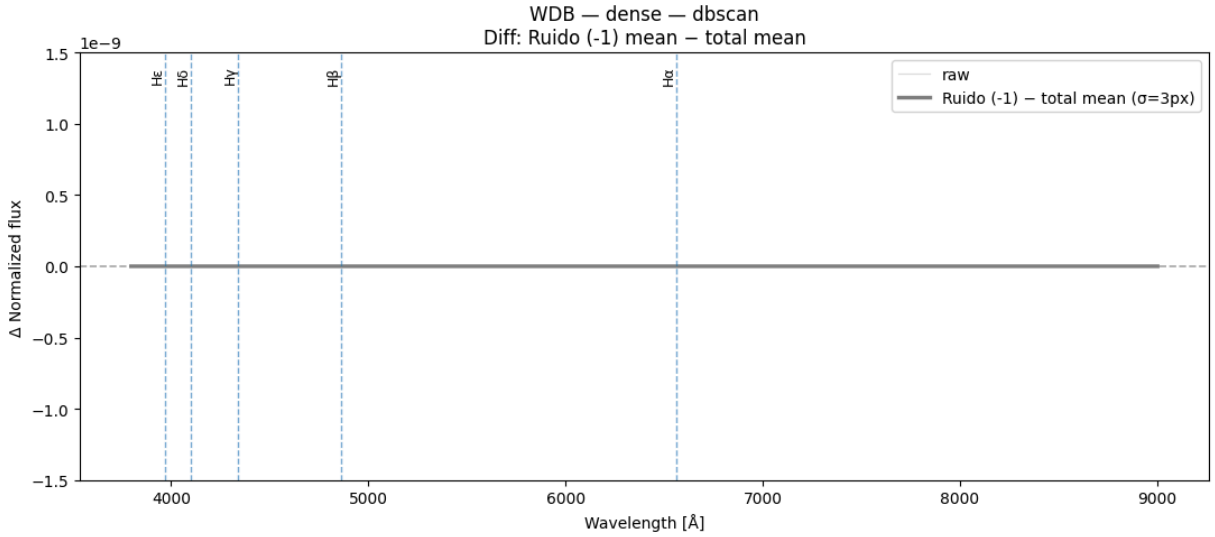
Espectros obtenidos de DBSCAN n = 1100

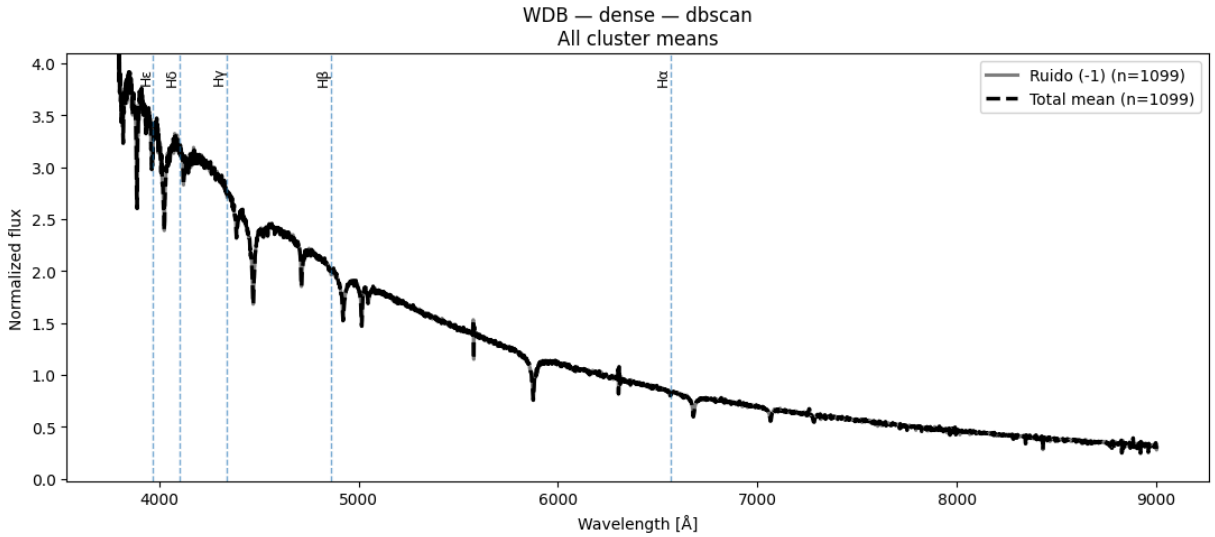
In [68]:
```
plot_spectra_dbscan(
    X_wdb_spec,
    labels_dbs_dense_wdb,
    W_wdb[0],
    class_name="WDB",
    layer_name="dense"
)
```

```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```


WDB — dense — dbscan
Ruido (-1) (n=1099)
Cluster -1 mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — dense — dbscan
Total mean spectrum
Total mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — dense — dbscan
Diff: Ruido (-1) mean − total mean
1e−9
raw
Ruido (-1) − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]

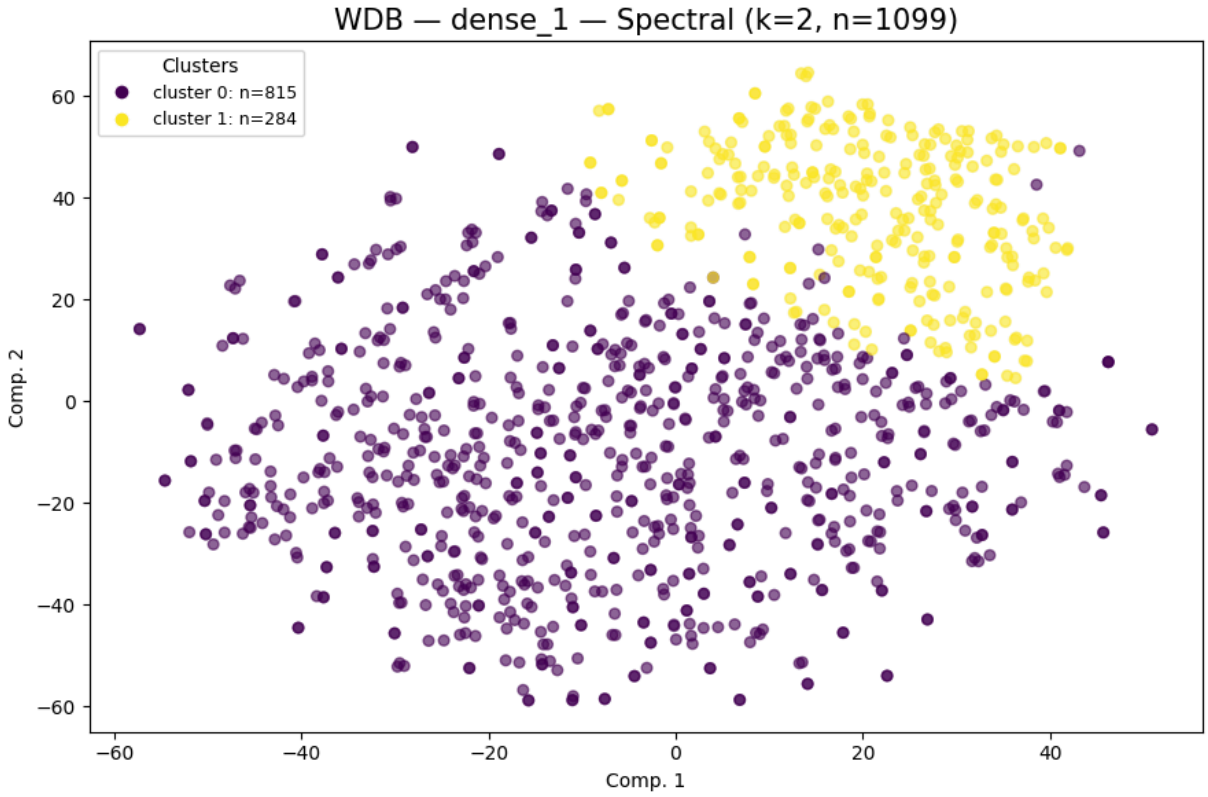


Out[68]:
{'total_mean': array([4.2840786 , 3.8304825 , 3.8275013 , ..., 0.3190189 , 0.27551222,
0.2985266], dtype=float32),
'group_means': {-1: array([4.2840786 , 3.8304825 , 3.8275013 , ..., 0.3190189 , 0.27551222,
0.2985266], dtype=float32)},
'unique_clusters': array([-1], dtype=int64)}

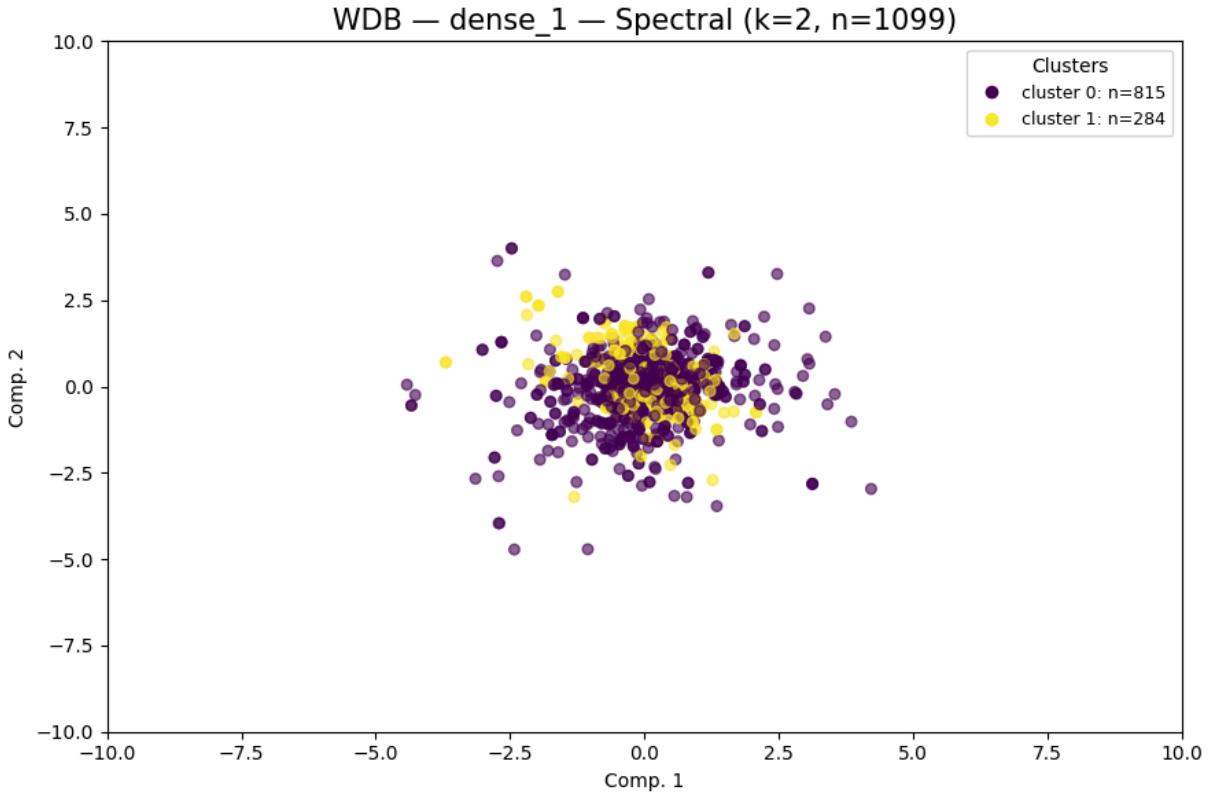
Resultados capa dense_1, con Spectral Clustering + tSNE y PCA para n=1100

Spectral Clustering k = 2

In [69]:
```
dense1_wdb_k2=cluster_plot_Spect(
            repr_wdb_32["dense_1"],
            "WDB",
            "dense_1",
            spectral_k=2,
            vis_method="tsne",
            container=cluster_labels_all
        )

labels_dense1_k2_wdb = dense1_wdb_k2["spectral"]
cluster_plot_Spect(
            repr_wdb_32["dense_1"],
            "WDB",
            "dense_1",
            spectral_k=2,
            vis_method="pca",
            xlim=(-10,10), ylim=(-10,10)
        )
```




```
WDB | dense_1 | spectral | k=2
clusters=2  noise=0  sil=-0.0278  DB=4.0860  CH=27.35
```

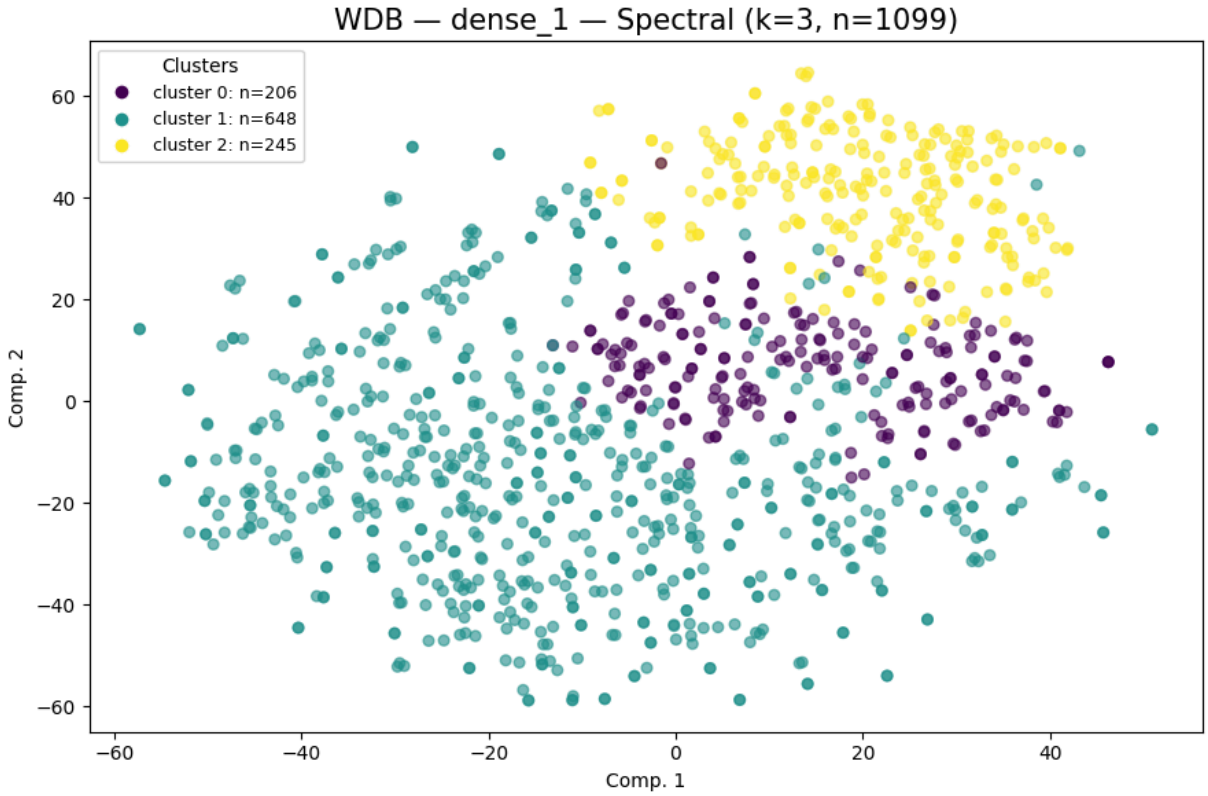


```
┌──────────────────────────────────────────────────────────────────────────┐
│  WDB | dense_1 | spectral | k=2                                          │
│  clusters=2  noise=0  sil=-0.0278  DB=4.0860  CH=27.35                   │
└──────────────────────────────────────────────────────────────────────────┘
```

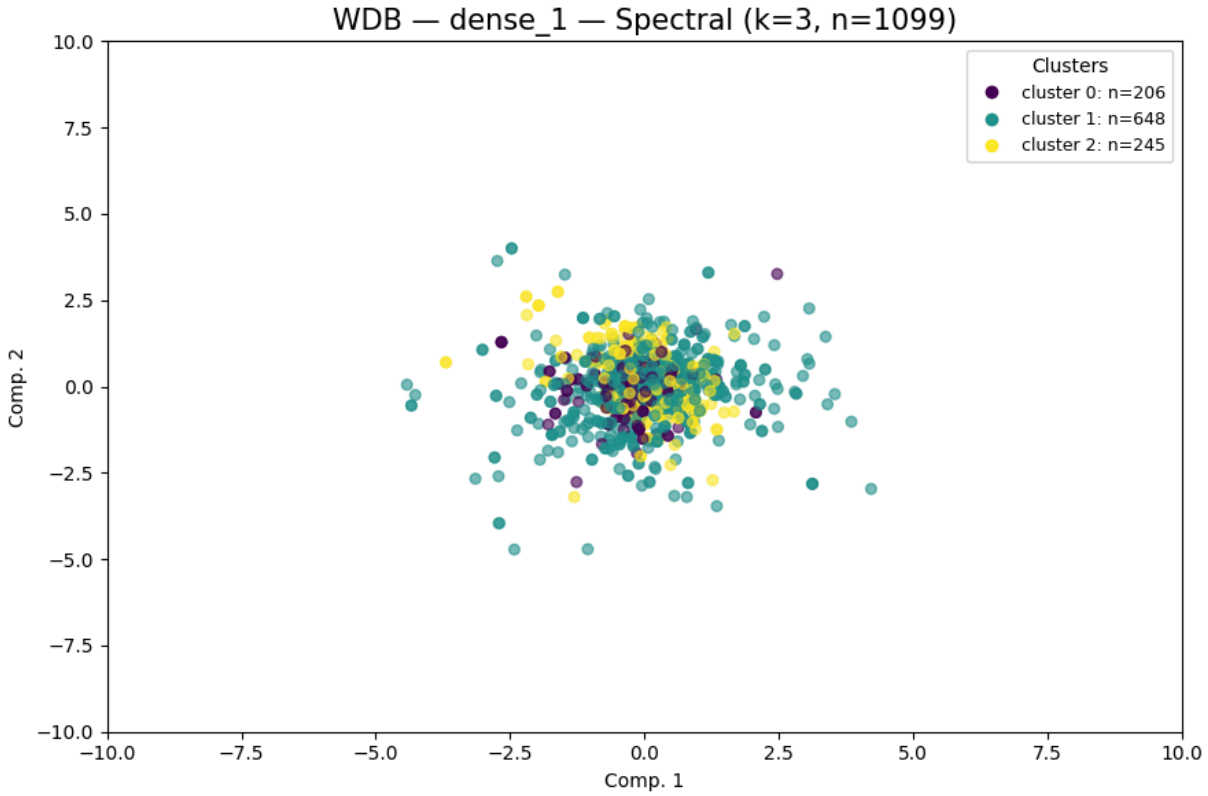
Out[69]:
```
{'spectral': array([0, 1, 1, ..., 0, 0, 0]),
 'metrics': {'n_clusters': 2,
  'n_noise': 0,
  'silhouette': -0.027791,
  'davies_bouldin': 4.085976,
  'calinski_harabasz': 27.3497}}
```

Spectral Clustering k = 3

In [70]:
```
dense1_wdb_k3=cluster_plot_Spect(
            repr_wdb_32["dense_1"],
            "WDB",
            "dense_1",
            spectral_k=3,
            vis_method="tsne",
            container=cluster_labels_all
        )

labels_dense1_k3_wdb = dense1_wdb_k3["spectral"]
cluster_plot_Spect(
            repr_wdb_32["dense_1"],
            "WDB",
            "dense_1",
            spectral_k=3,
            vis_method="pca",
            xlim=(-10,10), ylim=(-10,10)
        )
```




```
┌──────────────────────────────────────────────────────────────────────┐
│  WDB | dense_1 | spectral | k=3
│  clusters=3  noise=0  sil=-0.0564  DB=4.0825  CH=24.10
└──────────────────────────────────────────────────────────────────────┘
```

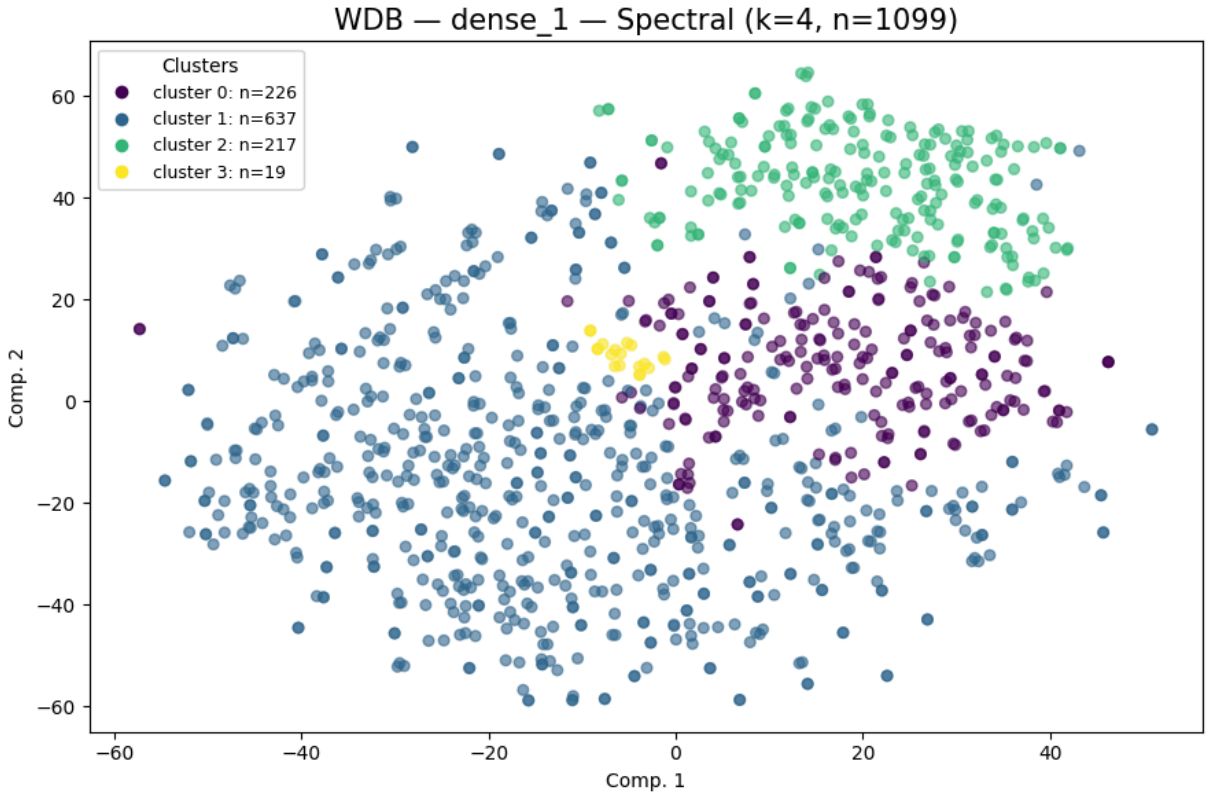


```
┌──────────────────────────────────────────────────────────────────────────┐
│  WDB | dense_1 | spectral | k=3
│  clusters=3  noise=0  sil=-0.0564  DB=4.0825  CH=24.10
└──────────────────────────────────────────────────────────────────────────┘
```

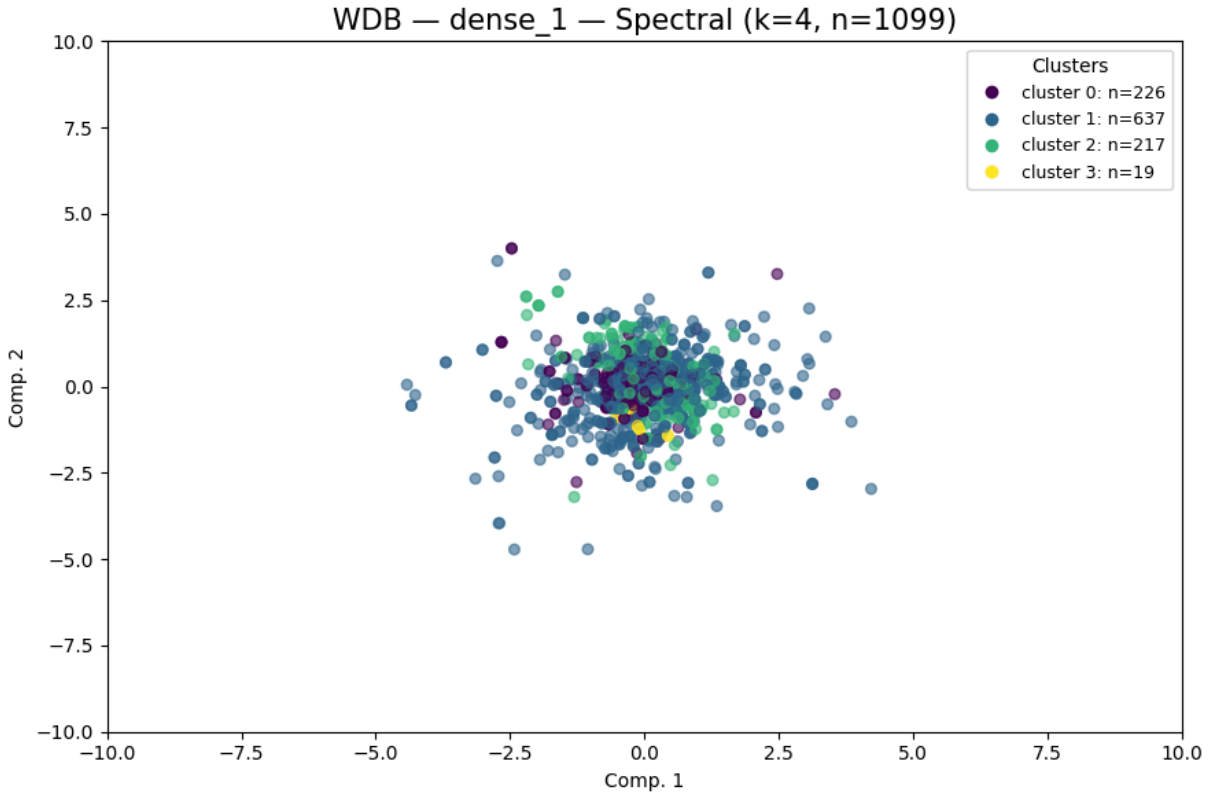
Out[70]:
```
{'spectral': array([1, 2, 0, ..., 0, 1, 1]),
 'metrics': {'n_clusters': 3,
  'n_noise': 0,
  'silhouette': -0.056364,
  'davies_bouldin': 4.082494,
  'calinski_harabasz': 24.0956}}
```

Spectral Clustering k = 4

In [71]:
```
dense1_wdb_k4=cluster_plot_Spect(
            repr_wdb_32["dense_1"],
            "WDB",
            "dense_1",
            spectral_k=4,
            vis_method="tsne",
            container=cluster_labels_all
        )

labels_dense1_k4_wdb = dense1_wdb_k4["spectral"]
cluster_plot_Spect(
            repr_wdb_32["dense_1"],
            "WDB",
            "dense_1",
            spectral_k=4,
            vis_method="pca",
            xlim=(-10,10), ylim=(-10,10)
        )
```




```
┌──────────────────────────────────────────────────────────────────────┐
│  WDB | dense_1 | spectral | k=4                                      │
│  clusters=4  noise=0  sil=-0.0713  DB=3.6420  CH=18.32               │
└──────────────────────────────────────────────────────────────────────┘
```

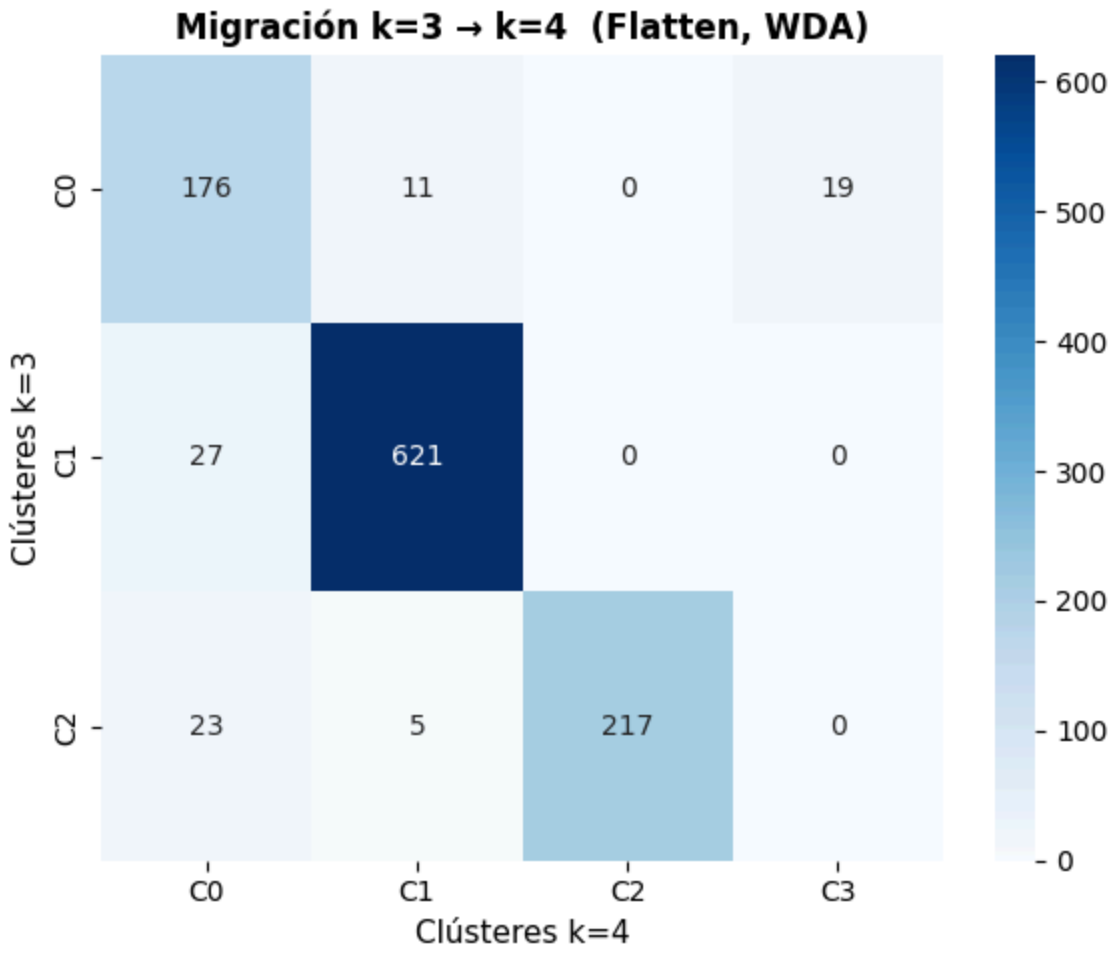


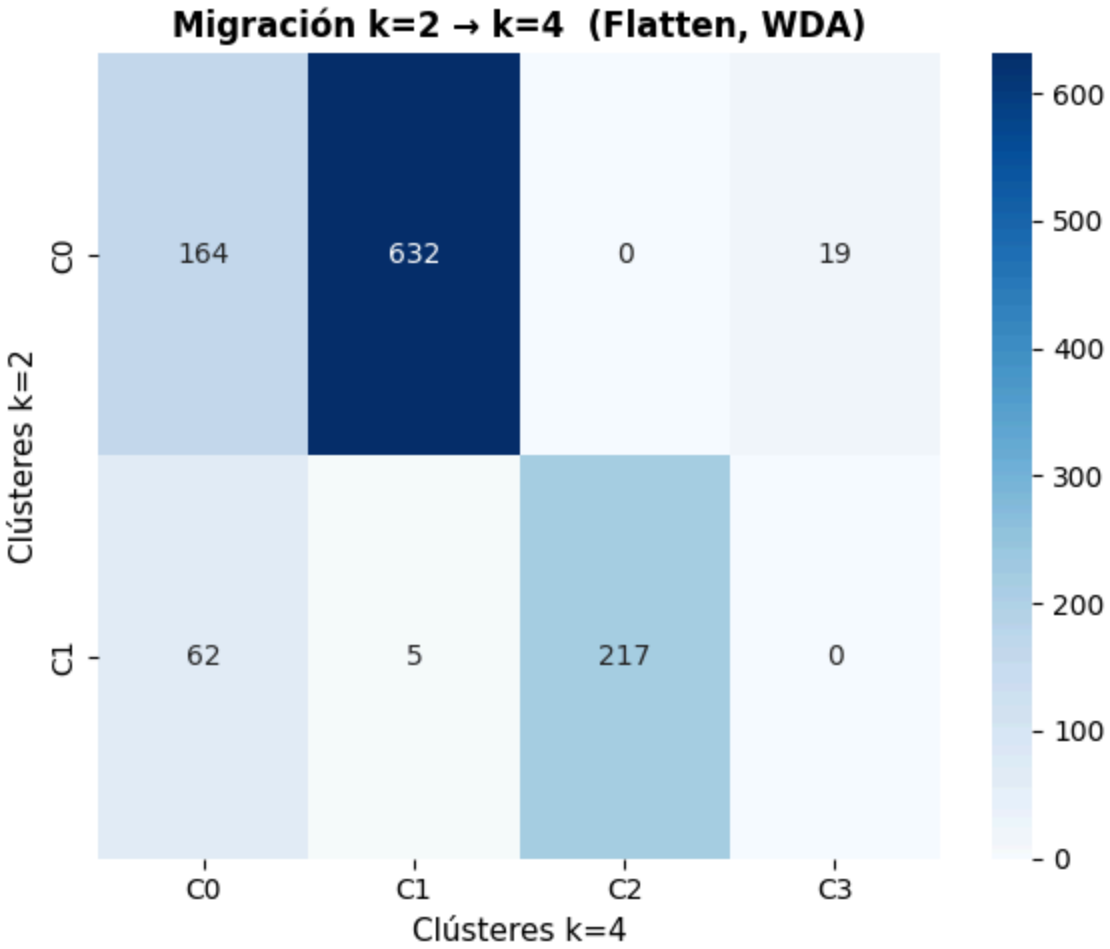
```
┌──────────────────────────────────────────────────────────────────────────┐
│  WDB | dense_1 | spectral | k=4
│  clusters=4  noise=0  sil=-0.0713  DB=3.6420  CH=18.32
└──────────────────────────────────────────────────────────────────────────┘
```

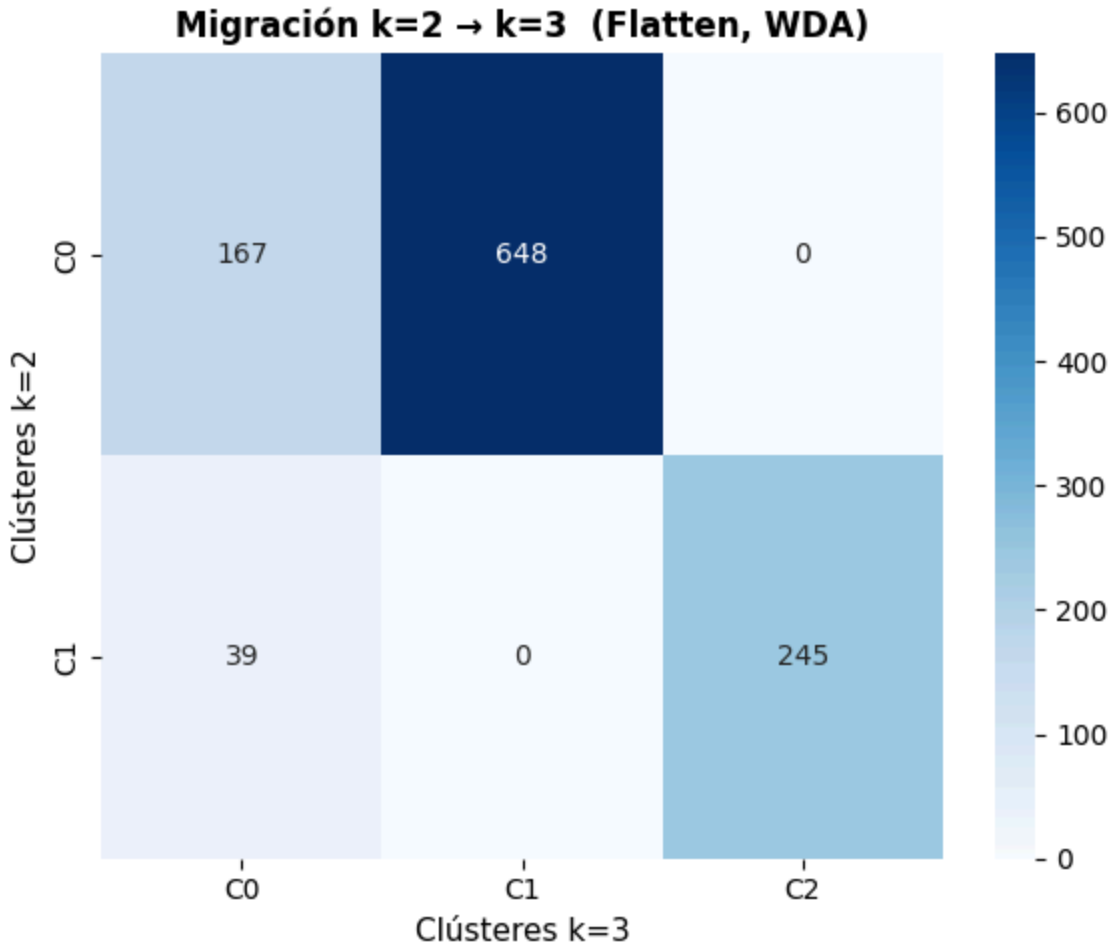
Out[71]:
```
{'spectral': array([1, 2, 0, ..., 0, 1, 1]),
 'metrics': {'n_clusters': 4,
  'n_noise': 0,
  'silhouette': -0.071346,
  'davies_bouldin': 3.641978,
  'calinski_harabasz': 18.3178}}
```

Análisis de migración de clústeres

In [72]:
```
plot_migration_matrix(labels_dense1_k3_wdb, labels_dense1_k4_wdb, ka=3, kb=4)
plot_migration_matrix(labels_dense1_k2_wdb, labels_dense1_k4_wdb, ka=2, kb=4)
plot_migration_matrix(labels_dense1_k2_wdb, labels_dense1_k3_wdb, ka=2, kb=3)
```



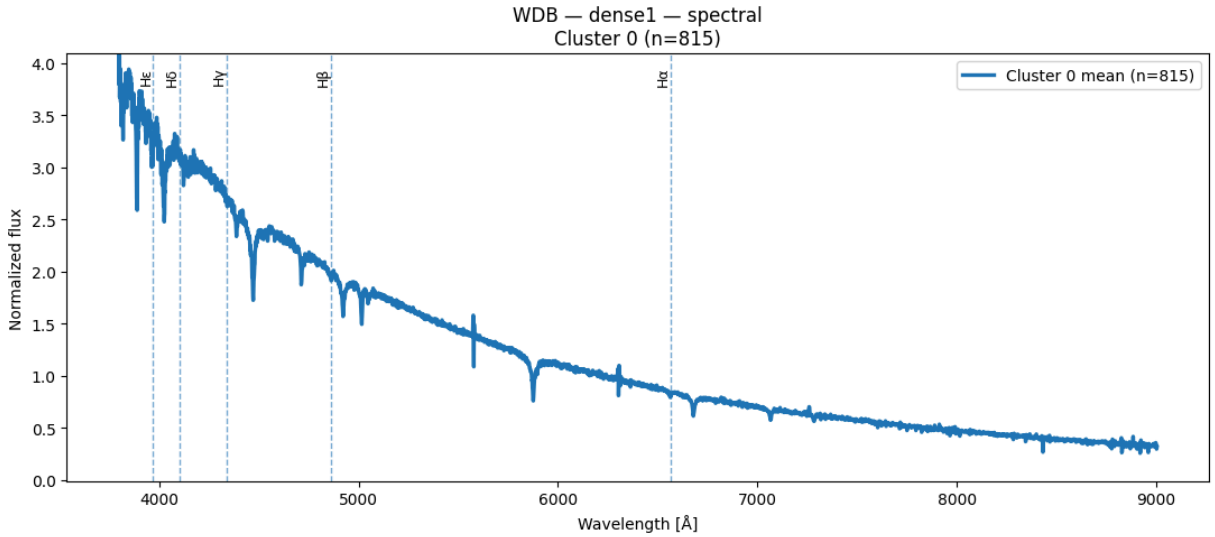

Migración k=2 → k=4 (Flatten, WDA)
Clústeres k=2
C0
C1
164
632
0
19
62
5
217
0
C0
C1
C2
C3
Clústeres k=4
600
500
400
300
200
100
0

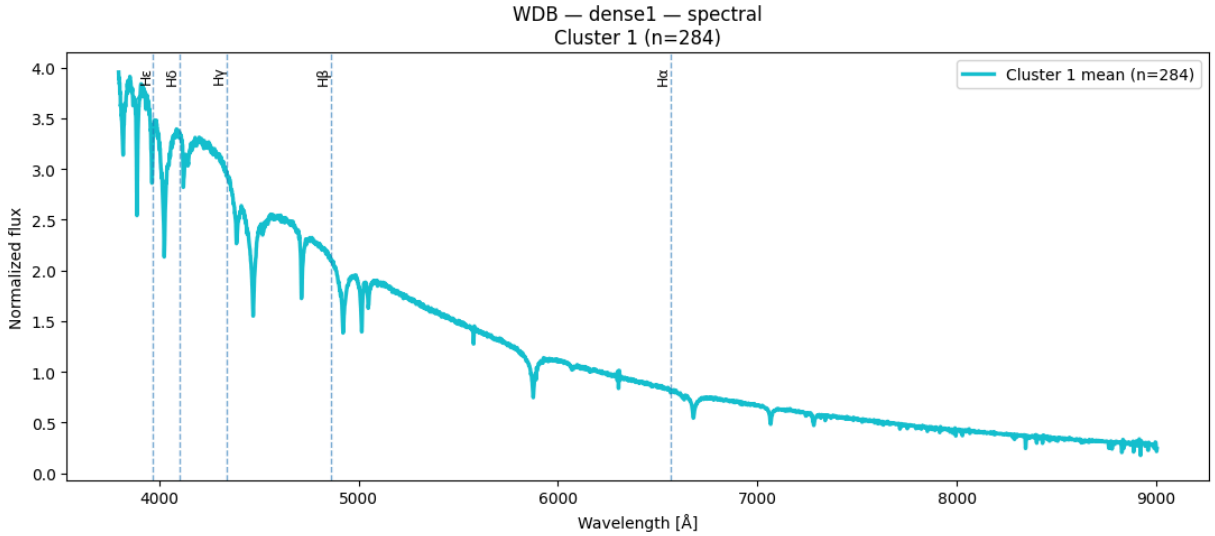


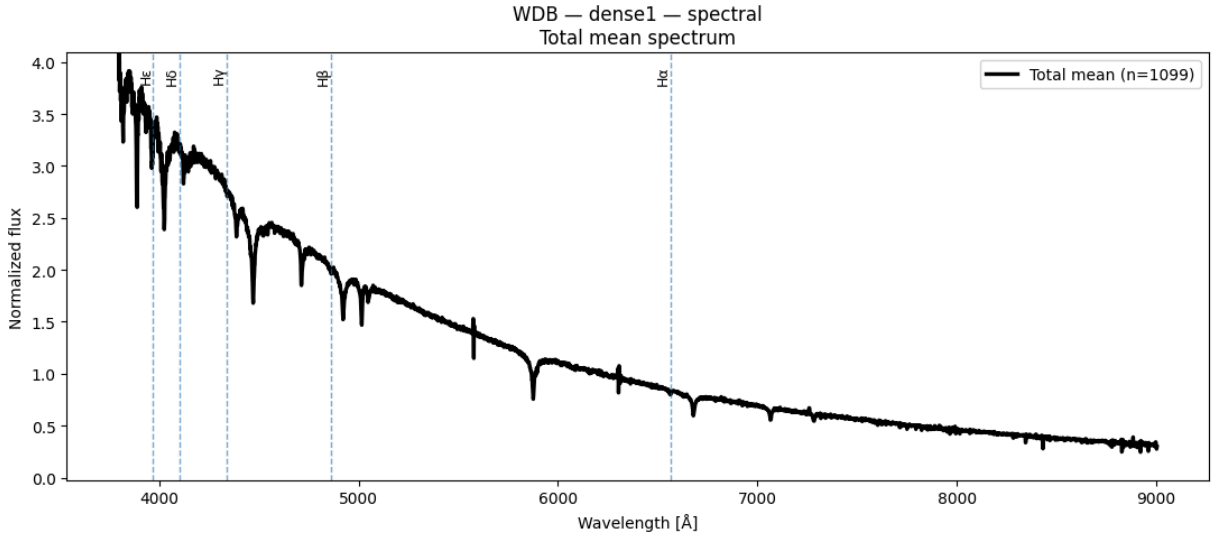
Espectros obtenidos de Spectral Clustering para k=2 y n=1100

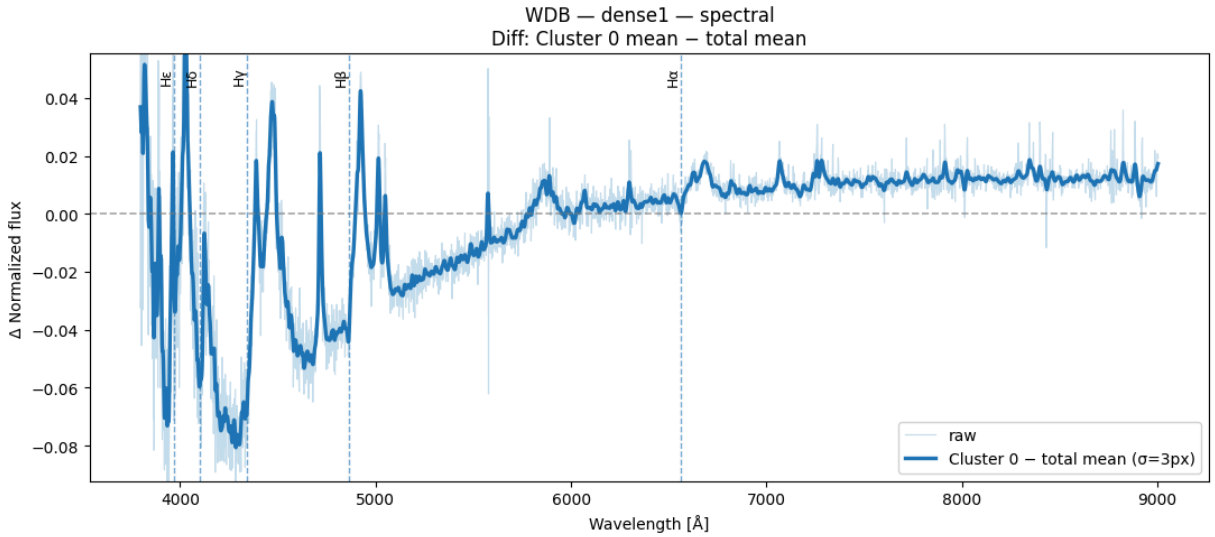
In [73]:
```
result = plot_spectra_spectral(X_wdb_spec, labels_dense1_k2_wdb, W_wdb[0],
                               class_name="WDB", layer_name="dense1")
```

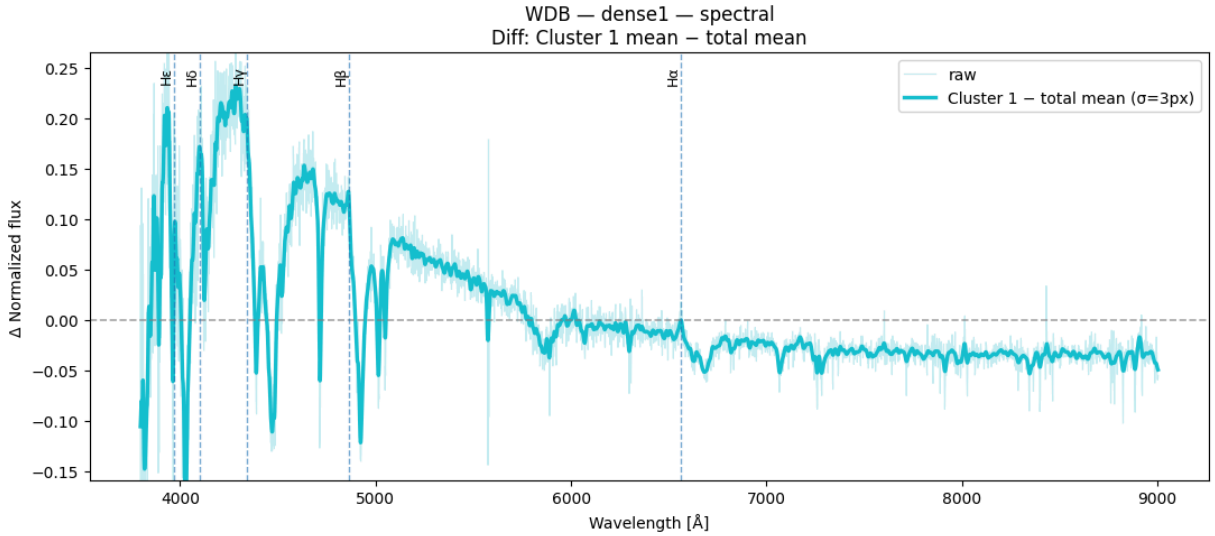
```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```

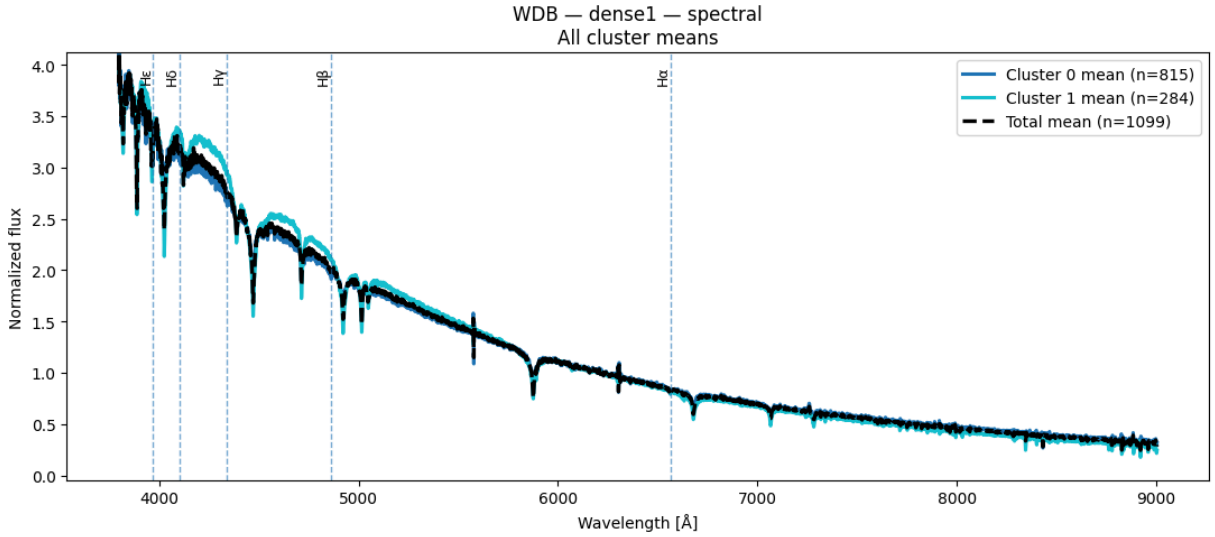

WDB — dense1 — spectral
Cluster 0 (n=815)
Cluster 0 mean (n=815)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — dense1 — spectral
Cluster 1 (n=284)
Cluster 1 mean (n=284)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — dense1 — spectral
Total mean spectrum
Total mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]



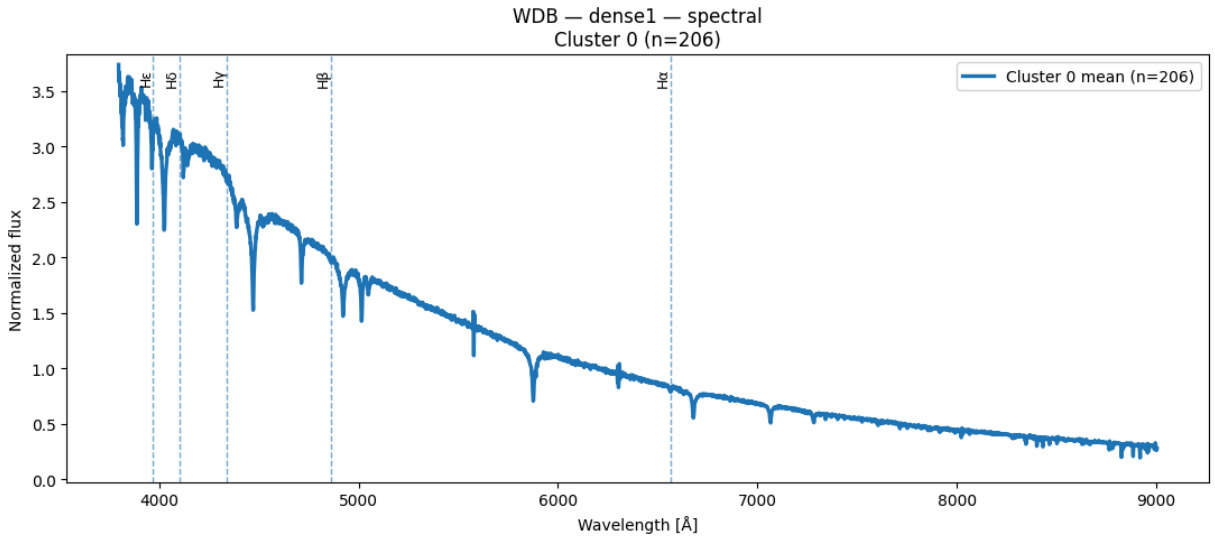


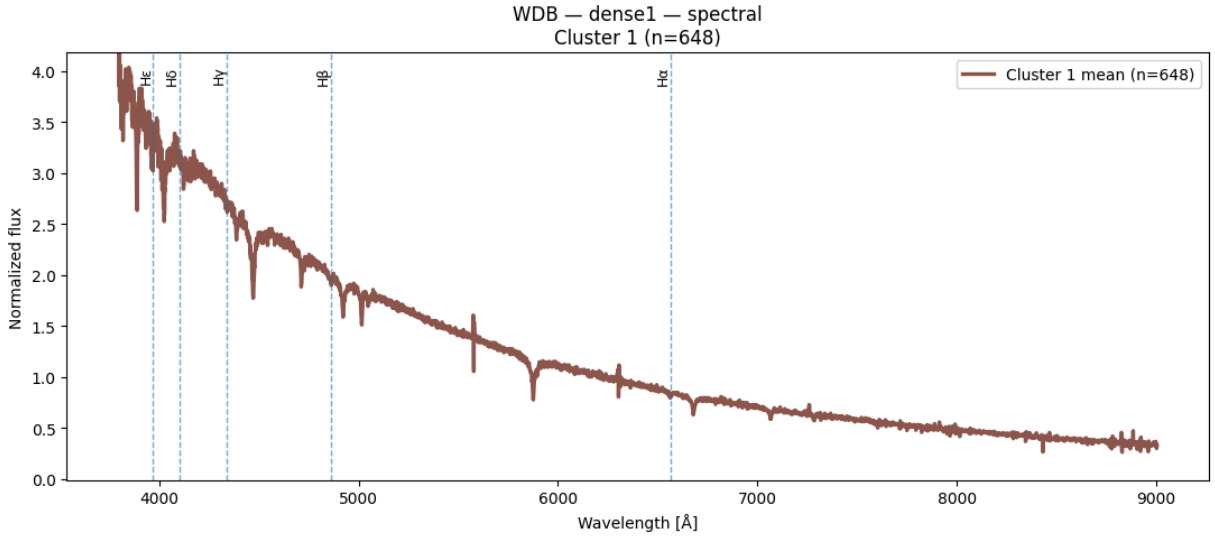


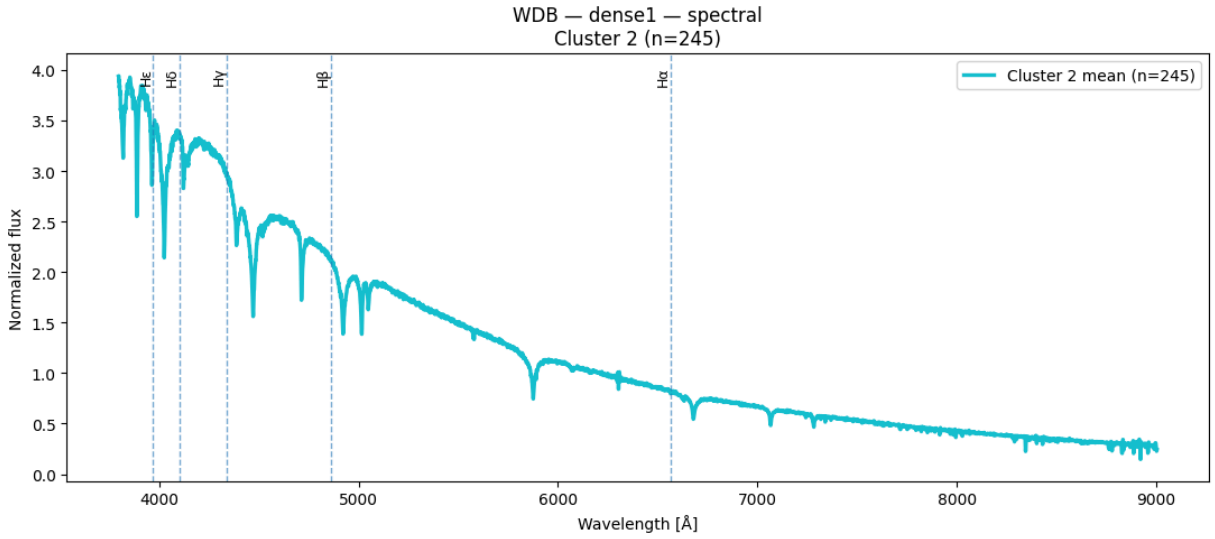
Espectros obtenidos de Spectral Clustering para k=3 y n=1100

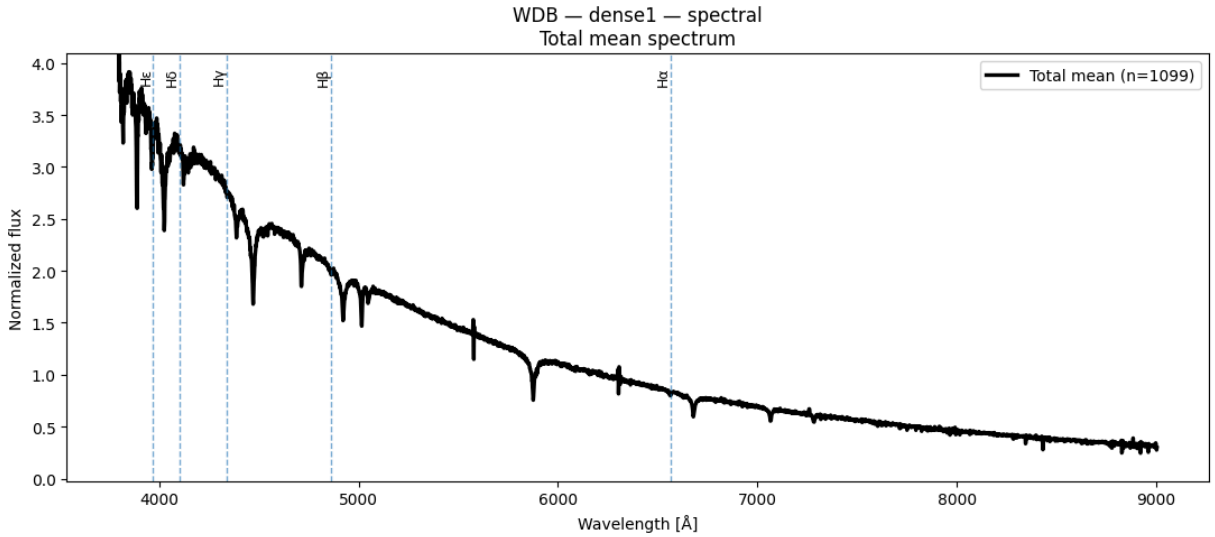
In [74]:
```
result = plot_spectra_spectral(X_wdb_spec, labels_dense1_k3_wdb, W_wdb[0],
                               class_name="WDB", layer_name="dense1")
```

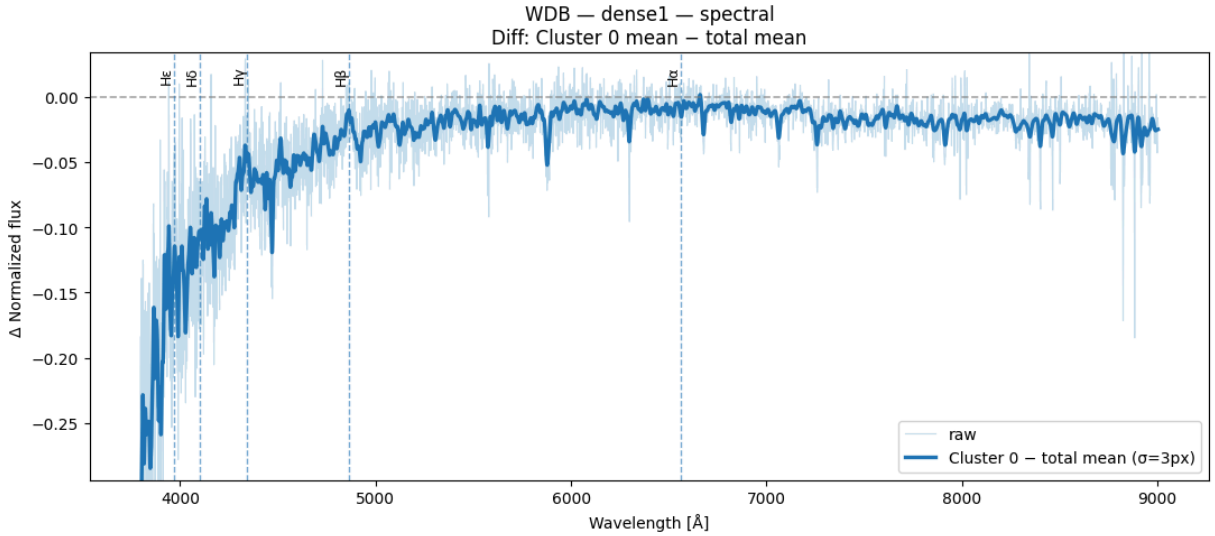
```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```

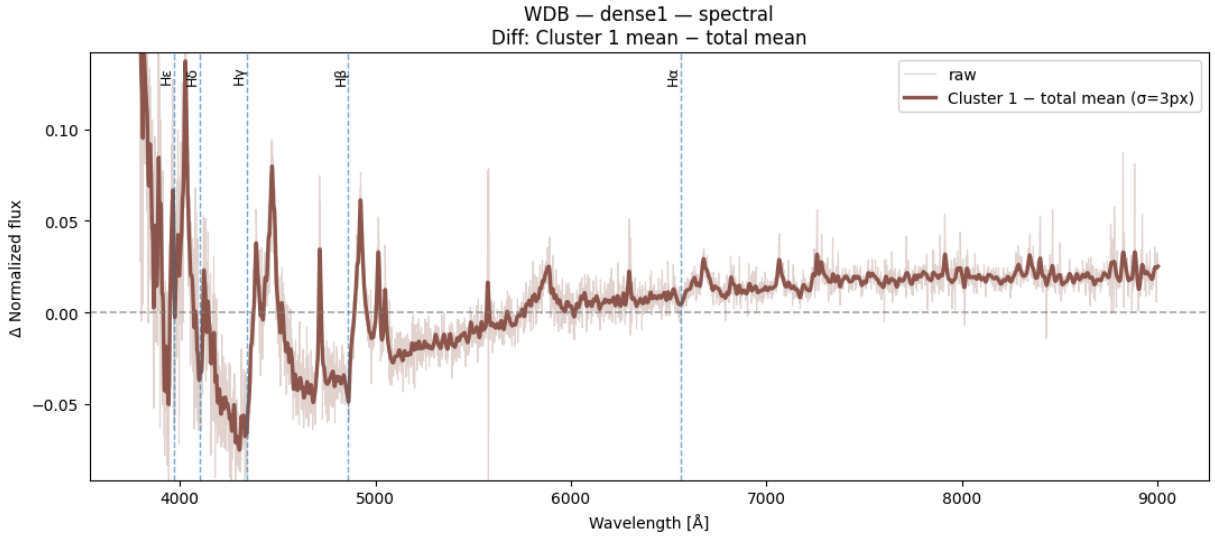


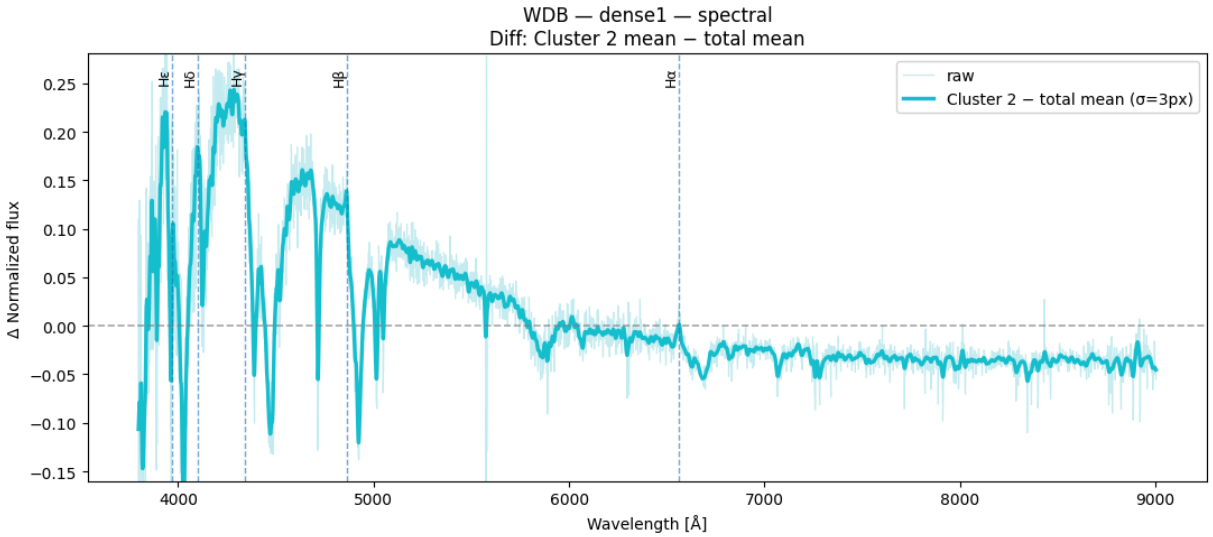


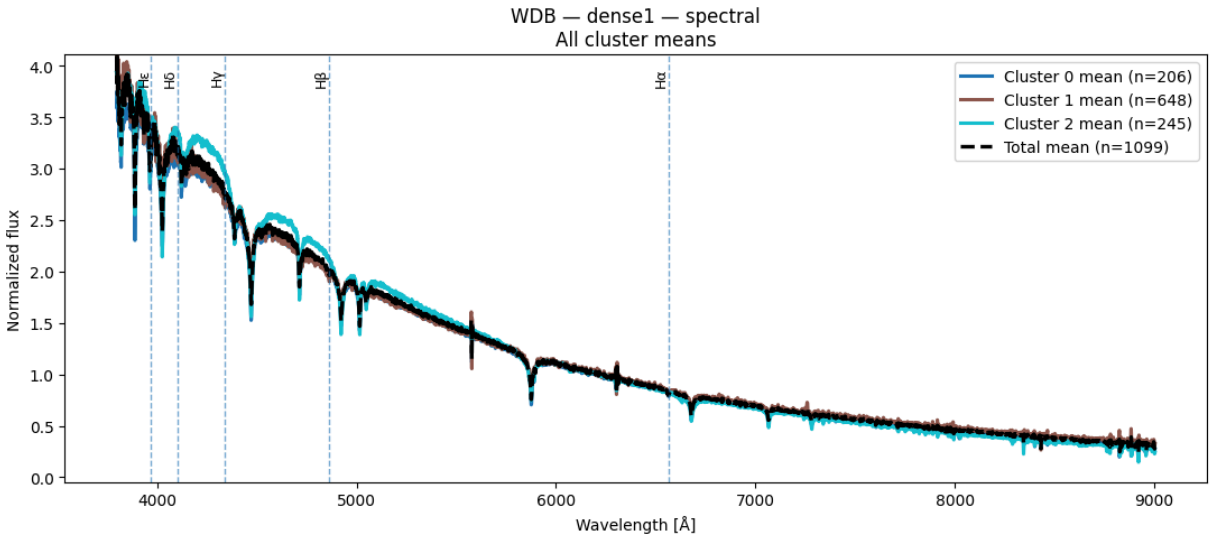



WDB — dense1 — spectral
Total mean spectrum
Hε
Hδ
Hγ
Hβ
Hα
Total mean (n=1099)
Normalized flux
Wavelength [Å]


WDB — dense1 — spectral
Diff: Cluster 0 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
raw
Cluster 0 − total mean (σ=3px)
Δ Normalized flux
Wavelength [Å]


WDB — dense1 — spectral
Diff: Cluster 1 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
raw
Cluster 1 − total mean (σ=3px)
Δ Normalized flux
Wavelength [Å]

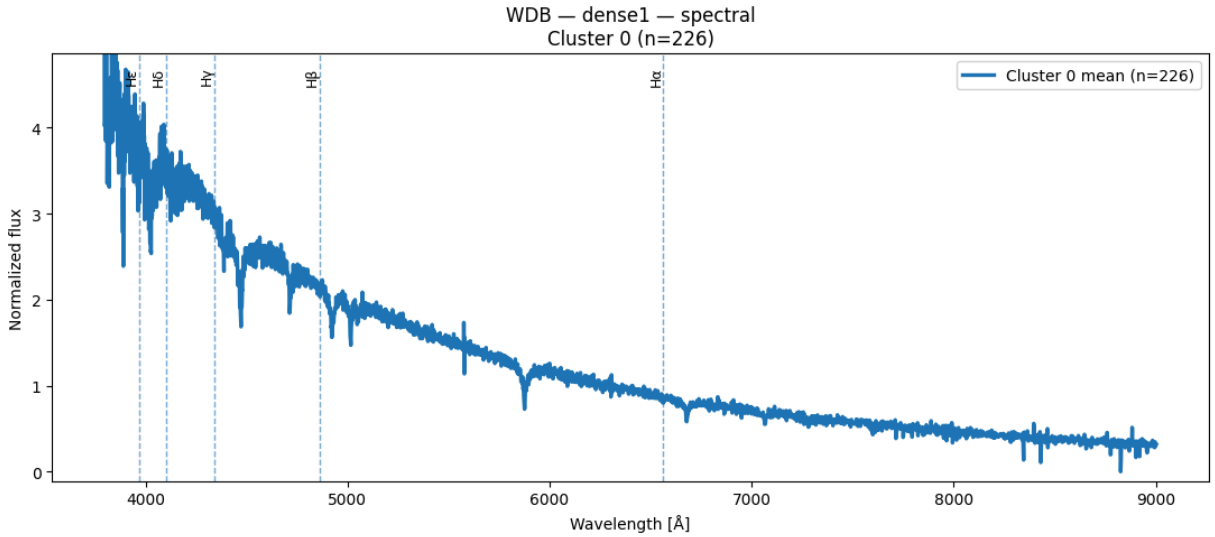


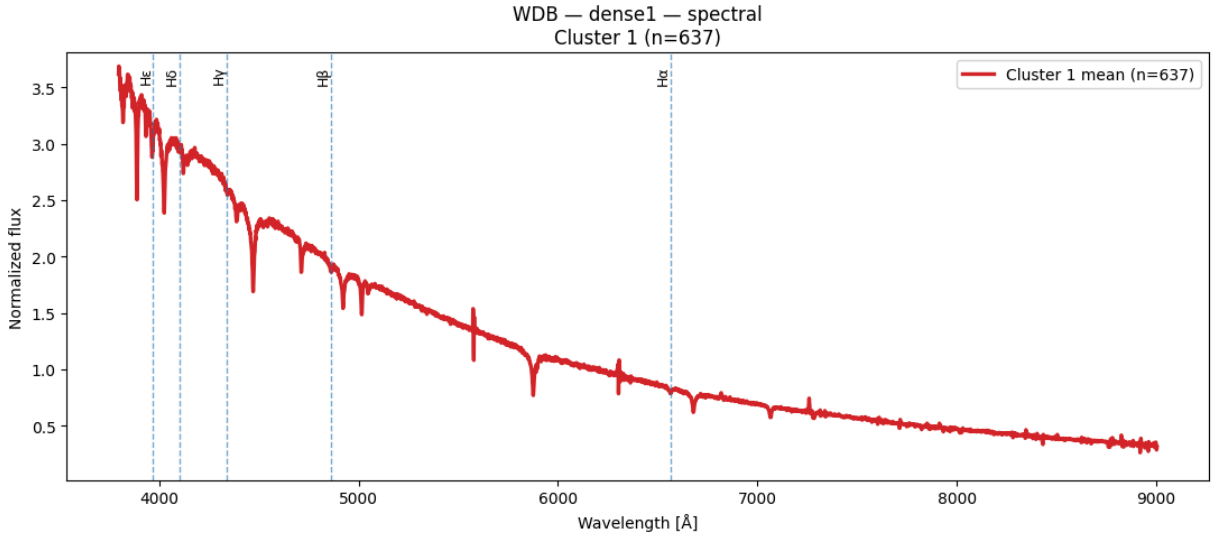


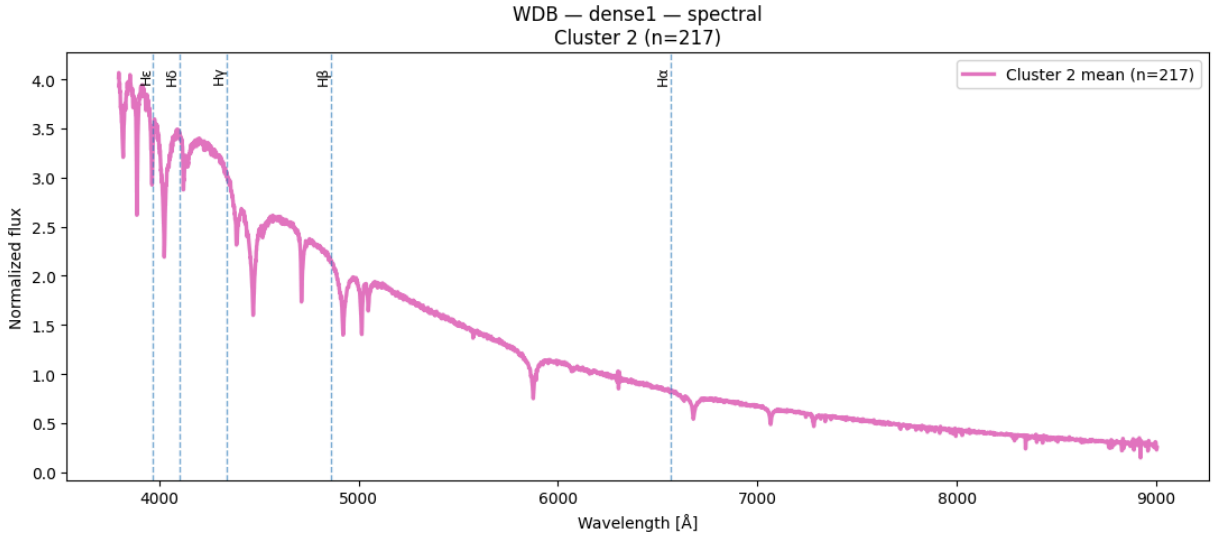
Espectros obtenidos de Spectral Clustering para k=4 y n=1100

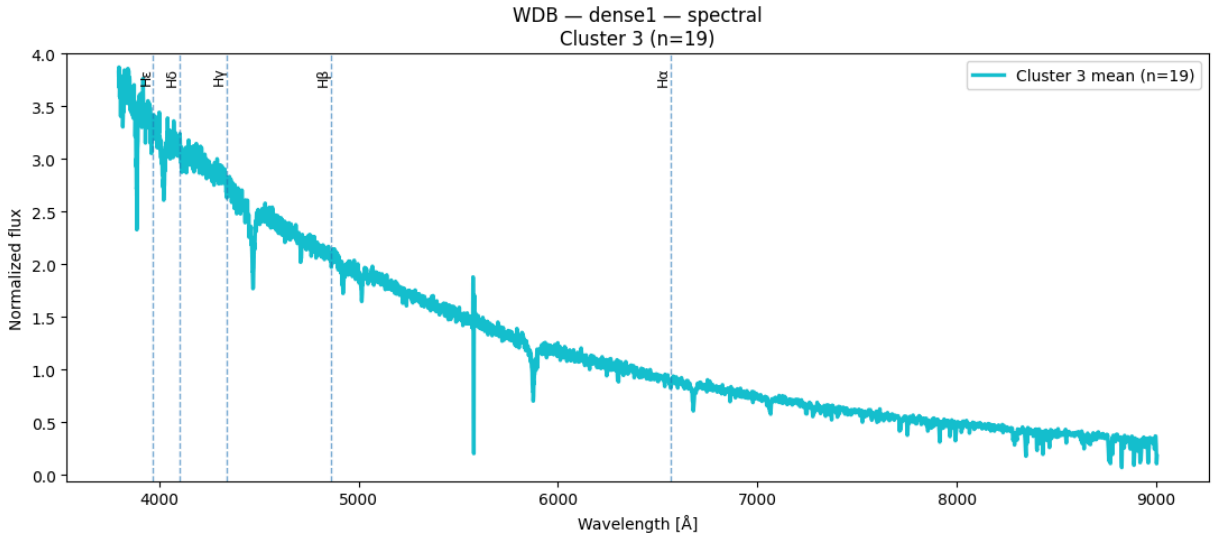
In [75]:
```
result = plot_spectra_spectral(X_wdb_spec, labels_dense1_k4_wdb, W_wdb[0],
                               class_name="WDB", layer_name="dense1")
```

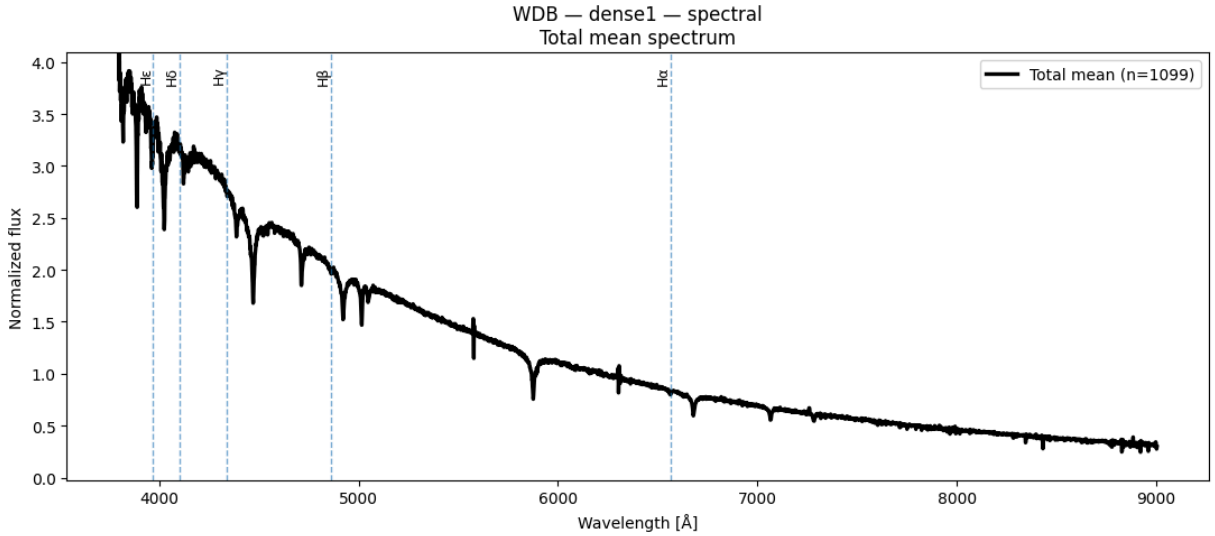
```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```

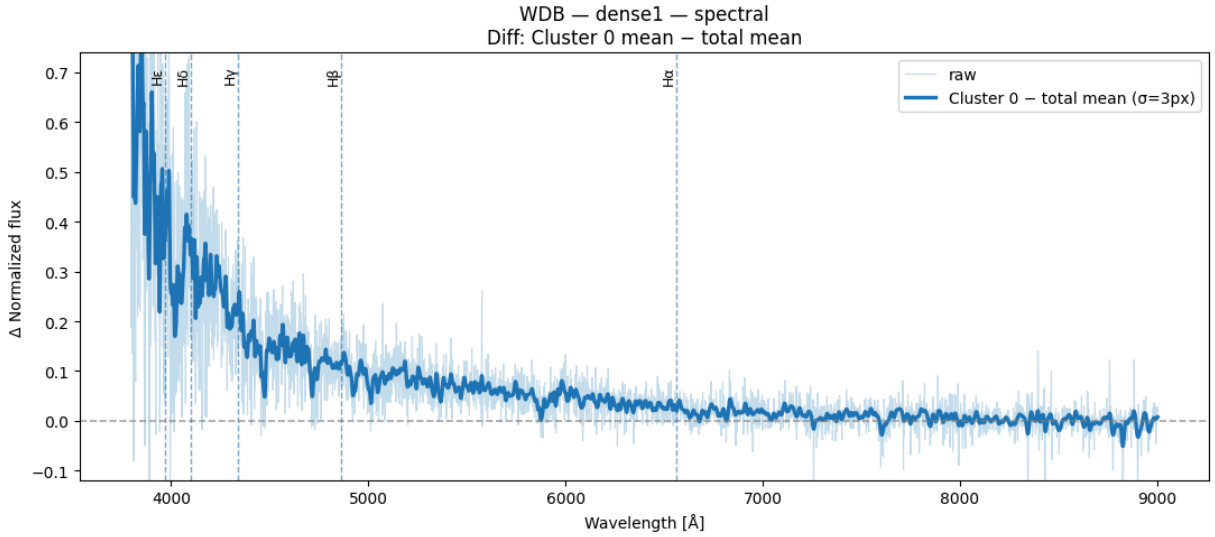

WDB — dense1 — spectral
Cluster 0 (n=226)
Cluster 0 mean (n=226)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

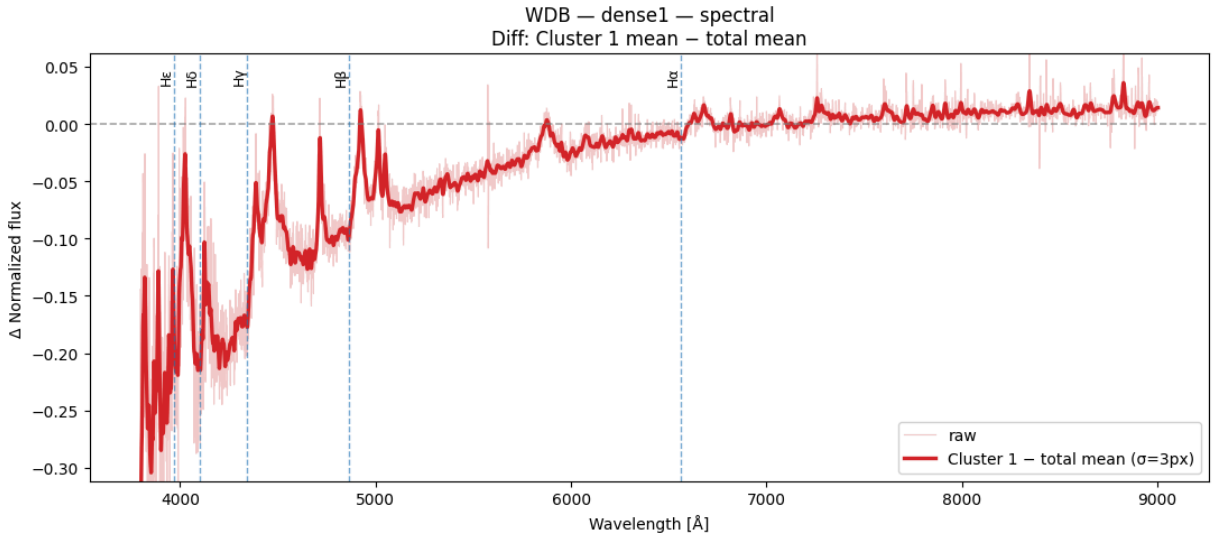

WDB — dense1 — spectral
Cluster 1 (n=637)
Cluster 1 mean (n=637)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

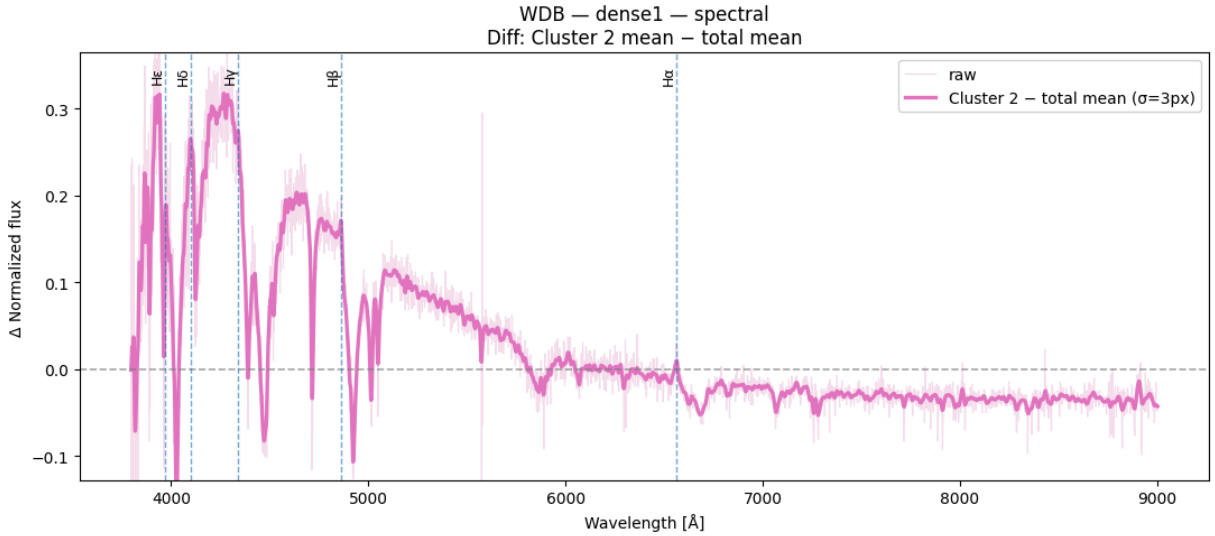

WDB — dense1 — spectral
Cluster 2 (n=217)
Cluster 2 mean (n=217)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

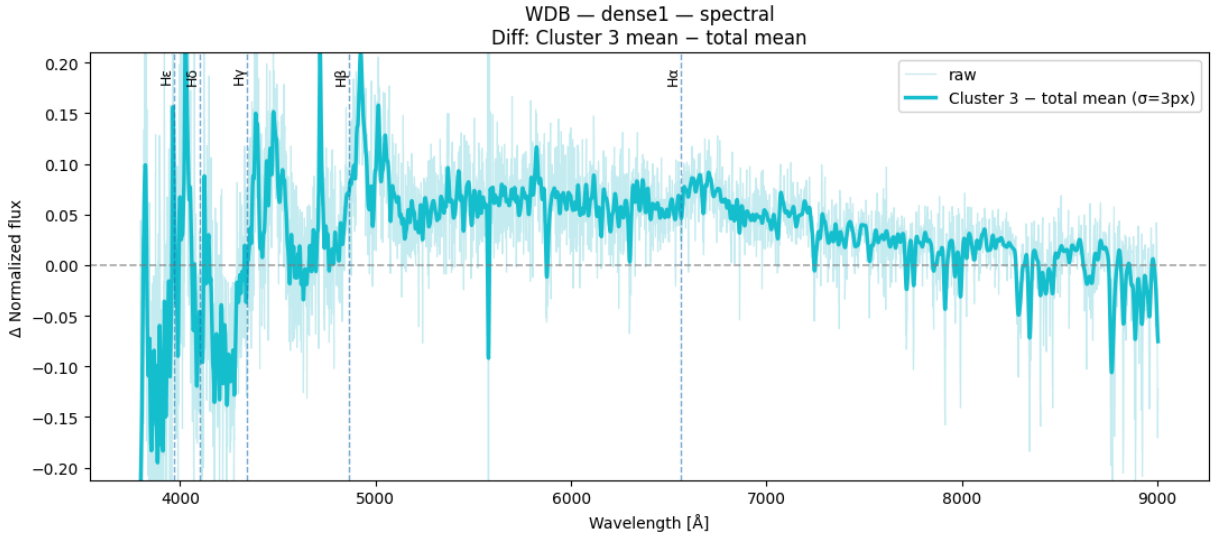

WDB — dense1 — spectral
Cluster 3 (n=19)
Cluster 3 mean (n=19)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

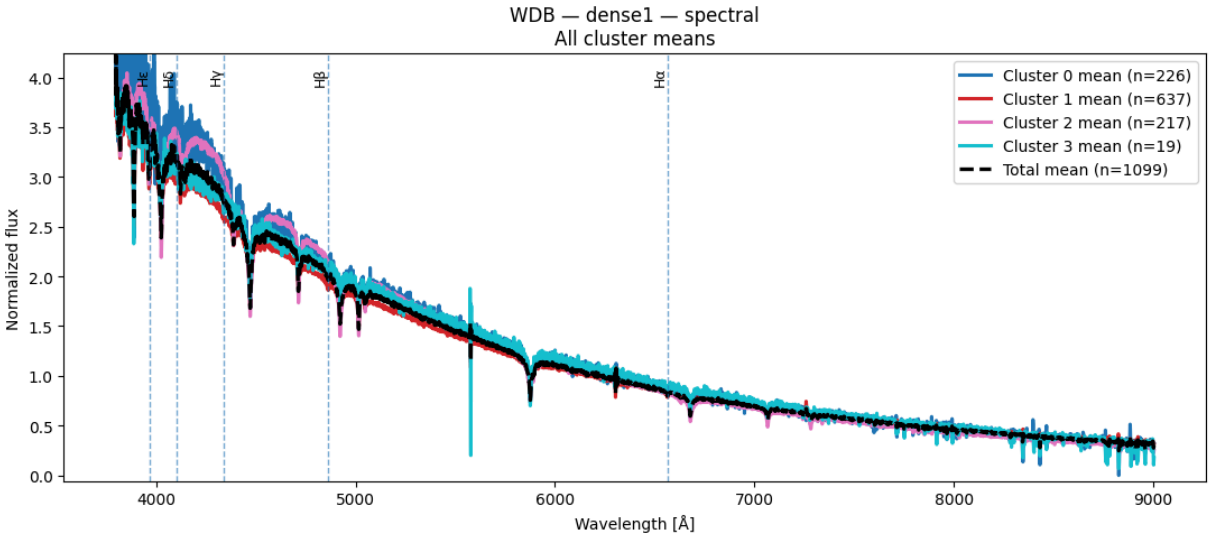

WDB — dense1 — spectral
Total mean spectrum
Total mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — dense1 — spectral
Diff: Cluster 0 mean − total mean
raw
Cluster 0 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]


WDB — dense1 — spectral
Diff: Cluster 1 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]
raw
Cluster 1 − total mean (σ=3px)

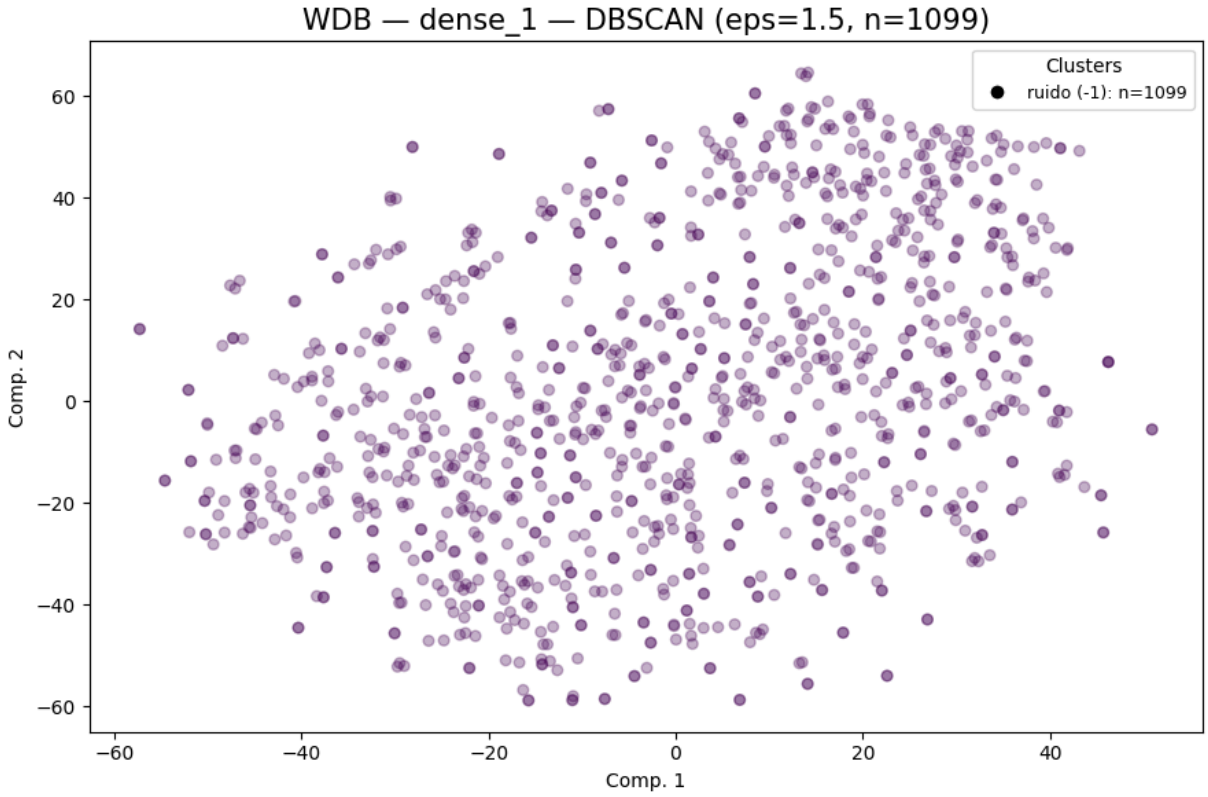

WDB — dense1 — spectral
Diff: Cluster 2 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]
raw
Cluster 2 − total mean (σ=3px)


WDB — dense1 — spectral
Diff: Cluster 3 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]
raw
Cluster 3 − total mean (σ=3px)

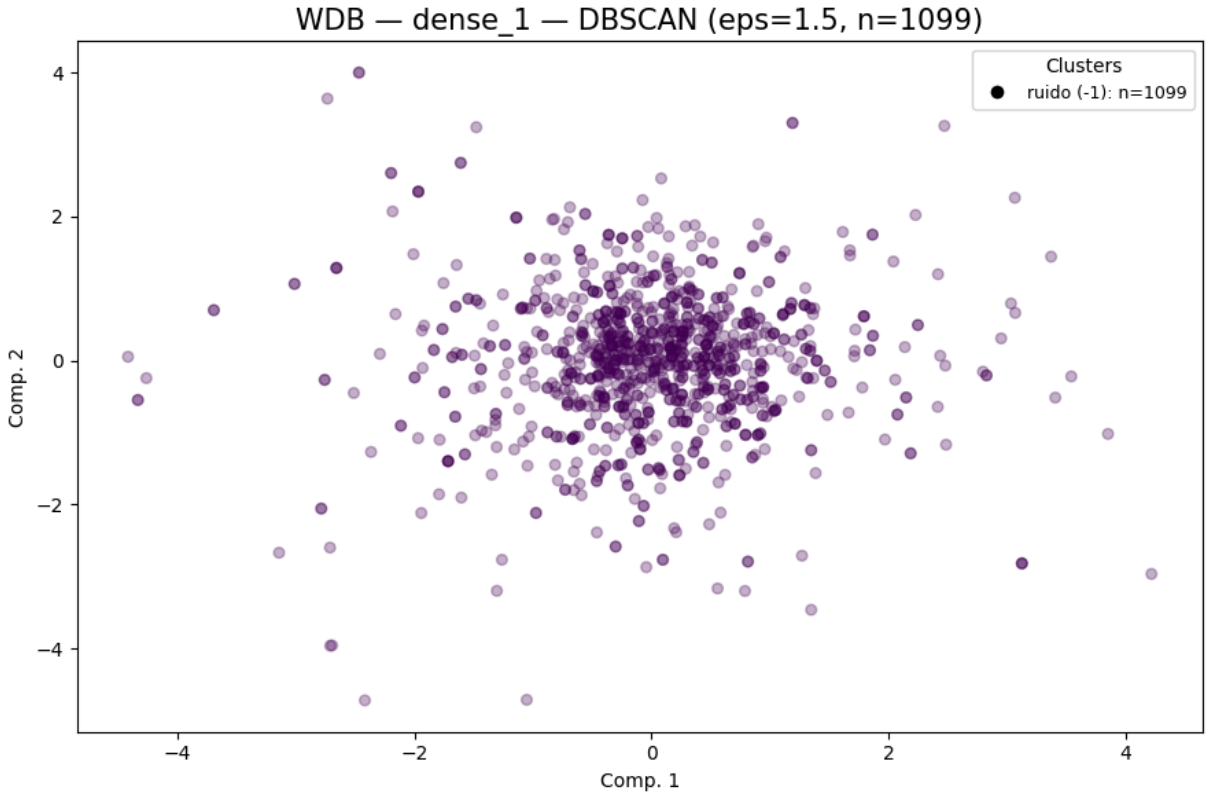


DBSCAN

In [76]:
```
dbs_dense1_wdb = cluster_plot_DBSCAN(
            repr_wdb_32["dense_1"],
            "WDB",
            "dense_1",
            dbscan_eps=1.5,
            dbscan_min_samples=10,
            vis_method="tsne",
            xlim=None,
            ylim=None,
            container=cluster_labels_all
        )
labels_dbs_dense1_wdb = dbs_dense1_wdb["dbscan"]
cluster_plot_DBSCAN(
            repr_wdb_32["dense_1"],
            "WDB",
            "dense_1",
            dbscan_eps=1.5,
            dbscan_min_samples=10,
            vis_method="pca",
            xlim=None,
            ylim=None,
            container=cluster_labels_all
        )
```




```
┌──────────────────────────────────────────────────────────────────────┐
│  WDB | dense_1 | dbscan | eps=1.5
│  clusters=0  noise=1099  sil=N/A  DB=N/A  CH=N/A
│  DBCV = N/A  (rango [-1,1], especifico DBSCAN)
└──────────────────────────────────────────────────────────────────────┘
```



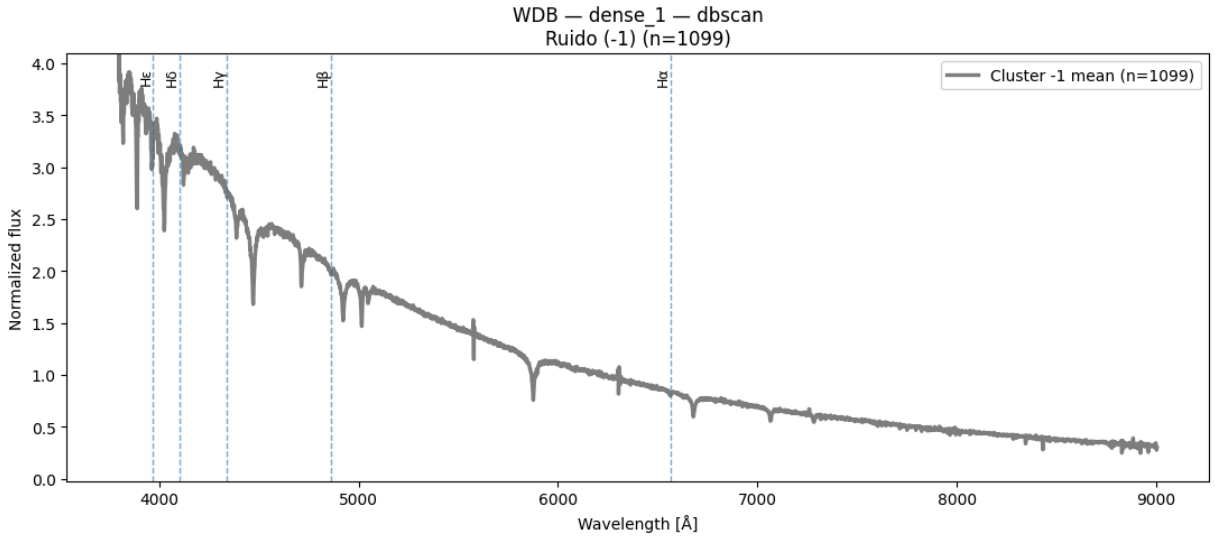
```
┌──────────────────────────────────────────────────────────────────────┐
│  WDB | dense_1 | dbscan | eps=1.5                                    │
│  clusters=0  noise=1099  sil=N/A  DB=N/A  CH=N/A                     │
│  DBCV = N/A  (rango [-1,1], especifico DBSCAN)                       │
└──────────────────────────────────────────────────────────────────────┘
```

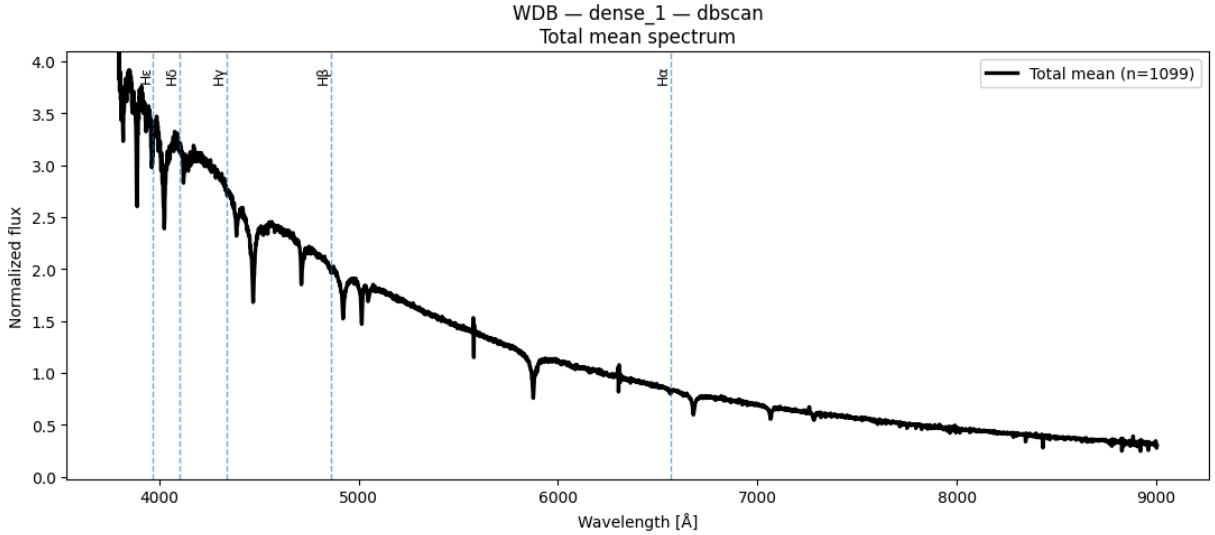
Out[76]:
```
{'dbscan': array([-1, -1, -1, ..., -1, -1, -1], dtype=int64),
 'metrics': {'n_clusters': 0,
  'n_noise': 1099,
  'silhouette': None,
  'davies_bouldin': None,
  'calinski_harabasz': None},
 'dbcv': nan}
```

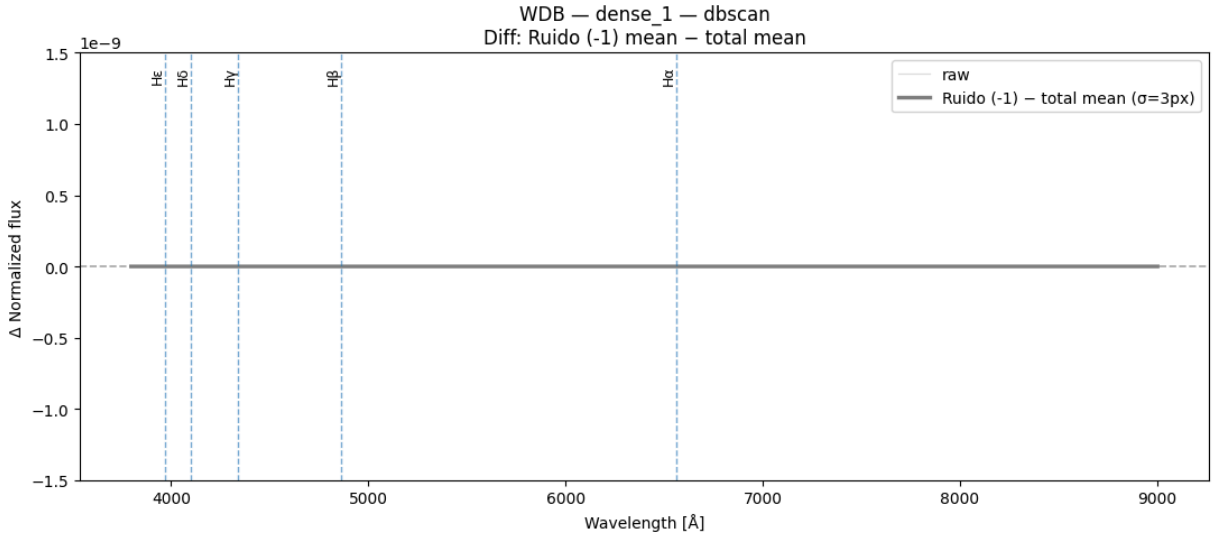
Espectros obtenidos de DBSCAN n = 1100

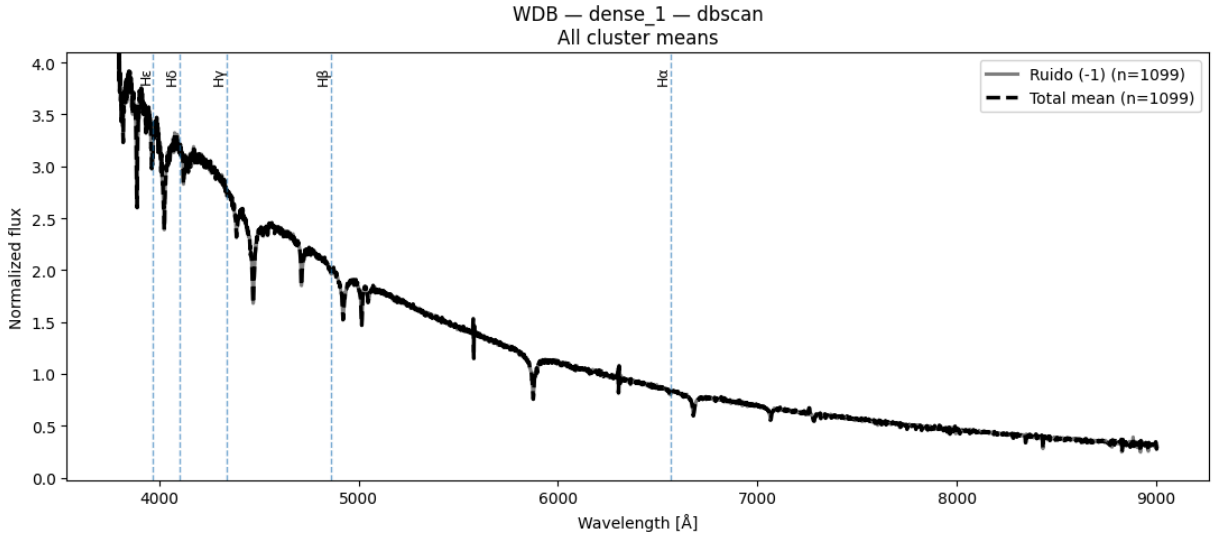
In [77]:
```
plot_spectra_dbscan(
    X_wdb_spec,
    labels_dbs_dense1_wdb,
    W_wdb[0],
    class_name="WDB",
    layer_name="dense_1"
)
```

```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```







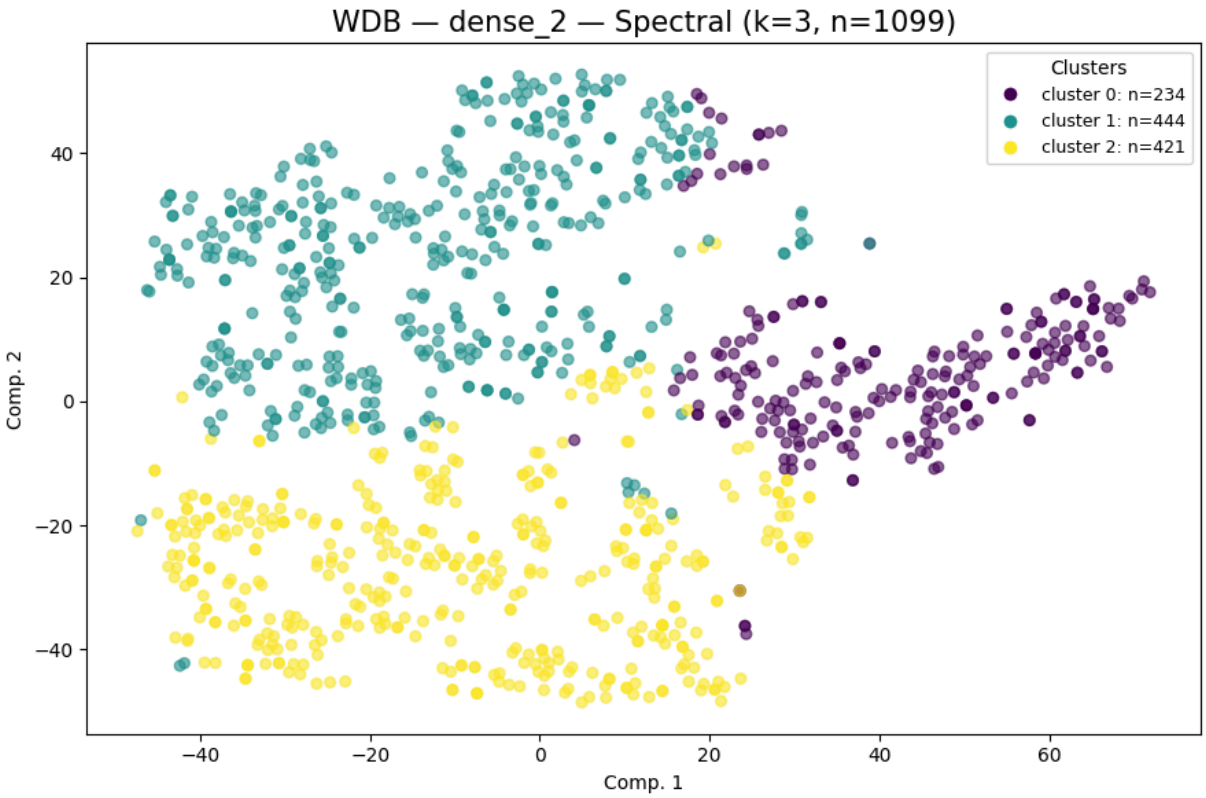


Out[77]:
```
{'total_mean': array([4.2840786 , 3.8304825 , 3.8275013 , ..., 0.3190189 , 0.27551
222,
       0.2985266 ], dtype=float32),
 'group_means': {-1: array([4.2840786 , 3.8304825 , 3.8275013 , ..., 0.3190189 ,
0.27551222,
        0.2985266 ], dtype=float32)},
 'unique_clusters': array([-1], dtype=int64)}
```

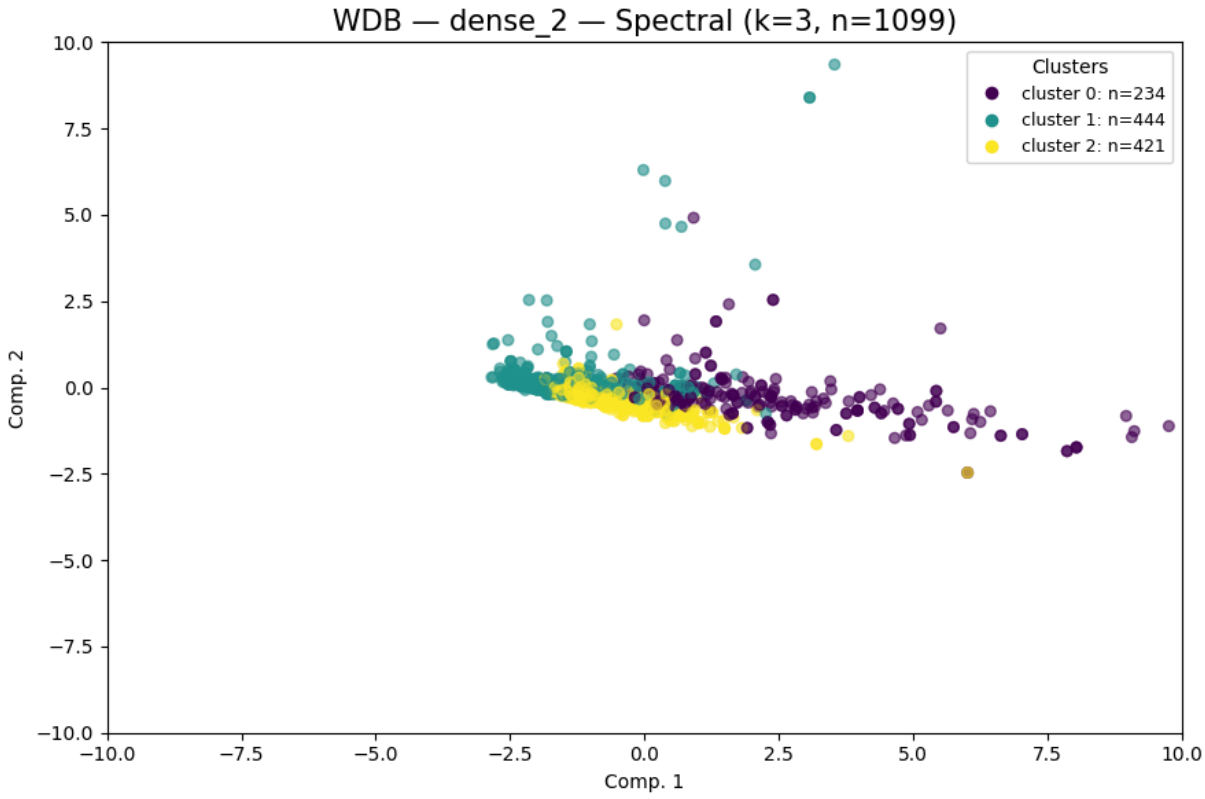
Resultados capa dense_2, con Spectral Clustering + tSNE y PCA para n=1100

Spectral Clustering k = 3

In [78]:
```
dense2_wdb_k3=cluster_plot_Spect(
            repr_wdb_32["dense_2"],
            "WDB",
            "dense_2",
            spectral_k=3,
            vis_method="tsne",
            container=cluster_labels_all
        )

labels_dense2_k3_wdb = dense2_wdb_k3["spectral"]
cluster_plot_Spect(
            repr_wdb_32["dense_2"],
            "WDB",
            "dense_2",
            spectral_k=3,
            vis_method="pca",
            xlim=(-10,10), ylim=(-10,10)
        )
```



```
┌──────────────────────────────────────────────────────────────────────────────┐
│  WDB | dense_2 | spectral | k=3
│  clusters=3  noise=0  sil=0.1530  DB=1.5104  CH=101.35
└──────────────────────────────────────────────────────────────────────────────┘
```



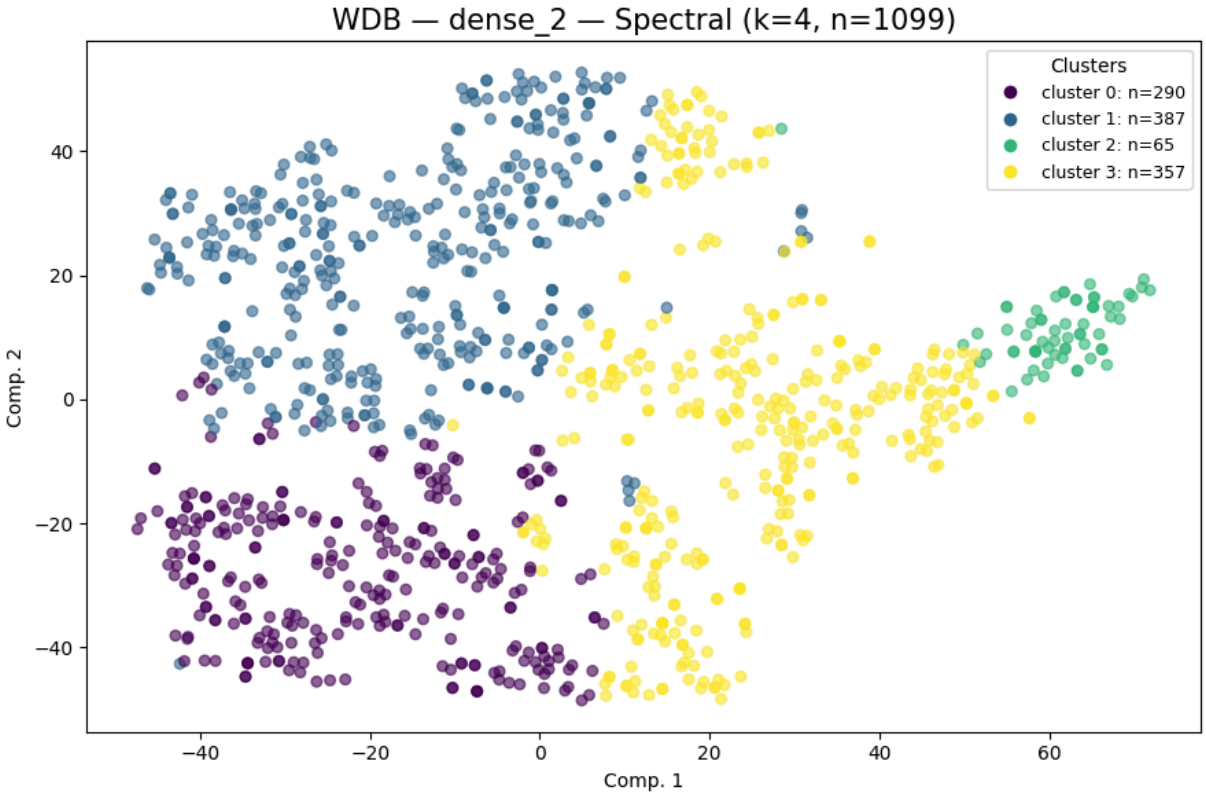

```
┌──────────────────────────────────────────────────────────────────────────────┐
│  WDB | dense_2 | spectral | k=3
│  clusters=3  noise=0  sil=0.1530  DB=1.5104  CH=101.35
└──────────────────────────────────────────────────────────────────────────────┘
```


Out[78]:
```
{'spectral': array([1, 2, 1, ..., 2, 2, 2]),
 'metrics': {'n_clusters': 3,
  'n_noise': 0,
  'silhouette': 0.152967,
  'davies_bouldin': 1.510379,
  'calinski_harabasz': 101.3525}}
```

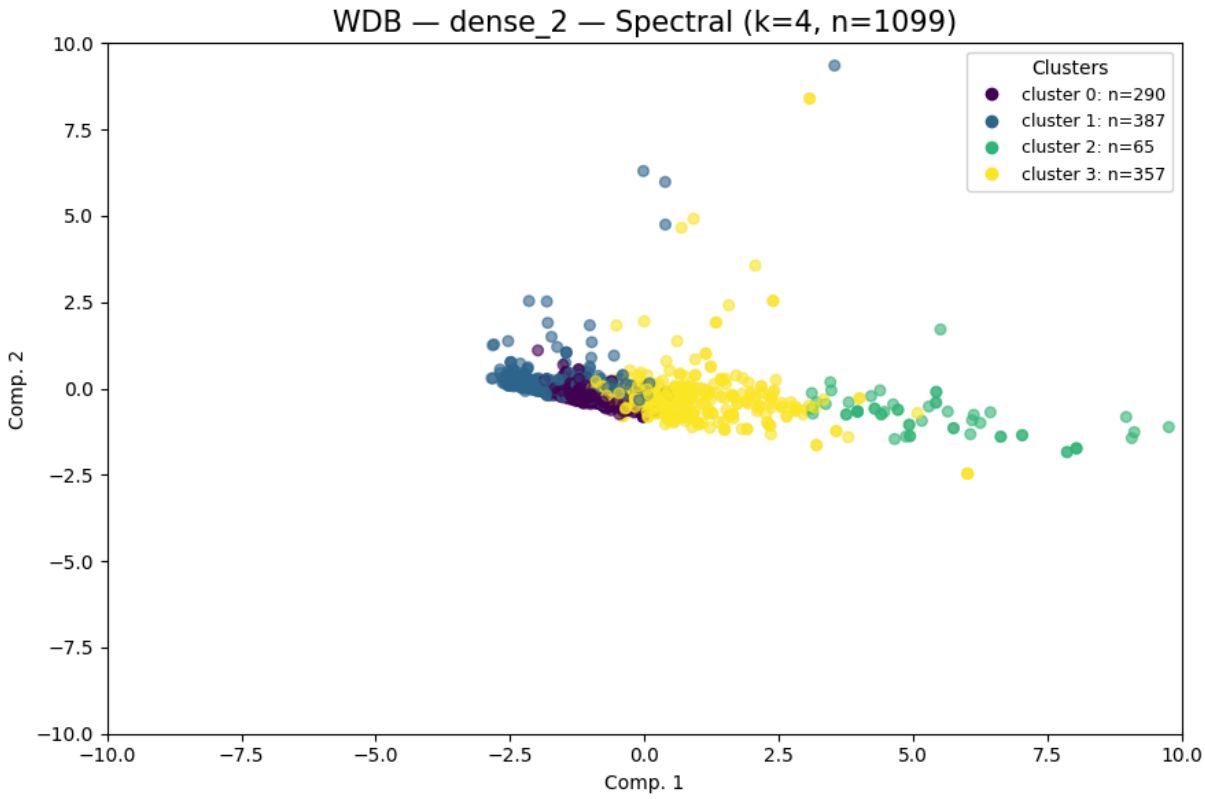
Spectral Clustering k = 4

In [79]:
```
dense2_wdb_k4=cluster_plot_Spect(
            repr_wdb_32["dense_2"],
            "WDB",
            "dense_2",
            spectral_k=4,
            vis_method="tsne",
            container=cluster_labels_all
        )

labels_dense2_k4_wdb = dense2_wdb_k4["spectral"]
cluster_plot_Spect(
            repr_wdb_32["dense_2"],
```

```
            "WDB",
            "dense_2",
            spectral_k=4,
            vis_method="pca",
            xlim=(-10,10), ylim=(-10,10)
        )
```




```
┌──────────────────────────────────────────────────────────────────────┐
│  WDB | dense_2 | spectral | k=4                                      │
│  clusters=4  noise=0  sil=0.1193  DB=1.3507  CH=110.47               │
└──────────────────────────────────────────────────────────────────────┘
```



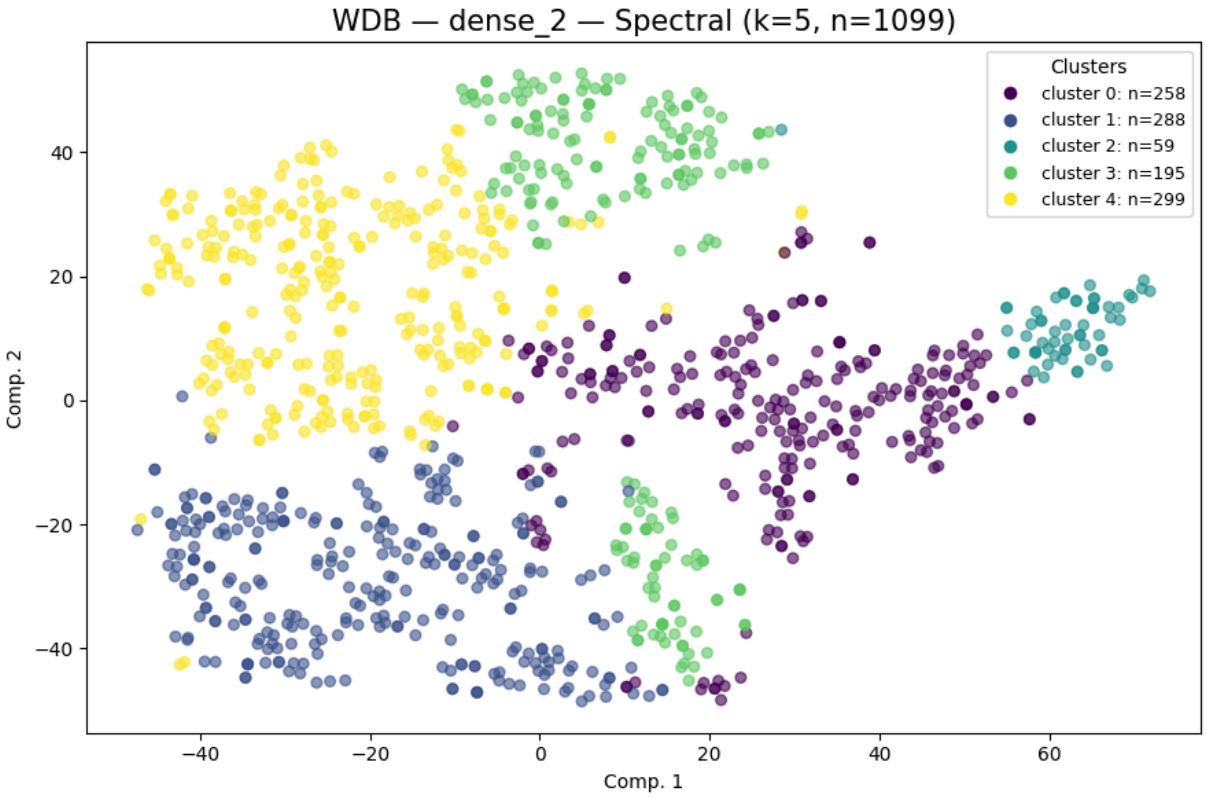

```
┌──────────────────────────────────────────────────────────────────────┐
│  WDB | dense_2 | spectral | k=4                                      │
│  clusters=4  noise=0  sil=0.1193  DB=1.3507  CH=110.47               │
└──────────────────────────────────────────────────────────────────────┘
```

Out[79]:
```
{'spectral': array([3, 0, 1, ..., 3, 3, 3]),
 'metrics': {'n_clusters': 4,
  'n_noise': 0,
  'silhouette': 0.119263,
  'davies_bouldin': 1.350695,
  'calinski_harabasz': 110.4732}}
```

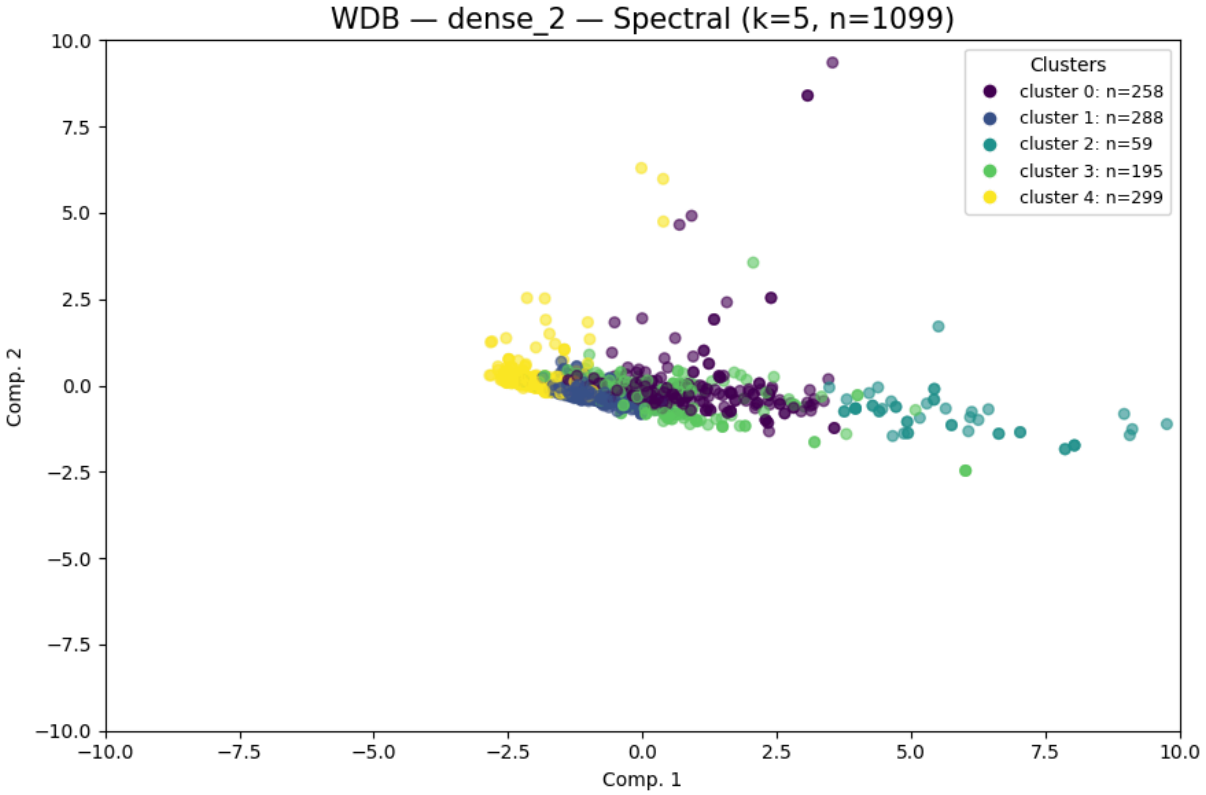
Spectral Clustering k = 5

In [80]:
```
dense2_wdb_k5=cluster_plot_Spect(
            repr_wdb_32["dense_2"],
            "WDB",
            "dense_2",
            spectral_k=5,
            vis_method="tsne",
            container=cluster_labels_all
        )

labels_dense2_k5_wdb = dense2_wdb_k5["spectral"]
cluster_plot_Spect(
            repr_wdb_32["dense_2"],
            "WDB",
            "dense_2",
            spectral_k=5,
            vis_method="pca",
            xlim=(-10,10), ylim=(-10,10)
```

```
)
```




```
┌──────────────────────────────────────────────────────────────┐
│  WDB | dense_2 | spectral | k=5
│  clusters=5  noise=0  sil=0.1367  DB=1.5547  CH=88.81
└──────────────────────────────────────────────────────────────┘
```

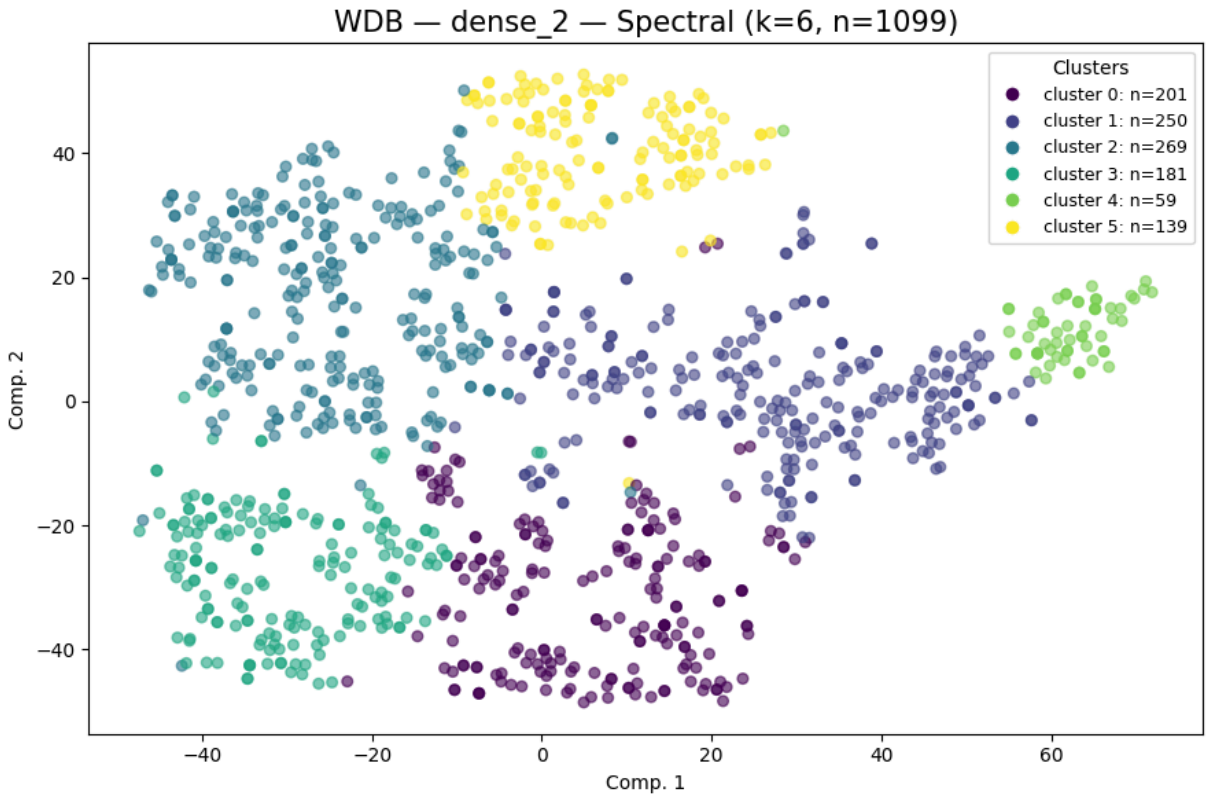


```
┌──────────────────────────────────────────────────────────────────────────┐
│  WDB | dense_2 | spectral | k=5
│  clusters=5  noise=0  sil=0.1367  DB=1.5547  CH=88.81
└──────────────────────────────────────────────────────────────────────────┘
```

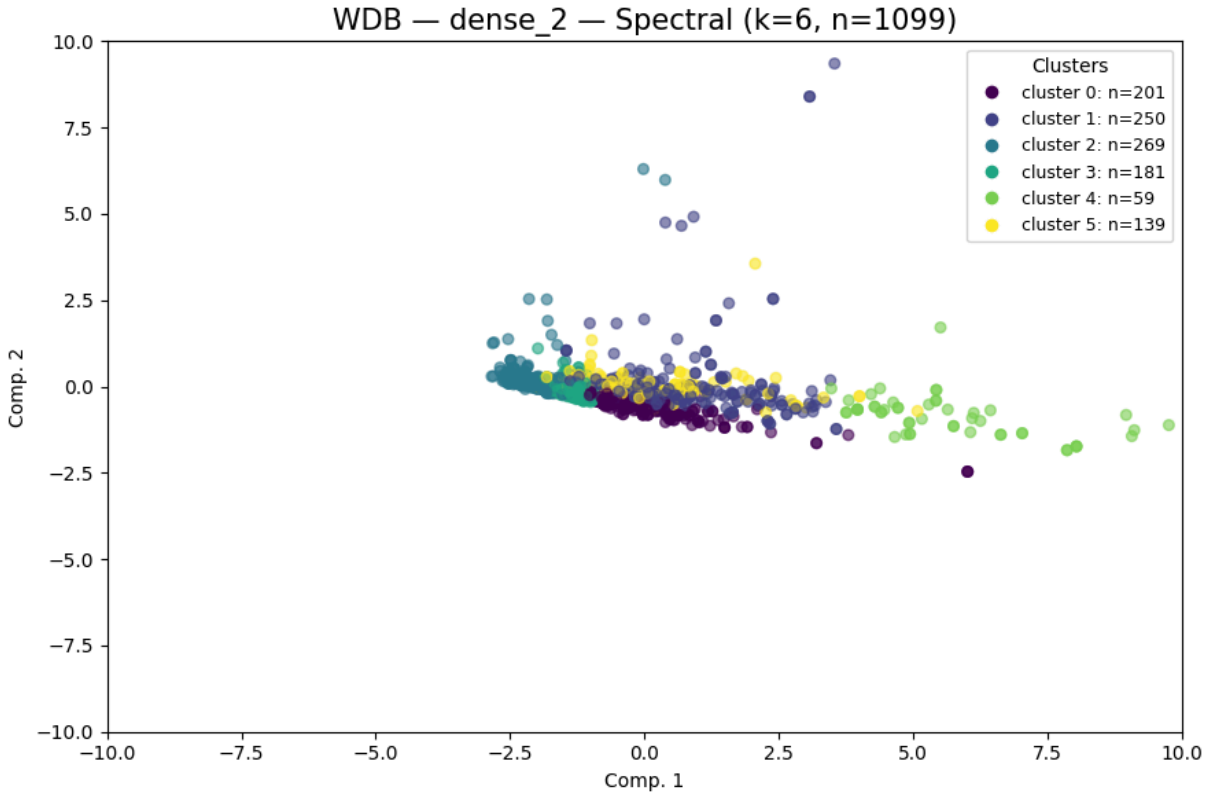
Out[80]:
```
{'spectral': array([0, 1, 4, ..., 0, 3, 0]),
 'metrics': {'n_clusters': 5,
  'n_noise': 0,
  'silhouette': 0.136683,
  'davies_bouldin': 1.554682,
  'calinski_harabasz': 88.8063}}
```

Spectral Clustering k = 6

In [81]:
```
dense2_wdb_k6=cluster_plot_Spect(
            repr_wdb_32["dense_2"],
            "WDB",
            "dense_2",
            spectral_k=6,
            vis_method="tsne",
            container=cluster_labels_all
        )

labels_dense2_k6_wdb = dense2_wdb_k6["spectral"]
cluster_plot_Spect(
            repr_wdb_32["dense_2"],
            "WDB",
            "dense_2",
            spectral_k=6,
            vis_method="pca",
            xlim=(-10,10), ylim=(-10,10)
        )
```




```
┌──────────────────────────────────────────────────────────────────────┐
│  WDB | dense_2 | spectral | k=6                                      │
│  clusters=6  noise=0  sil=0.0873  DB=1.5242  CH=75.04                │
└──────────────────────────────────────────────────────────────────────┘
```

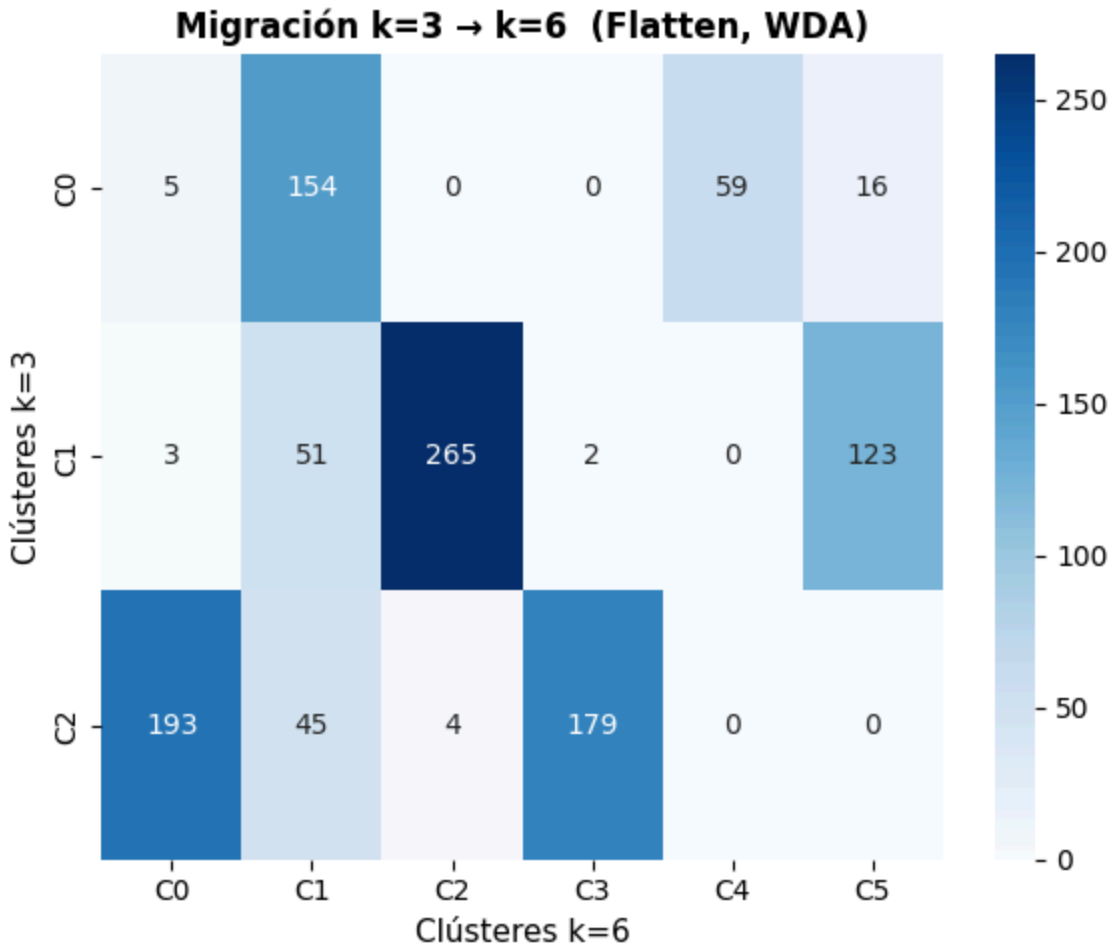


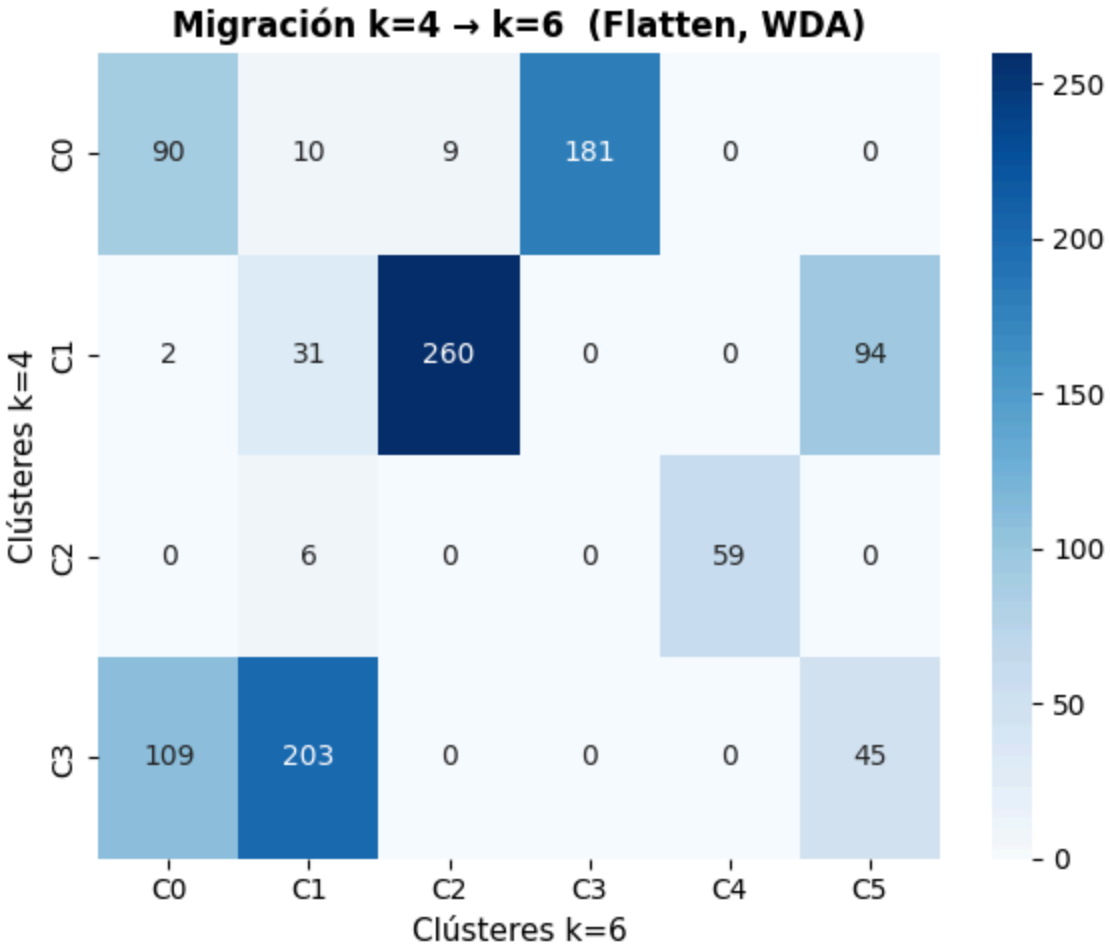
```
┌──────────────────────────────────────────────────────────────────────┐
│  WDB | dense_2 | spectral | k=6                                      │
│  clusters=6  noise=0  sil=0.0873  DB=1.5242  CH=75.04                │
└──────────────────────────────────────────────────────────────────────┘
```

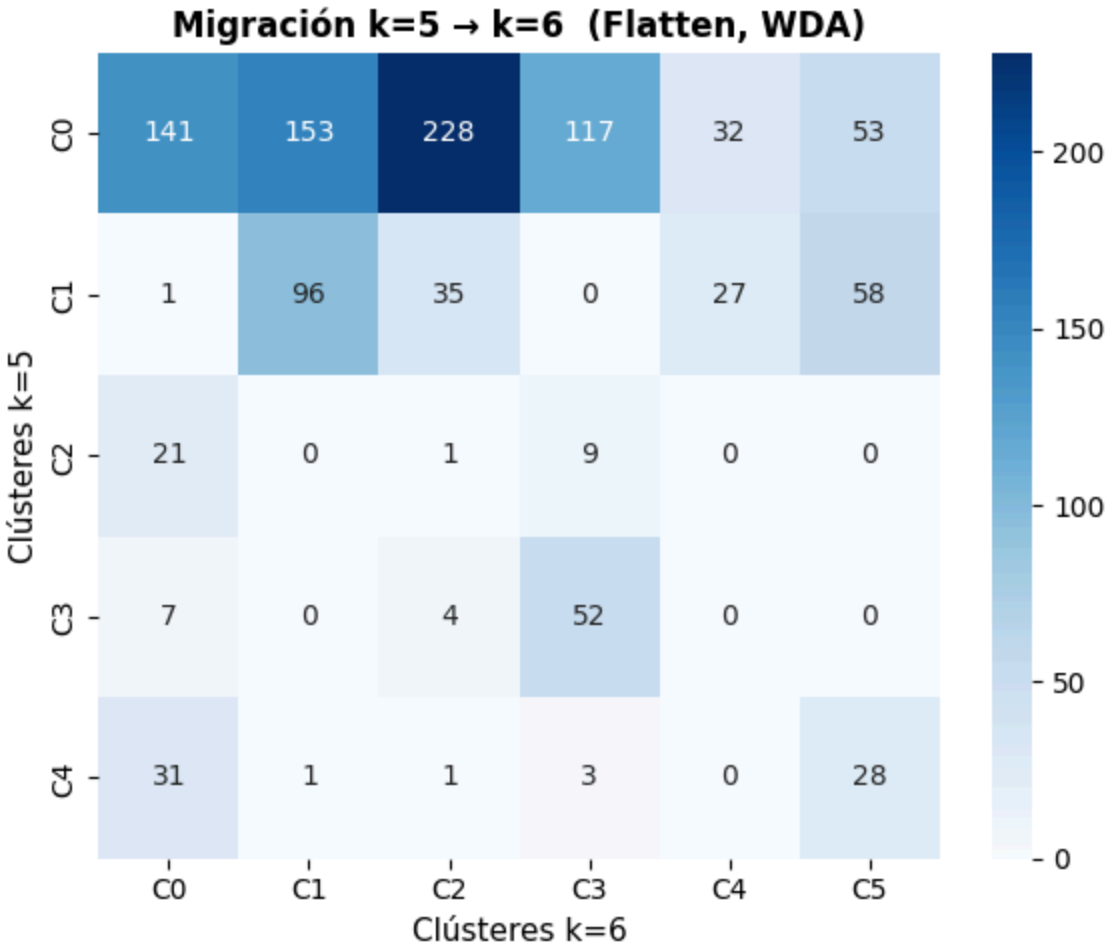
Out[81]:
```
{'spectral': array([1, 3, 2, ..., 0, 0, 1]),
 'metrics': {'n_clusters': 6,
  'n_noise': 0,
  'silhouette': 0.0873,
  'davies_bouldin': 1.524238,
  'calinski_harabasz': 75.0415}}
```

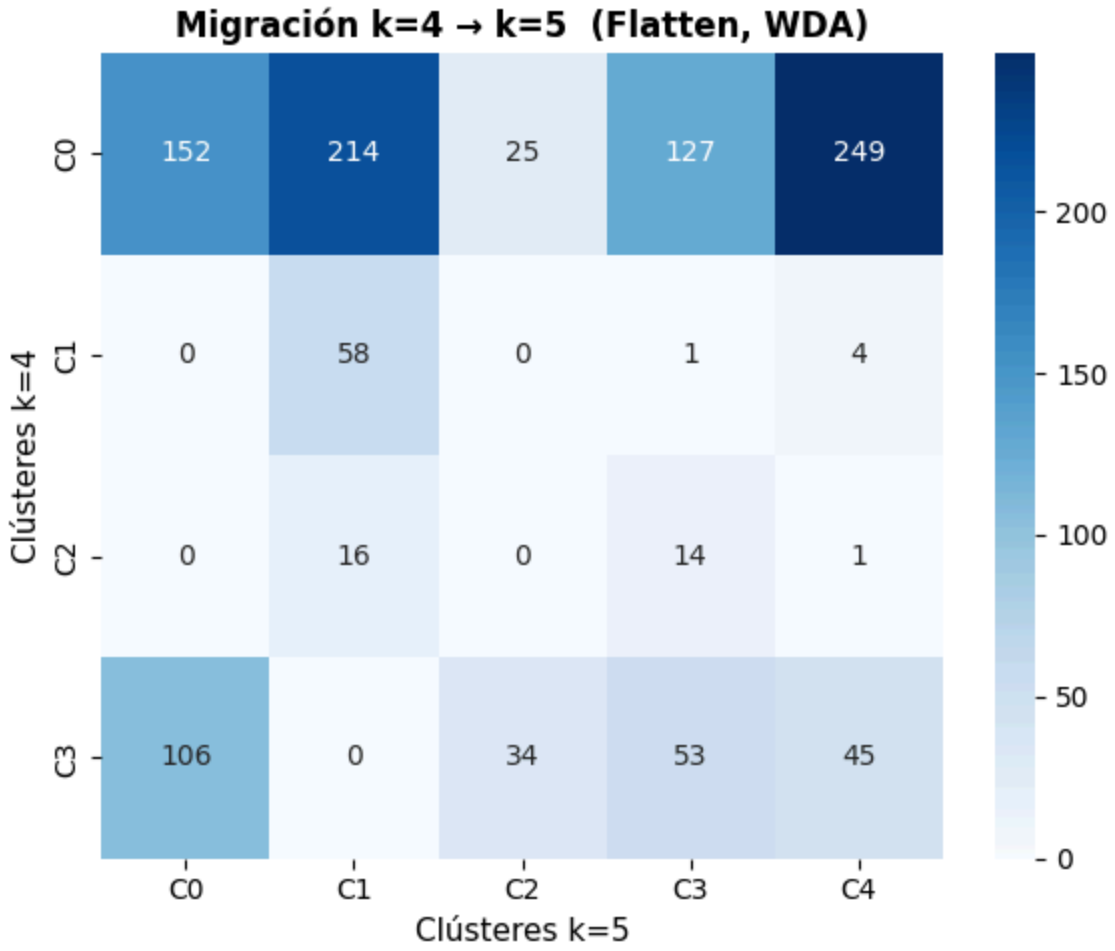
Análisis de migración de clústeres

In [82]:
```
plot_migration_matrix(labels_dense2_k3_wdb, labels_dense2_k6_wdb, ka=3, kb=6)
plot_migration_matrix(labels_dense2_k4_wdb, labels_dense2_k6_wdb, ka=4, kb=6)
plot_migration_matrix(labels_dense_k5_wdb, labels_dense2_k6_wdb, ka=5, kb=6)
plot_migration_matrix(labels_dense_k4_wdb, labels_dense2_k5_wdb, ka=4, kb=5)
```




Migración k=4 → k=6 (Flatten, WDA)
Clústeres k=4
C0
C1
C2
C3
90
10
9
181
0
0
2
31
260
0
0
94
0
6
0
0
59
0
109
203
0
0
0
45
C0
C1
C2
C3
C4
C5
Clústeres k=6
250
200
150
100
50
0

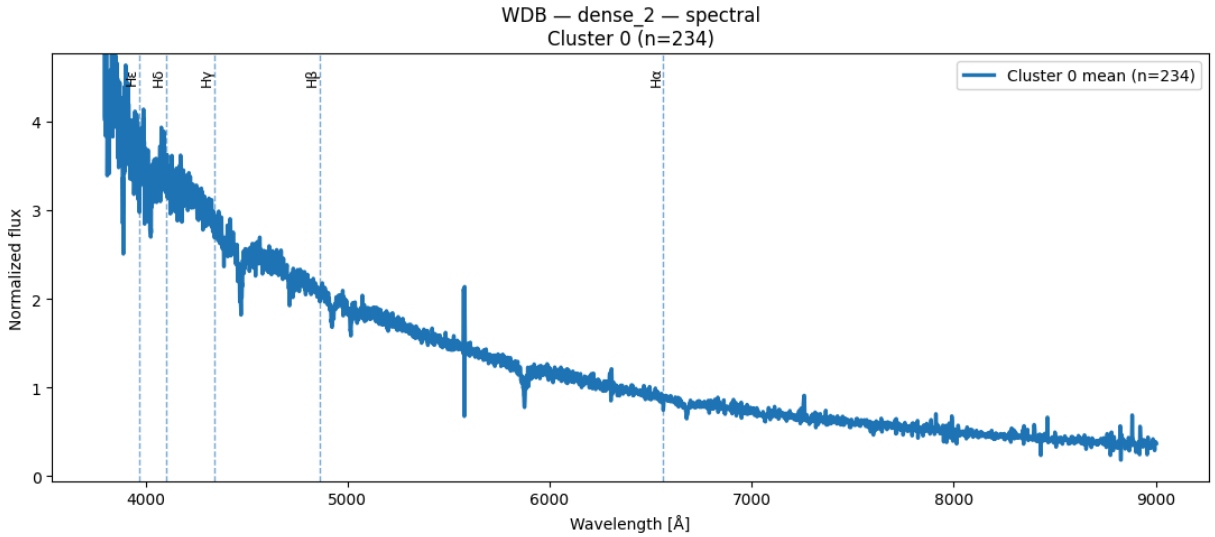

Migración k=5 → k=6 (Flatten, WDA)
Clústeres k=5
C0
C1
C2
C3
C4
141 153 228 117 32 53
1 96 35 0 27 58
21 0 1 9 0 0
7 0 4 52 0 0
31 1 1 3 0 28
C0
C1
C2
C3
C4
C5
Clústeres k=6
200
150
100
50
0

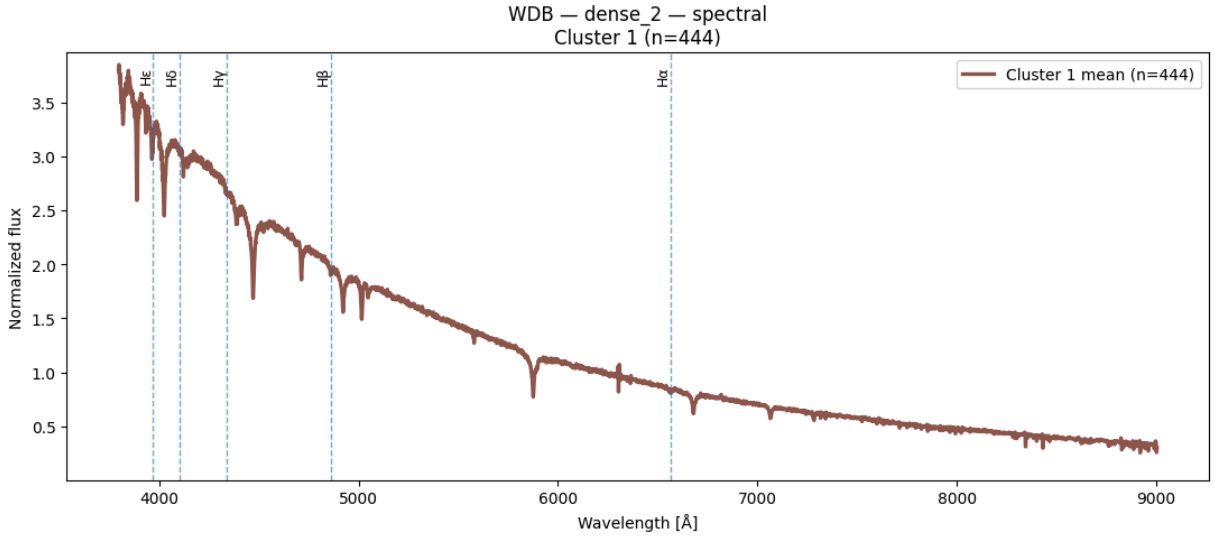


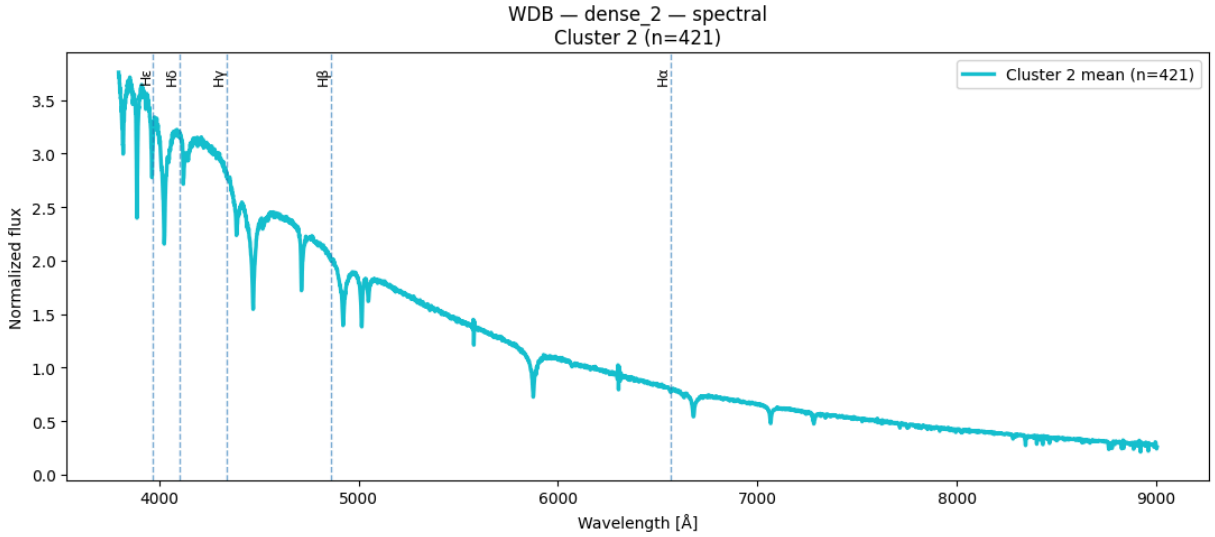
Espectros obtenidos de Spectral Clustering para k = 3 y n = 1100

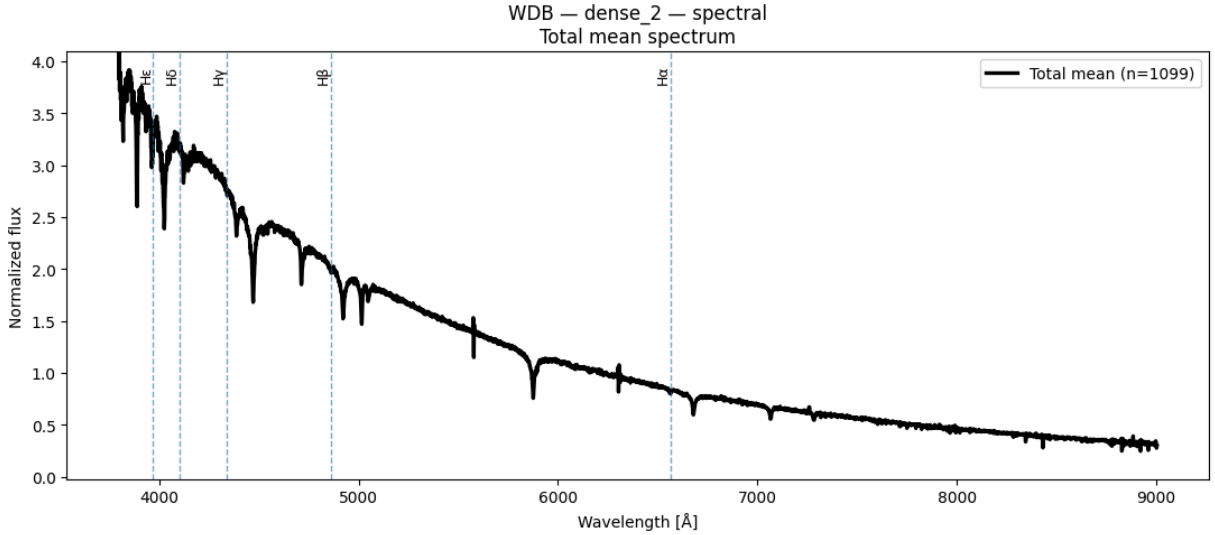
In [83]:
```
result = plot_spectra_spectral(X_wdb_spec, labels_dense2_k3_wdb, W_wdb[0],
                               class_name="WDB", layer_name="dense_2")
```

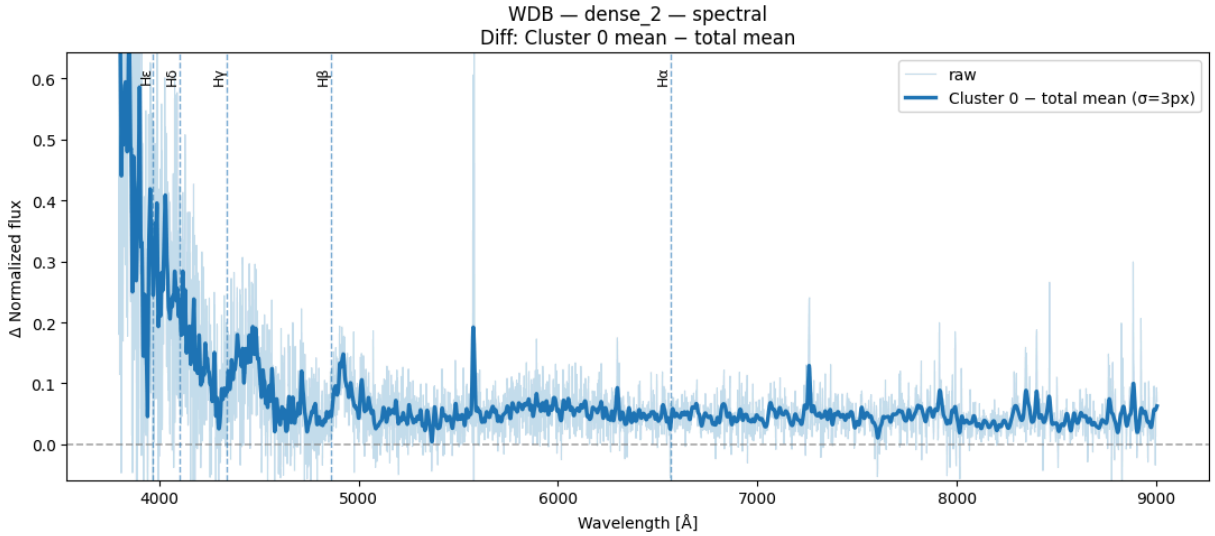
```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```

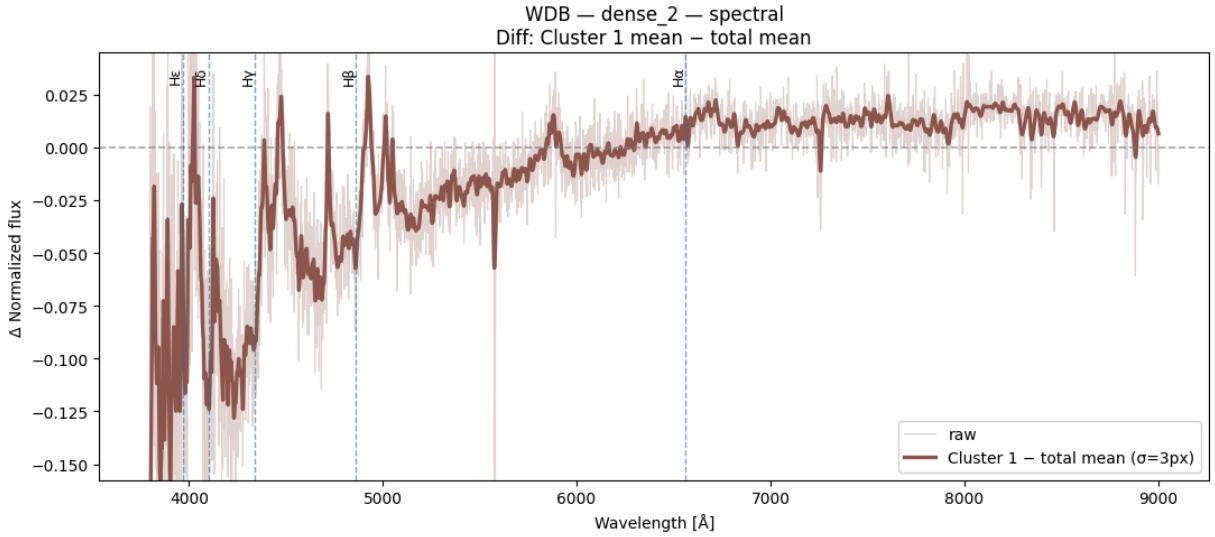

WDB — dense_2 — spectral
Cluster 0 (n=234)
Cluster 0 mean (n=234)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

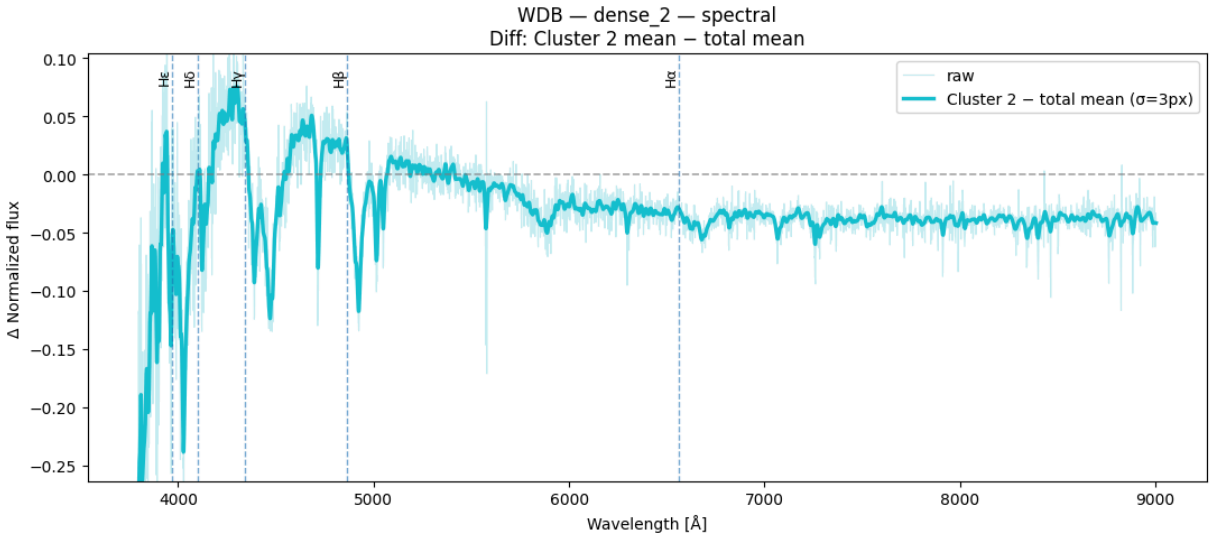

WDB — dense_2 — spectral
Cluster 1 (n=444)
Cluster 1 mean (n=444)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

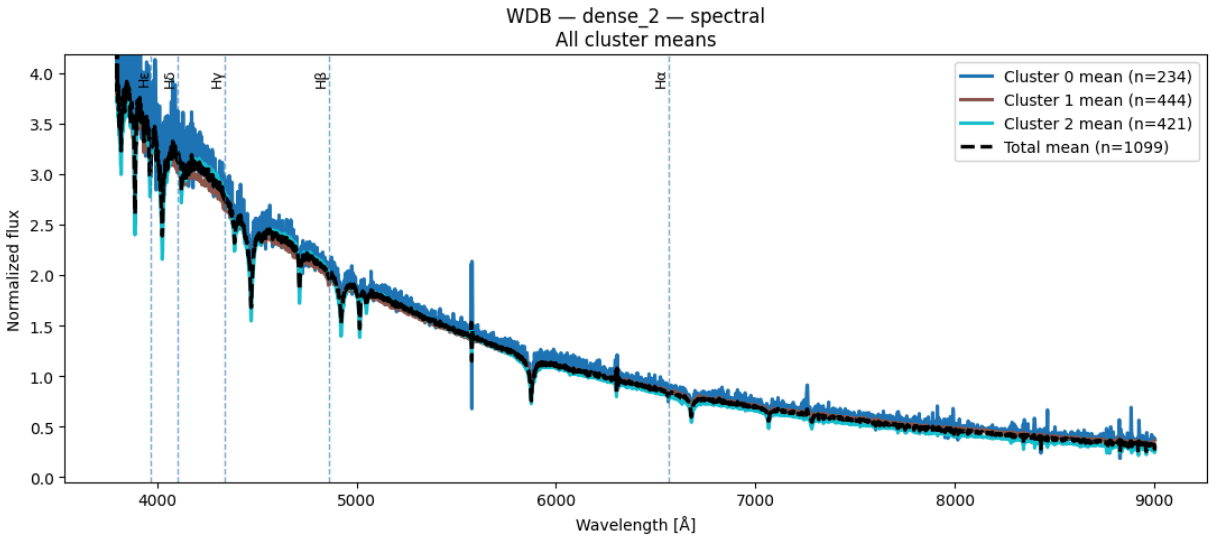

WDB — dense_2 — spectral
Cluster 2 (n=421)
Cluster 2 mean (n=421)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — dense_2 — spectral
Total mean spectrum
Total mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — dense_2 — spectral
Diff: Cluster 0 mean − total mean
raw
Cluster 0 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]


WDB — dense_2 — spectral
Diff: Cluster 1 mean − total mean
raw
Cluster 1 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]

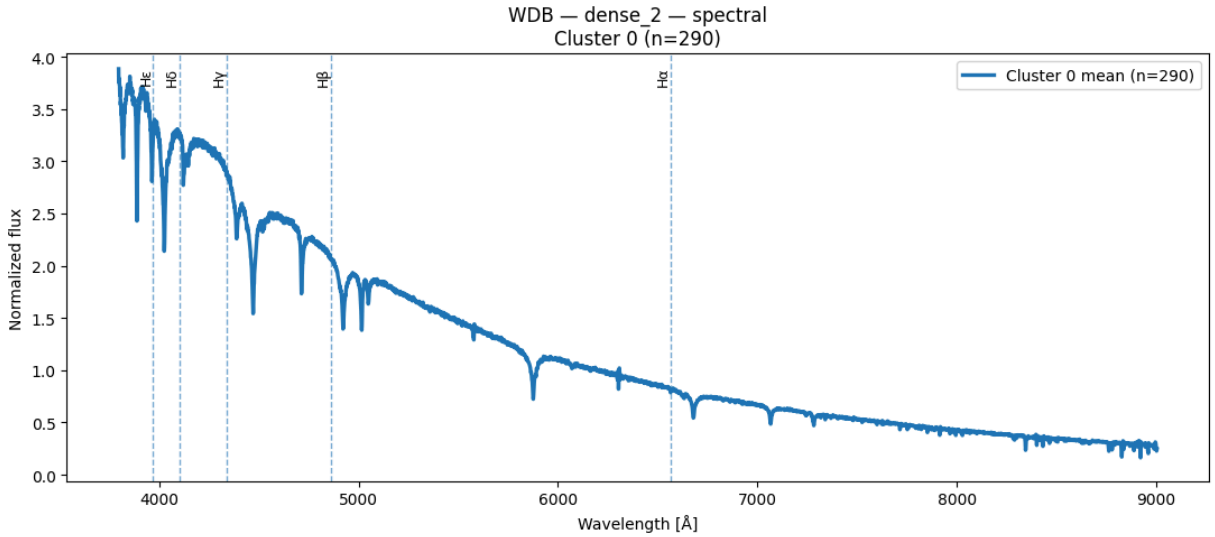


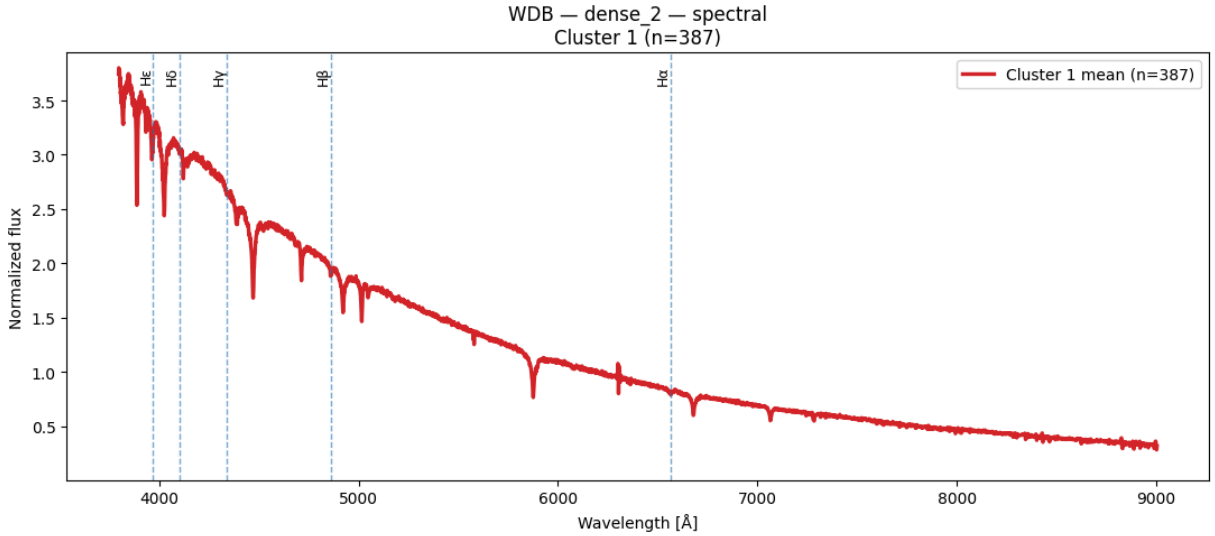


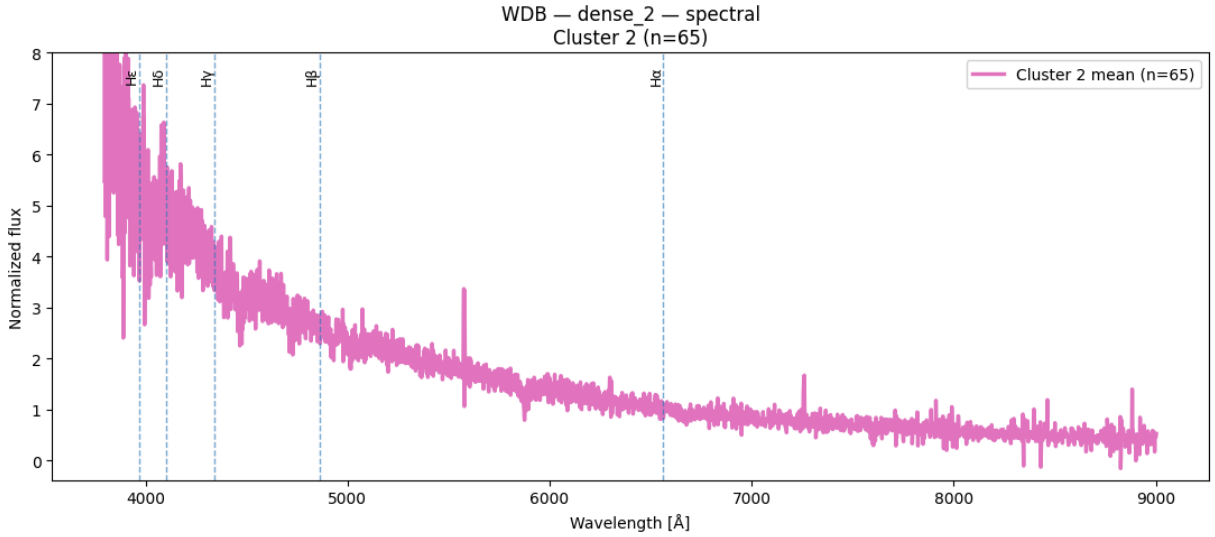
Espectros obtenidos de Spectral Clustering para k = 4 y n = 1100

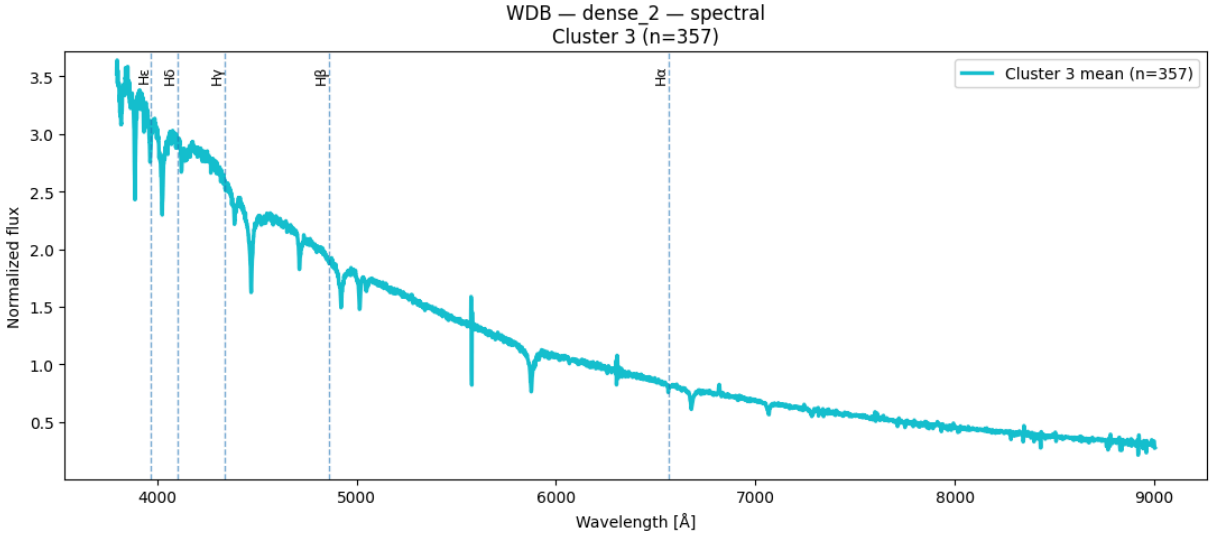
In [84]:
```
result = plot_spectra_spectral(X_wdb_spec, labels_dense2_k4_wdb, W_wdb[0],
                               class_name="WDB", layer_name="dense_2")
```

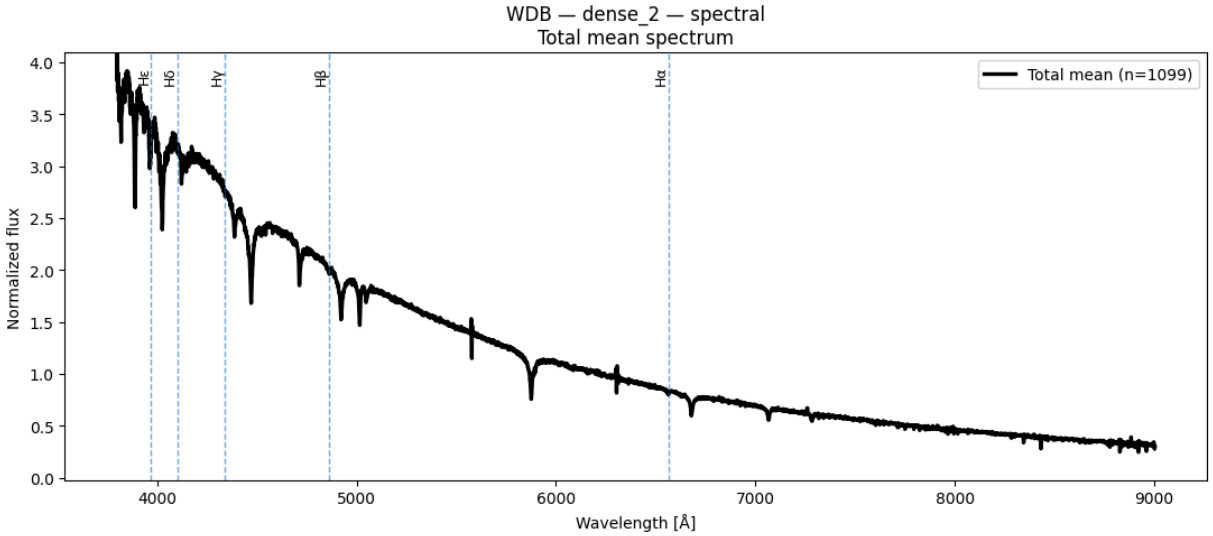
```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```

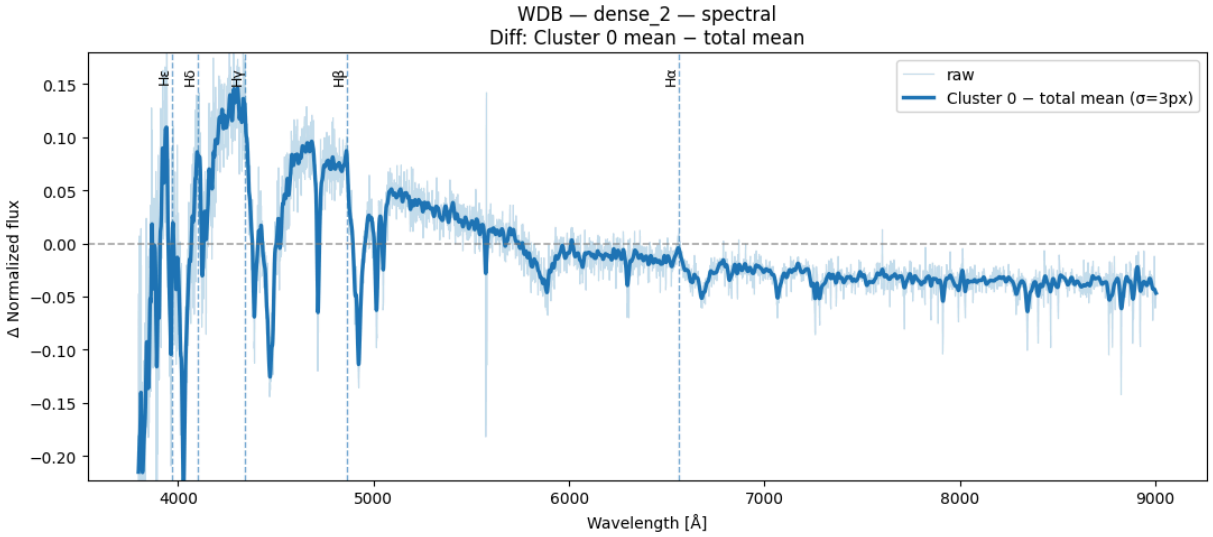

WDB — dense_2 — spectral
Cluster 0 (n=290)
Cluster 0 mean (n=290)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

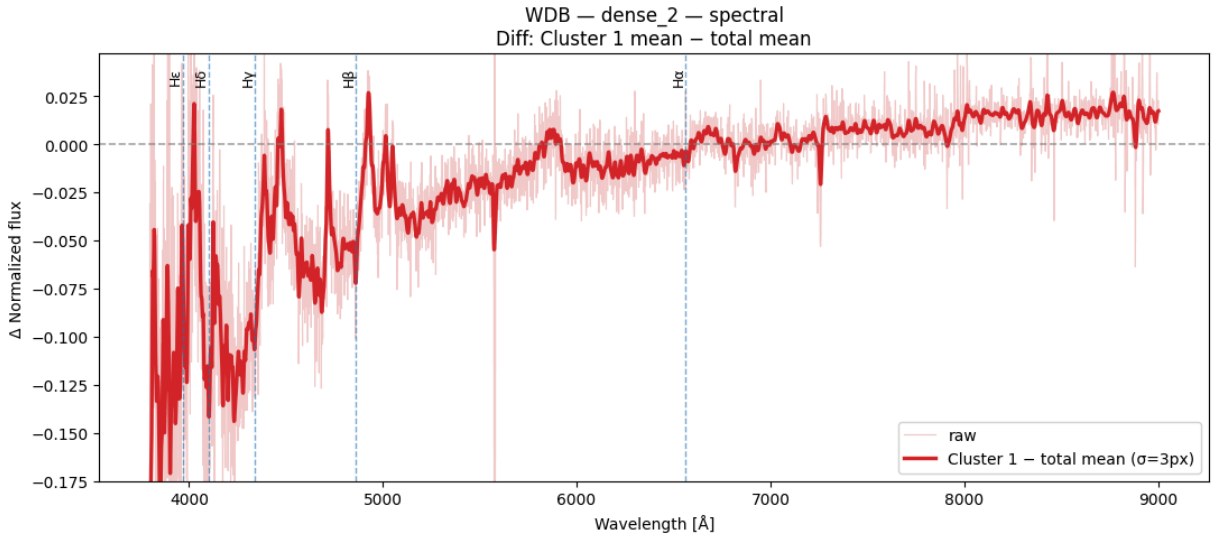

WDB — dense_2 — spectral
Cluster 1 (n=387)
Cluster 1 mean (n=387)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

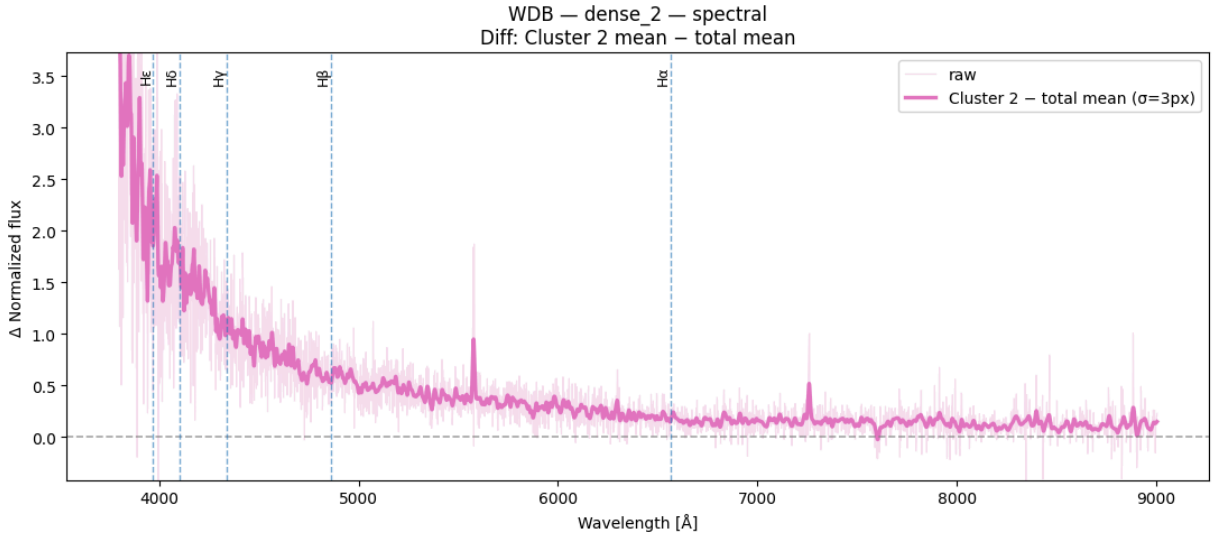

WDB — dense_2 — spectral
Cluster 2 (n=65)
Cluster 2 mean (n=65)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

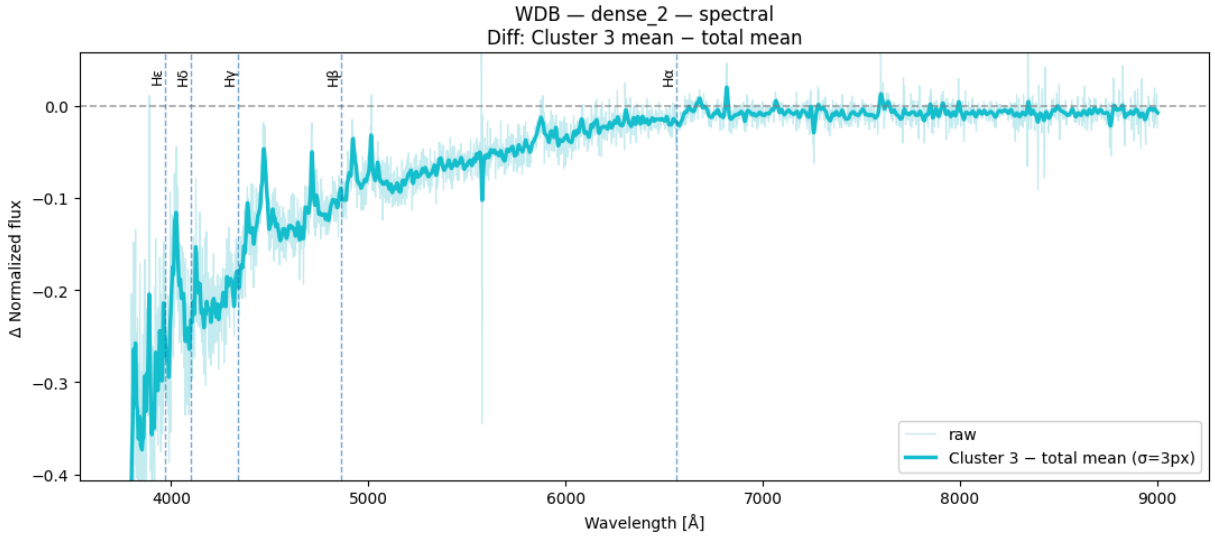

WDB — dense_2 — spectral
Cluster 3 (n=357)
Cluster 3 mean (n=357)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

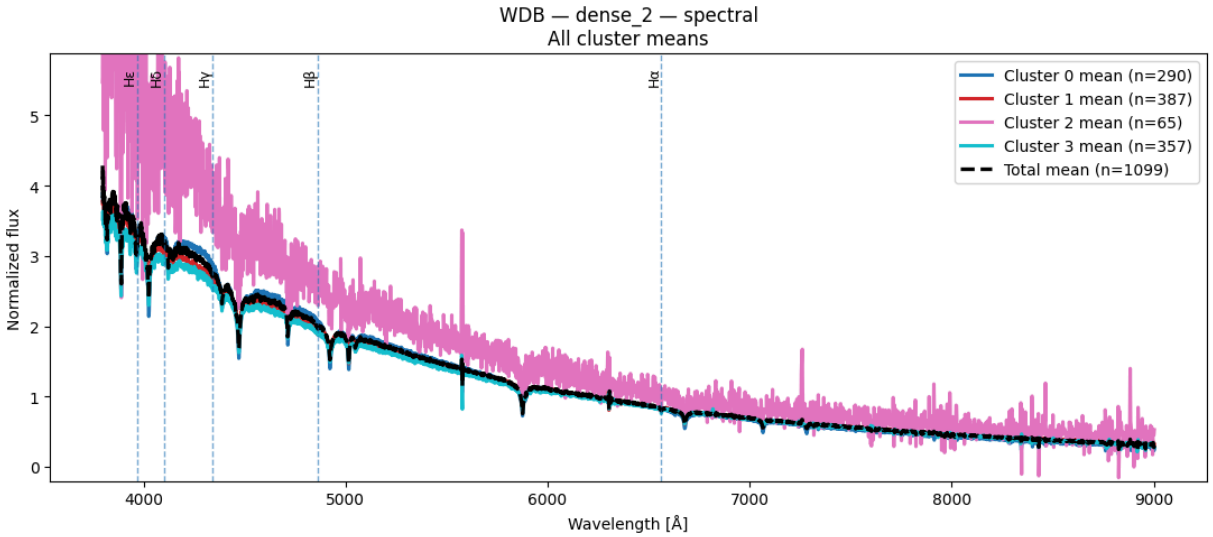

WDB — dense_2 — spectral
Total mean spectrum
Total mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]


WDB — dense_2 — spectral
Diff: Cluster 0 mean − total mean
raw
Cluster 0 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]


WDB — dense_2 — spectral
Diff: Cluster 1 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]
raw
Cluster 1 − total mean (σ=3px)


WDB — dense_2 — spectral
Diff: Cluster 2 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]
raw
Cluster 2 − total mean (σ=3px)

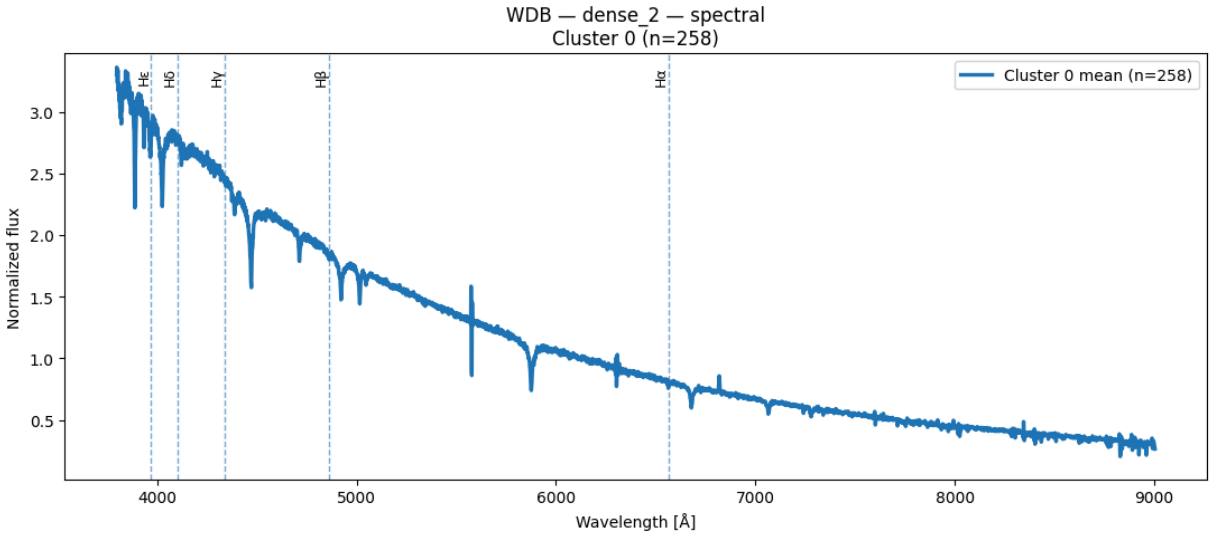

WDB — dense_2 — spectral
Diff: Cluster 3 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]
raw
Cluster 3 − total mean (σ=3px)

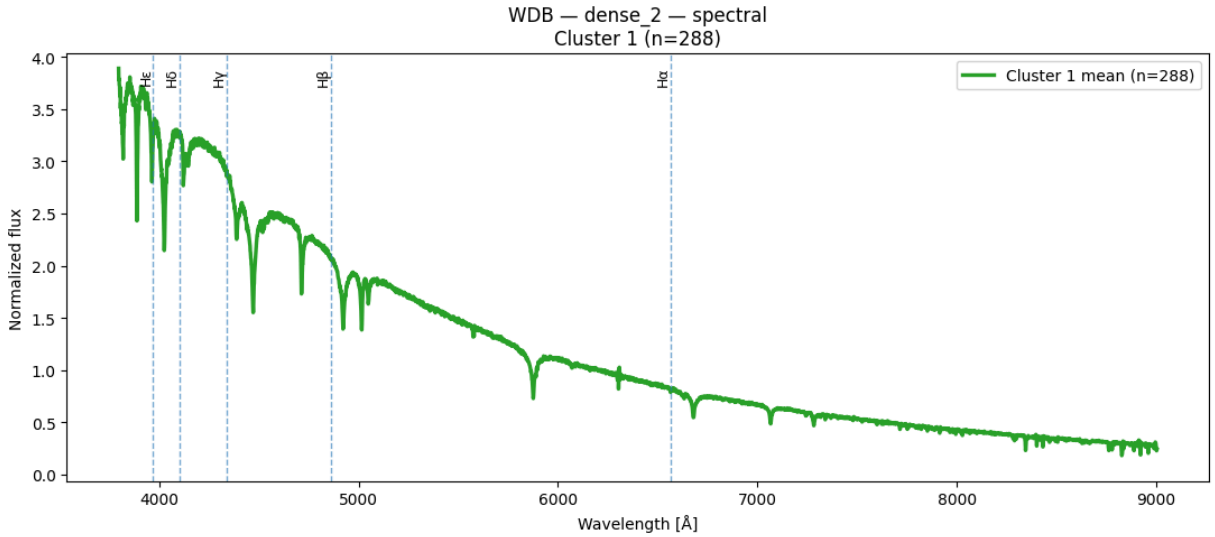


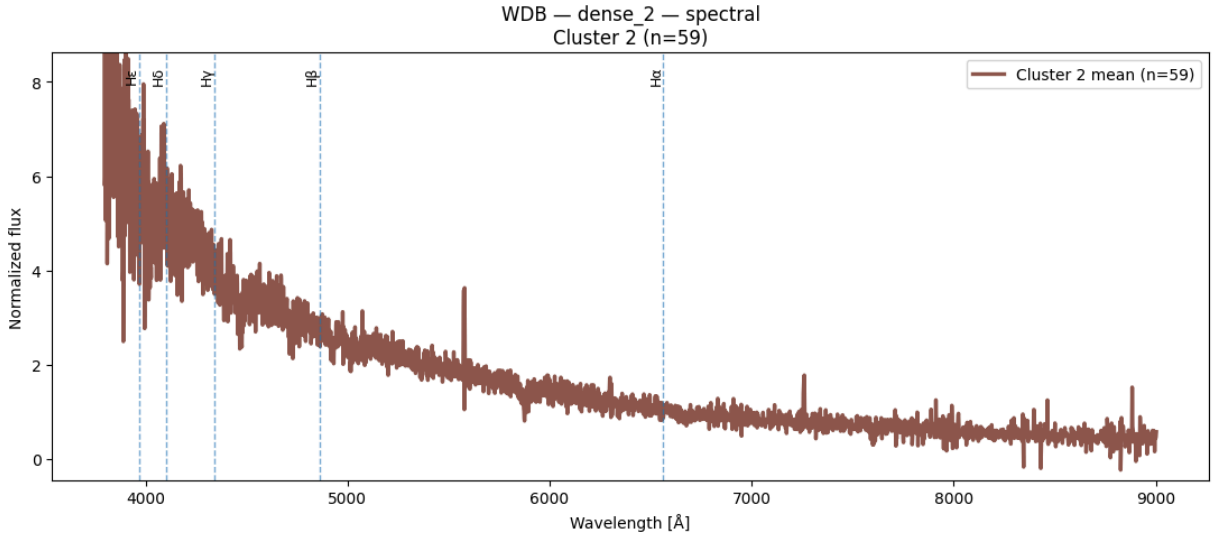
Espectros obtenidos de Spectral Clustering para k = 5 y n = 1100

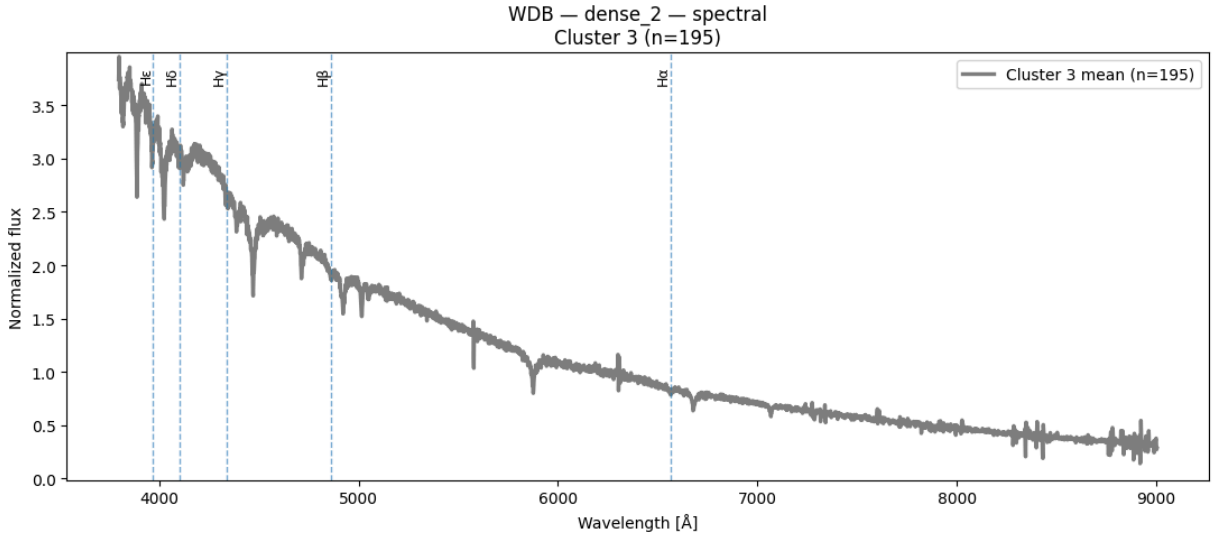
In [85]:
```
result = plot_spectra_spectral(X_wdb_spec, labels_dense2_k5_wdb, W_wdb[0],
                               class_name="WDB", layer_name="dense_2")
```

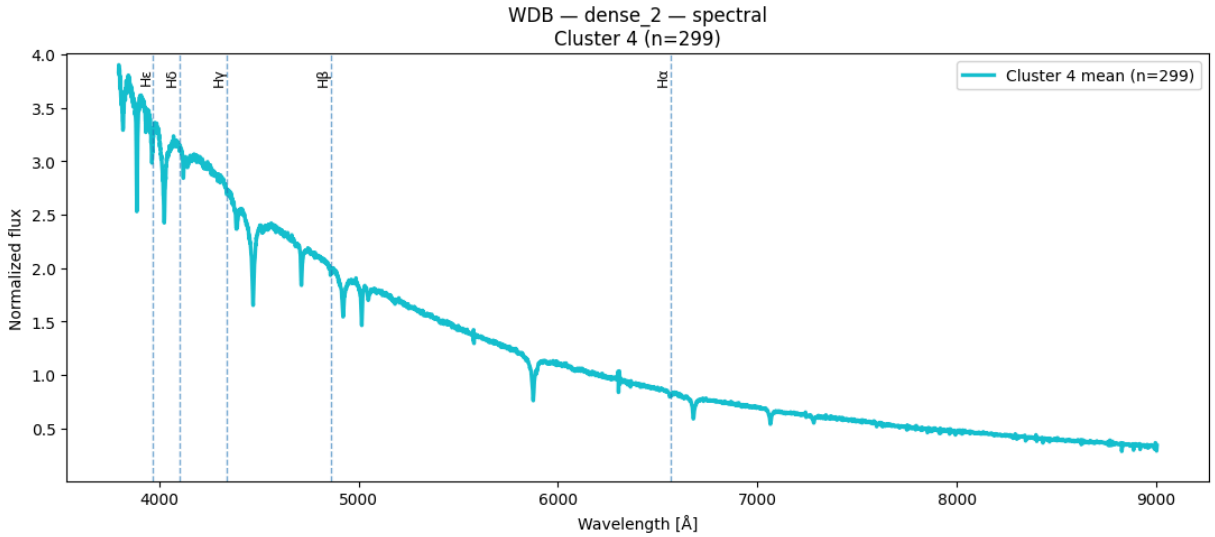
```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```

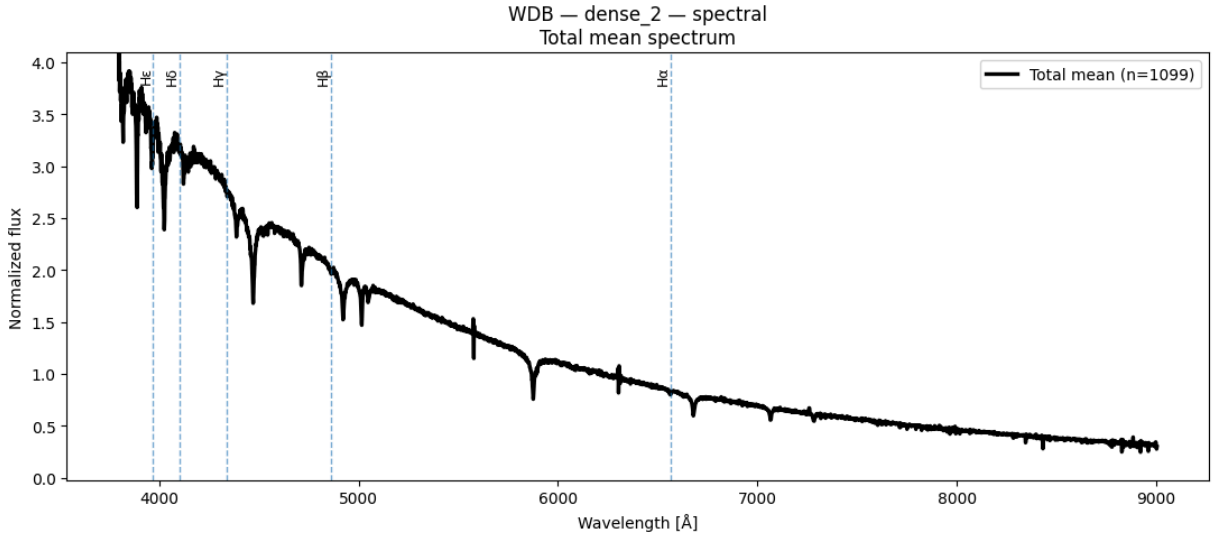


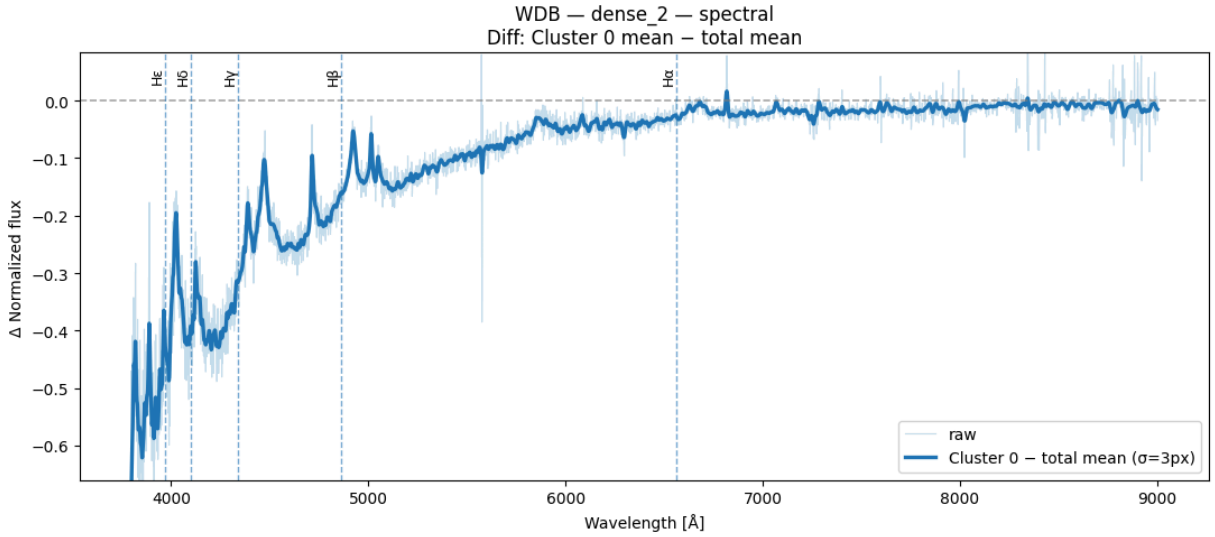

WDB — dense_2 — spectral
Cluster 1 (n=288)
Cluster 1 mean (n=288)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

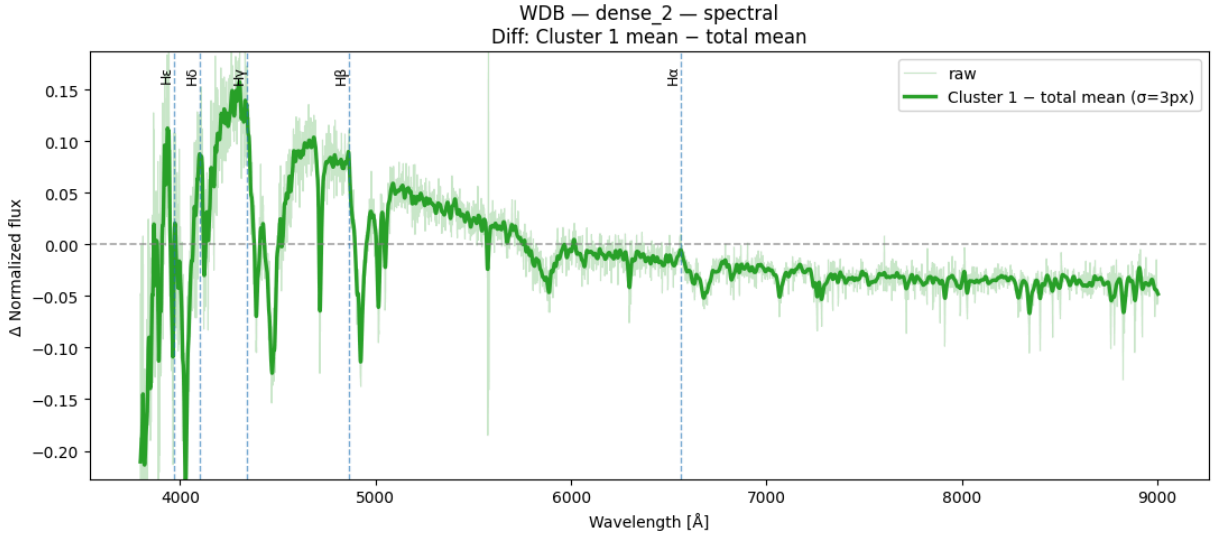

WDB — dense_2 — spectral
Cluster 2 (n=59)
Cluster 2 mean (n=59)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

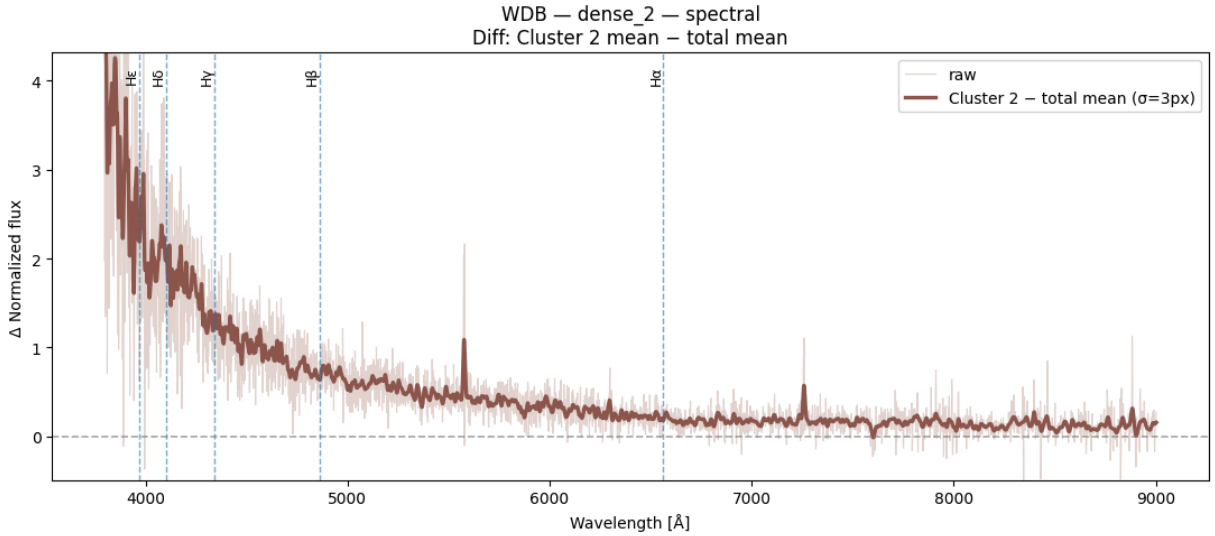

WDB — dense_2 — spectral
Cluster 3 (n=195)
Cluster 3 mean (n=195)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

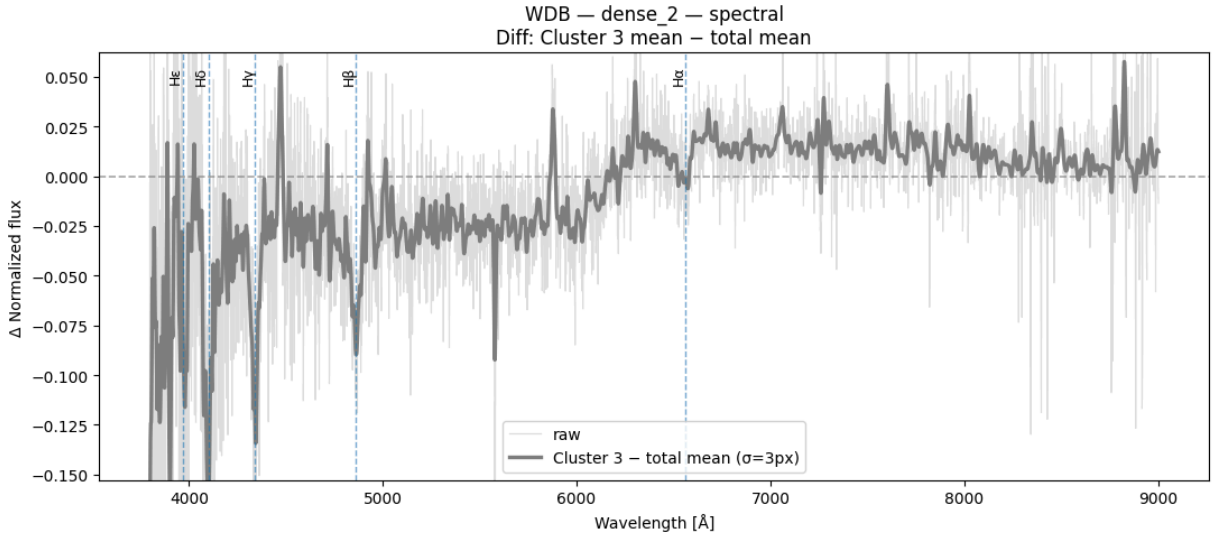

WDB — dense_2 — spectral
Cluster 4 (n=299)
Cluster 4 mean (n=299)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

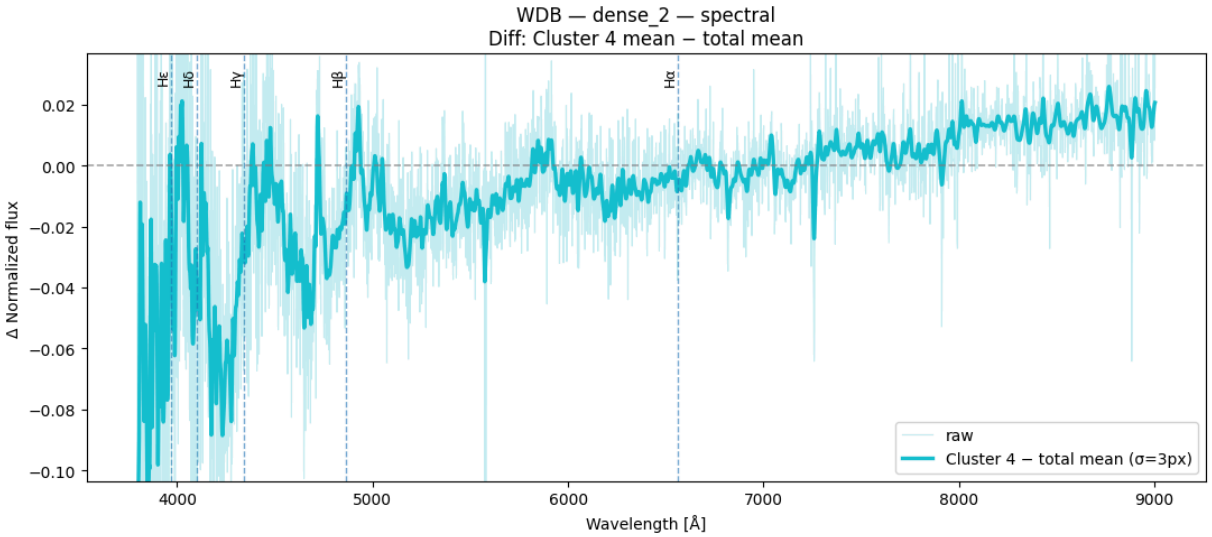

WDB — dense_2 — spectral
Total mean spectrum
Total mean (n=1099)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

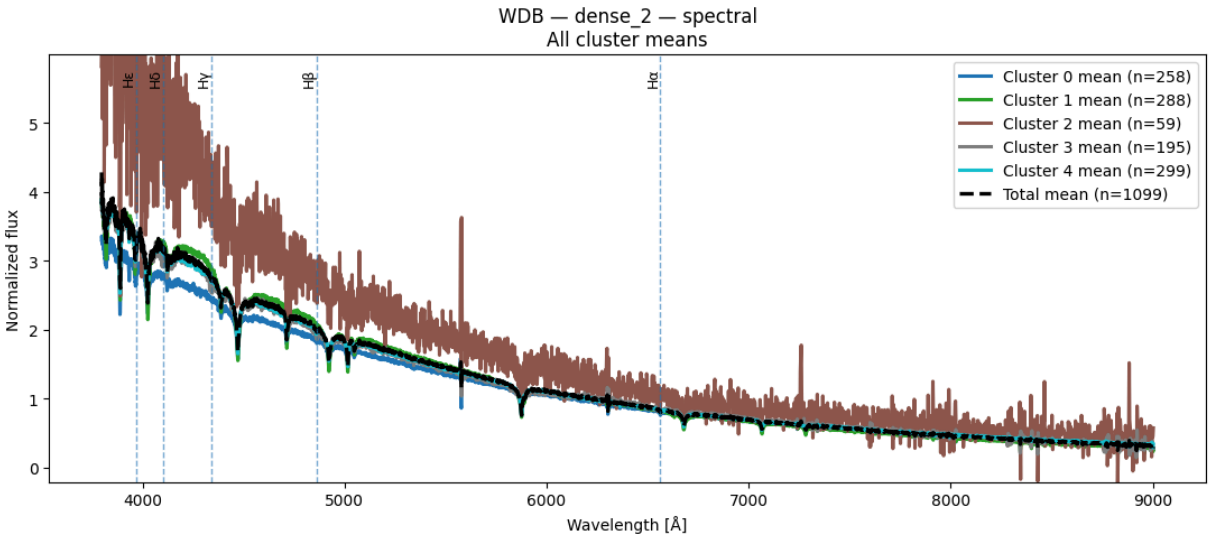

WDB — dense_2 — spectral
Diff: Cluster 0 mean − total mean
raw
Cluster 0 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]


WDB — dense_2 — spectral
Diff: Cluster 1 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
raw
Cluster 1 − total mean (σ=3px)
Δ Normalized flux
Wavelength [Å]


WDB — dense_2 — spectral
Diff: Cluster 2 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
raw
Cluster 2 − total mean (σ=3px)
Δ Normalized flux
Wavelength [Å]


WDB — dense_2 — spectral
Diff: Cluster 3 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
raw
Cluster 3 − total mean (σ=3px)
Δ Normalized flux
Wavelength [Å]

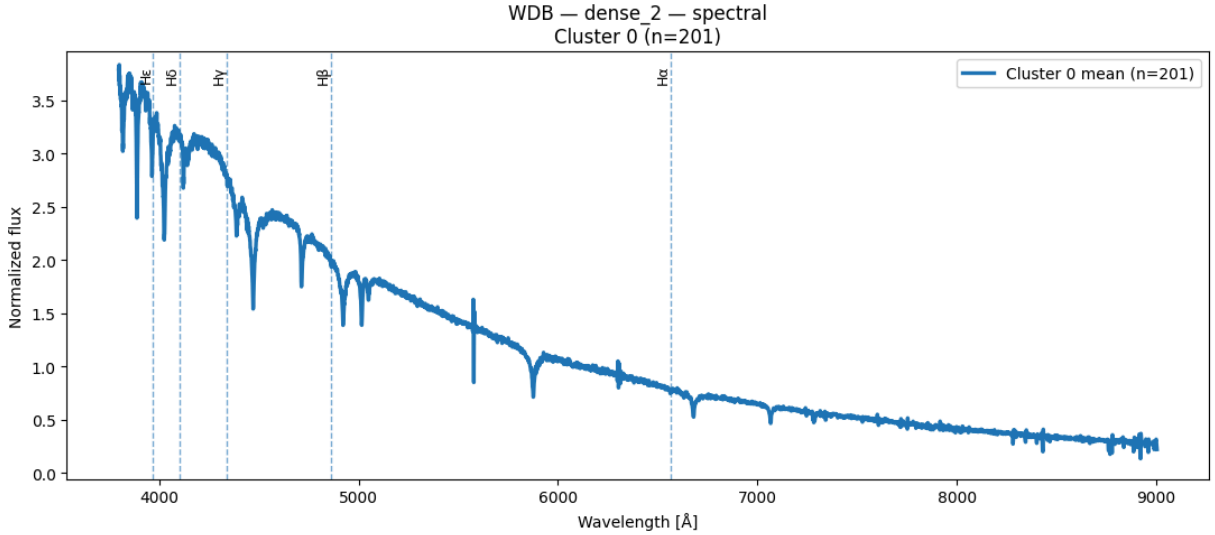


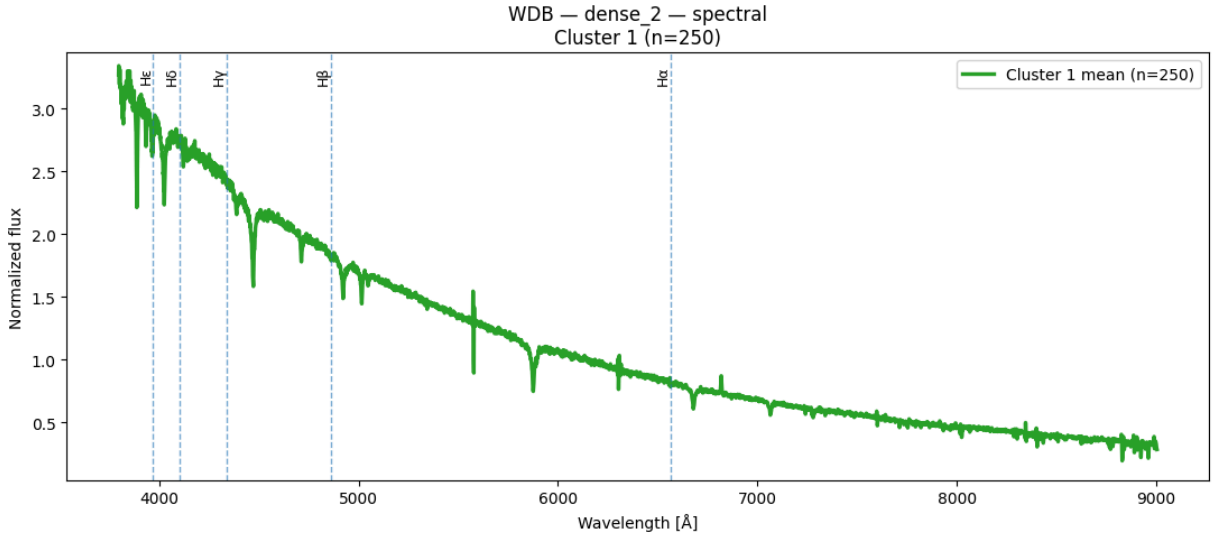


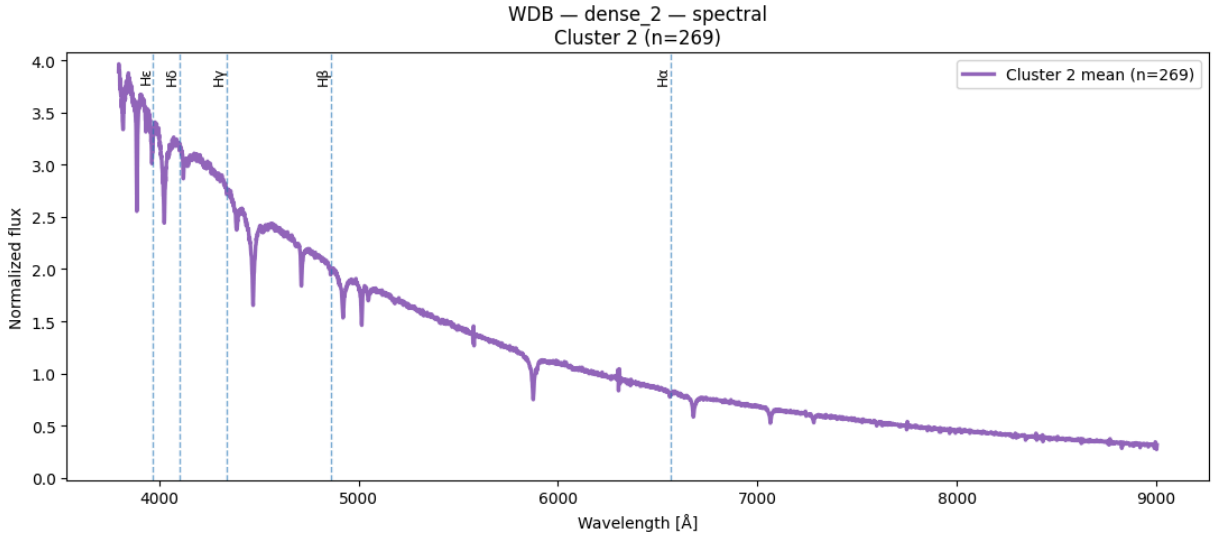
Espectros obtenidos de Spectral Clustering para k = 6 y n = 1100

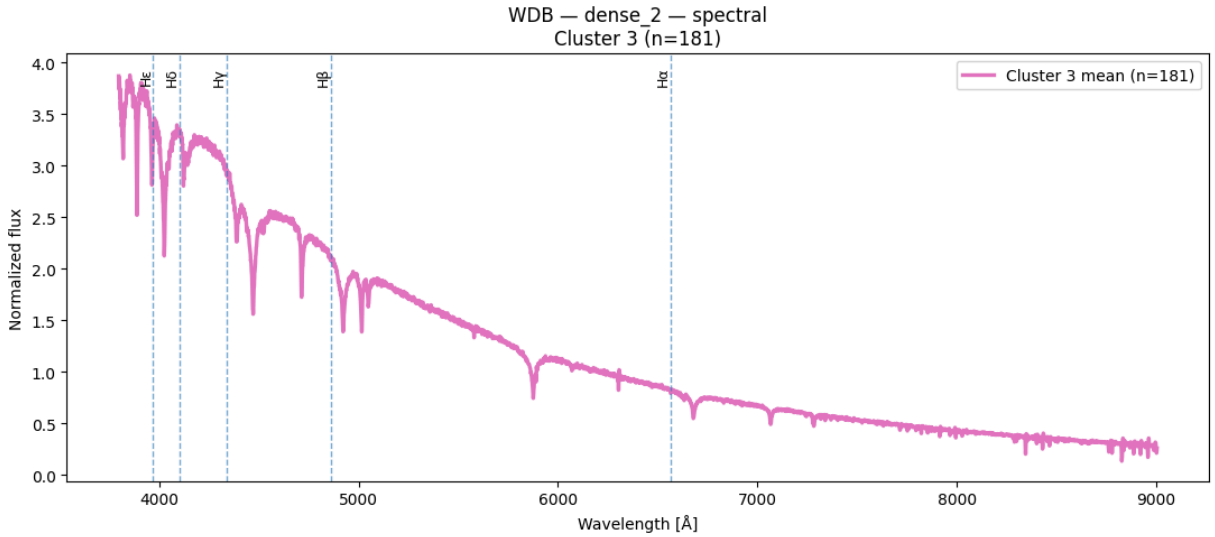
In [86]:
```
result = plot_spectra_spectral(X_wdb_spec, labels_dense2_k6_wdb, W_wdb[0],
                               class_name="WDB", layer_name="dense_2")
```

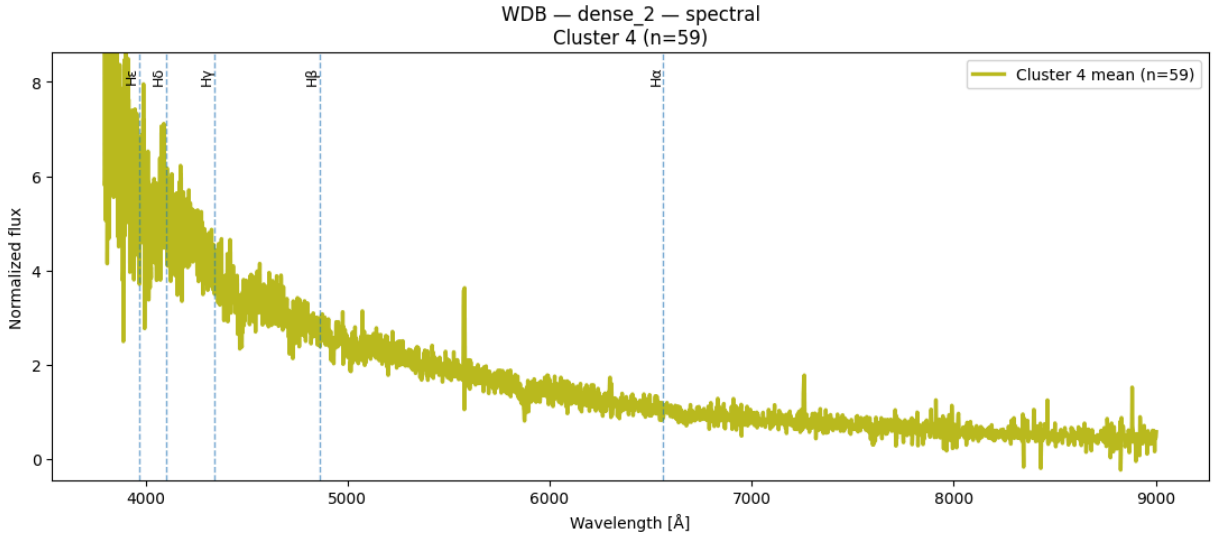
```
C:\Users\javip\AppData\Local\Temp\ipykernel_24108\531566132.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or ``matplotlib.colormaps.get_cmap()`` or ``pyplot.get_cmap()`` instead.
  cmap       = plt.cm.get_cmap("tab10", n_clusters)
```

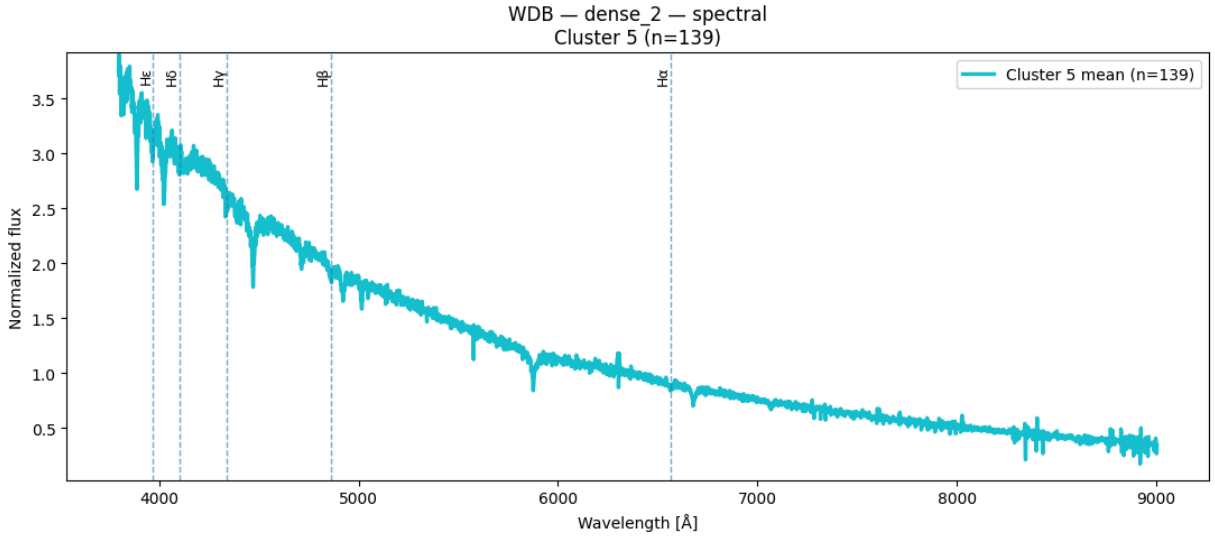

WDB — dense_2 — spectral
Cluster 0 (n=201)
Cluster 0 mean (n=201)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

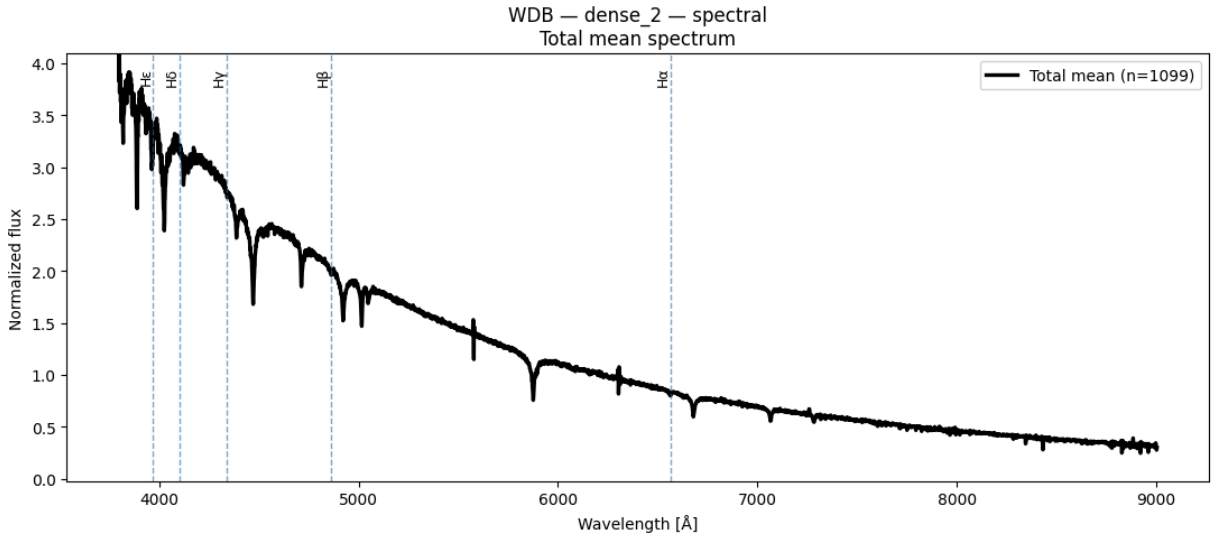

WDB — dense_2 — spectral
Cluster 1 (n=250)
Cluster 1 mean (n=250)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

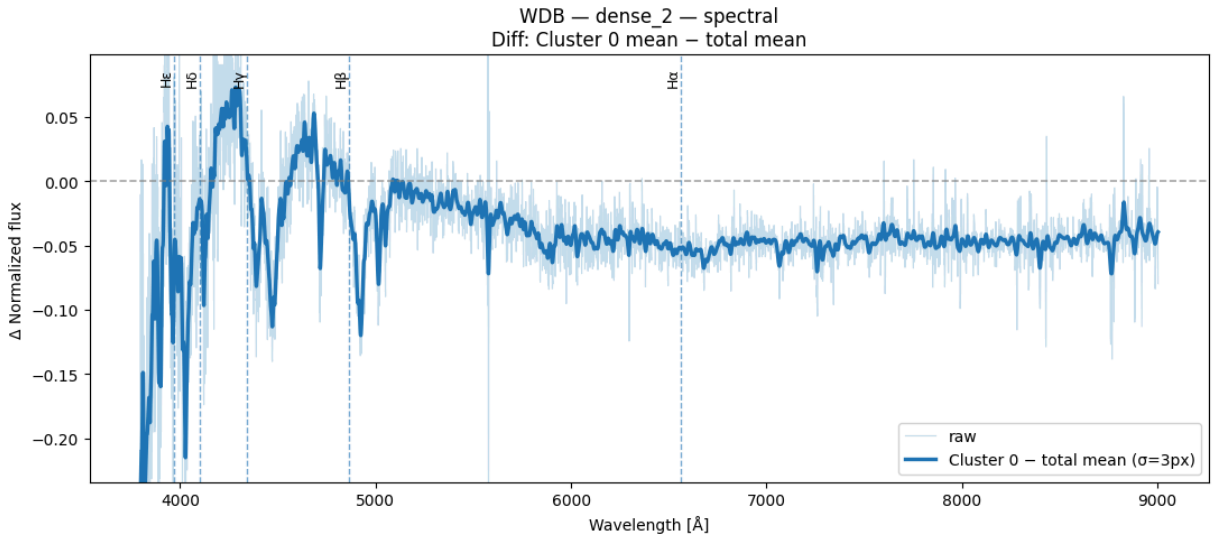

WDB — dense_2 — spectral
Cluster 2 (n=269)
Cluster 2 mean (n=269)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

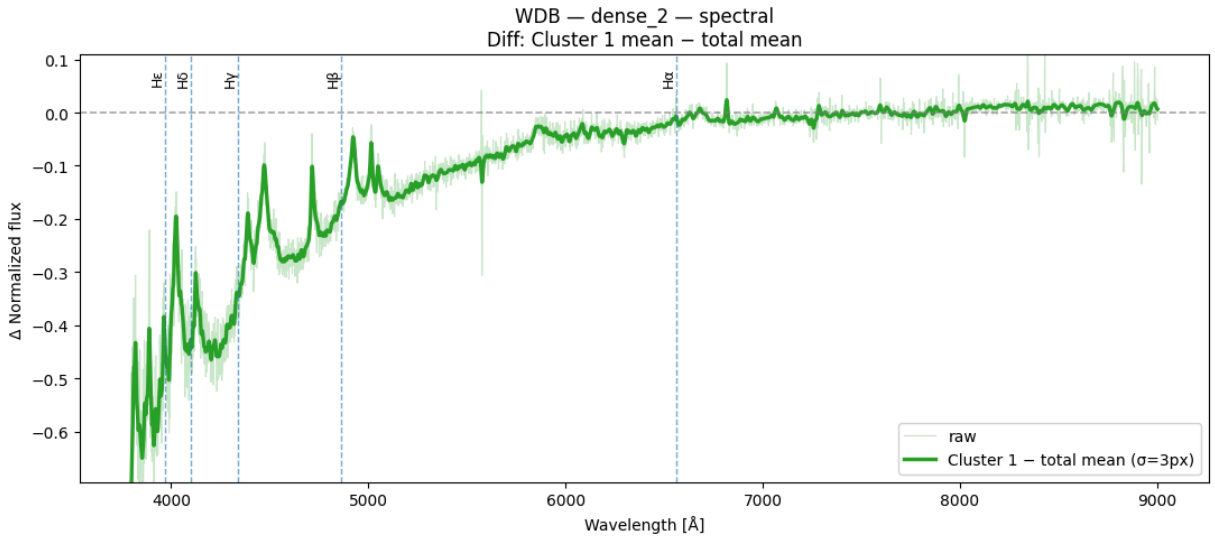

WDB — dense_2 — spectral
Cluster 3 (n=181)
Cluster 3 mean (n=181)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

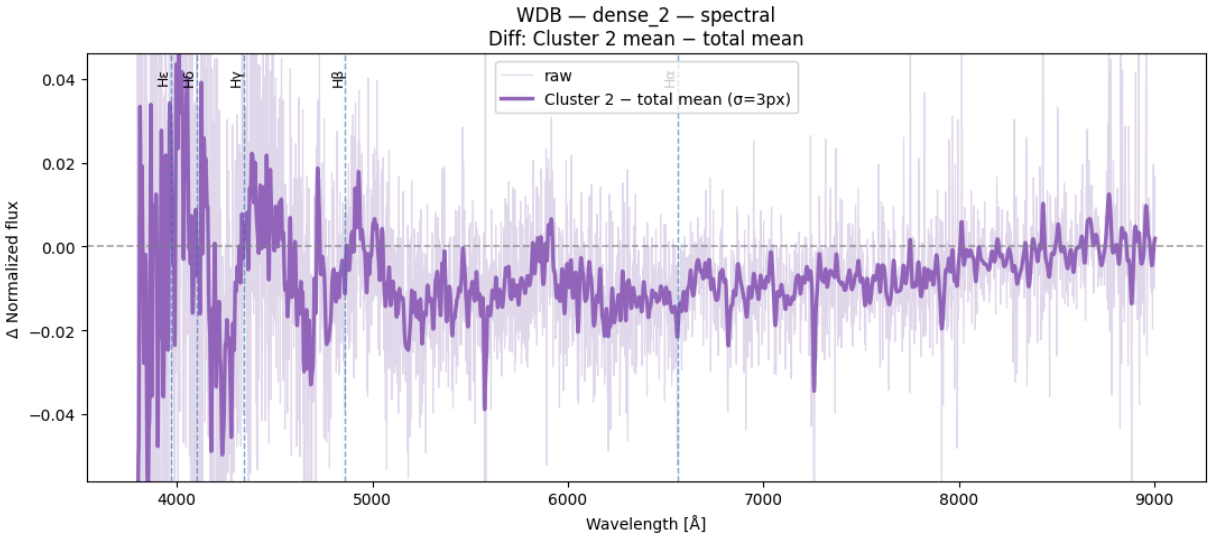

WDB — dense_2 — spectral
Cluster 4 (n=59)
Cluster 4 mean (n=59)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

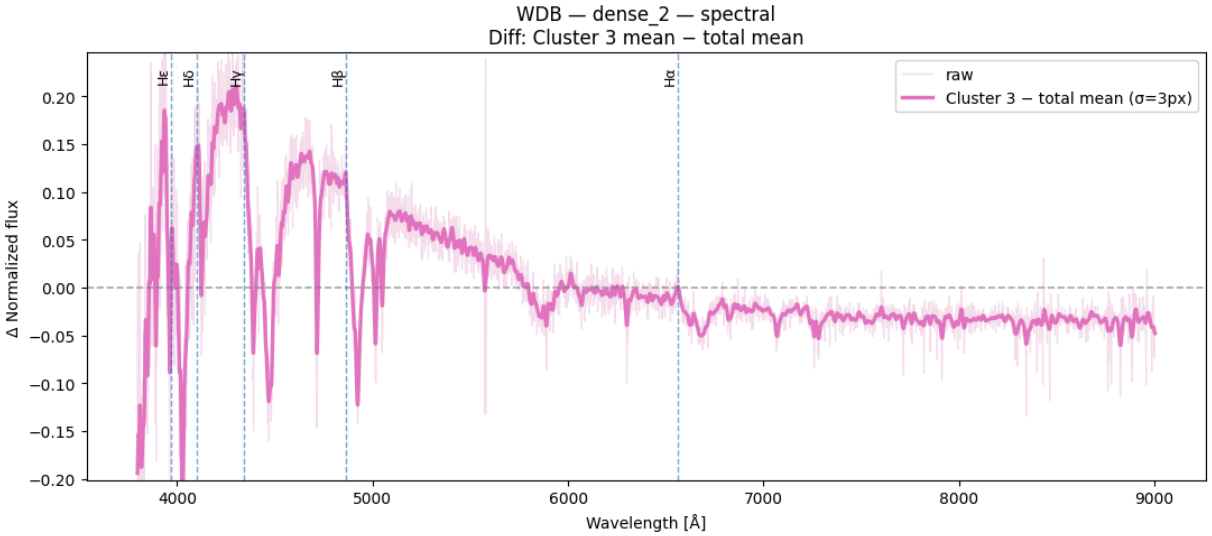

WDB — dense_2 — spectral
Cluster 5 (n=139)
Cluster 5 mean (n=139)
Hε
Hδ
Hγ
Hβ
Hα
Normalized flux
Wavelength [Å]

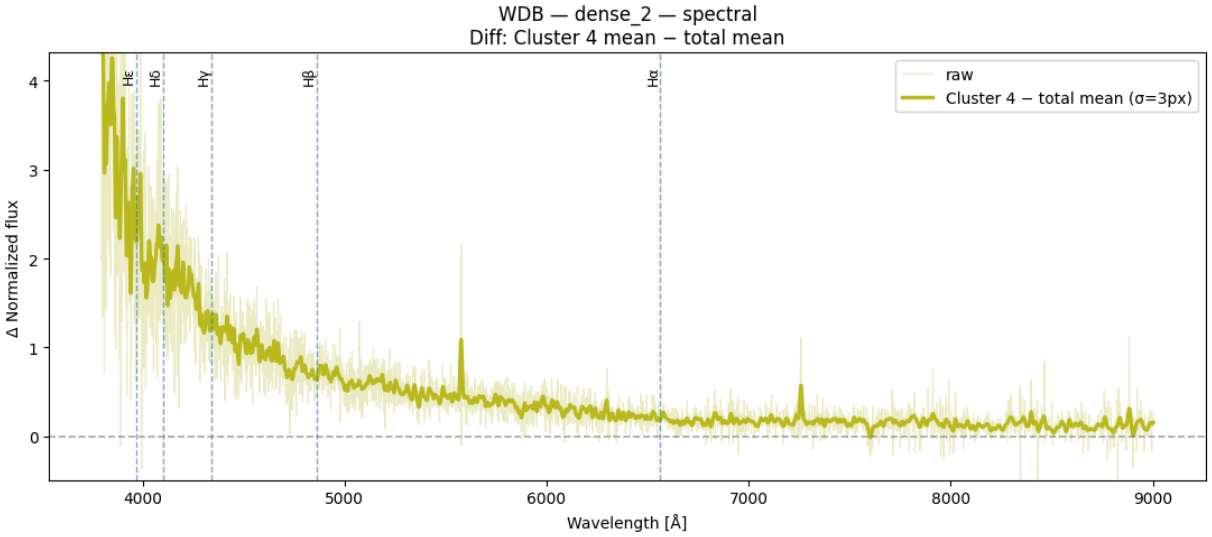

WDB — dense_2 — spectral
Total mean spectrum
Hε
Hδ
Hγ
Hβ
Hα
Total mean (n=1099)
Normalized flux
Wavelength [Å]

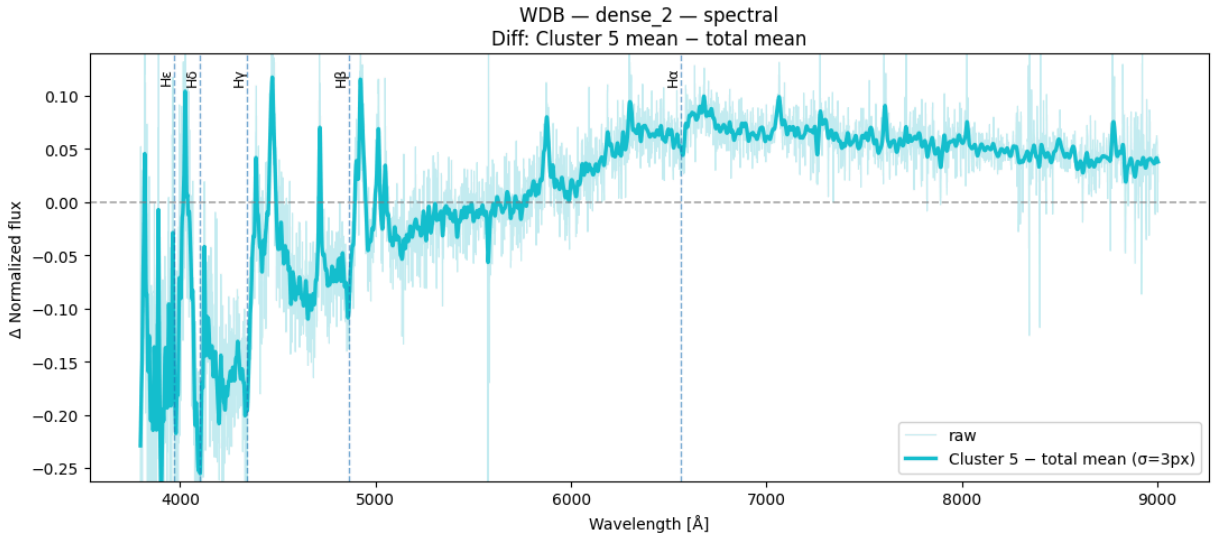

WDB — dense_2 — spectral
Diff: Cluster 0 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
raw
Cluster 0 − total mean (σ=3px)
Δ Normalized flux
Wavelength [Å]


WDB — dense_2 — spectral
Diff: Cluster 1 mean − total mean
Hε
Hδ
Hγ
Hβ
Hα
raw
Cluster 1 − total mean (σ=3px)
Δ Normalized flux
Wavelength [Å]


WDB — dense_2 — spectral
Diff: Cluster 2 mean − total mean
raw
Cluster 2 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]


WDB — dense_2 — spectral
Diff: Cluster 3 mean − total mean
raw
Cluster 3 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]


WDB — dense_2 — spectral
Diff: Cluster 4 mean − total mean
raw
Cluster 4 − total mean (σ=3px)
Hε
Hδ
Hγ
Hβ
Hα
Δ Normalized flux
Wavelength [Å]



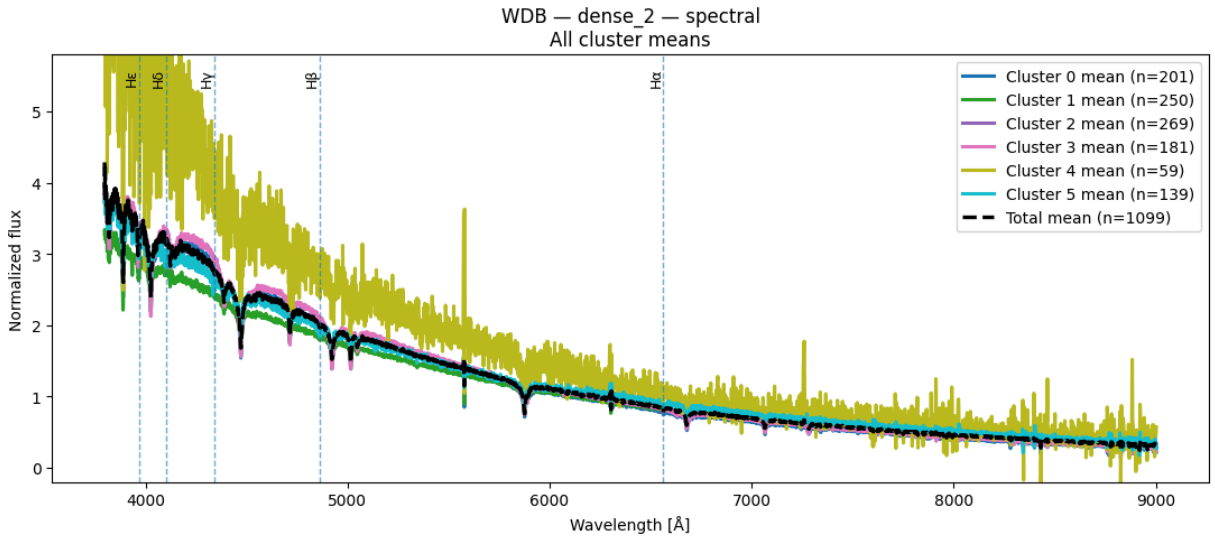


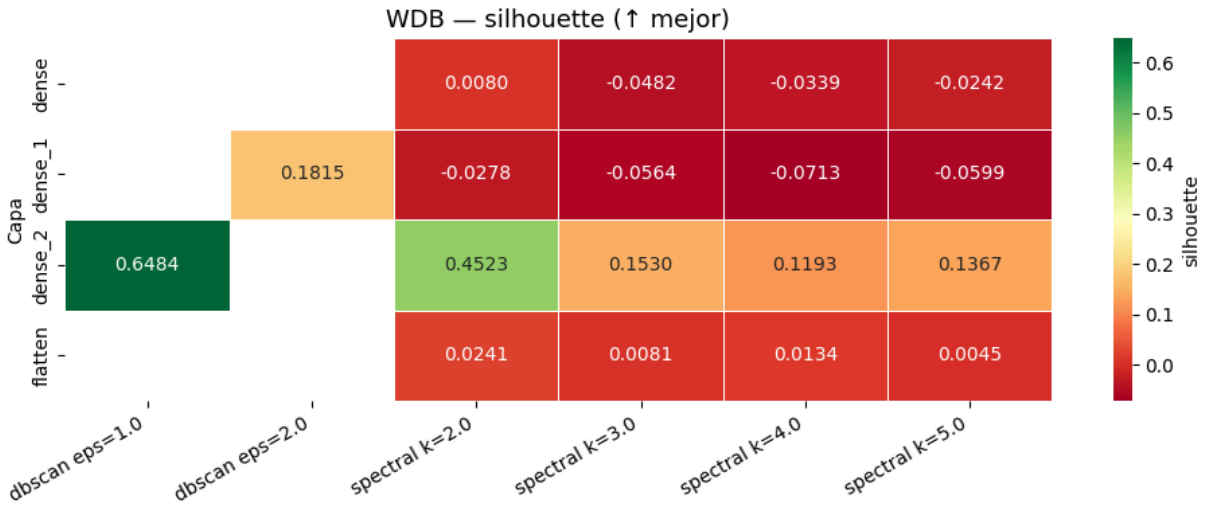
Tabla de evaluación de métricas

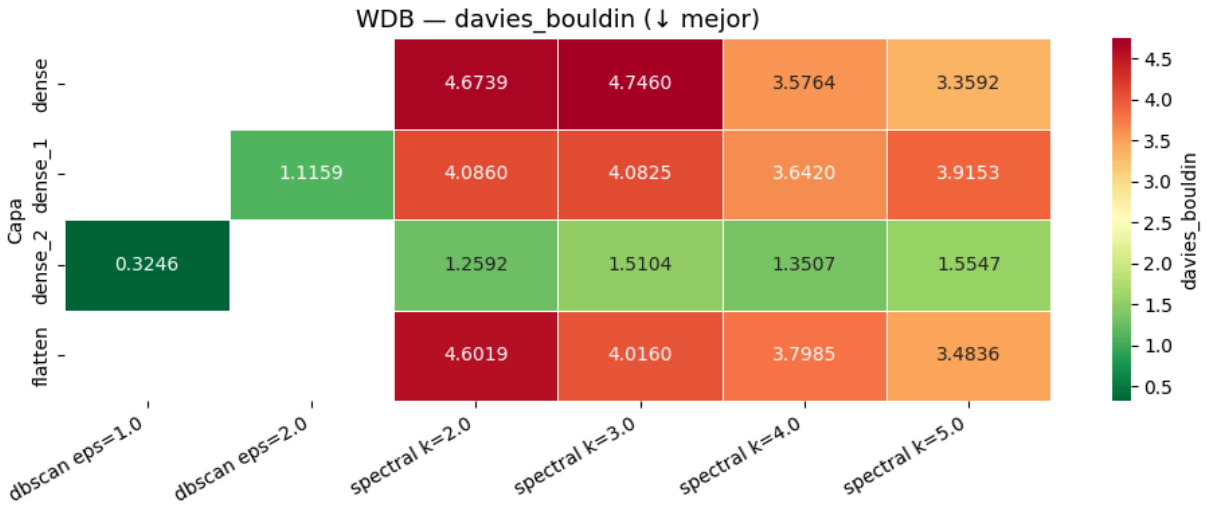
In [87]:
```
# construir tabla para WDB con n=1100
metrics_wdb = build_metrics_table(
    representations_32   = repr_wdb_32,
    class_name           = "WDB",
    spectral_k_values    = [2, 3, 4, 5],
    dbscan_eps_values    = [0.5, 1.0, 1.5, 2.0, 2.5, 3.0],
    dbscan_min_samples   = 10
)

print(metrics_wdb.to_string(index=False))

# ver las 5 mejores configuraciones
top_configs = get_best_configs(metrics_wdb, top_n=5)
print(top_configs.to_string(index=False))

# heatmaps de cada métrica
plot_metrics_heatmap(metrics_wdb, metric="silhouette",         class_name="WDB")
```

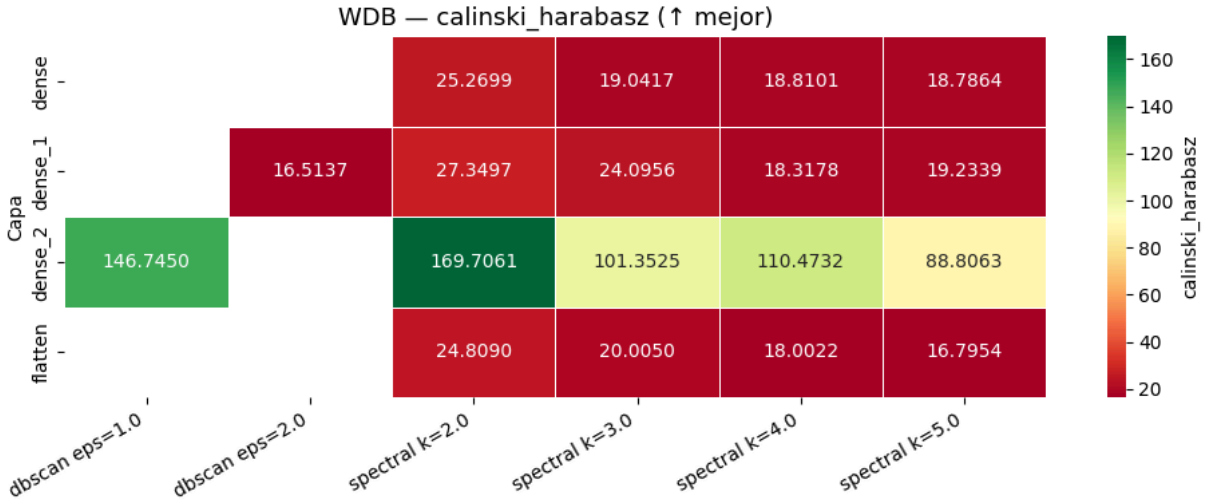
```
plot_metrics_heatmap(metrics_wdb, metric="davies_bouldin",     class_name="WDB")
plot_metrics_heatmap(metrics_wdb, metric="calinski_harabasz",  class_name="WDB")
```

class	layer	method	param	param_value	n_clusters	n_noise	silhouette	davies_bouldin	calinski_harabasz
WDB	flatten	spectral	k	2.0	2	0	0.024117	4.601948	24.8090
WDB	flatten	spectral	k	3.0	3	0	0.008147	4.015992	20.0050
WDB	flatten	spectral	k	4.0	4	0	0.013448	3.798504	18.0022
WDB	flatten	spectral	k	5.0	5	0	0.004542	3.483588	16.7954
WDB	flatten	dbscan	eps	0.5	0	1099	NaN	NaN	NaN
WDB	flatten	dbscan	eps	1.0	0	1099	NaN	NaN	NaN
WDB	flatten	dbscan	eps	1.5	0	1099	NaN	NaN	NaN
WDB	flatten	dbscan	eps	2.0	1	897	NaN	NaN	NaN
WDB	flatten	dbscan	eps	2.5	1	679	NaN	NaN	NaN
WDB	flatten	dbscan	eps	3.0	1	493	NaN	NaN	NaN
WDB	dense	spectral	k	2.0	2	0	0.007989	4.673864	25.2699
WDB	dense	spectral	k	3.0	3	0	-0.048228	4.746015	19.0417
WDB	dense	spectral	k	4.0	4	0	-0.033931	3.576440	18.8101
WDB	dense	spectral	k	5.0	5	0	-0.024244	3.359229	18.7864
WDB	dense	dbscan	eps	0.5	0	1099	NaN	NaN	NaN
WDB	dense	dbscan	eps	1.0	0	1099	NaN	NaN	NaN
WDB	dense	dbscan	eps	1.5	0	1099	NaN	NaN	NaN
WDB	dense	dbscan	eps	2.0	1	938	NaN	NaN	NaN
WDB	dense	dbscan	eps	2.5	1	683	NaN	NaN	NaN
WDB	dense	dbscan	eps	3.0	1	497	NaN	NaN	NaN
WDB	dense_1	spectral	k	2.0	2	0	-0.027791	4.085976	27.3497
WDB	dense_1	spectral	k	3.0	3	0	-0.056364	4.082494	24.0956
WDB	dense_1	spectral	k	4.0	4	0	-0.071346	3.641978	18.3178
WDB	dense_1	spectral	k	5.0	5	0	-0.059925	3.915260	19.2339
WDB	dense_1	dbscan	eps	0.5	0	1099	NaN	NaN	NaN
WDB	dense_1	dbscan	eps	1.0	0	1099	NaN	NaN	NaN
WDB	dense_1	dbscan	eps	1.5	0	1099	NaN	NaN	NaN

```
  WDB dense_1   dbscan   eps          2.0           2      823    0.181462        1.
115932           16.5137
  WDB dense_1   dbscan   eps          2.5           1      575         NaN
NaN                NaN
  WDB dense_1   dbscan   eps          3.0           1      408         NaN
NaN                NaN
  WDB dense_2 spectral     k          2.0           2        0    0.452322        1.
259234           169.7061
  WDB dense_2 spectral     k          3.0           3        0    0.152967        1.
510379           101.3525
  WDB dense_2 spectral     k          4.0           4        0    0.119263        1.
350695           110.4732
  WDB dense_2 spectral     k          5.0           5        0    0.136683        1.
554682            88.8063
  WDB dense_2   dbscan   eps          0.5           1      628         NaN
NaN                NaN
  WDB dense_2   dbscan   eps          1.0           2      228    0.648392        0.
324572           146.7450
  WDB dense_2   dbscan   eps          1.5           1      101         NaN
NaN                NaN
  WDB dense_2   dbscan   eps          2.0           1       65         NaN
NaN                NaN
  WDB dense_2   dbscan   eps          2.5           1       37         NaN
NaN                NaN
  WDB dense_2   dbscan   eps          3.0           1       33         NaN
NaN                NaN
class   layer   method  param_value  n_clusters  n_noise  silhouette  davies_bouldin
calinski_harabasz  score_compuesto
  WDB dense_2   dbscan          1.0           2      228    0.648392        0.324572
146.7450         1.333333
  WDB dense_2 spectral          2.0           2        0    0.452322        1.259234
169.7061         2.000000
  WDB dense_2 spectral          4.0           4        0    0.119263        1.350695
110.4732         4.333333
  WDB dense_2 spectral          3.0           3        0    0.152967        1.510379
101.3525         4.333333
  WDB dense_2 spectral          5.0           5        0    0.136683        1.554682
88.8063          5.333333
```







interactive -> Spectral Clustering (tSNE + PCA)

In [93]:
```
spectral_explorer(repr_wdb_32, class_name="WDB", suffix="wdb")
```

```
interactive(children=(Dropdown(description='Capa:', options=('flatten', 'dense', 'de
nse_1', 'dense_2'), value=…
```

interactive —> DBSCAN (tSNE + PCA)

In [92]:
```
dbscan_explorer(repr_wdb_32, class_name="WDB", suffix="wdb")
```

```
interactive(children=(Dropdown(description='Capa:', options=('flatten', 'dense', 'de
nse_1', 'dense_2'), value=…
```

Explorador de espectros por cluster (Spectral + DBSCAN)

Visualiza los espectros promedio de cada cluster a partir de experimentos ya ejecutados anteriormente

Controles:

- **Método** / **Capa** / **k** (Spectral) o **eps** + **min_samples** (DBSCAN)
- **Media clase** — muestra/oculta el espectro promedio de la clase en el gráfico combinado
- **Balmer** — superpone las líneas de Balmer
- **±1σ individual** — banda de desviación estándar en los gráficos individuales

In [91]:
```
spectra_viewer(X_wdb_spec, W_wdb[0], class_name="WDB", suffix="wdb")
```

```
VBox(children=(HBox(children=(Dropdown(description='Metodo:', layout=Layout(width='1
90px'), options=('spectral…
```